6 Routh Criteria for a Polynomial

Objective

After completing this chapter, the user will be able to understand

1. The concept of BIBO stability
2. Routh stability criteria
3. How to calculate the number of poles of the closed-loop system in left half s-plane, right half s-plane and on the imaginary axis.
4. How to calculate the range of open-loop system gains for the stability of the closed-loop system.
5. How to determine the frequency of oscillations when the closed-loop system is marginally stable.
6. The analysis of the relative stability of a control system.

Introduction

1. Among the many forms of performance specifications used in the design of a control system, the most important requirement is that the system must be stable at all times.
2. Stability in a system implies that small changes in inputorinitial conditions or system parameters do not result in large changes in system output.
3. A stable system, if disturbed from its equilibrium position, returns to its state eventually after disturbance is removed.
4. For example, when a pendulum is disturbed from its original position by a small amount, it exhibits damped oscillations about the mean position and finally settles down to the original mean position.

Stability of LTI System

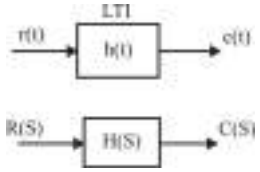


Fig. 6.1

$$h(t) \xleftrightarrow{L.T.} H(S)$$

For an LTI system to be stable following techniques can be used:

1. For every bounded input, the output must be bounded.

   Bounded input: Inputs having finite amplitude for

   $-\infty < t < \infty$

   $|x(t)| \le M < \infty \qquad -\infty < t < \infty$

   E.g.: Sin(t), u(t), cos(t)

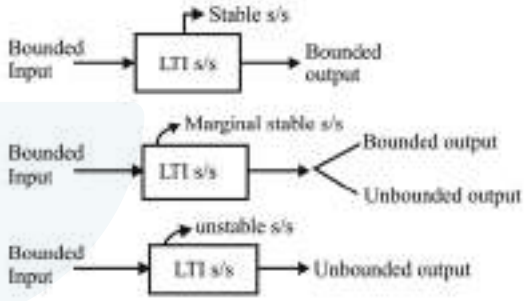


Fig. 6.2

2. For an LTI system to be stable, unit impulse response [h(t)] must be absolutely integrable.

   $$\int_{-\infty}^{\infty} |h(t)|\,dt \rightarrow Finite$$

   # Comment on stability of following LTI system having h(t) as

   $h(t) = u(t) \rightarrow unstable\ LTI\ s/s$

   $h(t) = \delta(t) \rightarrow stable\ LTI\ s/s$

   $h(t) = e^{-2t}u(t) \rightarrow stable\ LTI\ s/s$

3. For an LTI system to be stable, ROC (Region of convergence) of H(s) must include jω-axis.

   ROC: $e^{-at}u(t) \leftrightarrow \frac{1}{s+a}$

   Pole: s = –a

   Re {s} = –a

   Re {s} > –a

$$-e^{-\alpha t}u(-t) \leftrightarrow \frac{1}{s+\alpha}$$

pole: s = −a

$\text{Re}\{s\} = -\alpha \rightarrow \text{Re}\{s\} < -\alpha$

Example 1: Comment on the stability of following LTI system having h(t) as shown:

Solution:

1. $h(t) = \sin 2tu(t) \leftrightarrow H(s)$

   $= \frac{2}{s^2+4}$ $s = \sigma + j\omega$

   Poles: $s = \pm 2j$

   Re{s} = 0

   $\text{Re}\{s\} > 0 \rightarrow \sigma > 0$

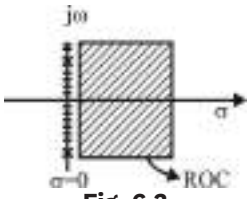


Fig. 6.3

Unstable system.

2. $h(t) = e^{-2t}\sin 3tu(t) \leftrightarrow H(S) = \frac{3}{(S+2)^2+9}$

   Pole: S = −2 ± 3j

   Re{S} = −2

   Roc: Re {S} > −2

   $\sigma > -2$

   $j\omega$-axis included

   Stable system

Points To Remember

1. For a causal LTI system to be stable, all the poles of H(S) must be strictly in LHP.
2. In control system, we have to deal with causal + LTI system.

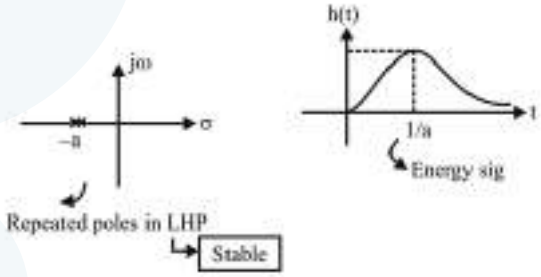


Fig. 6.5

Location of Poles vs Stability of Causal LTI System

1. $H(S) = \frac{1}{S+a} \rightarrow h(t)$

   $= e^{-\alpha t}u(t)$

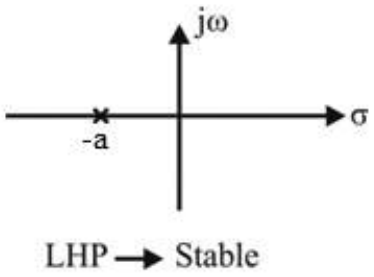


Fig. 6.4

   a) Energy signal with exponential
   b) *Absolutely* int*egrable*
   c) LTI system → Sable

2. $H(S) = \frac{1}{(S+a)^2} \leftrightarrow h(t)$

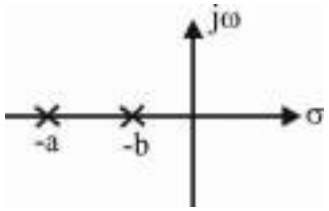
   $= te^{-\alpha t}u(t)$

3. $H(S) = \frac{1}{(S+a)(S+b)} \leftrightarrow h(t)$

   $= c_1e^{-\alpha t}u(t) + c_2e^{-bt}u(t)$

   a> 0 Energy signal

   b > 0



   Non-repeated poles in LHP → Stable

4. $H(S) = \frac{b}{(S+a)^2+b^2} \leftrightarrow h(t)$

   $= e^{-\alpha t}\sin bt\,u(t)$

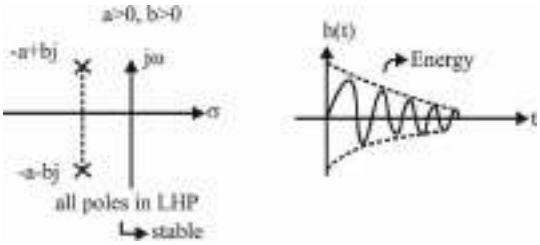


Fig. 6.6

All poles in LHP → Stable.

5. $H(S) = \dfrac{b}{S^2 + b^2} \leftrightarrow h(t)$

$= \sin bt\, u(t) \rightarrow$ *power signal*

Marginal stable

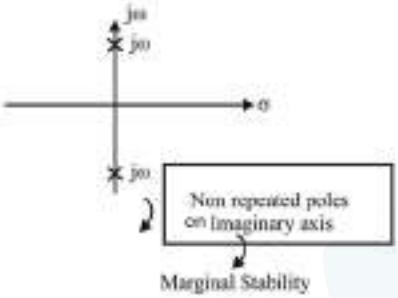


Fig. 6.7

6. $H = \dfrac{1}{S} \leftrightarrow h(t) = u(t) \rightarrow$ *power signal*

Marginal stable

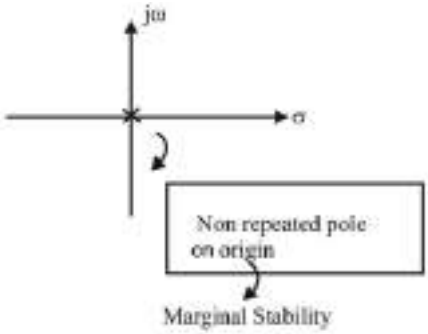


Fig. 6.8

7. $\dfrac{1}{(S^2 + a^2)(S^2 + b^2)} \leftrightarrow h(t)$

$= C_1 \sin at\, u(t) + C_2 \sin bt\ u(t)$

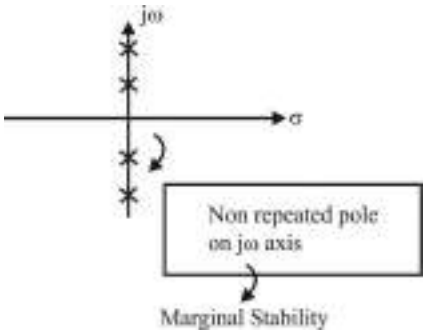


Fig. 6.9

8. $H(S) = \dfrac{1}{S^2} \leftrightarrow h(t) = tu(t)$

NENP but not Impulse

unstable

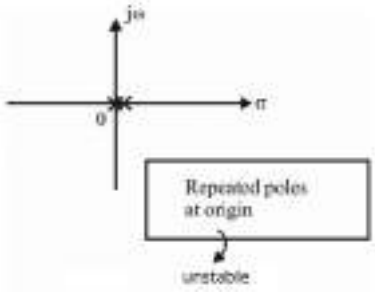


Fig. 6.10

9. $H(S) = \dfrac{b}{(S^2 + b^2)^2} \leftrightarrow h(t) = kt \sin bt\, u(t)$

N.E.N.P.

un stable

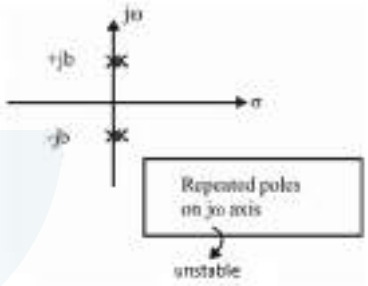


Fig. 6.11

10. $H(S) = 1 \leftrightarrow h(t) = \delta(t)$

NENP

Stable

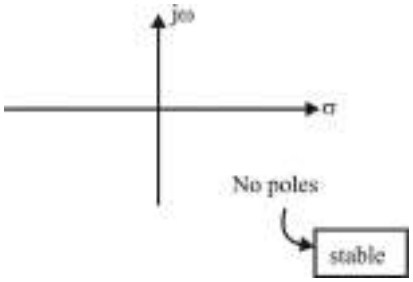


Fig. 6.12

11. $H(S) = \dfrac{1}{S - a}\, a > 0 \leftrightarrow h(t) = e^{at}u(t)$

NENP

un stable

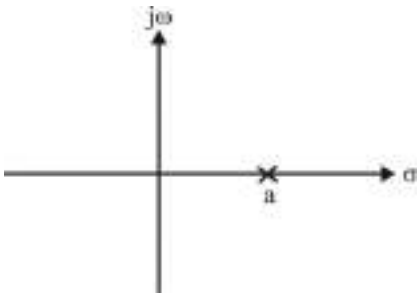


Fig. 6.13

Even if single pole lies in RHP → Unstable

Points To Remember

Location of Poles in H(s)	Stability of Causal LTI s/s
1. Repeated or non-repeated poles in LHP	Stable
2. Non repeated poles on $j\omega$-axis	Marginally stable
3. Repeated poles on $j\omega$-axis	Unstable
4. Single pole at origin	Marginally stable
5. Repeated poles at origin	Unstable
6. Pole in RHP	Unstable
7. No pole	Stable

Table 6.1

Image Location vs Imaginary Axis

Imaginary (or) $j\omega$-axis:

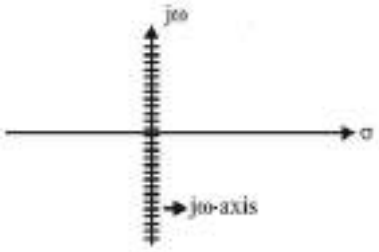


Fig. 6.1

Poles on imaginary axis:

[Complex Roots occurs in conjugate pair only when coefficients of the polynomial are real]

'Default'

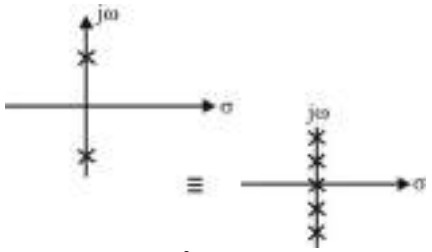


Fig. 6.14

Image location: Location symmetric with origin.

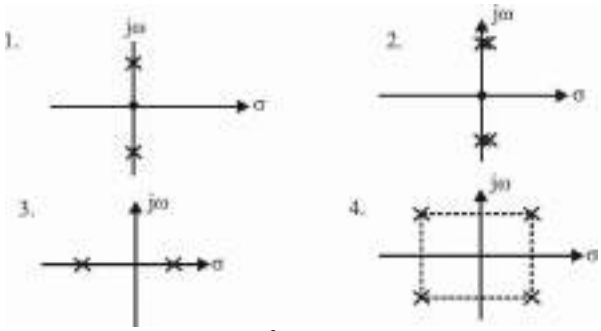


Fig. 6.15

Poles on image location:

1. $H(S) = \dfrac{1}{S^2 - 4}$

2. $H(S) = \dfrac{1}{(S^2 + 4)^2}$

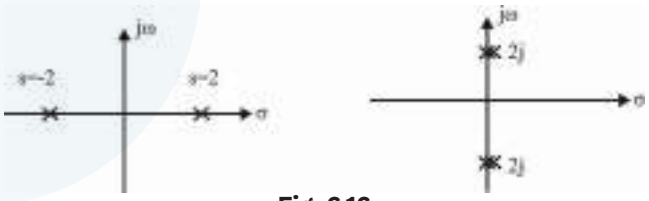


Fig. 6.16

3. $H(S) = \dfrac{1}{s^4 + 1}$

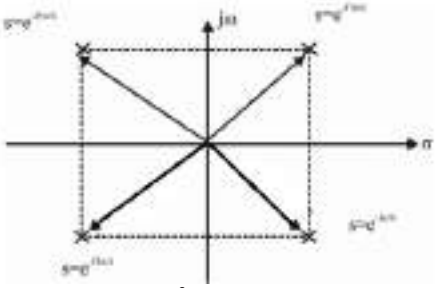


Fig. 6.17

$S^4 = -1 = e^{-jk\pi}$

$S^4 = e^{-jk\pi} \rightarrow k\ odd\ \ S^4 = e^{+j(2k-1)\pi}$

$S = e^{\frac{-jk\pi}{4}} \qquad k = 0, 1, 2, 3$

$S = e^{\frac{-j\pi}{4}}$

$S = e^{\frac{-j3\pi}{4}}$

$S = e^{\frac{-j5\pi}{4}}, \qquad S = e^{\frac{-j7\pi}{4}}$

Calculation of the Roots of a Polynomial

$$T(S)=\frac{N(S)}{D(S)}$$

$D(S) \rightarrow$ Denominator polynomial:

Roots of D(S) = Poles of T(S)

First order polynomial:

If all the coefficients of 1st order polynomial have the same sign, then all the roots are in LHP

$T(S)=S+2 \quad T(S)=-S-2$

$S+2=0 \qquad -S-2=0$

$S=-2 \qquad S=-2$

Second order polynomial:

If all the coefficients of a 2nd order polynomial have the same sign, then all the roots will be in LHP

$T(S)=S^2+3S+2 \qquad T(S)=-S^2-3S-2$

$T(S)=0 \qquad T(S)=0$

$S=-2 \qquad S=-1$

$S=-1 \qquad S=-2$

Third order polynomial:

If all the coefficients of a 3rd order polynomial are of the same sign, then:

Real roots $\rightarrow$ Always in LHP 3 Real

Complex roots: Cannot be predicted

1 Real + 2 Complex

$$D(S)=S^3+4S^2+6S+4=0 \rightarrow -2,-1\pm j$$

$$D(S)=S^3+S^2+5S+125=0 \rightarrow -5,2\pm 4.5i$$

$$D(S)=S^3-S^2+5S+125$$

$$=0 \rightarrow -4.37,\ 2.65\pm j4.6$$

At least 1 Root will be RHP

Methodsto Calculate the Location of Roots for HigherOrderPolynomials

Routh–Hurwitz method:

The criterion is based on ordering the coefficients of the characteristic equation into an array called Routh array.

$$D(S)=a_0S^6+a_1S^5+a_2S^4+a_3S^3+a_4S^2+a_5S+a_6$$

S^6	a_0	a_2	a_4	a_6
S^5	a_1	a_3	a_5	0
S^4	$b_1=\frac{a_1a_2-a_0a_3}{a_1}$	$b_2=\frac{a_1a_4-a_0a_5}{a_1}$	a_6	0
S^3	$c_1=\frac{b_1a_3-b_2a_1}{b_1}$	$c_2=\frac{b_1a_5-a_1a_6}{b_1}$	0	
S^2	$d_1=\frac{c_1b_2-b_1c_2}{c_1}$	a_6	0	
s^1	$\frac{d_1c_2-c_1a_6}{d_1}$	0		
s^0	a_6			

E.g.: $3S^4+S^3+S^2+2S+1=0$

S^4	3	1	1	
S^3	1	2	0	Odd row
S^2	$\frac{-5}{1}$	1		
S^1	$\frac{-11}{-5}=\frac{11}{5}\rightarrow 11$	$0\rightarrow 0$		Odd row
S^0	1			

Table 6.2

Important observations:

1. If any row of the RH-table is multiplied or divided by a positive constant, then the result does not change.
2. If all the elements of the first column have the same sign, then no roots will be in RHP
3. If all the coefficients of the 1st column have the same sign and there is no ROZ (Row of zeros) $\rightarrow$ odd row zero $\rightarrow$ Rows corresponding to odd powers of s.

   Then all the roots will be in LHP
4. The no. of sign changes in the first column indicates the no. of roots in RHP

Key points:

If any power of s is missing in the polynomial, then the following cases can exist.

1. One or more than one root may exist on RHP

   $D(S) = S^2 - 1 = S^2 + 0S^1 - 1S^0 = 0$

   $S = \pm 1$

1. Root lies in RHP
2. Non-repeated roots may exist on the $j\omega$ -axis.

   $D(S) = S^2 + 1 \Rightarrow S^2 + 1 = 0$

   $S = \pm j$

3. Repeated roots may exist on the $j\omega$ -axis.

   $D(S) = S^4 + 2S^2 + 1$

   $= (S^2 + 1)^2 = 0 \quad S = \pm j, \pm j$

4. There may be complex roots in RHP
5. If a polynomial has only even powers of "S", then roots will be at the image location.

   E.g.: 1. $D(S) = S^2 + 1$

   Roots $S = \pm j$

   Image location

   Imaginary axis.

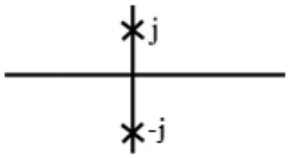


Fig. 6.18

   E.g.: 2. $D(S) = S^2 - 1$

   Roots: $S = \pm 1$

   Image location

   Image location is not at the imaginary axis.

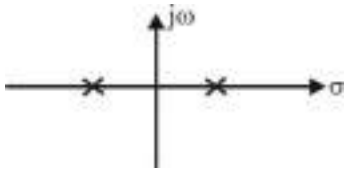


Fig. 6.19

   E.g.: 3. $D(S) = S^4 + 2S^2 + 1$

   $D(S) = (S^2 + 1)^2$

   Roots $S = \pm j$

   $S = \pm j$

   Poles are at the image location.

   Image location is on the imaginary axis.

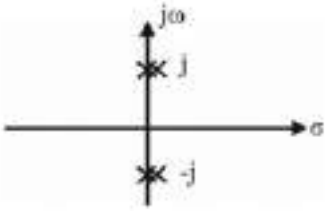


Fig. 6.20

   E.g.: 4. $D(S) = S^4 + 1$

   Roots $= S = e^{j(2k-1)\frac{\pi}{4}}$

   k = 0, 1, 2, 3

   Image location

   Not imaginary axis

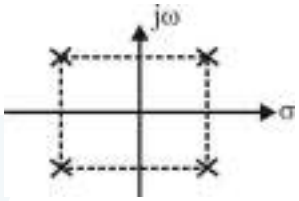


Fig. 6.21

6. If a polynomial has only odd powers of 'S', then a minimum of one root will be at origin, and the remaining roots will be at the image location.

   E.g.: $D(S) = S^3 + S = \underset{\text{Root at origin}}{S} \underset{\text{Root at Image location}}{\{S^2 + 1\}}$

7. If any odd row in RH-table becomes zero or odd ROZ is found, then
   a) Few roots of the polynomial will be at the image location.
   b) Image location becomes imaginary axis if all the signs of elements in 1st column are the same.
8. If no odd row is zero in the RH-table, then no roots will be at the image location, thereby no roots on the imaginary axis.

Example 2: $D(S) = a_1 s + a_0$

Solution: RH-table:

$S^1 \quad a_1$

$S^0 \quad a_0$

Case 1: $a_1 > 0; \quad a_0 > 0 \rightarrow$ *No Roots in RHP* $\rightarrow$ *odd ROZ* $\rightarrow$ *DNE* (*Does not exist*)

All roots in LHP

Special Cases in the RH-Table

Case 1: If 1st element of any row becomes zero while all other elements of that row are not all zero.

Example 7: $D(S) = S^5 + 2S^4 + S^3 + 2S^2 + 3S + 10$

Solution:

+	S^5	1	1	3
+	S^4	2	2	10
+	S^3	0	–2	0
+	S^2	$\frac{2\varepsilon+4}{\varepsilon}$	10	0
–	S^1	$\frac{-2\left\{\frac{2\varepsilon+4}{\varepsilon}\right\}-10\varepsilon}{\frac{(2\varepsilon+4)}{\varepsilon}}$	0	0
+	S°	10		

$$\lim_{\varepsilon\to 0^+} \frac{2\varepsilon+4}{\varepsilon} = +ve$$

$$\lim_{\varepsilon\to 0^+} -2-\frac{10\varepsilon^2}{(2\varepsilon+4)} = -2 = -ve$$

1. 2 poles lies in RHP
2. 3 poles lies in LHP.

Example 8: $D(S) = 3S^4 + S^3 + 3S^2 + S + 2$

Solution:

+	S^4	3	3	2
+	S^3	1	1	0
+	S^2	0	2	0
–	S^1	$\frac{\varepsilon-2}{\varepsilon}$	0	0
+	S°	2		

Sign

$$\lim_{\varepsilon\to 0^+} \frac{\varepsilon-2}{\varepsilon} = 1-\frac{2}{\varepsilon} = \text{- ve}$$

2 roots in RHP

2 roots in LHP

Case 2: When all the elements of the odd row becomes zero

or

The row corresponding to odd powers of S is zero.

or

Odd ROZ (Row of zeros)

Let polynomial is D(S)

1. It suggests that roots will be at the *image location*.

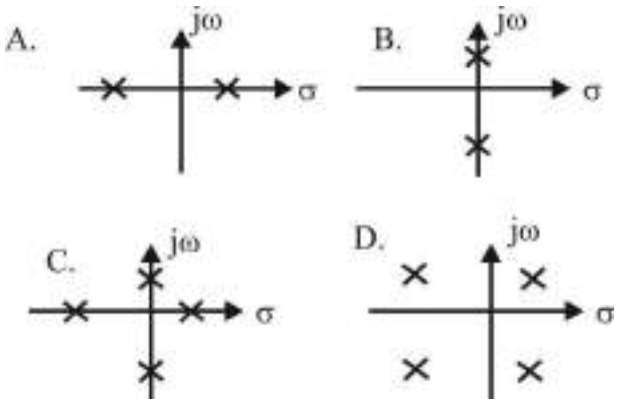


Fig. 6.22

2. To proceed the RH-table forward, the auxiliary characteristic equation is formed, which is an even row just above the odd row: A(S) = 0
3. Then A(S) is differentiated:

$$B(S) = \frac{dA(S)}{ds} = 0$$

Replace the elements of odd row with coefficient of B(S)

4. Roots of A(S) = 0 will also be roots of original polynomial D(S) = 0

$$\frac{D(S)}{A(S)} \to \text{Remainder} = 0$$

$$\frac{D(S)}{A(S)} = P(S) \to P(S)$$

$= 0 \to$ *Calculate* Remaining *Roots*.

5. In this case, along with the location of roots, exact values of roots can also be calculated.

If odd row becomes zero once:

1. Roots of AE will give the image locations.
2. Roots of A(S) will be non-repeated in nature.
3. If A(S) is of 2nd order and roots of A(S) are on the $j\omega$-axis then the roots

will represent the un-damped natural frequency of oscillations.

a) D(S)

b) A(S) = 0 $\rightarrow$ Roots $\Rightarrow$ Image locations

c) $\frac{D(S)}{A(S)} = P(S) \Rightarrow$ Remaining roots

If odd row becomes zero twice:

1. In this case, two AE's will be obtained.



Fig. 6.23

2. Roots of $A_2(S)$ will automatically covered by $A_1(S)$.
3. Roots of $A_2(S)$ will represent roots of D(S).
4. Roots of $A_1(S)$ will be at the image location.
   a) $D(S)$
   b) $A_1(S)$ & $A_2(S)$
   c) $\frac{D(S)}{A_1(S)} = P(S) \rightarrow$ Remaining roots

Note: Since AE contains only even powers of 'S', so roots of AE has to be at the image location.

Important formula:

No. of roots of D(S) on the imaginary axis = (Highest order of AE) – 2 × (No. of sign changes below the highest order AE) (i)

No. of roots of D(S) in RHP = No. of sign changes in 1st columnof the RH-table (ii)

No. of roots of D(S) in LHP = Order of D(S)-(i)-(ii)

Example 9: $D(S) = S^4 + S^3 - 3S^2 - S + 2 = 0$

Solution:

+	S^4	1	–3	2
+	S^3	1	–1	0
–	S^2	–2	2	0
	S^1	0	0	0
0	S^1	–4	0	0
+	S^0	2		

$\rightarrow$ *Few roots will be at image location*

$A(S) = -2S^2 + 2 \rightarrow S = \pm 1$

$\frac{dA(S)}{dS} = -4S^1$

Concept 1: Location of roots:

2 roots on RHP

No of roots on the imaginary axis = 2 – 2 × (1) = 0

No of roots on LHP = $4 - 2 = 2$

Concept 2: $A(S) = -2S^2 + 2 = 0 \rightarrow S = \pm 1 \rightarrow$ Image location

$P(S) = \frac{D(S)}{A(S)} = -S^2 - S + 2$

$P(S) = 0$

$S^2 + S - 2 = 0$

$S = +1, -2$

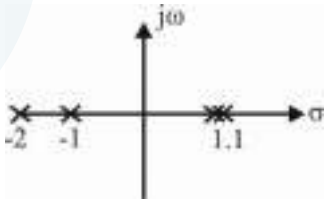


Fig. 6.24

1. $A(S) \rightarrow 2nd\ order$
2. $A(S) = 0$

Roots not on $j\omega$-axis, so ω_n cannot be calculated

Note:

1. Since the odd row is zero only once, the image location roots will be non-repeated.
2. A(S) $\rightarrow$ Order = 4 $\rightarrow$ ω_n cannot be calculated

Example 11: $D(S) = S^3 + 8S^2 + 4S + 32 = 0$

Solution:

+	S^3	1	4
+	S^2	8	32
+	S^1	0	0
+	S^1	2	0
+	S^0	4	

$\rightarrow$ *Roots will be at image location*

$A(S) = S^2 + 4 \rightarrow A(S) = 0$

$S = \pm 2j$

$\frac{dA(S)}{dS} = 2S$

Concept 1:

Zero roots on RHP

Imaginary axis = $2 - 2 \times (0) = 2$

1 pole on LHP

Concept 2:

$A(S) = S^2 + 4 = S = \pm 2j \rightarrow$ Image location

$\frac{D(S)}{A(S)} = P(S) = (S + 8) = 0$

$S = -8$

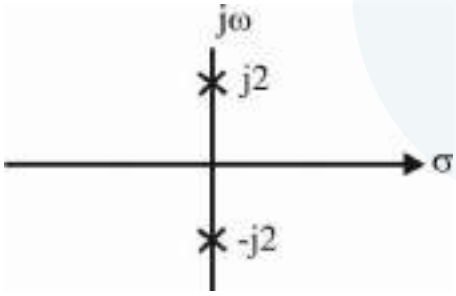


Fig. 6.25

$A(S) \rightarrow 2nd\ order$

$Roots \rightarrow$ Imaginary

$S = \pm j\omega_n \qquad \omega_n = 2$

System is marginally stable:

Important Points

Solving Quadratic Inequality:

1. $f(x) = ax^2 + bx + c \qquad D=b^2 - 4ac$
   a) a> 0
   b) D < 0 → No real roots

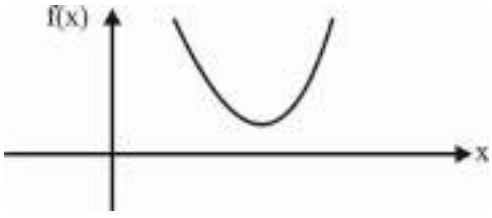


Fig. 6.26

$\Rightarrow ax^2 + bx + c > 0$

2. $f(x) = ax^2 + bx + c$
   a> 0
   D = 0

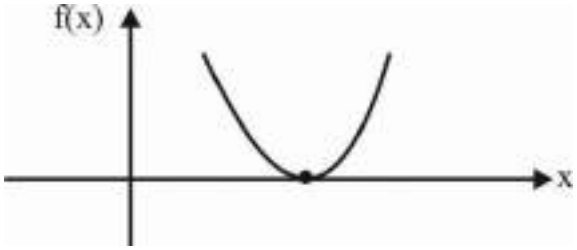


Fig. 6.27

$f(x) \geq 0$

3. $f(x) = ax^2 + bx + c$
   a> 0
   D > 0

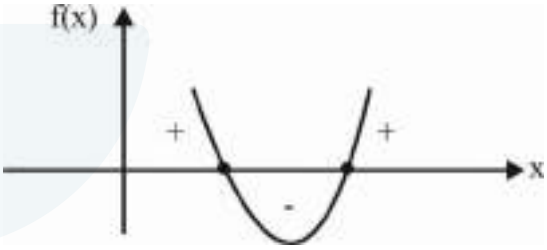


Fig. 6.28

$f(x) \gtrless 0$

1. $ax^2 + bx + c$
   a< 0
   D < 0

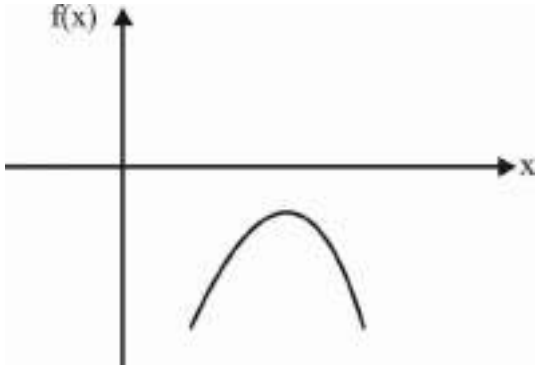


Fig. 6.29

f(x) < 0

5. $ax^2 + bx + c$
   a< 0
   D = 0

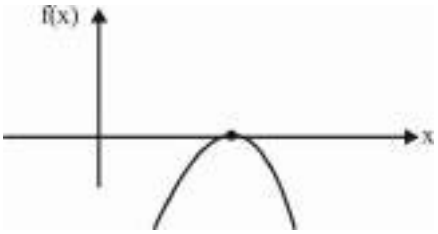


Fig. 6.30

$f(x) \leq 0$

2. $f(x) = ax^2 + bx + c$

   a< 0

D > 0

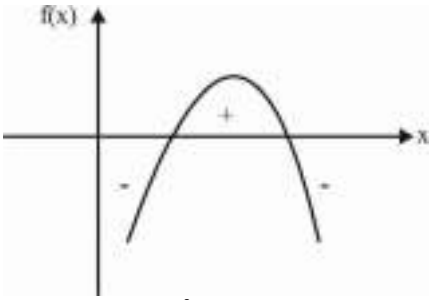


Fig. 6.31

$f(x) \gtrless 0$

Types of Problems

Type 1: Range of certain parameters for the system to be stable.

S.1 Form RH-table using denominator polynomial of that system.

S.2 Make sure that all the elements of the 1st column have similar signs. $\Rightarrow$ Wavy curve

Type 2: Value of certain parameters such that the system is marginal stable.

S. 1 Form RH-table from denominator polynomial of the corresponding system.

S. 2 Sign of all elements of 1st column should be same.

S. 3 Make sure that the odd row is zero.

a) only once a) $\cap$ c)

b) Roots of AE are non-repeated jω-axis.

Type 3: The system oscillates with a constant frequency.

or

The system has a sustained frequency of oscillations

or

The system has an un-damped natural frequency.

Process:

1. Form the RH-table form denominator polynomial of the corresponding system.
2. Make sure that the sign of all elements of the 1st column is the same.
3. Odd Row zero once and A(s) is having 2nd order equation with roots on jω-axis.

Note: Sometimes,maximisation or minimisation of certain parameters is asked.

Use AOD 'Maxima and Minima'

Example 15:

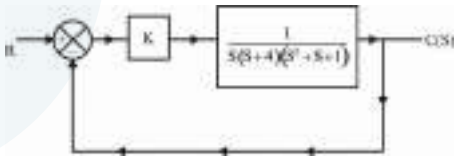


Fig. 6.32

1. Determine the stability of OLS
2. Determine the range of K for CLS to be stable.

Solution: OLS

$$G(S)H(S) = \frac{K}{S(S+4)(S^2+S+1)}$$

Poles: S = 0

S = -4

$$S = \frac{-1}{2} \pm j\frac{\sqrt{3}}{2}$$

OLS $\rightarrow$ Marginal stable

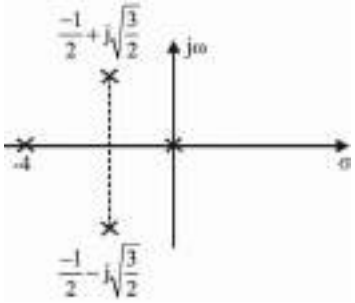


Fig. 6.33

CLS: $\frac{C(S)}{R(S)} = \frac{K}{S(S+4)(S^2+S+1)+K}$

$D(S) = S\left[S^3 + S^2 + S + 4S^2 + 4S + 4\right] + K$

$D(S) = S^4 + 5S^3 + 5S^2 + 4S + K$

+ve	S^4	1	5	K
+ve	S^3	5	4	0
+ve	S^2	$\frac{21}{5}$	K	0
	S^1	$\frac{\frac{84}{5} - 5K}{\frac{21}{5}}$	0	0
	S^0	K		

a) $\left(\frac{\frac{84}{5} - 5K}{\frac{21}{5}}\right) > 0$

b) K > 0 a ∩ b

c) $\frac{84}{5} - 5K > 0$

$5K < \frac{84}{5}$

$K < \frac{84}{25}$

1. CLS will be stable if $0 < K < \frac{84}{25}$
2. Maximum value of K up to which the system remains stable is

$K_{max} = \frac{84}{25}$

b) K > 0

$o < K < \frac{84}{25}$

Let $K = \frac{84}{25}$

Odd row zero once:

$A(S) = \frac{21}{5}S^2 + K = 0$

$S = \pm j\frac{2}{\sqrt{5}}$ Imaginary axis

CLS will become marginal stable.

Limitation of the Routh table method:

1. D(S) must be polynomial in nature.
2. $D(S) = S^3 + \sin s\ S^2 + e^5 S^1 + 3 = 0$

$D(S) = S^3 + \cos s\ S^2 + \frac{S}{(S-1)} + 4 = 0$

RH-table method is invalid.

3. RH-table is valid only when order of thepolynomial is finite:

$D = 1 + e^{-S} = 1 + 1 + S + \frac{S^2}{2!} + \frac{S^3}{3!} + \ldots$

Transportation LAG System

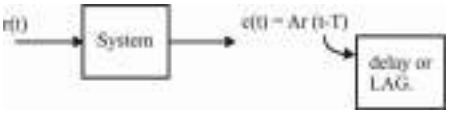


Fig. 6.34

$H(S) = \frac{C(S)}{R(S)} = Ae^{-ST}$

$h(t) = A\delta(t-T)$ **Transportation LAG system**

Example 21:

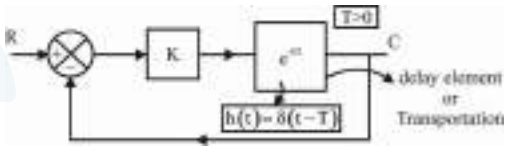


Fig. 6.35

Stability of above closed-loop system depends on

1. K,T
2. T
3. K
4. Neither K nor T

Solution:

D(S) = 1 + G(S) H (S) = 0

$D(S) = 1 + Ke^{-ST} = 0$

1. RH criteria fails
2. $e^{-ST} \approx 1 - ST \rightarrow$ *Not allowed*
3. $1 + Ke^{-ST} = 0 \rightarrow$ *Infinite roots*

$e^{-ST} = \frac{-1}{K} \rightarrow e^{-ST} = \frac{+1}{K}e^{j(2n+1)\pi}$

If x is very small, $e^{-x} = 1 - x$

Method is wrong

$-1 = e^{j(2n+1)\pi}$

$n \in I$

$e^{-ST} = \frac{-1}{K} \rightarrow e^{-ST} = \frac{+1}{K} e^{j(2n+1)\pi}$

Apply "ln" on both sides

$-ST = \ell n\left(\frac{1}{K}\right) + \ell n\left[e^{j(2n+1)\pi}\right]$

$S = -\frac{1}{T}\ell n\left\{\frac{1}{K}\right\} - \frac{1}{T}\left(j(2n+1)\pi\right)$

$S = \frac{1}{T}\ell n(K) - j\left(\frac{1}{T}(2n+1)\pi\right) \quad n \in I$

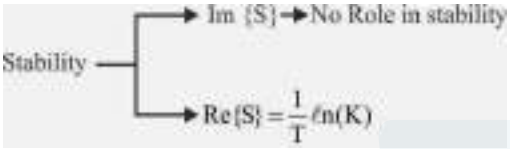


Fig. 6.36

$\text{Re}\{S\} = \frac{1}{T}\ell n(K)$

$\frac{1}{T} > 0$

$0 < K < 1 \rightarrow \text{Re}\{S\} = -ve \rightarrow stable$

$K = 1 \rightarrow \text{Re}\{S\} = 0 \rightarrow M.S.$

$K > 1 \rightarrow \text{Re}\{S\} = +ve \rightarrow unstable$

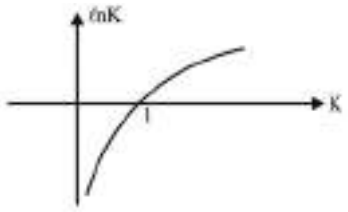


Fig. 6.37

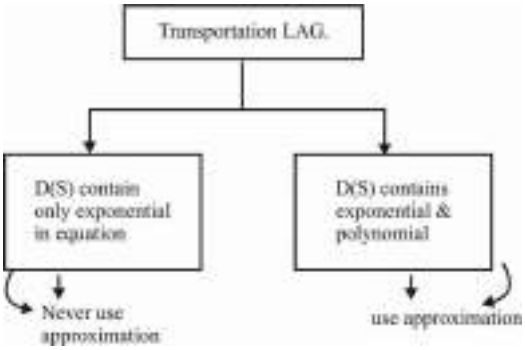


Fig. 6.38

Relative stability and shift of origin:

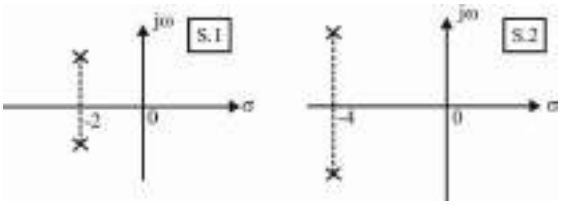


Fig. 6.39

S-2 is relatively more stable than S-1

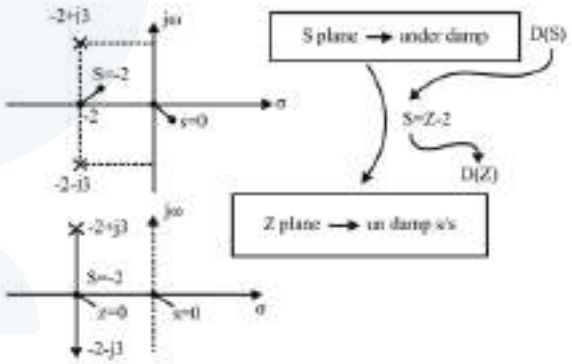


Fig. 6.40

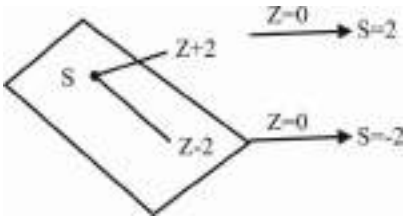


Fig. 6.41

Exercise Problems

1. A unity feedback system that uses a proportional + integral (PI) controller is shown in the figure.

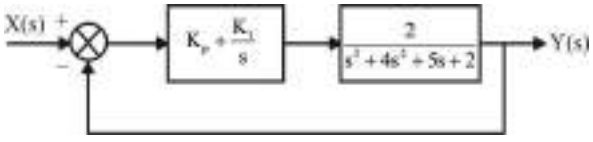


Fig. 6.42

The stability of the overall system is controlled by tuning the PI controller parameters K_p *and* K_I. The maximum value of K_I that can be chosen so as to keep the overall system stable or in the worst case, marginally stable (round off to three decimal places) is _____.

Solution: 3.125

2. The loop transfer function of a unity negative feedback system is

$$G(s)H(s)=\frac{K(s+11)}{s(s+2)(s+8)}.$$

The value of K, for which system is marginally stable, is _____.

Solution: 160

3. Consider a unity feedback system as shown in the figure with an integral compensator K/s and open-loop transfer function

$$G(s)=\frac{1}{s^2+3s+2}$$

where, K > 0. The positive value of K for which there are exactly two poles of the unity feedback system on the $j\omega$- axis is equal to _____ (rounded off to 2 decimal places).

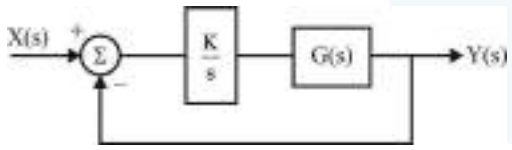


Fig. 6.43

Solution: 6

4. Consider $p(s)=s^2+a_2s^2+a_1s+a_0$ with all real coefficients. It is known that its derivative $p'(s)$ has no real roots. The number of real roots of p(s) is

(A) 0 (B) 1

(C) 2 (D) 3

Solution: (B)

5. The first two rows in the Routh table for the characteristic equation of a certain closed-loop control system are given as

s^3	1	(2K + 3)
s^2	2K	4

The range of K for which the system is stable is

(A) –2.0 < K < 0.5

(B) 0 < K < 0.5

(C) 0 < K <∞

(D) 0.5 < K <∞

Solution: (D)

6. The transfer function of a linear time invariant system is given by

$$H(s)=2s^4-5s^3+5s-2$$

The number of zeros in the right half of the s-plane is _____. **[GATE Set-01]**

Solution: 3

7. A plant transfer function is given as $G(s)=\left(K_p+\frac{K_I}{s}\right)\frac{1}{s(s+2)}$. When the plant operates in a unity feedback configuration, the condition for the stability of the closed-loop system is **[GATE Set-01]**

(A) $K_P>\frac{K_I}{2}>0$ (B) $2K_I>K_P>0$

(C) $2K_I<K_P$ (D) $2K_I>K_P$

Solution: (A)

Exercise Problems

1. (3.125)

Given the unity feedback system is shown below:

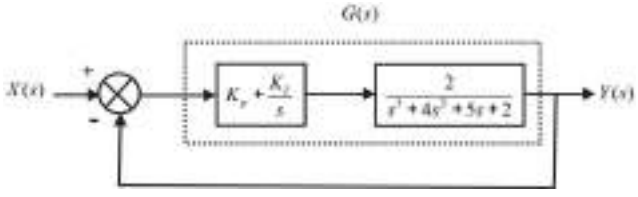


Fig. 6.44

The characteristic equation of unity feedback system is

$$1+G(s)=0$$

$$1+\left(K_p+\frac{K_I}{s}\right)\left(\frac{2}{s^3+4s^2+5s+2}\right)=0$$

$$1+\frac{K_ps+K_I}{s}\frac{2}{s^3+4s^2+5s+2}=0$$

$$s(s^3 + 4s^2 + 5s + 2) + 2sK_p + 2K_I = 0$$

$$s^4 + 4s^3 + 5s^2 + 2s + 2sK_p + 2K_I = 0$$

$$s^4 + 4s^2 + 5s^2 + (2 + 2K_p)s + 2K_I = 0$$

For keeping all coefficients, the necessary (but not sufficient) condition for the system to be stable is positive.

$K_p > -1$, $K_I > 0$

Routh tabulation,

s^4	1	5 $\quad 2K_I$
s^3	4	$2 + 2K_p$
s^2	$\frac{18 - 2K_p}{4}$	$2K_I$
s^1	$\frac{\left(\frac{18 - 2K_p}{4}\right)(2 + 2K_p) - 8K_I}{\frac{18 - 2K_p}{4}}$	
s^0	$2K_I$	

For the system to be stable, all the roots must be on the left–half of thes–plane, thus all the coefficients in the first column must be positive in same sign or positive.

1. $\frac{18 - 2K_p}{4} > 0$

$K_p < 9$

Thus, K_p varies from $-1 < K_p < 9$

2. $\frac{\left(\frac{18 - 2K_p}{4}\right)(2 + 2K_p) - 8K_I}{\frac{18 - 2K_p}{4}} = 0$

$(18 - 2K_p)(2 + 2K_p) - 4 \times 8K_I = 0$

$4[(9 - K_p)(1 + K_p)] - 4 \times 8K_I = 0$

$(9 - K_p)(1 + K_p) = 8K_I$

$K_I = \frac{(9 - K_p)(1 + K_p)}{8}$

Thus, K_I is function of K_p.

From maxima or minima, $\frac{d}{dK_p} K_I = 0$

$$\frac{d}{dK_P}\left[\frac{(1 + K_p)(9 - K_p)}{8}\right] = 0$$

$$\frac{1}{8}[(1 + K_p)(-1) + 9 - K_p] = 0$$

$9 - K_p = 1 + K_p$

$2K_p = 8$

$K_P = 4$

Now, double derivative of K_I is,

$$\frac{d^2}{dK_P^2} K_I = -\frac{1}{4} < 0$$

It means K_I has maxima at $K_p = 4$.

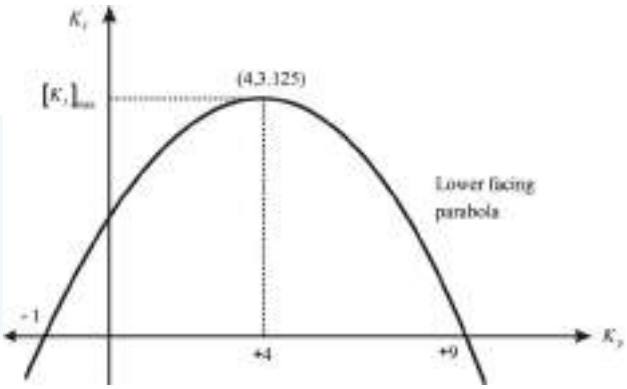


Fig. 6.45

Thus maximum value of K_I at $K_p = 4$ will be,

$$[K_I]_{max} = \frac{(1 + 4)(9 - 4)}{8} = \frac{25}{8} = 3.125$$

Hence, the correct answer is 3.125.

2. (160)

Given,

For a negative feedback system

$$G(s)H(s) = \frac{K(s + 11)}{s(s + 2)(s + 8)}$$

The characteristic equation is given as,

$1 + G(s)H(s) = 0$

$s[s^2 + 10s + 16] + Ks + 11K = 0$

$s^3 + 10s^2 + (16 + K)s + 11K = 0$

Method 1:

Preparing RH-table,

s^3	1	$16+K$	0
s^2	10	$11K$	0
s^1	$\frac{(160+10K)-11K}{10}$	0	0
s^0	$11K$	0	0

At $K = K_{marginal}$, RH-table should have row of zeros at odd power of s other than the last row.

So, $$\frac{(160+10K_{marginal})-11K_{marginal}}{10} = 0$$

$$160 - K_{marginal} = 0$$

$$\therefore \quad K_{marginal} = 160$$

Hence, the correct answer is 160.

Method 2:

From the characteristic equation,

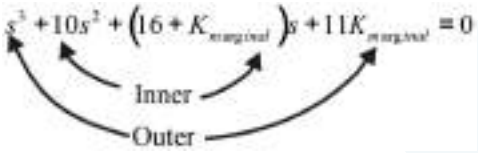


Fig. 6.46

At $K = K_{marginal}$,

Inner product = Outer product

$$1 \times 11K_{marginal} = 10(16 + K_{marginal})$$

$$11K_{marginal} - 10K_{marginal} = 160$$

$$\therefore \quad K_{marginal} = 160$$

Hence, the correct answer is 160.

3. (6)

Given,

Integral constant = $\frac{K}{s}$

$$G(s) = \frac{1}{s^2+3s+2}$$

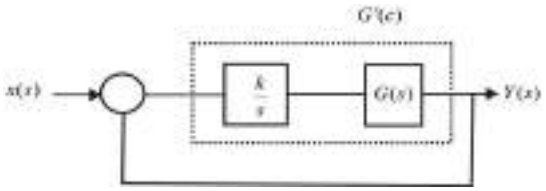


Fig. 6.47

$$G'(s) = \frac{K}{s} \times G(s) = \frac{K}{s} \times \frac{1}{s^2+3s+2}$$

$$G'(s) = \frac{K}{s(s^2+3s+2)}$$

The characteristics polynomial equation,

$$C_f = 1 + G'(s) = 1 + \frac{K}{s(s^2+3s+2)}$$

$$\Rightarrow \quad s^3 + 3s^2 + 2s + K = 0$$

So, it is given that exactly two poles on $j\omega$ axis that means the marginal stable system

Third order cubic and all the signs are the same.

So, EP = IP

K = 6

Hence, the value of K is 6.

4. (B)

Given

1. $p(s) = s^3 + a_2 s^2 + a_1 s + a_0$
2. Derivative of p(s) has no real roots.

$$p(s) = s^3 + a_2 s^2 + a_1 s + a_0$$

$$p'(s) = 3s^2 + 2a_2 s + a_1$$

Routh Tabulation:

s^2	3	a_1
s^1	$2a_2$	0
s^0	a_1	0

Since derivative of p(s) has no real roots, i.e., it has imaginary roots.

To have imaginary roots, ROZ must occur in the Routh's tabulation.

This leads to the following conditions:

$$2a_2 = 0 \Rightarrow a_2 = 0$$

Auxiliary equation is given by,

$$A(s) = 3s^2 + a_1 = 0$$

$$s^2 = \frac{-a_1}{3} \Rightarrow s = \pm j\sqrt{\frac{a_1}{3}}$$

Therefore, a_1 must be positive to have imaginary roots.

$$p(s) = s^3 + a_2 s^2 + a_1 s + a_0$$

$p(s) = s^3 + 0s^2 + a_1 s + a_0$

Routh tabulation:

s^3	1	a_1
s^2	0	a_0
s^1	$-\infty$	0
s^0	a_0	0

Number of sign changes in first column = Number of roots in the right half of s –plane =2

These poles may be positive real roots or complex conjugate pair which is symmetrical about imaginary axis.

Example: Let

1. $a_1 = 5,\ a_0 = 9$

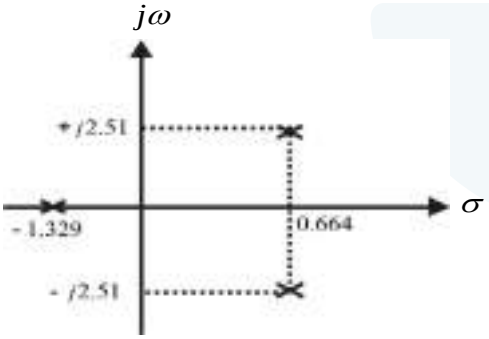


Fig. 6.48

Then $s = -1.329, 0.664 \pm j2.51$

2. $a_1 = 15, a_0 = 33$

Then $s = -1.806,\ 0.903 \pm j4.177$

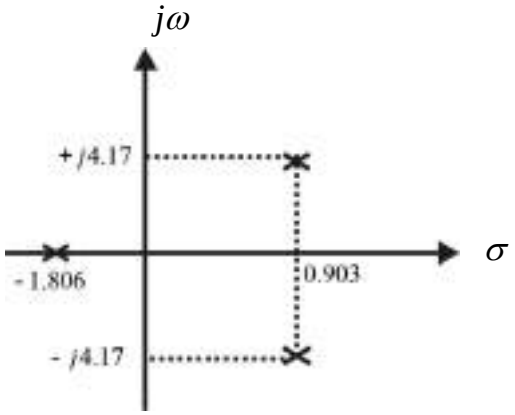


Fig. 6.49

3. $a_1 = 126,\ a_0 = 196$

Then $s = -1.527,\ 0.763 \pm j11.302$

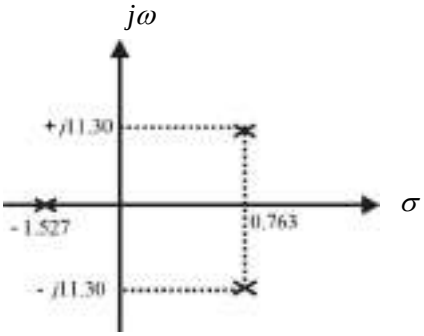


Fig. 6.50

From the above example it is clear that the right-hand side roots are complex conjugate. Hence, the number of real roots of p(s)is 1.

Hence, the correct option is (B).

Note: If a polynomial of s has n number of real roots, then its derivative will have atleast (n-1) real roots.

5. (D)

From the Routh table, Routh array can be written as,

Routh tabulation:

s^3	1	(2K+3)
s^2	2K	4
s^1	$\frac{4K^2 + 6K - 4}{2K}$	0
s^0	4	0

For the system to be stable, all the roosts must be in the left half of the s-plan; thus, all the coefficient in the first column of Routh's tabulation must have the same sign. Therefore, the coefficient of first column of the Routh's table should be positive.

This leads to the following conditions:

1. $2K > 0 \quad \Rightarrow \quad K > 0$ ...(i)

2. $\frac{4K^2 + 6K - 4}{2K} > 0$

$2K^2 + 3K - 2 > 0$

$(2K - 1)(K + 2) > 0$

$K > \frac{1}{2}$ or $K > -2$ ...(ii)

On combining equations(i) and (ii)

$$K > \frac{1}{2}$$

Thus, from options, 0.5<K<∞.

Hence, the correct option is (D).

6. (3)

To find the number of zeros in the right half of s–plane, applying Routh–Hurwitz criteria,

$$2s^4 - 5s^3 + 5s - 2 = 0$$

Routh Tabulation:

$$2s^4 - 5s^3 + 0.s^2 + 5.s - 2 = 0$$

s^4	2	0	-2
s^3	-5	5	0
s^2	2	-2	0
s^1	0	0	0

Since a row of zeros appears, we form the auxiliary equation using the coefficients of s^2 row,

$$A(s) = 2s^2 - 2$$

The derivative of $A(s)$ with respect to s is,

$$\frac{d}{ds}[2s^2 - 2] = 4s$$

From which the coefficients 4 replace the zeros in the s^1 row of the original tabulation. The remaining portion of the Routh's tabulation is,

s^1	4	0
s^0	-2	0

Number of sign change in first column

= Number of Roots (zeros) of characteristic equation in right half of s – plane =3

Hence, the number of zeros in the right half of the s–plane is 3.

7. (A)

Given, $G(s) = \left(K_p + \frac{K_I}{s}\right)\frac{1}{s(s+2)}$

$$G(s) = \frac{(sK_p + K_I)}{(s^3 + 2s^2)}$$

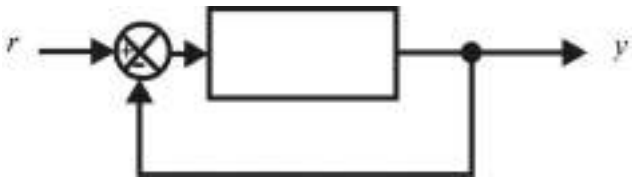


Fig. 6.51

The characteristic equation is given by,

$$1 + G(s) = 0$$

$$s^3 + 2s^2 + K_p s + K_I = 0$$

Routh tabulation:

s^3	1	K_p
s^2	2	K_I
s^1	$\frac{(2K_p - K_I)}{2}$	0
s^0	K_I	0

For the system to be stable, all the roots must be in the lefthalf of the s–plane; thus, all the coefficients in the first column of Routh's tabulation must have the same sign. Therefore, the coefficient of first column of the Routh's table should be positive.

This leads to the following conditions:

1. $K_I > 0$
2. $\left(\frac{2K_p - K_I}{2}\right) > 0$

$$K_p > \frac{K_I}{2}$$

On combining above two conditions,

$$K_p > \frac{K_I}{2} > 0$$

Hence, the correct option is (A).

7 Root Locus Technique

Objective

After completing this chapter, the user will be able to understand

- How to plot the Root locus for a given transfer function by the varying gain of the system.
- The analysis of the stability of the system from the Root locus plot.
- How to determine all parameters related to theroot locus plot.
- How to plot complementary root locus for negative values of gain.
- How to plot root contours by varying multiple parameters.

Introduction

- The transient response of a closed-loop system is dependent upon the location of closed-loop poles.
- IF the system has a variable gain, then the location of closed-loop poles is dependent on the value of gain.
- It is, therefore, necessary that we must know how the location of closed-loop poles varies with the change in the value of loop gain.
- In control system design, it may help to adjust the gain to move the closed-loop poles at the desired location. So, the Root locus plot gives the location of closed-loop poles as system gain K is varied.

Root locus: The portion of root loci when K is assumed to vary in the positive range: that is $0 \le K < \infty$

Complementary root locus: The portion of root loci when K is assumed to vary in the negative range: that is $-\infty < K \le 0$

Root contour: It is a root locus when more than one system parameter varies.

Root Locus

- Locus of roots of CE
- Or locus of zeros of CE
- Or locus of poles of C.L.S

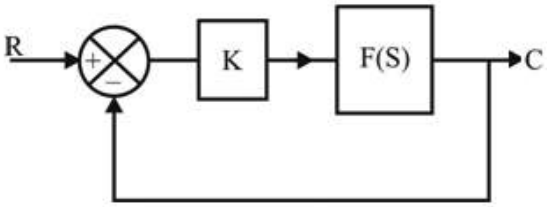


Fig. 7.1

1. $$\frac{C}{R} = \frac{KF(S)}{1+KF(S)} = \frac{G(S)}{1+G(S)H(S)}$$

CE $\rightarrow$ 1 + G(S) H(S) = 1 + KF(S) = 0 $\rightarrow$ Plot of roots of 1 + G(S) H(S) is called Root locus.

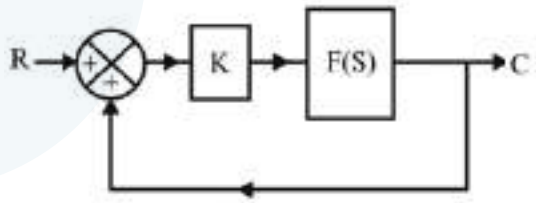


Fig. 7.2

$$\frac{C}{R} = \frac{KF(S)}{1-KF(S)} = \frac{G(S)}{1-G(S)H(S)}$$

CE $\rightarrow$ 1 – G(S)H(S) = 1 – KF(S) = 0 $\rightarrow$ Plot of roots of 1 – G(S)H(S) = 0 is called as Root locus

Case 1:

$$G(S)H(S) = KF(S)$$

- $0 < K < \infty$
- Negative feedback

Root Locus

Locus of roots of 1 + G(S)H(S) on S-plane as K varies $0 < K < \infty$

CE $\rightarrow$ 1 + G(S) H(S) = 0

1 + KF(S) = 0

KF(S) = –1 $\rightarrow$ Root locus $\rightarrow$ 'Direct Root locus (DRL)'

where K is positive.

Case 2:

$G(S)H(S) = -KF(S)$

- $0 < K < \infty$
- Negative feedback

Root Locus

Locus of roots of CE 1 + G(S) H(S) = 0 on S-plane as K varies $0 < K < \infty$.

$C.E. \rightarrow 1 + G(S)H(S) = 0$

$1 - KF(S) = 0$

$KF(S) = 1 \rightarrow Root\ Locus \rightarrow$ Complimentary Root Locus or Inverse Root locus

(CRL. or I.RL)

Case 3:

$G(S)H(S) = KF(S) \quad 0 < K < \infty$

Positive Feedback

Root Locus

Locus of roots of CE 1 – G(S)H(S) = 0 on S-plane as K varies $0 < K < \infty$

CE $1 - G(S)H(S) = 0$

$G(S)H(S) = 1$

$KF(S) = 1 \rightarrow Root\ locus \rightarrow$ Complimentary Root locus or Inverse Root locus

Case 4:

$G(S)H(S) = -KF(S) \quad 0 < K < \infty$

Positive Feedback

Root Locus

Locus of roots of CE1 – G(S)H(S) = 0 on S-plane as K varies $0 < K < \infty$

CE $1 - G(S)H(S) = 0$

1 + KF(S) = 0

$KF(S) = -1 \rightarrow Root\ locus \rightarrow Direct\ Root\ locus$

Case 5:

$G(S)H(S) = KF(S) \quad -\infty < K < 0$

Negative Feedback

Root Locus

Locus of roots of CE 1 + G(S)H(S) = 0 on S-plane as K varies $-\infty < K < 0$

CE $1 + G(S)H(S) = 0$

1 + KF (S) = 0

KF(S) = –1

$K = -K_1$

$-\infty < K < 0$

$0 < K_1 < \infty$

$-K_1F(S) = -1 \rightarrow K_1F(S) = +1 \rightarrow C.R.L.\ or\ I.R.L.$

Case 6:

$G(S)H(S) = -KF(S) \quad -\infty < K < 0$

Negative Feedback

$1 + G(S)H(S) = 0$

$1 - KF(S) = 0$

$KF(S) = 1$

$K = -K_1$

$-\infty < K < 0$

$0 < K_1 < \infty$

$-K_1F(S) = 1$

$-K_1F(S) = 1 \quad 0 < K_1 < \infty \rightarrow DRL$

Criterion for Root Loci

Magnitude and angle condition:

1. G(S) H(S) = KF(S) $0 < K < \infty$
   Negative feedback
   CE 1 + G(S) H(S) = 0

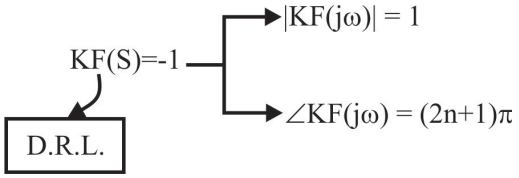


Fig. 7.3

2. G(S) H(S) = –KF(S) $0 < K < \infty$
   Positive feedback
   CE 1 – G(S) H(S) = 0
   1 + KF(S) = 0

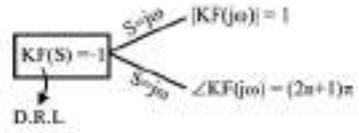


Fig. 7.4

3. G(S) H(S) = KF(S) $0 < K < \infty$
   Positive feedback
   CE 1 – G(S) H(S) = 0
   1 – KF(S) = 0

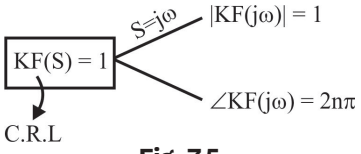


Fig. 7.5

4. G(S) H(S) = –KF(S) $0 < K < \infty$
   Negative feedback
   CE 1 + G(S) H(S) = 0
   1 - KF(S) = 0

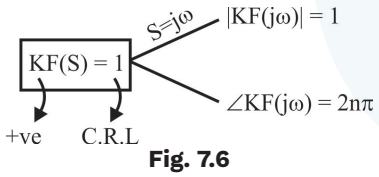


Fig. 7.6

Summary:

G(S)H(S)	K	Feedback	Locus
KF(S)	$0 < K < \infty$	Negative	DRL
-KF(S)	$0 < K < \infty$	Negative	CRL
KF(S)	$0 < K < \infty$	Positive	CRL
-KF(S)	$0 < K < \infty$	Positive	DRL

Table 7.1

Direct Root Locus

Locus of closed-loop poles on S-plane as the gain K varies under desired feedback

K → Proportional control gain

→ Gain present in the forward path

2nd : - ∞< K < 0

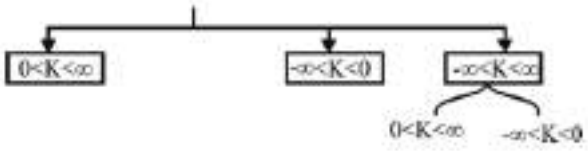


Fig. 7.7

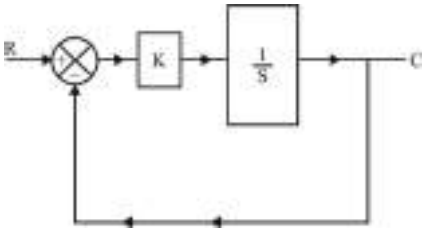


Fig. 7.8

$0 < K < \infty$

Plot Root locus?

1. Root Locus → DRL
2. Locus of roots of 1 + G(S) H(S) = 0 as $0 < K < \infty$

$$1 + \frac{K}{S} = 0 \quad \text{or} \quad \frac{C}{R} = \frac{K}{S+K} = \frac{G}{1+GH}$$

S + K = 0

$S = -K \rightarrow plot\ 0 < K < \infty$

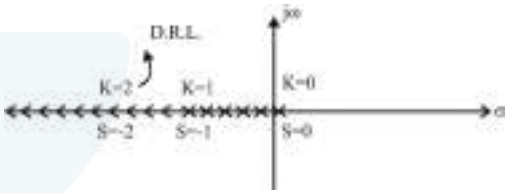


Fig. 7.9

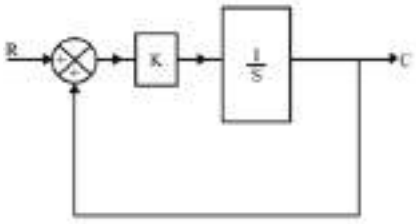


Fig. 7.10

$0 < K < \infty$

Plot root locus

1. CRL
2. Locus of roots of 1 – G(S) H(S) = 0 as $0 < K < \infty$

$$\frac{C}{R} = \frac{K}{S-K} = \frac{G}{1-GH} \qquad S - K = 0$$

$S = K \quad 0 < K < \infty$

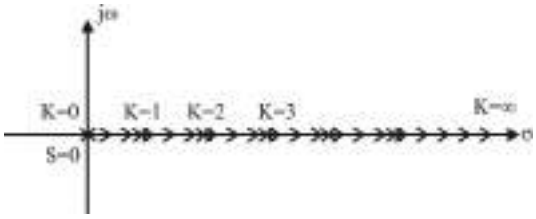


Fig. 7.11

Example 1:

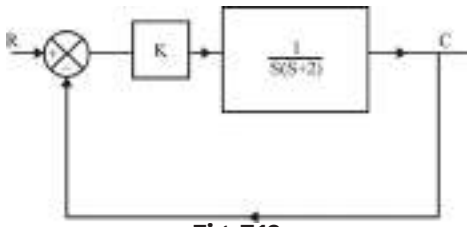


Fig. 7.12

- $0 < K < \infty$
- Plot root locus
  a) DRL
  b) $\frac{C}{R} = \frac{K}{S^2 + 2S + K}$

  Plot roots of $S^2 + 2S + K = 0$ *as* $0 < K < \infty$

Solution:

K = 0	S = 0, S = –2
$K = \frac{1}{2}$	S = –0.29, S = –1.7
K = 1	S = –1, S = –1
K = 1.5	S = –1 ± j0.7
K = 2	S = –1 ± j
K = 4	S = –1 ± j$\sqrt{3}$

Table 7.2

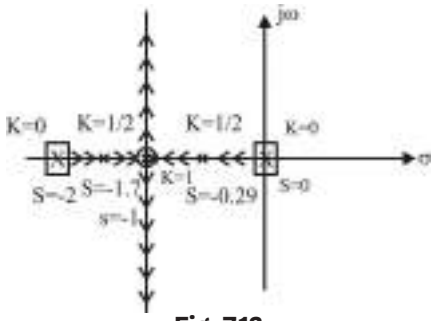


Fig. 7.13

DRLPlotting Technique

Given: $G(S)H(S) = KF(S)$

$0 < K < \infty$ & – *ve feedback*

$G(S)H(S) = -KF(S)$ $0 < K < \infty$ & + *ve feedback*

- OLTF is given
- IdentIFy IF DRL will be the Root locus

Rule 1:

- Root locus always originates from open-loop poles.
- These open-loop poles may be present at the finite value of S or infinite value of S.
- IF the open-loop pole is on a real axis, then Root locus will originate in the direction where the Root locus exists.
- IF open-loop poles are at the complex location or imaginary location, the Root locus will originate in the direction provided by 'Angle of Departure'.
- IF the open-loop pole is at the infinite location,then Root locus will originate in the direction provided by

  'Angle of Asymptotes'.
- At open-loop poles, value of K = 0.

Rule 2:

- Root locus terminates at open-loop zeros.
- The open-loop zeros may be present at finite value of S or infinite value of S.
- IF open-loop zeros are on real axis then Root locus will terminate in the direction where Root locus exist.
- IF open-loop zeros are at complex or imaginary location then Root locus will terminate in the direction provided by 'Angle of Arrival'.
- IF the open-loop zeros are at infinite location then Root locus will terminate in the direction provided by the 'Angle of Asymptotes'.
- At open-loop zero, the value of K is ∞.

Ex:

$$G(S)H(S) = \frac{KS^3}{(S+2)(S^2+S+1)}$$

By default $0 < K < \infty$

–ve feedback D.R.L

Poles:

$$S = -2, \quad S = \frac{-1}{2} \pm j\frac{\sqrt{3}}{2}$$

Zeros: S = 0, 0, 0

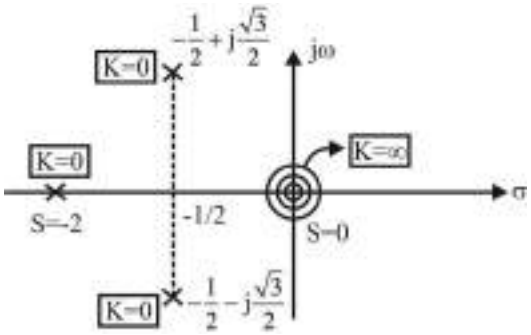


Fig. 7.14

Rule 3:

Existence of Root locus on 'Real Axis'.

Segment of the real axis where poles of the closed-loop system will exist as K varies.

Method 1: Angle condition for DRL

$$\angle F(S) = (2n+1)\pi$$

'Angle subtended by all open-loop finite poles and zeros must be odd multiples of π in that region of real axis'.

Method 2: No. of open-loop finite poles and zeros must be odd in number towards its right.

Example 2: Indicate real axis loci for the loop transfer function,

$$G(S) = \frac{K}{S(S+2)} \qquad 0 < K < \infty$$

–ve feedback

Solution:

Method 1:

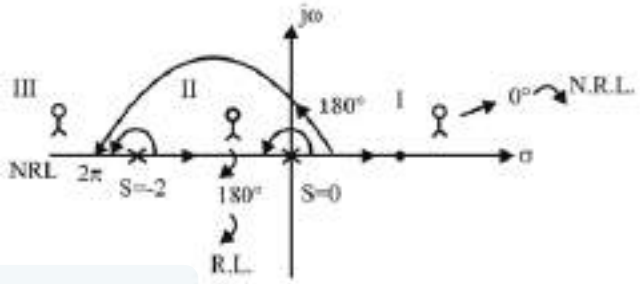


Fig. 7.15

Points To Remember

Ist quadrant: $(a,b) \rightarrow \theta = \tan^{-1}\frac{b}{a} \pm 360°$

IInd quadrant: $(-a,b) \rightarrow \theta = \tan^{-1}\frac{b}{-a} \pm 180°$

IIIrd quadrant: $(-a,-b) \rightarrow \theta = \tan^{-1}\left(\frac{-b}{-a}\right) \pm 180°$

IVth quadrant: $(a,-b) \rightarrow \theta = \tan^{-1}\left(\frac{-b}{a}\right) \pm 360°$

Common mistake:

User usually makes the mistake for 2nd and 3rdquadrant; they miss to add or subtract the 180 degrees into the angle.

Example 6: $G(S)H(S) = \frac{K}{(S+1)^4}$

S = –3 + 4j is part of Root locus?

Solution: $0 < K < \infty$

Negative F / B

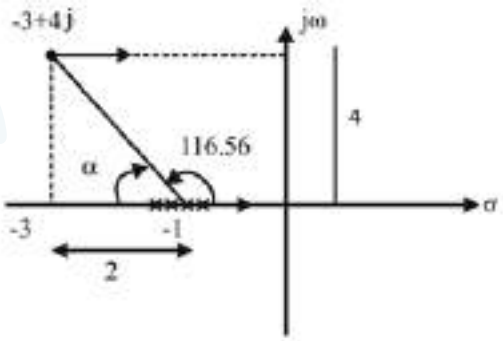


Fig. 7.16

$$\tan\alpha = \frac{4}{2}$$

$$\alpha = 63.43$$

Total angle = 4*(180-63.43) = 466.26 ≠ $(2n+1)\pi$

'S = –3 + 4j is not part of root locus'.

Calculation of proportional controller gain K at S = S_0, 'when S = S_0 is part of Root locus'.

$$G(S)H(S) = \frac{K(S+a)(S+b)}{(S+c)(S+d)(S+e)} \qquad 0 < K < \infty$$

–ve F / B

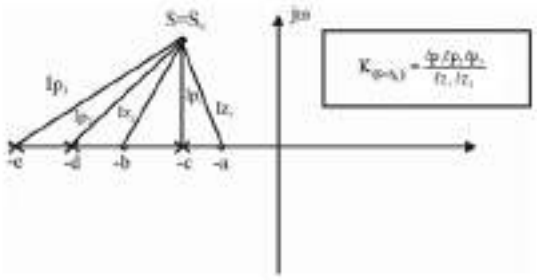


Fig. 7.17

$1 + G(S)H(S) = 0$

$$S = S_0 \rightarrow 1 + G(S_0)H(S_0) = 0 \qquad 0 < K < \infty$$

$$\frac{K(S_0 + a)(S_0 + b)}{(S_0 + c)(S_0 + d)(S_0 + e)} = -1 \qquad |A\ B| = |A||B|$$

$$|K| = \frac{|S_0 + c||S_0 + b||S_0 + e|}{|S_0 + a||S_0 + d|} \left|\frac{A}{B}\right| = \frac{|A|}{|B|}$$

$$K = \frac{|S_0 - (-c)||S_0 - (-d)||S_0 - (-e)|}{|S_0 - (-a)||S_0 - (-b)|} = \frac{\ell p_1 \ell p_2 \ell p_3}{\ell z_1 \ell z_2}$$

Example 8: $G(S)H(S) = \dfrac{K}{(S+1)^4}$

Determine the value of K at S = –3 + 4j for the above loop transfer function.

Solution:

$0 < K < \infty$

–ve F / B

Since S = –3 + 4i is not part of RL So, no such K exist.

Rule 4: Calculation of angle of asymptotes:

Angle of asymptotes need to be calculated for those branches of Root locus which originates or terminates at infinity.

No. of asymptotes =

P = No. of finite open-loop poles

Z = No. of finite open-loop zeros.

1. $G(S)H(S) = \dfrac{K}{S(S+1)}$

$S_p = 0, -1 \quad P = 2$

$S_Z = No \quad Z = 0$

$P - Z = 2$

P – Z = 2 → '2 branches of root locus will originate from 2 finite open-loop poles, and they will be terminated at zeros present at infinity using angle of asymptotes'.

2. $G(S)H(S) = K(S+1)(S+2)(S+3)$

$S_p = No \quad P = 0$

$S_Z = -1, -2 - 3 \quad Z = 3$

$Z - P = 3$

$Z - P = 3 \Rightarrow 3$ branches of root locus will originate from ∞ using angle of asymptotes, and they will terminate at finite open-loop zeros.

3. $G(S)H(S) = \dfrac{K(S+1)}{(S+2)(S+3)}$

$S_p = -2, -3 \quad P = 2$

$S_Z = -1 \quad Z = 1$

$P - Z = 1$

$P - Z = 1 \rightarrow$ '2 Branches of root locus will originate from 2 finite open-loop poles out of which one will terminate at finite open-loop zero and other will terminate at infinity by using Angle of Asymptotes'.

4. $G(S)H(S) = \dfrac{K(S+1)(S+2)}{(S+3)}$

$S_p = -3 \quad P = 1$

$S_Z = -1, -2 \quad Z = 2$

$Z - P = 1$

$Z - P = 1 \rightarrow$ 2 Branches of root locus will originate one from open-loop finite pole and another from ∞ by using Angle of Asymptotes: Both will terminate at finite open-loop zeros.

5. $G(S)H(S) = \dfrac{K(S+1)(S+2)}{(S+3)(S+4)}$

$S_p = -3, -4 \quad P = 2$

$S_Z = -1, -2 \quad Z = 2$

$P - Z = 0 \rightarrow$ 2 Branches of root locus will originate from finite open-loop poles and they will terminate at finite open-loop zeros.

Note: Calculation of angle of asymptotes is not needed for P – Z = 0.

Angle of asymptotes:

P = No. of finite open-loop poles

Z = No. of finite open-loop zeros.

S.1 $P - Z \neq 0$

S.2

$P > Z$ $\qquad$ $Z > P$

$$\theta_K = \frac{(2K+1)}{(P-Z)} \times 180° \qquad \theta_K = \frac{(2K+1)}{(Z-P)} \times 180°$$

$K = 0, 1, 2, \ldots P - Z - 1$ $\qquad$ $K = 0, 1, 2, \ldots Z - P - 1$

(P-Z) or (Z-P)	θ_K
1	180°
2	90°, 270°
3	60°, 180°, 300°
4	45°, 135°, 225°, 315°

Table 7.3

Note: θ_K is measured from Real axis σ in A.C.W. direction.

Centroid: (σ)

- The angle of asymptotes are always drawn from a point on real axis know as centroid.
- The origination point of Asymptotes is a point on real axis know as centroid.
- All the Asymptotic lines (A. O.A.) meet at common point on real axis known as Centroid.
- Centroid is always present on real axis.
- Centroid may or may not be part of Root locus.
- Value of centroid may be $\begin{cases} +ve \\ -ve \\ 0 \end{cases}$

$$\sigma = \frac{\sum P - \sum Z}{P - Z} = \frac{\sum Z - \sum P}{Z - P}$$

$\sigma \in R \rightarrow$ Real No.

Note: IF P = Z $\rightarrow$ 'No need to calculate centroid'.

Rule 5: 'Break Point (BP)':

- It is the location in S-plane, where 2 or more than 2 poles of a closed-loop system coincide simultaneously.
- It is the part of Root locus, where 2 or more than 2 poles of C.L.S. coincide simultaneously.

$$G(S)H(S) = \frac{K}{S(S+2)} \longrightarrow$$

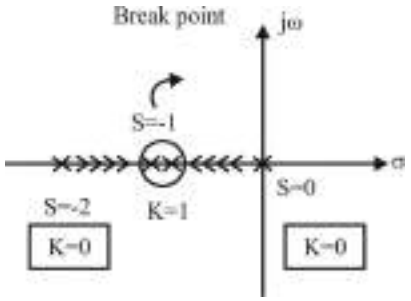


Table 7.18

Types of BP's

Break away point:

- 'It is the location in S-plane where 2 or more than 2 poles of closed-loop system coincide simultaneously'
- After the BA point, Root locus breaks, and it can't remain on the real axis. It breaks into dIFferent parts in the complex conjugate plane.
- Break away point means shIFting of Root locus from real axis to complex conjugate plane.

Ex. $G(S)H(S) = \frac{K}{S(S+2)}$

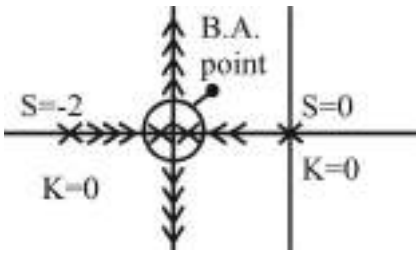


Fig. 7.19

Note: At BA points, K = K_{min} is the boundary
K<K_{min} $\rightarrow$ RL is on real axis
K>K_{min} $\rightarrow$RL is enters into complex plane

Break-in point:

- It is the location in S-plane, where 2 or more than 2 poles coincide simultaneously.
- After the break-in point, root locus breaks and enters into the real axis from the complex plane.

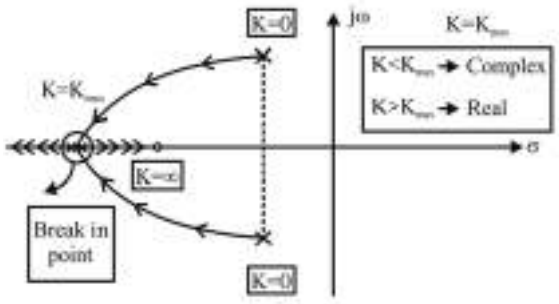


Fig. 7.20

General break points:

- It is the location in S-plane where 2 or more than 2 poles of C.L.S. coincide simultaneously.
- In this case, poles coincide on complex plane, and they break into dIFferent parts on the complex plane itself.

Mathematical calculation:

Method 1: Most General method:

S.1 From CE $\rightarrow$ 1 + G(S) H(S) = 0

1 + KF(S) = 0

$$K = \frac{-1}{F(S)}$$

$$\frac{dk}{dS} = 0 \rightarrow S = S_1, S = S_2, S = S_3 \ldots \rightarrow \textit{possible B.P.}$$

S.2 (validity check) S = S_1 is part of RL $\rightarrow$ S = S_1 is valid BP

S = S_2 is not part of RL $\rightarrow$ S = S_2 is invalid BP

$$\left.\frac{d^2k}{ds^2}\right|_{S=S_1} > 0 \rightarrow \textit{Break in}$$

$$\left.\frac{d^2k}{ds^2}\right|_{S=S_1} < 0 \rightarrow \textit{Break} - \textit{Away}$$

Method 2: When OLTF has non-repeated, real poles and zeros.

$$G(S) = \frac{K(S+a)(S+b)}{(S+c)(S+d)}$$

S.1 Solve $\frac{1}{(S+a)} + \frac{1}{(S+b)} = \frac{1}{(S+c)} + \frac{1}{(s+d)}$

S.2 $\rightarrow$ Check validity

Method 3: Only whenreal non-repeated poles and no zeros are present.

$$G(S) = \frac{K}{(S+p_1)(S+p_2)} = \frac{K}{D(S)}$$

S.1 $D(S) = (S+p_1)(S+p_2) \rightarrow \frac{dD(S)}{dS}$

$= 0 \rightarrow S = ?$

S. 2 $\rightarrow$ Validity check

Properties of break points:

1. The branches of Root locus either approaching or leaving the BP starts or ends at an initial angle of $\pm\frac{180^\circ}{n}$ where n is no. of branches approaching or leaving the BP

Example 12:

$$G(S)H(S) = \frac{K}{S(S+2)} \qquad 0 < K < \infty$$

$$-ve\ F\ /\ B$$

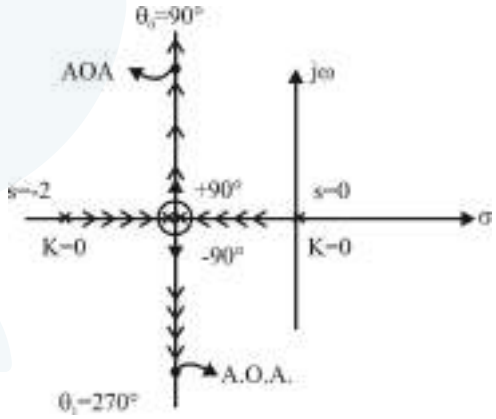


Fig. 7.21

$$\pm\frac{180^\circ}{n} \rightarrow \frac{\pm 180^\circ}{2} = \pm 90^\circ$$

$\theta_K = 90^\circ, 270^\circ$

$\sigma = -1$

2. Whenever there are two adjacent open-loop poles on real axis and segment between them is part of root locus then there will be at least 1 break away point between them.

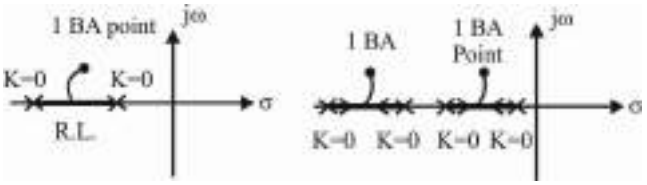


Fig. 7.22

3. Whenever there are two zeros of open-loop systems are adjacent on the real axis, and a segment between them is part of the Root locus, then there will be at least 1 break-in point between them.

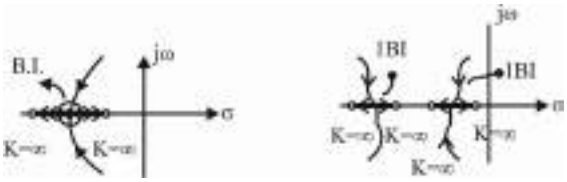


Fig. 7.23

4. IF 2 poles of an open-loop system coincide at K = 0; then the pole value itself behaves as Break Away point

$$G(S)H(S)=\frac{K}{(S+1)^2}$$

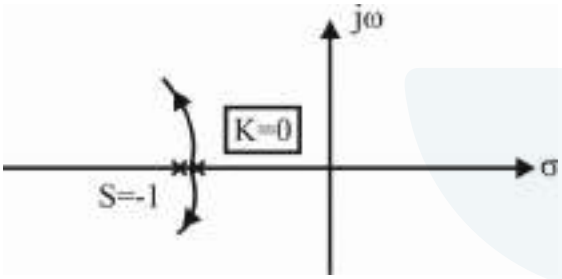


Fig. 7.24

5. IF 2 open-loop zeros coincide at $K=\infty$; then the zero value itself behaves as a break-in point

$$G(S)H(S)=K(S+1)^2$$

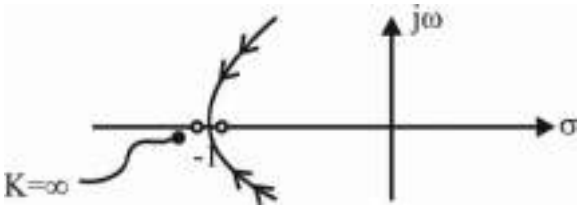


Fig. 7.25

E.g. $G(S)H(S)=\frac{K(S+1)^2}{(S+2)^2}$

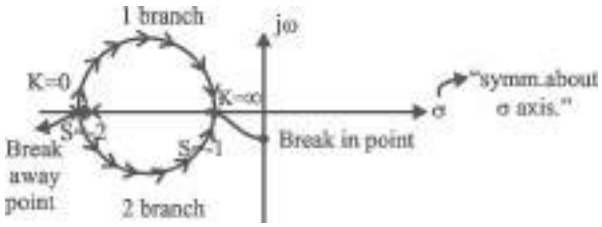


Fig. 7.26

Eg. $G(S)H(S)=\frac{K(S+2)^2}{(S+1)^2}$

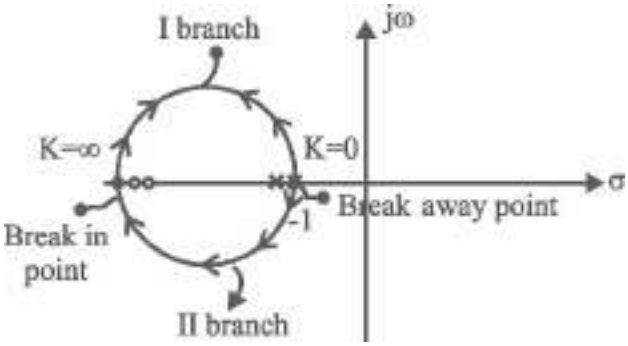


Fig. 7.27

Rule 6: Angle of departure:

- It is calculated only when open-loop pole is complex.
- It gives the origination direction to branch of RL

$$\theta_d=180^\circ-\left[\phi_p-\phi_z\right]$$

$\phi_p \rightarrow$ Angle subtended by remaining O.L. poles towards desired complex pole.

$\phi_z \rightarrow$ Angle subtended by all O.L. zeros towards desired complex pole.

Rule 7: Angle of arrival: (θ_a)

- It is calculated only when open-loop zeros are complex.
- It gives the termination direction to the branch of root locus.

$$\theta_a=180^\circ-\left[\phi_z-\phi_p\right]$$

$\phi_z \rightarrow$ Angle subtended by remaining O.L. zeros towards the desired complex zero.

$\phi_p \rightarrow$ Angle subtended by all O.L. poles towards desired complex zero.

Advantages of root locus:

- The Root locus is a powerful technique as it gives the complete dynamic response of the system.
- As Root locus is a graphical technique, with the help of approximate Root locus designer can easily visualize the effect of various system parameters and root locations.
- The Root locus also provides a measure of sensitivity of roots to the variation in parameters being considered.

- It may further be pointed out here that Root locus technique is applicable to single as well as multiple loop system.

'Rules to plot CRL. or I.RL'

Rule 1: Origination point: Same as D. R. L.

Rule 2: Termination point: Same

Rule 3: Existence of Root on Real axis:

1. Only that section of Real axis is part of Root locus where angle subtended by all the open-loop poles and zeros are even multiples of $\pi = 2n\pi$ n = 0, 1, ...
2. Or no. of open-loop poles and zeros towards its right is even in number.

Common point to DRL and CRL.

Intersection with j-axis: 'This means identIFication of Root locus with j-axis'

Step 1: Form CE and use R.H. table

Step 2:

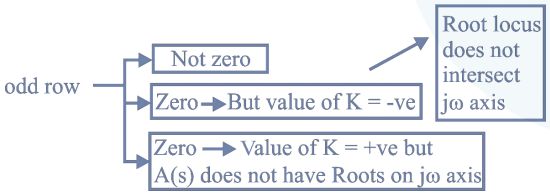


Fig. 7.28

Step 3: Odd row zero → K = +ve → Roots of A(S) =

jω axis ⇒ Root locus intersects jω-axis.

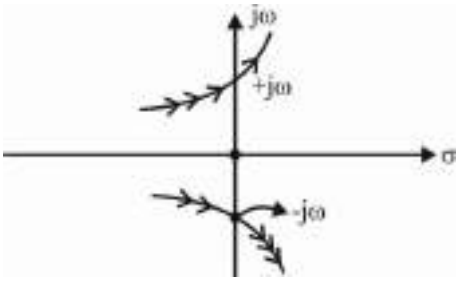


Fig. 7.29

Limitation: Above method gives intersection with j ω-axis on complex location.

Example 20: $G(S)H(S) = \dfrac{-K(S+1)^2}{(S+2)^2}$

1. $0 < K < \infty$
2. –ve feedback

Solution: CRL

S.1 $S_p = -2, -2 \rightarrow No\ \theta_d$

$S_z = -1, -1 \rightarrow No\ \theta_a$

S.2 $P = 2$

$Z = 2$ $\qquad P = Z$

S.3 Asymptotes: Not needed

S.4

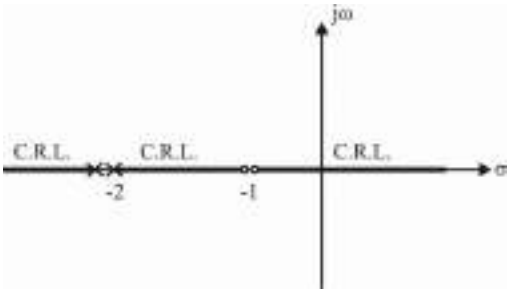


Fig. 7.30

S.5 break points:

$S = -1 \qquad K_{(S=-1)} = \infty$

$S = -2 \qquad K_{(S=-2)} = 0$

S.6

$S^2(1-K) + S(4-2K) + (4-K) = 0$

$= 1 + G(S)H(S)$

$S^2(1-K) \qquad (4-K)$

$S^1(4-2K) \rightarrow odd\ row \quad K = 2, A(S) = -S^2 + 2 = 0$

$S^0 \quad 4-K \qquad\qquad S = \pm\sqrt{2}$

'No intersection with jω-axis at complex location'

$S^0 \rightarrow (4-K) = 0$

$K = 4 \rightarrow S = 0\ or\ origin \rightarrow$ Root locus will pass from origin at K = 4

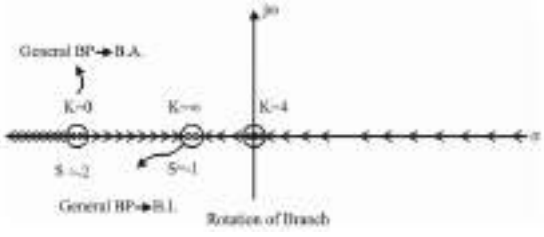


Fig. 731

1 + G(S)H(S) = 0

$K = 2 \quad S = \sqrt{+2}, S = -\sqrt{2}$

$K = 4 \quad S = 0, -1.33$

$K = 5 \quad S = -1.33, -0.19$

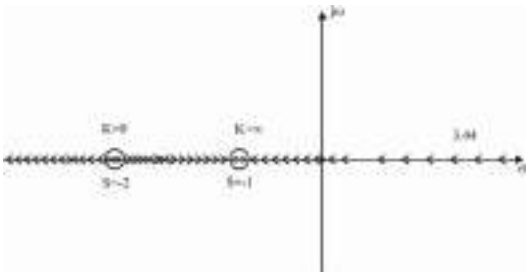


Fig. 7.32

$1 + G(S)H(S) = 0$

K		Roots
$K = 0.5$	$\rightarrow$	$-1.58, -4.41$
$K = 0.8$	$\rightarrow$	$-1.52, -10.47$
$K = 0.9$	$\rightarrow$	$-1.51, -20.48$
$K = 1$	$\rightarrow$	$-1.5 \quad S = -\infty$
$K = 1.5$	$\rightarrow$	$-1.44 \quad 3.44$
$K = 2$	$\rightarrow$	$-\sqrt{2} \quad +\sqrt{2}$
$K = 4$	$\rightarrow$	$0 \quad -1.33$
$K = 5$	$\rightarrow$	$-1.33 \quad -0.19$

Example 23: Draw the Root locus diagram for $G(S)H(S) = \dfrac{KS^3}{(S+1)(S+2)}$

- $0 < K < \infty$
- $-ve\ F\ /\ B$

Solution:

S.1 $S_z = 0, 0, 0 \rightarrow No\ \theta_a$

$S_P = -1 - 2 \rightarrow No\ \theta_d$

S.2

$P = 2$

$Z = 3 \qquad Z - P = 1$

S.3 Angle of asymptote, $\theta = 180^0$

Centroid, $\sigma = +3 = \dfrac{\sum Z - \sum P}{Z - P}$

S.4

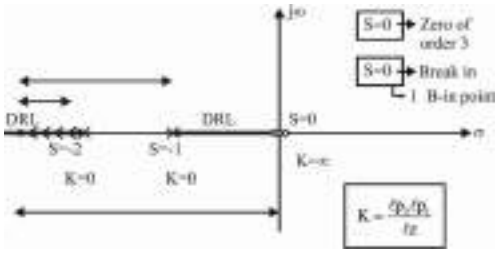


Fig. 7.33

S.5 $1 + G(S)H(S) = 0 \quad -K \rightarrow -\dfrac{dK}{dS} = 0$

$S = 0$

$S = -4.37 \qquad S = 1.26 \rightarrow Invalid$

$valid \rightarrow S = -4.37 \rightarrow K_{[S=-4.37]} = 0.096$

$S = 0 \rightarrow K = \infty$

2 Break points

$S = 0 \rightarrow Break\ in\ point \rightarrow \left(\dfrac{d^2K}{dS^2}\right)_{S=0} > 0$

$\underset{Breakaway\ point}{S = -4.37} \quad \left(\dfrac{d^2K}{dS^2}\right)_{S=-4.37} < 0$

S.6 Intersection with $j\omega$-axis:

$1 + G(S)H(S) = 0$

$(S+1)(S+2) + KS^3 = 0$

$KS^3 + S^2 + 3S + 2 = 0$

S^3	K	3	
S^2	1	2	$\rightarrow A(S) = S^2 + 2 = S = \pm j\sqrt{2}$
S^1	$3 - 2K$	0	$\rightarrow odd\ row = 0 \qquad \boxed{K = 1.5}$
S^0	2		

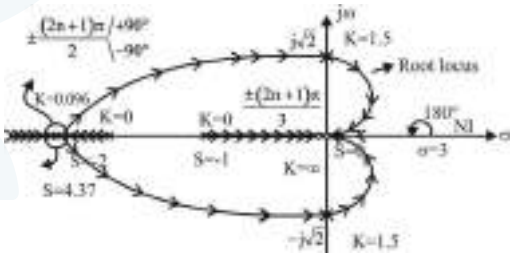


Fig. 7.34

Termination angle(at B.P):

$$\frac{\pm(2n+1)\pi}{n \rightarrow \text{no of branches terminating}}$$

Origination angle(at B.P):

$$\frac{\pm(2n+1)\pi}{n \rightarrow \text{no of branches originating}}$$

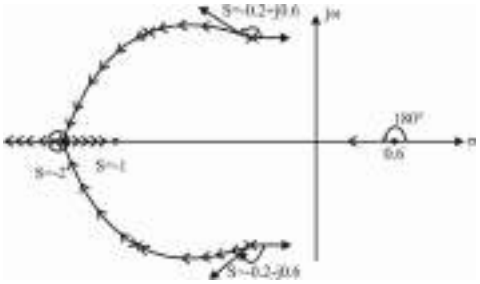


Fig. 7.35

Mode Based root locus:

Mode 1:

$$G(S)H(S) = \frac{K(S+1)}{S^2(S+\alpha)} \qquad 0 < K < \infty \\ -ve\,F\,/\,B \rightarrow DRL$$

Case 1: $\alpha = 20$

$$G(S)H(S) = \frac{K(S+1)}{S^2(S+20)}$$

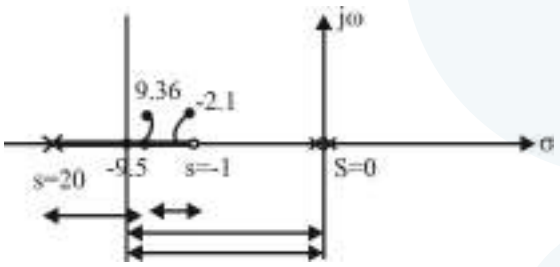
1. $S_p = 0, 0, -20 \rightarrow No\ \theta_d$
   $S_Z = -1 \rightarrow No\ \theta_a$
2. $P = 3 \quad P - Z = 2$
   $Z = 1$
3. $\theta = 90^0, 270^0$
   $$\sigma = \frac{\sum P - \sum Z}{(P-Z)} = -9..5$$
4. 

   Fig. 7.36
5. **Break point:** 1 + G(S)H(S) = 0
   $$-K \rightarrow \frac{-dK}{dS} = 0$$
   Validity: $S = 0$
   $S = -9.36 \rightarrow valid$
   $S = -2.13$
   $K_{[S=0]} = 0 \qquad K_{[S=-2.13]} = 71.76$
   $K_{[S=-9.36]} = 111.5$
6. **Intersection with jω-axis:**
   $$1 + G(S)H(S) \Rightarrow S^3 + 20S^2 + KS + K = 0$$

S^3	1	K
S^2	20	K
S^1	$\frac{19K}{20}$	0
S^0	K	

$$A(S) = 20S^2 + 0 = 0$$

$S = 0$

$K = 0 \Rightarrow j\omega\ axis\ avoid$

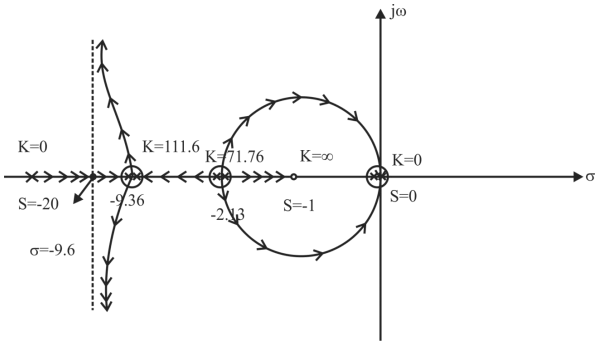


Fig. 7.36

Case 2:

$$G(S)H(S) = \frac{K(S+1)}{S^2(S+9)} \qquad 0 < K < \infty \Rightarrow DRL \\ -ve\,F\,/\,B$$

$\alpha = 9$

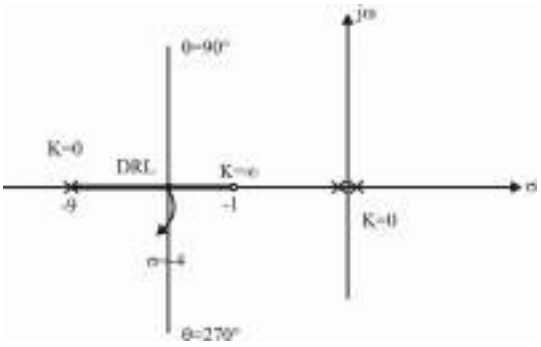
1. $S_p = 0, 0 - 9 \rightarrow No\ \theta_d$
   $S_Z = -1 \rightarrow No\ \theta_a$
2. P = 3
   Z = 1
   P-Z = 2
3. $\theta_d = 90^0, 270^0$
   $\sigma = -4$
4. 

   Fig. 7.37
5. 1 + G(S)H(S) = 0
   $$\frac{-dK}{dS} \Rightarrow S = 0, -3 - 3$$
   $S = 0 \rightarrow K_{[S=0]} = 0$
   $S = -3.3 \rightarrow K_{[S=-3]} = 27$

$S(S+3)^2 = 0$

$S = 0$

$(S+3)^2 = 0 \Rightarrow S = -3, -3$

$\frac{-dK}{dS} = 0 \rightarrow (S+3) = 0 \quad S = -3\ BP \Rightarrow$

2 Branches will coincide

$\frac{-dK}{dS} = 0 \rightarrow (S+3)^2 = 0 \quad S = -3\,BP \Rightarrow$

3 Branches will coincide

$\frac{-dK}{dS} = 0 \rightarrow (S+3)^n = 0 \quad S = -3 \Rightarrow$

$(n+1)$ *Branches will coincide*

6. Intersection with jω-axis:

$1 + G(S)H(S) = S^3 + 9S^2 + KS + K = 0$

No.intersection

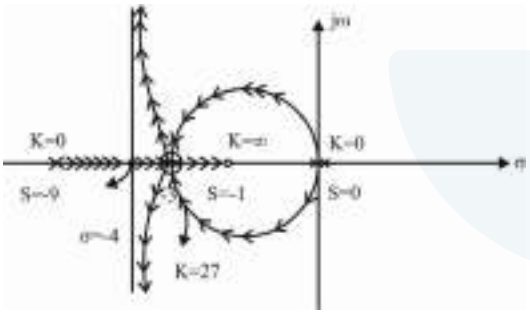


Fig. 7.38

Case 3:

$G(S)H(S) = \frac{K(S+1)}{S^2(S+2)} \quad 0 < K < \infty \quad -ve\,F/B \rightarrow DRL$

S.1 $S_P = 0, 0, -2 \rightarrow No\ \theta_d \quad \alpha = 2$

$S_Z = -1 \rightarrow No\ \theta_a$

S.2 P = 3 P – Z = 2

Z = 1

S.3 $\sigma = -0.5$

$\theta = 90^0, 270^0$

S.4

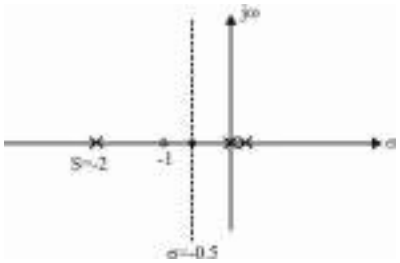


Fig. 7.39

S.5 Break points: S = 0 → valid

$S = -1.25 \pm j0.66 \rightarrow Invalid$

S.6 Intersection with jω-axis → No

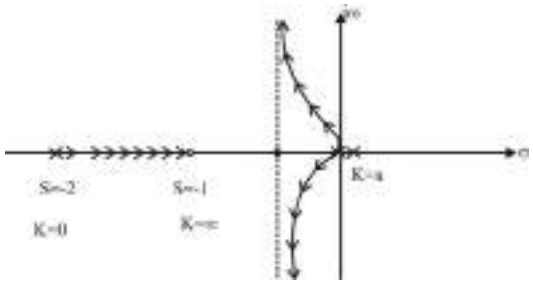


Fig. 7.40

Case 4:

$G(S)H(S) = \frac{K(S+1)}{S^2(S+0.1)} \quad 0 < K < \infty \quad -ve\,F/B \rightarrow DRL$

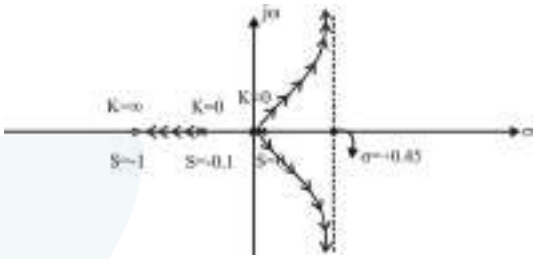


Fig. 7.41

Model 2: 'Problems in which some part of Root locus is circle'

Case 1: $G(S)H(S) = \frac{K(S+b)}{S(S+a)}$

$0 < K < \infty$

$-ve\,F/B$

$1 + G(S)H(S) = 0$

$\frac{K(S+b)}{S(S+a)} = -1$

$S = \sigma + j\omega$

a> 0

b> 0

a < b

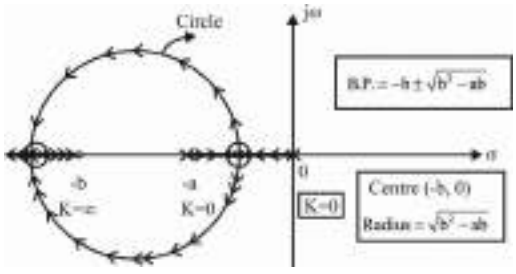


Fig. 7.42

$$G(S)H(S) = \frac{K(s+b)}{S(s+a)} = T(S)$$

$$\frac{1}{a+jb} \Rightarrow -\tan^{-1}\frac{b}{a}$$

$$a+jb \Rightarrow \tan^{-1}\frac{b}{a}$$

$$T(\sigma+j\omega) = \frac{(\sigma+j\omega+b)}{(\sigma+j\omega)(\sigma+j\omega+a)}$$

$$\angle T = \tan^{-1}\frac{\omega}{\sigma+b} - \tan^{-1}\frac{\omega}{\sigma} - \tan^{-1}\frac{\omega}{\sigma+a} = 180^0$$

$$\tan^{-1}\frac{\omega}{\sigma+b} - \tan^{-1}\frac{\omega}{\sigma} = 180^0 + \tan^{-1}\frac{\omega}{\sigma+0}$$

$$\tan^{-1}\left[\frac{\omega\sigma - \omega\sigma - \omega b}{\sigma(\sigma+b)+\omega^2}\right] = 180^0 + \underbrace{\tan^{-1}\left(\frac{\omega}{\sigma+a}\right)}_{\phi}$$

$$\frac{-\omega b}{\omega^2+\sigma(\sigma+b)} = \frac{\omega}{\sigma+a}$$

$$-b\sigma - ab = \omega^2 + \sigma^2 + \sigma b$$

$$\omega^2 + \sigma^2 + 2b\sigma + b^2 = b^2 - ab$$

$$(\sigma+b)^2 + (\omega-o)^2 = \left(\sqrt{b^2-ab}\right)^2$$

Note: $\tan(180+\phi)$

$$\tan\phi = \tan\left[\tan^{-1}\frac{\omega}{\sigma+a}\right] = \frac{\omega}{\sigma+a}$$

E.g.

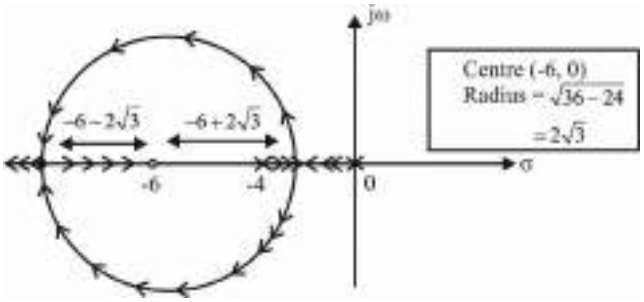


Fig. 7.43

Case 2:

$$G(S)H(S) = \frac{-K(S-b)}{S(S+a)} \qquad \begin{matrix} 0 < K < \infty \\ -ve\,F/B \end{matrix} \rightarrow CRL$$

a> 0

b> 0

Break point = $b \pm \sqrt{b^2+ab}$

Centre = (b, 0)

Radius = $\sqrt{b^2+ab}$

$j\omega = \pm j\sqrt{ab}$

at K = a

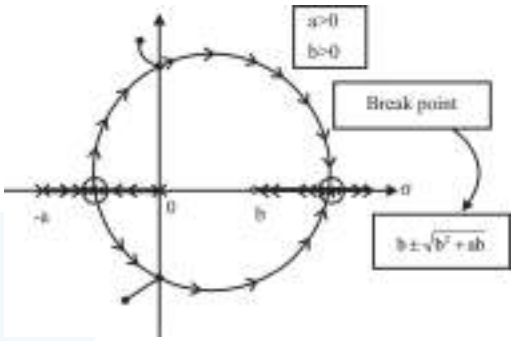


Fig. 7.43

$$T(S) = \frac{-K(S-b)}{S(S+a)} \quad -ve\,F/B \quad 1+G(S)H(S) = 0$$

$$1 - K\frac{(S-b)}{S(s+a)} = 0$$

$$\frac{K(S-b)}{S(S+a)} = 1 \rightarrow CRL$$

$$\frac{K(\sigma+j\omega-b)}{(\sigma+j\omega)(\sigma+j\omega+a)} = 1$$

$$\tan^{-1}\frac{\omega}{\sigma-b} - \tan^{-1}\frac{\omega}{\sigma} - \tan^{-1}\frac{\omega}{\sigma+a} = 0^0$$

Case 3:

$$G(S)H(S) = \frac{KS}{(S-a)(S-b)} \qquad \begin{matrix} 0 < K < \infty \\ -ve\,F/B \end{matrix} \rightarrow DRL$$

a **>** 0 b > a

b >0

$BP = 0 \pm \sqrt{ab}$

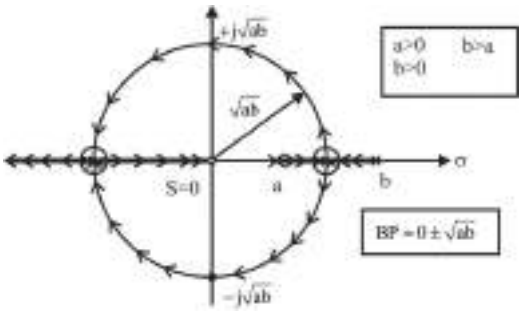


Fig. 7.44

$$1+G(S)H(S)=0$$

$$1+\frac{KS}{(S-a)(S-b)}=0$$

$$\frac{KS}{(S-a)(S-b)}=-1$$

$$\frac{K(\sigma+j\omega)}{(\sigma-a+j\omega)(\sigma-b+j\omega)}=-1$$

$$\underbrace{\tan^{-1}\frac{\omega}{\sigma}-\tan^{-1}\frac{\omega}{\sigma-a}-\tan^{-1}\frac{\omega}{\sigma-b}=180^{0}}_{Circle}$$

Case 4:

$$G(S)H(S)=\frac{-KS}{(S+a)(S+b)}\quad \begin{matrix}0<K<\infty\\ -ve\,F\,/\,B\end{matrix}\rightarrow CRL$$

b> a

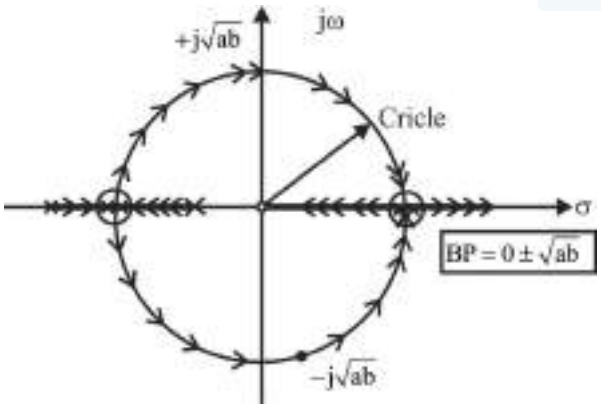


Fig. 7.45

Centre = (0, 0)

Radius = $\sqrt{ab}$

$j\omega=\pm j\sqrt{ab}$

at K = a + b

$C.E.\rightarrow -ve\,F\,/\,B\quad 1+G(S)H(S)=0$

$$1-\frac{KS}{(S+a)(S+b)}=0$$

$$\frac{KS}{(S+a)(S+b)}=1$$

$$\frac{K(\sigma+j\omega)}{(\sigma+j\omega+a)(\sigma+j\omega+b)}=1$$

$$\tan^{-1}\frac{\omega}{\sigma}-\tan^{-1}\frac{\omega}{\sigma+a}-\tan^{-1}\frac{\omega}{\sigma+b}=0$$

Case 5:, $G(S)H(S)=\dfrac{K(S+a)^{2}}{(S+b)^{2}}$

$0<K<\infty\quad -ve\,F\,/\,B$

a > 0

b < 0

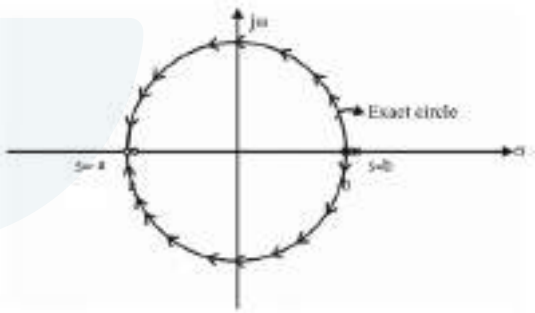


Fig. 7.46

DRL

$$\angle T(\sigma+j\omega)=\pi$$

$$2\tan^{-1}\frac{\omega}{\sigma+a}-2\tan^{-1}\frac{\omega}{\sigma+b}=\pi$$

$$\tan^{-1}\left[\frac{\frac{\omega}{\sigma+a}-\frac{\omega}{\sigma+b}}{1+\frac{\omega^{2}}{(\sigma+a)(\sigma+b)}}\right]=\frac{\pi}{2}$$

$$(\sigma+a)(\sigma+b)+\omega^{2}=0$$

$$\sigma^{2}+\sigma(a+b)+ab+\omega^{2}=0$$

$$\sigma^{2}+\sigma(a+b)+\left(\frac{a+b}{2}\right)^{2}+\omega^{2}=\left(\frac{a+b}{2}\right)^{2}-ab$$

$$\left\{\sigma+\left(\frac{a+b}{2}\right)\right\}^{2}+\omega^{2}=\left(\sqrt{\frac{a^{2}+b^{2}}{2}}\right)^{2}$$

Centre = $\left(\frac{-(a+b)}{2}, 0\right)$

Radius = $\sqrt{\frac{a^2+b^2}{2}}$

Model 3:

$$G(S)H(S) = \frac{K}{S(S+a)(S^2+bS+C)} \quad \underbrace{-ve\,F\,/\,B}_{DRL} \quad o < K < \infty$$

Case 1:

$$G(S)H(S) = \frac{K}{S(S+2)(S^2+2S+2)} \quad \underbrace{-ve\,F\,/\,B}_{DRL} \quad 0 < K < \infty$$

S.1 $S_P = 0, -2, -1 \pm i \rightarrow \theta_d$

$S_Z \rightarrow No$ zero = No θ_a

S.2

$P = 4$
$Z = 0$ $\rightarrow P - Z = 4$

S.3 Centroid, $\sigma = -1$

$\theta = 45^0, 135^0, 225^0, 315^0$

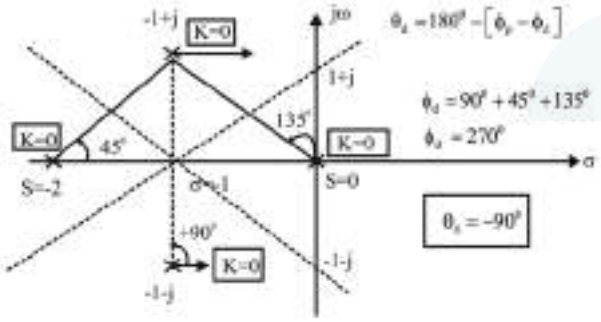


Fig. 7.47

S.5 Break point: $\frac{-dK}{dS} = 0$

$\frac{-dK}{dS} = (S+1)^3 = 0$

$\underbrace{S = -1}_{valid} \rightarrow$ *Branches wil meet up*

$K_{(S=-1)} = 1$

S.6 Intersection with j-axis $\rightarrow S = \pm j$
K at BP = 5

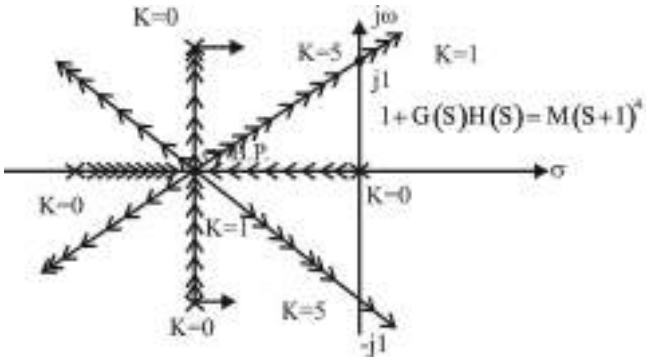


Fig. 7.48

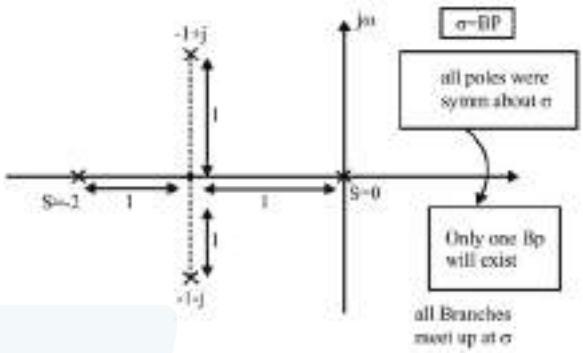


Fig. 7.49

E.g

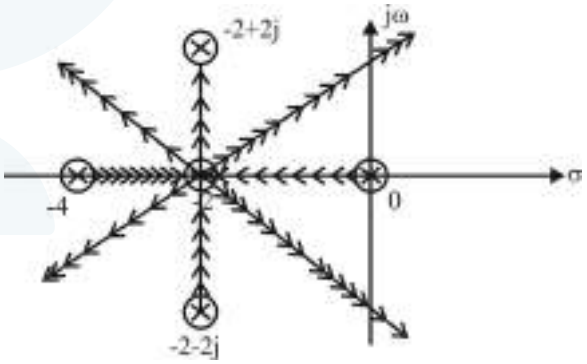


Fig. 7.50

Case 2:

$$G(S)H(S) = \frac{K}{S(S+2)(S^2+2S+1.25)}$$

$0 < K < \infty$

$\underbrace{-ve\,F\,/\,B}_{DRL}$

1. $S_P = 0, -2, -1 \pm j0.5 \rightarrow \theta_d$
$S_Z = 0$

2. P – Z = 4

3. $\sigma = -1$

$\theta = 45^0, 135^0, 225^0, 315^0$

4.

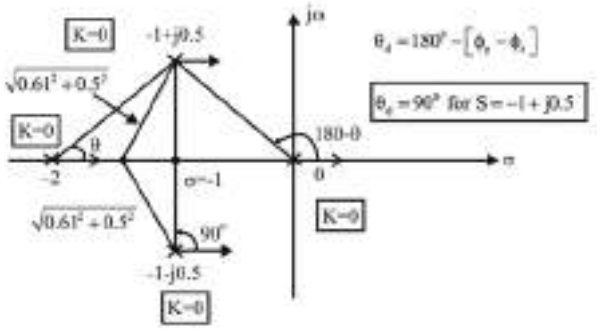


Fig. 7.51

5. Break points:

$S = -0.387$

$S = -1 \quad \rightarrow$ *All terms are valid*

$S = -1.61$

$(S + 0.387)(S + 1)(S + 1.61) = 0$

$S = 2$ *Branches*

$K_{[S=-0.387]} = 0.39$

$K_{[S=-1]} = \frac{1}{4}$

$K_{[S=-1.61]} = 0.5$

6. Intersection with jω-axis:

$S = \pm j0.8$

At $K = 2.89$

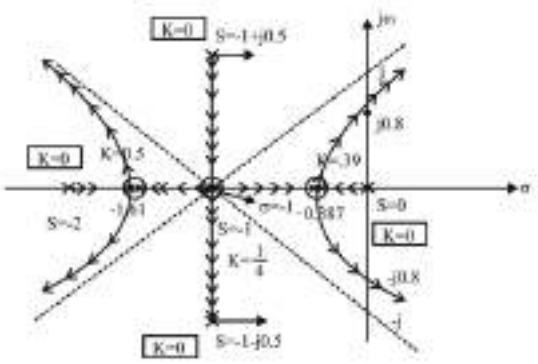


Fig. 7.52

Note:

$\sigma = BP$

Complex poles were closer to σ

3 real break points

Case 3:

$$G(S)H(S) = \frac{K}{S(S+2)(S^2+2S+10)} \qquad \begin{matrix} 0 < K < \infty \\ -ve\ F\ /\ B \end{matrix}$$

S.1 $S_p = 0, -2, -1 \pm 3j$

$S_z = No$ zero

S.2 P – Z = 4

S.3 $\sigma = -1$

$\theta = 45^0, 135^0, 225^0, 315^0$

S.4

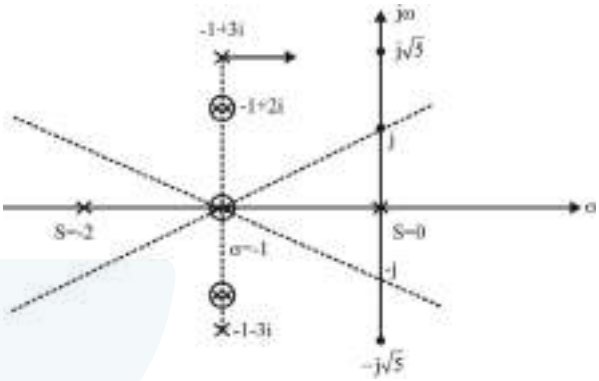


Fig. 7.43

S.5 Break points:

$S = -1 \rightarrow$ *valid*

$S = -1 \pm 2j \rightarrow$ *valid*

$K_{[S=-1]} = 9$

$K_{[S=-1\pm 2j]} = 25$

S.6 Intersection with jω-axis:

$S = \pm j\sqrt{5}$

$K_{[S=j\sqrt{5}]} = 45$

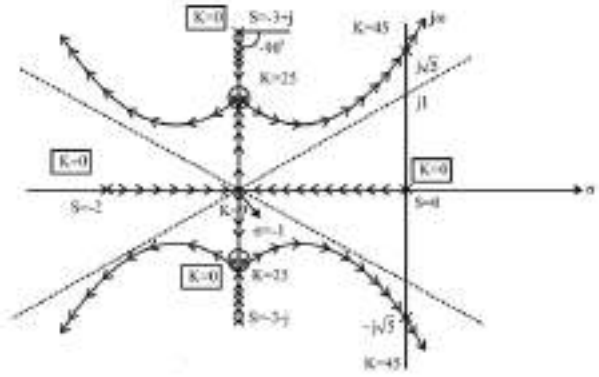


Fig. 7.54

Note:

$\sigma = BP$

Complex poles were far away from σ compared to real poles

$3\,Break\ point \begin{cases} 1\,Real \\ 2\,Complex \end{cases}$

Mode 3: Conclusions:

$$G(S)H(S) = \frac{K}{S(S+a)(S^2+bS+C)}$$

$S^2 + bS + C \rightarrow Complex\ Roots$

Case 1:

1. $\sigma = BP$
2. Real poles are symmetric WRT $\sigma \rightarrow$ 1 Real BP
3. Complex poles are symmetric WRT σ
4. $\text{Re}\{pole\} = \sigma = BP$

Case 2:

1. $\sigma =$ One of the BP = Real BP
2. Real poles are symmetric WRT σ 3 Real BP
3. Complex poles are symmetric WRT σ (Near to centroid compared to real poles)
4. $\text{Re}\{pole\} = \sigma$ = One of the Real BP

Case 3:

1. $\sigma =$ One of the BP = Real BP
2. Real poles are symmetric WRT $\sigma \rightarrow 3BP = \begin{matrix} 1\text{Re}al \\ 2\,Complex \end{matrix}$
3. Complex poles are symmetric WRT σ (For away from centroid compared to real poles)
4. $\text{Re}\{pole\} = \sigma$ = One of the real BP

Case 4: IF all poles or some of the poles are not symmetric about σ.

$$G(S)H(S) = \frac{K}{S(S+a)(S^2+bs+C)}$$

1. $a \neq b \rightarrow$ 1 Real BP will exist
2. a = b $\rightarrow$ 3 BP will exist

$$\frac{ax}{2} = C \begin{cases} x \geq 5 \rightarrow 1\text{Re}al\ 2\,complex \\ x < 5 \rightarrow 3\text{Re}al \end{cases}$$

Root Locus of Non-Minimum Phase System

Minimum phase system: All poles and zeros of the I.F. should be in L.H.P.

Non-minimum phase system: Which is not the minimum phase.

Maximum phase system: All poles and zeros of the T.F. should be in R.H.P.

1. $T(S) = \frac{(1+S)}{(2+S)}$ $\qquad T(S = j\omega) = \frac{1+j\omega}{2+j\omega}$

   Frequency response

   $$\angle T(S = j\omega) = \tan^{-1}\omega - \tan^{-1}\frac{\omega}{2}$$

2. $T(S) = \frac{(1-S)}{(2+S)}$

   $$T(S = j\omega) = \frac{1-j\omega}{2+j\omega} = -\tan^{-1}\omega - \tan^{-1}\frac{\omega}{2}$$

Note: All the rules of DRL/CRL. are valid only when the coefficient of Sare



Eg. $G(S)H(S) = \frac{K}{S+2}$ $\qquad G(S)H(S) = \frac{2K}{S+2}$

$s = -(2+K)$ $\qquad S = -2-2K = -2(1+K)$

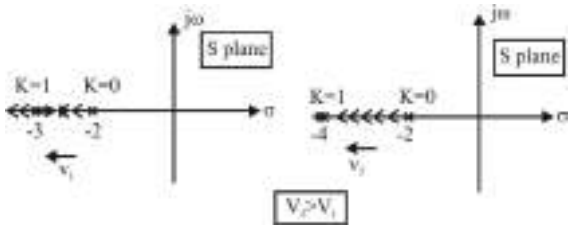


Fig. 7.55

Example 26: The CLTF of a unity gain negative feedback system is $\frac{C(S)}{R(S)} = \frac{1}{S^2+(4-K)S+3}$, where $0 < K < \infty$. Obtain OLTF and plot Root locus diagram.

Solution:

$$\frac{C(S)}{R(S)} = \frac{1}{S^2+4S+3-KS}$$

$$\frac{C(S)}{R(S)} = \frac{\frac{1}{S^2+4S+3}}{1+\left\{\frac{-KS}{S^2+4S+3}\right\}}$$

$$G(S)H(S) = \frac{-KS}{(S+1)(S+3)} \quad \begin{matrix} 0 < K < \infty \\ -ve\,F\,/\,B \end{matrix} \rightarrow CRL$$

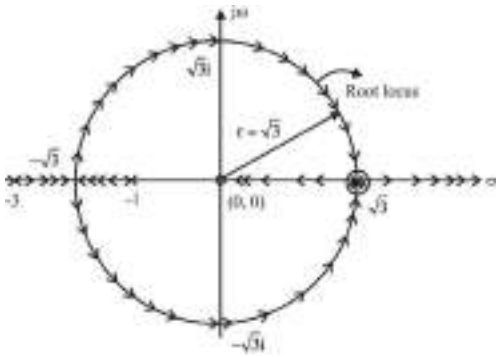


Fig. 7.55

Points To Remember

- To plot Root locus we always need a format.
- $G(S)H(S) = Variable\ parameter \dfrac{N(S)}{D(S)}$
- IF the above format is not achieved, then write CE (1 ± G(S)H(S)). Regenerate the G(S)H(S) in the above format.

Root contour:

1. It is the Root locus when multiple parameters vary in OLTF
2. Root contour is collection of multiple Root locus diagrams.
3. When multiple parameters are variable then one parameter is considered at a time by choosing proper constant value for other parameters and individual Root locus are plotted.

Root Locus of Transportation LAG System

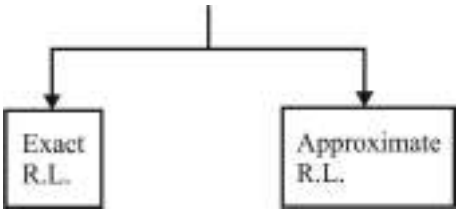


Fig. 7.56

$e^{-x} = 1 - x$

Non-minimum phase $\{-(x-1) \Rightarrow proceed\}$

Example 27:

$$G(S)H(S) = \frac{Ke^{-S}}{S(S+8)} \quad \begin{matrix} 0 < K < \infty \\ -ve\,F\,/\,B \end{matrix}$$

Solution: $G(S)H(S) = \dfrac{K(1-S)}{S(S+8)}$

$$G(S)H(S) = \frac{-K(S-1)}{S(S+8)} \rightarrow C.R.L.$$

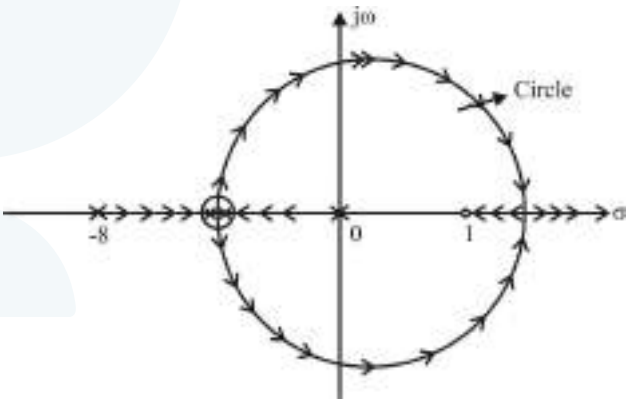


Fig. 7.57

Important Points

While calculating BP:

K = F(S)

$$\frac{dK}{dS} = 0 \Rightarrow \frac{dF}{dS} = 0 \Rightarrow \frac{dF(S)}{dS} = 0$$

$$\frac{dF(S)}{dS} = G(S)$$

Critical points:

$S = S_1\ \ S_2\ \ S_3$

$S = S_1 \rightarrow Valid\ BP$

Wavy curve of G(S):

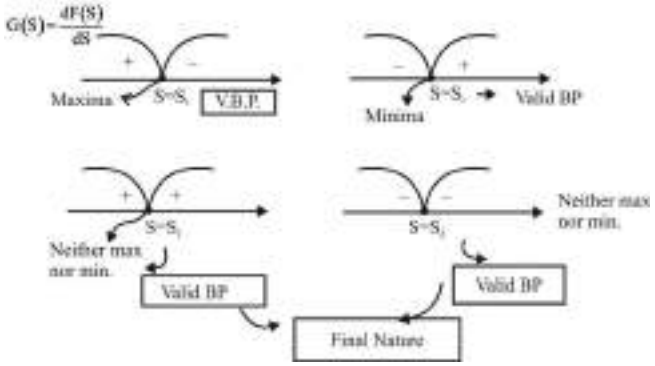


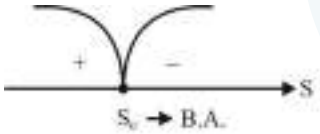
Fig. 7.58

K= F(S)

1st order derivable test:

$$\left(\frac{d^2K}{dS^2}\right)_{S=S_0} < 0 \Rightarrow S = S_0 \underset{\text{max ima}}{\rightarrow} B.A.\ point$$

$$\left(\frac{d^2K}{dS^2}\right)_{S=S_0} > 0 \Rightarrow S = S_0 \underset{\text{minima}}{\rightarrow} B.In\ point$$



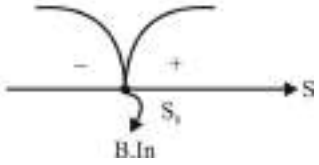


Fig. 7.59

2nd order derivable test:

$$\left(\frac{d^2K}{dS^2}\right)_{S=S_0} < 0 \Rightarrow S = S_0 \underset{\text{max ima}}{\rightarrow} B.A.\ point$$

$$\left(\frac{d^2K}{dS^2}\right)_{S=S_0} > 0 \Rightarrow S = S_0 \underset{\text{minima}}{\rightarrow} B.I.\ point$$

Example 29:

$$G(S)H(S) = \frac{K}{S(S+2)(S^2+2S+1.25)} \quad 0 < K < \infty$$

$-ve\ F\ /\ B$

Solution: $\frac{dK}{dS} = -(S+2)\ (S+.387)(S+1.61)$

$S = -2 \rightarrow B.A.$

$S = -1.61 \rightarrow B.In$

$S = -0.387 \rightarrow B.A.$

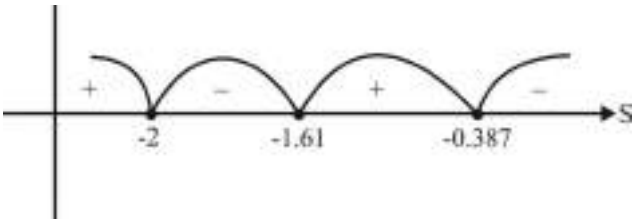


Fig. 7.60

Important notes:

1. 'Root locus branches are symmetrical about real axis'.
2. 'Total no. of branches originating will always be the order of CE'
3. For K = 0: Location of open-loop poles itself becomes the location of closed-loop poles.
4. For K = ∞: Location of open-loop zeros itself becomes the location of C.L. poles
5.

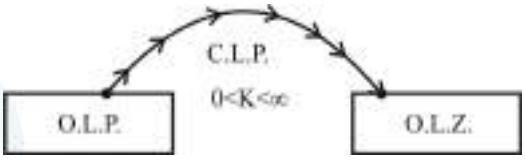


Fig. 7.61

Common mistakes:

- Sometimes students directly proceed with given closed-loop system, kindly convert closed-loop into open-loop before drawing Root locus diagram.
- Carefully observe the nature of feedback and analyse the type of Root locus (direct or complementary).
- Carefully identIFy break in and break away points.
- Carefully identIFy the poles existing on Root locus.

Exercise Problems

1. The characteristic equation of a system is

$$s^3 + 3s^2 + (K+2)s + 3K = 0$$

In the Root locus plot for the given system, as K varies from 0 to ∞, the break-away or break-in point(s) lie within

(A) $-\infty, -3$ (B) $-3, -2$

(C) $-1, 0$ (D) $-2, -1$

Solution: (C)

2. A linear-time invariant (LTI) system with the transfer function

$$G(s) = \frac{K(s^2 + 2s + 2)}{(s^2 - 3s + 2)}$$ is connected in

unity feedback configuration, as shown in the figure.



For the closed-loop system shown, the Root locus for $0 < K < \infty$ intersects the imaginary axis for K = 1.5. The closed-loop system is stable for **[GATE Set-01]**

(A) K > 1.5

(B) 1 < K < 1.5

(C) 0 < K < 1

(D) no positive value of K

Solution: (A)

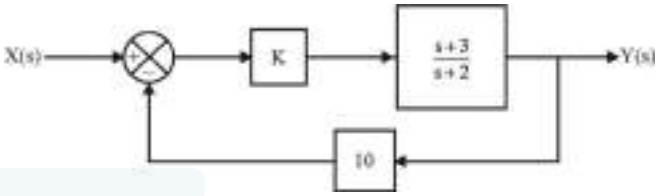
3. The forward-path transfer function and the feedback-path transfer function of a single loop negative feedback control system are given as $G(s) = \frac{K(s+2)}{s^2 + 2s + 2}$ *and* $H(s) = 1$, respectively. IF the variable parameter K is real positive, then the location of the break-away point on the Root locus diagram of the system is _____. **[GATE Set-03]**

Solution: -3.41

4. The open-loop transfer function of a unity-feedback control system is $G(s) = \frac{K}{s^2 + 5s + 5}$. The value of K at the break-away point of the feedback control system root-locus plot is **[GATE Set-01]**

Solution: 1.25

5. For the system shown in the figure, s = –2.75 lies on the Root locus IF K is _____. **[GATE Set-03]**



Solution: 0.3

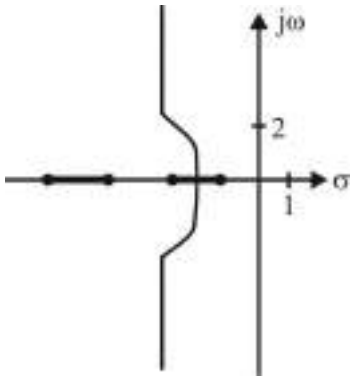
6. A unity negative feedback system has the open-loop transfer function

$$G(s) = \frac{K}{s(s+1)(s+3)}$$

The value of gain K(> 0) at which the Root locus crosses the imaginary axis is **[GATE Set-01]**

Solution: 12

7. In the Root locus plot shown in the figure, the pole/zero marks and the arrows have been removed. Which one of the following transfer functions has this Root locus? **[GATE Set-03]**



(A) $\frac{(s+1)}{(s+2)(s+4)(s+7)}$

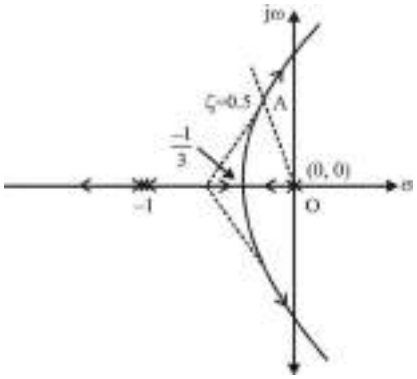
(B) $\frac{(s+4)}{(s+1)(s+2)(s+7)}$

(C) $\frac{(s+7)}{(s+1)(s+2)(s+4)}$

(D) $\frac{(s+1)(s+2)}{(s+7)(s+4)}$

Solution: (B)

8. The characteristic equation of a unity negative feedback system is 1 + kG(s) = 0. The open-loop transfer function G(s) has one pole at 0 and two poles at –1. The Root locus of the system for varying K is shown in the figure. **[GATE Set-04]**



The constant damping ratio line, for $\zeta = 0.5$, intersects the Root locus at point A. The distance from the origin to point A is given as 0.5. The value of K at point A is _____.

Solution: 0.375

Exercise Problems Solution

1. Solution: (C)

Given,

The characteristic equation is,

$$C.E = s^3 + 3s^2 + (2+K)s + 3K = 0$$

Converting to standard form,

$$s^3 + 3s^2 + 2s + Ks + 3K = 0$$

$$s^3 + 3s^2 + 2s + K(s+3) = 0$$

$$1 + \frac{K(s+3)}{s^3 + 3s^2 + 2s} = 0$$

$$1 + \frac{K(s+3)}{s(s^2 + 3s + 2)} = 0$$

$$1 + \frac{K(s+3)}{s(s+1)(s+2)} = 0$$

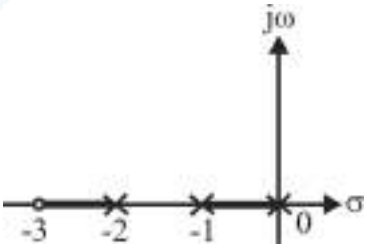
The above equation is of the standard form,

$$1 + G(s)H(s) = 0$$

Where G(s)H(s) is the open loop transfer function.

$$G(s)H(s) = \frac{K(s+3)}{s(s+1)(s+2)}$$

Drawing the pole-zero diagram,



The thick lines lie on the root locus, and thin lines do not lie on the root locus. From the prediction of break-away or break-in point, IF there are consecutively placed poles on the real axis, with the section of the real axis between them as a part of RL, then there exists at least one break-away or break-in point between them. Since there are poles located at -1 and 0, the break-away point will lie between -1 and 0.

Hence, the correct option is (C).

2. Solution: (A)

Given,

Open loop transfer function for a unity feedback system is given by,

$$G(s) = \frac{K(s^2 + 2s + 2)}{(s^2 - 3s + 2)}$$

The characteristics equation is given by,

$1+G(s)=0$

$$1+\frac{K\left(s^2+2s+2\right)}{s^2-3s+2}=0$$

$$s^2-3s+2+K\left(s^2+2s+2\right)=0$$

$$s^2(1-K)+s(-3+2K)+(2+2K)=0$$

Routh tabulation:

s^2	$1+K$	$2+2K$
s^1	$-3+2K$	0
s^0	$2+2K$	0

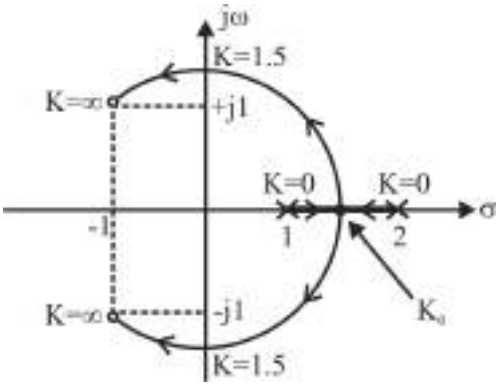
For the system to be stable, all the roots must be in the left-half of the s-plane; thus, all the coefficients in the first column of Routh's tabulation must have the same sign. Therefore, the coefficient of first column of the Routh's table should be positive.

This leads to the following conditions:

1. $-3+2K>0 \Rightarrow K>1.5$
2. $2+2K>0$

   $1+K>0 \Rightarrow K>-1$

   On combining the above two conditions, Thus, for overall stability, K >1.5, the root locus diagram is shown below,



Hence, the correct option is (A).

3. Solution: -3.41

Given, $G(s)=\dfrac{K(s+2)}{s^2+2s+2}$

$H(s)=1$

The characteristics equation is given by,

$1+G(s)H(s)=0$

$$1+\frac{K(s+2)}{s^2+2s+2}=0$$

$$K=\frac{-\left(s^2+2s+2\right)}{(s+2)}$$

Break-away/Break-in point:

$$\frac{dK}{ds}=0$$

$$\frac{d}{ds}\left[\frac{(2s+2)(s+2)-\left(s^2+2s+2\right)}{(s+2)^2}\right]=0$$

$$(2s+2)(s+2)-\left(s^2+2s+2\right)=0$$

$$2s^2+6s+4-s^2-2s-2=0$$

$$s^2+4s+2=0$$

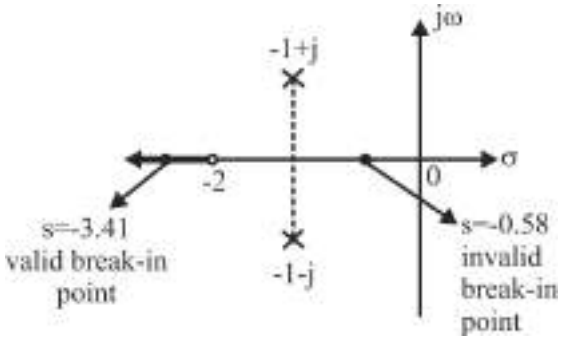
$$s=\frac{-4\pm\sqrt{16-8}}{2}=\frac{-4\pm2\sqrt{2}}{2}$$

$$s=-2\pm\sqrt{2}$$

$s=-0.58, s=-3.41$

Since point s = -3.41 is lying between two adjacent zeros (one physical zero and one virtual zero) on the root locus branch of the real axis, hence it is a valid break-in point.

Since, point s = -0.58 is not lying on the root locus branch of real axis, hence it is an invalid break-in point.



Hence, the location of the break-in point is -3.41.

4. Solution: 1.25

Given,

The open-loop transfer function of a unity-feedback control system is,

$$G(s)=\frac{K}{s^2+5s+5}$$

The characteristics equation is,

$$1+G(s)=0$$

$$1+\frac{K}{s^2+5s+5}=0$$

$$K=-s^2-5s-5$$

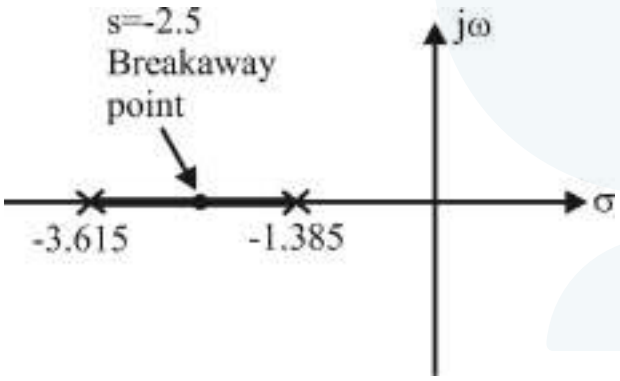
Break-away/Break-in point:

$$\frac{dK}{ds}=0$$

$$\frac{dK}{ds}=-2s-5=0$$

$$s=-2.5$$

Since point s = -2.5 is lying between two adjacent poles on the root locus of the real axis, hence it is a break-away point.



Since the break-away point will lie on the root locus. Hence, from magnitude condition,

$$\left|G(s)\right|_{s=-2.5}=1 \qquad ...1$$

Magnitude of G(s) at s = -2.5 is given by,

$$\left|G(s)\right|_{s=-2.5}=\frac{K}{\left|(-2.5)^2+5\times(-2.5)+5\right|}$$

$$\left|G(s)\right|_{s=-2.5}=\frac{K}{1.25} \qquad ...2$$

From the equations 1 and 2,

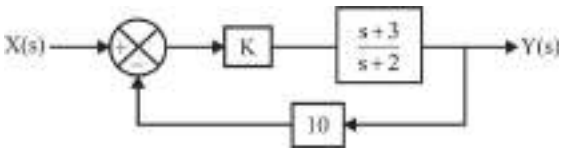
$$\frac{K}{1.25}=1$$

$$K=1.25$$

Hence, the value of K at the break-away point is 1.25.

5. Solution: 3

Given,

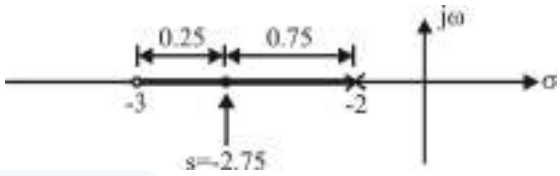


Method 1:

Graphical calculation of K:

$$G(s)H(s)=\frac{10K(s+3)}{(s+2)}=\frac{K'(s+3)}{(s+2)}$$

Where, K' = 10 K



$$K'=\frac{\text{Product of vector lengths drawn from the poles of G(s)H(s) to s}}{\text{Product of vector lengths drawn from the zeros of G(s)H(s) to s}}$$

$$K'=\frac{0.75}{0.25}=3$$

$$10K=3$$

$$K=0.3$$

Hence, the value of K is 0.3

Method 2:

From magnitude criteria,

$$\left|G(s)H(s)\right|_{s=-2.75}=1$$

$$\frac{K\times 10\left|-2.75+3\right|}{\left|-2.75+2\right|}=1$$

$$K=\frac{0.75\times 1}{0.25\times 10}=0.3$$

Hence, the value of K is 0.3.

6. Solution: 12

Given, the open loop transfer function of a unity negative feedback system is,

$$G(s)=\frac{K}{s(s+1)(s+3)}$$

Method 1:

The value of K at the point where the root locus branch crosses the imaginary axis is determined by applying Routh Hurwitz criterion to the characteristic equation.

The characteristics equation is given by,

$$1+G(s)H(s)=0$$

$$1+\frac{K}{s(s+1)(s+3)}=0$$

$$s^2+4s^2+3s+K=0$$

Routh tabulation:

S^3	1	3
S^2	4	K
S^1	$\frac{12-K}{4}$	0 ← Row of zeros
S^0	K	0

The intersection of the root locus plot with imaginary axis is given by the value of K, obtained by solving the following equation:

$$\frac{12-K}{4}=0 \Rightarrow K=12$$

Hence, the value of gain K (>0) at which the root locus crosses the imaginary axis is 12.

Method 2:

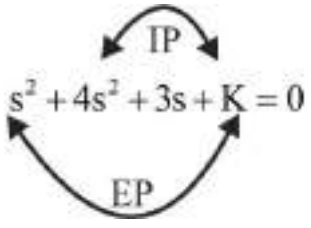
The characteristics equation is given by,

$$1+G(s)H(s)=0$$

$$1+\frac{K}{s(s+1)(s+3)}=0$$

$$s^3+4s^2+3s+K=0$$

For the third order system to be marginally stable,



Internal product (IP) = External product (EP)

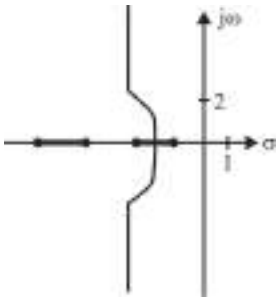
$$4\times 3=1\times K$$

K = 12

Hence, the value of gain K (> 0) at which the root locus crosses the imaginary axis is 12.

7. Solution: (B)

The given root locus plot is shown below,



1. For options (A), (B) and (C)

Since P > Z branches will start from each open loop pole and terminate at open loop zeros. The remaining P – Z branches will approach to infinity. The angle of asymptotes is given by,

$$\angle A=\frac{(2\alpha+1)\times 180^0}{P-Z}$$

Where $\alpha=0,1,2,.....(P-Z-I)$

$\alpha=0,1$

$$\angle A=90^0,270^0$$

Root locus branches will approach to infinity with angles $90^0$and 270^0.

The intersection point of asymptotes on the real axis is called centroid.

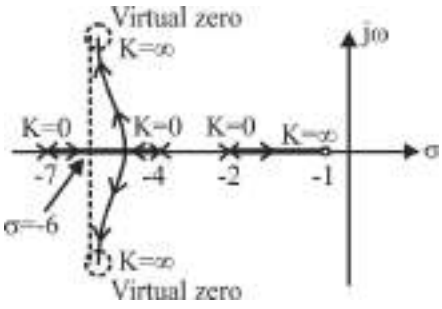
Centroid is given by,

$$\sigma=\frac{\sum \mathrm{Re}(Poles)-\sum \mathrm{Re}(Zeros)}{P-Z}$$

2. For option (A),

$$\sigma=\frac{-2-4-7+1}{3-1}=-6$$

The root locus can be drawn as shown below,

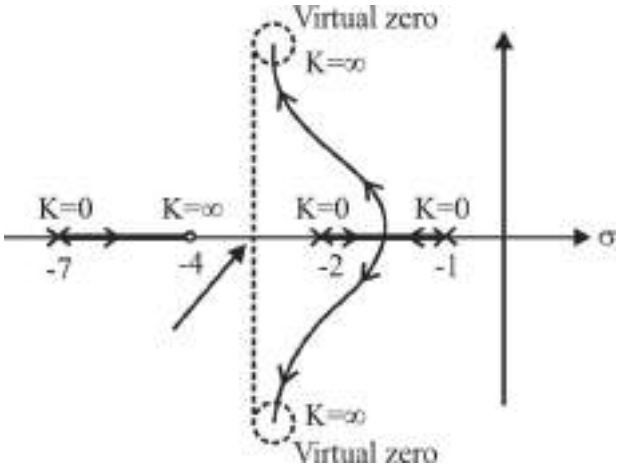


Thus, option (A) is incorrect.

3. For option (B)

$$\sigma = \frac{-1-2-7+4}{3-1} = -3$$

The root locus can be drawn as shown below,

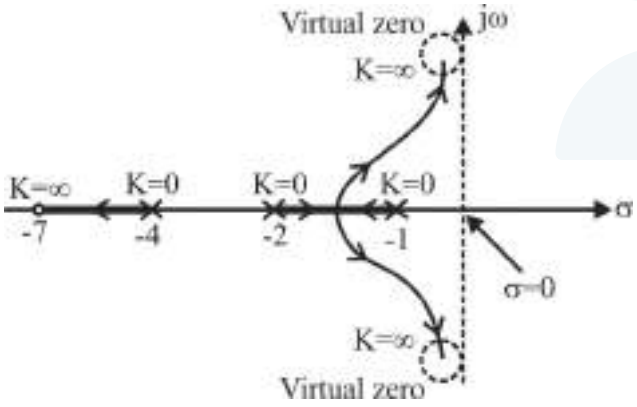


Thus, option (B) is correct.

4. For option (C),

$$\sigma = \frac{-1-2-4+7}{3-1} = 0$$

The root locus can be drawn as shown below,



Thus, option (C) is incorrect.

5. For option (D),

Number of poles, P = 2

Number of zeros, Z = 2

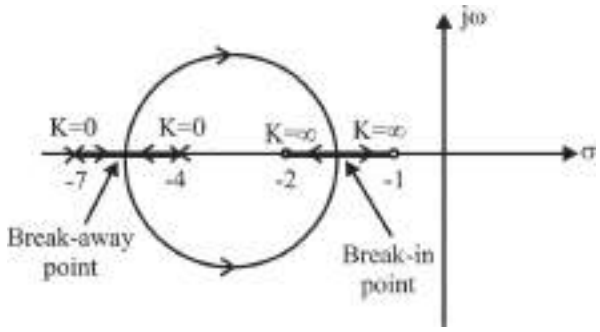
Number of branches, B = 2

Number of asymptotes, A = 0

Angle of asymptotes does not exist.

Centroid does not exist.

The root locus can be drawn as shown below,



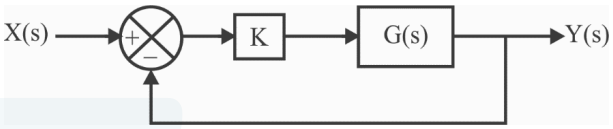
Thus, option (D) is incorrect.

Hence, the correct option is (B).

8. Solution: 0.375

Given,

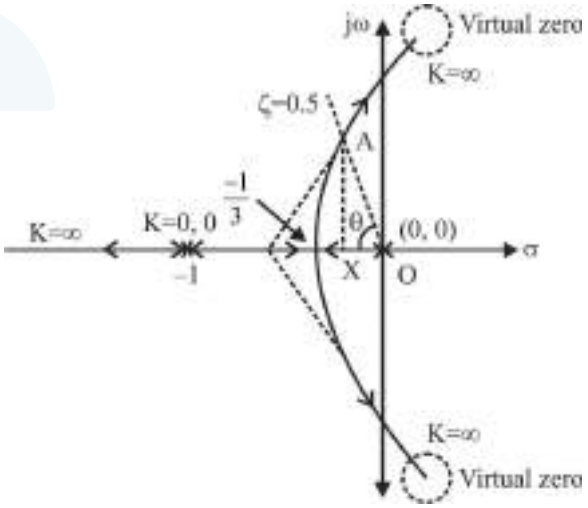
1. Open loop transfer function G(s) has one pole at 0 and two poles at -1.
2. $\xi = 0.5,\ 0A = 0.5$



$$G'(s) = KG(s) = \frac{K}{s(s+1)^2}$$

Method 1:

The given root locus for varying K is shown below,



$$\theta = \cos^{-1}\xi \Rightarrow \theta = 60^0$$

$$\text{In } \Delta AXO,\ \cos\theta = \frac{OX}{OA}$$

$$OX = OA\cos 60^0 = 0.5 \times \frac{1}{2} = \frac{1}{4}$$

$$\sin\theta = \frac{AX}{AO}$$

$$AX = AO\sin 60^0 = 0.5 \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$$

Hence, the coordinate of A is $\frac{-1}{4} + j\frac{\sqrt{3}}{4}$.

At $s_l = \frac{-1}{4} + j\frac{\sqrt{3}}{4}$

$$G'(s_l) = \frac{K}{\left(\frac{-1}{4} + j\frac{\sqrt{3}}{4}\right)\left[\left(\frac{-1}{4} + j\frac{\sqrt{3}}{4}\right) + 1\right]^2}$$

$$G'(s_l) = \frac{K}{\left(\frac{-1}{4} + j\frac{\sqrt{3}}{4}\right)\left[\left(\frac{3}{4} + j\frac{\sqrt{3}}{4}\right)\right]^2}$$

$$\left|G'(s_l)\right| = \frac{K}{\left(\frac{1}{16} + \frac{3}{16}\right)^{\frac{1}{2}}\left(\frac{9}{16} + \frac{3}{16}\right)^{2\times\frac{1}{2}}}$$

$$\left|G'(s_l)\right| = \frac{K}{\frac{1}{2} \times \frac{12}{16}} = \frac{16K}{6} \quad \ldots 1$$

Since the point s_l is lying on the root locus, hence from the magnitude condition,

$$\left|G'(s_l)\right| = 1 \quad \ldots 2$$

Note: There is no need to check the angle condition.

From equations1and 2,

$$\frac{16K}{6} = 1 \Rightarrow K = \frac{3}{8} = 0.375$$

Hence, the value of K at point A is 0.375.

8 Exact Frequency Response Analysis

Objectives

After completing this chapter, the user will be able to understand

- The frequency domain specifications like resonant peak, resonant frequency, and bandwidth
- How to find the output of a system for a sinusoidal input
- The frequency domain plots

Introduction

- When any system is subjected to sinusoidal input, the output is also sinusoidal in nature,having different magnitude and phase angles but the same frequency ω rad/sec.

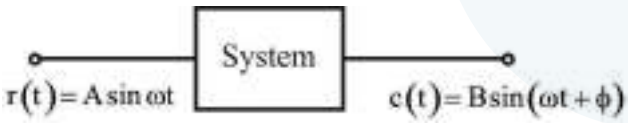


Fig. 8.1

- Frequency response analysis implies varying ω from 0 to ∞ and observing corresponding variations in the magnitude and phase angle of the response.
- For a system with very large time constants, the frequency test is cumbersome to perform as the time required for the outputs to reacha steady-state for each frequency of the test signal is excessively long.
- Therefore, frequency response analysis is not recommended for a system with large time constants.

Advantages of Frequency Response Analysis

1. Analysis is easy even for higher-order systems as there exists a number of plots which are not limited by the size of systems.
2. Time-domain approaches like the Routh criterion establish the stability of the system easily, but relative stability is very difficult to determine. Whereas relative stability is much easier to determine using frequency domain analysis.
3. Systems noise sensitivity can be analysed with frequency response.

Frequency Response Analysis

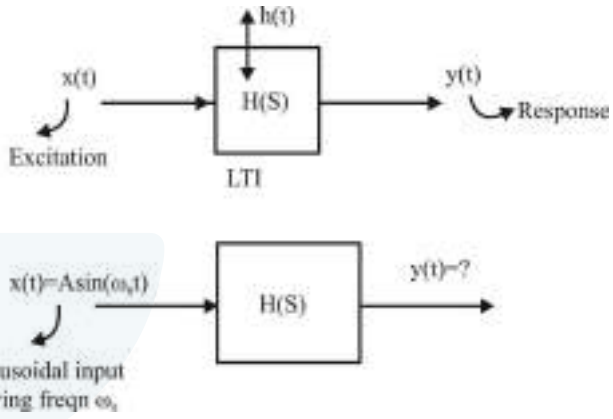


Fig. 8.2

Let H(S): Rational function

$$H(S)=\frac{b'_1}{(S+S_1)}+\frac{b'_2}{(S+S_2)}+\ldots\frac{+b'_n}{(S+S_n)}$$

"Stable system".

$$Y(S)=X(S)H(S)=\frac{A\omega_0}{S^2+\omega_0^{\ 2}}\times H(S)$$

$$\frac{a_1}{S+j\omega_0}+\frac{a_2}{S-j\omega_0}+\frac{b_1}{(S+S_1)}+\frac{b_2}{(S+S_2}$$

$$\frac{a_2}{S-j\omega_0}+\frac{b_1}{(S+S_1)}$$

$$y(t)=\underbrace{a_1e^{-j\omega_0t}+a_2e^{j\omega_0t}}_{\text{Steady state responise}}+\underbrace{b_1e^{-S_1t}+b_2e^{-S_2t}}_{\text{Transient responise}}+\ldots\ldots\ldots$$

$$\text{state}:\{y(t)\}_{ss}=\{y(t)\}_{t\to\infty}=\left[a_1e^{-j\omega_0t}\right.$$

$$\{t)\}_{ss}=\{y(t)\}_{t\to\infty}=$$

$$e^{j\omega_0t}=\cos\omega_0t+j\sin\omega_0t$$

$$e^{j\omega_0(\infty)}=\cos\omega_0\infty+j\sin\omega_0\infty\to Finite$$

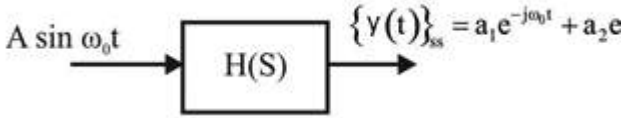


Fig. 8.3

- L.T.I. system
- Stable system

$$a_1 = \lim_{S \to -j\omega_0} \left\{ \frac{A\omega_0}{(S+j\omega_0)(S-j\omega_0)} \times \frac{}{(S+j\omega_0) \times H(S)} \right\} = \frac{-AH(-j\omega_0)}{2j}$$

$$a_2 = \lim_{S \to +j\omega_0} \left\{ \frac{A\omega_0}{(S+j\omega_0)(S-j\omega_0)} \times \frac{}{(S-j\omega_0) \times H(S)} \right\} = \frac{AH(j\omega_0)}{2j}$$

$$\{y(t)\}_{ss} = \frac{-A}{2j} e^{-j\omega_0 t} H(-j\omega_0) + \frac{A}{2j} e^{j\omega_0 t} H(j\omega_0)$$

$$h(t) \xleftrightarrow{F.T.} h(j\omega) = |H(j\omega)| e^{j\angle H(j\omega)}$$

$$h(t) : Real \rightarrow |H(j\omega)| \rightarrow Even$$

$$\rightarrow |H(j\omega)| = |H(-j\omega)|$$

$$\underset{phase\ spectrum}{\angle H(j\omega)} = odd \rightarrow \angle H(j\omega) = -\angle H(-j\omega)$$

$$H(j\omega) = |H(j\omega)| e^{j\angle H(j\omega)}$$

$$H(-j\omega) = |H(-j\omega)| e^{+j\angle H(-j\omega)} = |H(j\omega)| e^{-j\angle H(j\omega)}$$

$$\{y(t)\}_{ss} = \frac{A}{2j} e^{j\omega_0 t} |H(j\omega_0)| e^{j\angle H(j\omega_0)} -$$

$$\frac{A}{2j} e^{-j\omega_0 t} |H(j\omega_0)| e^{-j\angle H(j\omega_0)}$$

$$\{y(t)\}_{ss} = A|H(j\omega_0)| \frac{\left[e^{j(\omega_0 t + \angle H(j\omega_0))} - e^{-j(\omega_0 t + \angle H(j\omega_0))} \right]}{2j}$$

$$\{y(t)\}_{ss} = A|H(j\omega_0)| \sin(\omega_0 t + \angle H(j\omega_0))$$

1.

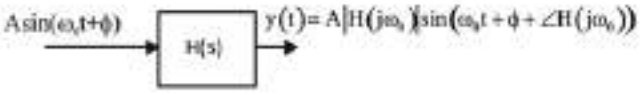


Fig. 8.4

y(t) is final the value or steady-state value of output

or

y(t) is the frequency response of the L.T.I.system.

$$H(j\omega) \left\langle \begin{matrix} |H(j\omega_0)| \\ \angle H(j\omega_0) \end{matrix} \right.$$

h(t) : Real

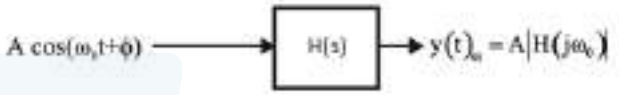
$$h(t) \leftrightarrow H(j\omega)$$

h(t) : Real

L.T.I. system

Stable system

2.



$$\cos(\omega_0 t + \phi + \angle H(j\omega_0))$$

Fig. 8.5

y(t) is the frequency response of the L.T.I.system.

$$H(j\omega) \left\langle \begin{matrix} |H(j\omega_0)| \\ \angle H(j\omega_0) \end{matrix} \right.$$

$$h(t) \leftrightarrow H(j\omega)$$

h(t) : Real

L.T.I. system

Stable system

For Frequency Response Analysis

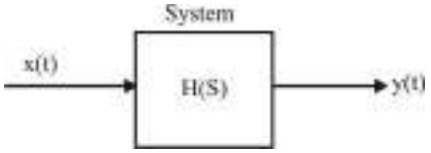


Fig. 8.6

$H(S) \rightarrow Transfer\ \ Function$

$Put\ \ S = j\omega$

F.T. of h(t)

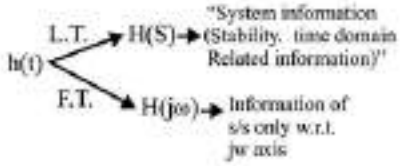


Fig. 8.7

Band width → Frequency related information

$$h(t) = e^{-2t}u(t) \leftrightarrow \underset{\substack{\text{complete S-plane} \\ \text{information}}}{H(S)} = \underbrace{\frac{1}{S+2}}_{\substack{\text{"pole-zero} \\ \text{analysis"}}} \rightarrow \textit{No plot}$$

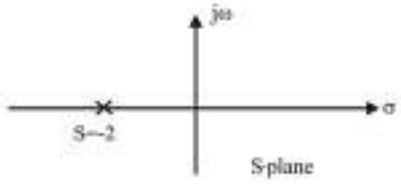


Fig. 8.8

S domain analysis → Time domain analysis

$$h(t) = e^{-2t}u(t) \xleftrightarrow{F.T.} H(j\omega) = \frac{1}{2 + j\omega}$$

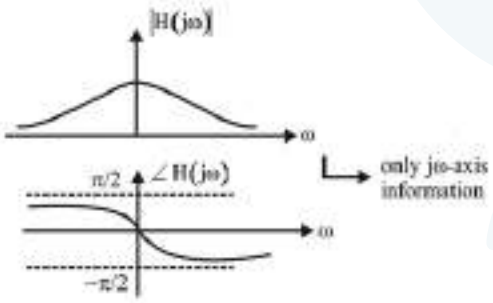


Fig. 8.9

$$\left|H(j\omega)\right| = \frac{1}{\sqrt{4+\omega^2}}$$

$$\angle H(j\omega) = -\tan^{-1}\omega/2$$

$S = j\omega \Rightarrow$ *Frequency line Analysis* → *F.T.*
Freq^n Domain Analysis

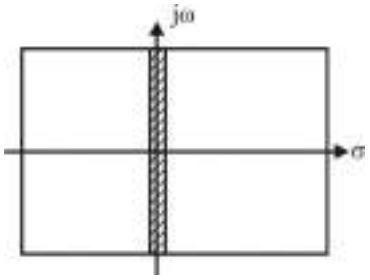


Fig. 8.10

Complete S–plane analysis → L.T. ⇒ Time-Domain Analysis.

$$G(S)H(S) = \frac{K(S)}{(S-1)(S+2)} \xrightarrow{\text{Root Locus}}$$

$S = \sigma + j\omega \rightarrow$ *Time Domain*

$$G(S)H(S) = \frac{K}{S\left(\frac{S}{a}+1\right)} \xrightarrow{\text{Bode plot}}$$

$S = j\omega \rightarrow$ *Frequency Domain Analysis*

1.

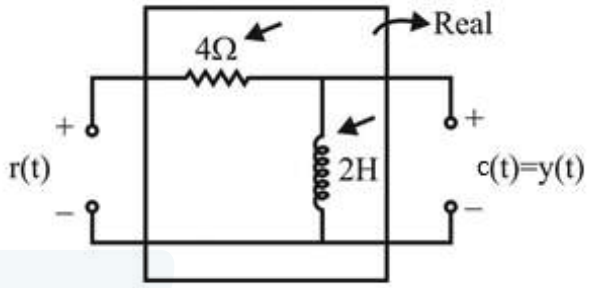


Fig. 8.11

Find the output of the network given above, if input to the network is r(t) = 2 sin 2t → $\omega_0 = 2$

Solution:

$$\xleftrightarrow{\text{"}h(t)\text{:Real"}} H(S) = \frac{S}{S+2} \leftrightarrow H(j\omega)$$

$$= \frac{j\omega}{j\omega+2} = \frac{j2}{2+j2} = \frac{1}{\sqrt{2}}\angle 45^0$$

$$Y(S) = H(S)X(S) = \frac{4}{S^2+4} \times \frac{S}{S+2}$$

$$= \frac{a_0}{(S+2j)} + \frac{a_1}{(S-2j)} + \frac{a_2}{(S+2)}$$

$$Y(S) \leftrightarrow y(t)$$

$$\{y(t)\}_{ss} = 2\left|H(j\omega_0)\right|\sin\left(2t + \angle H(j\omega_0)\right)$$

$$= 2\left|H(j2)\right|\sin\left(2t + \angle H(j2)\right) = \frac{2}{\sqrt{2}}\sin\left(2t + \frac{\pi}{4}\right)$$

Frequency Domain Analysis of 1st Order System

1. "Frequency domain analysis of 1st order system".

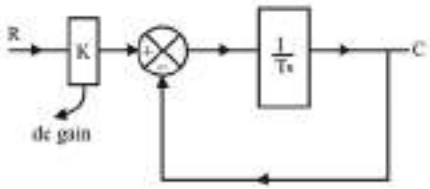


Fig. 8.12

$$\frac{C(S)}{R(S)} = T(S) = \frac{K\frac{1}{T}}{S+\frac{1}{T}}$$

For positive frequency line put

$S = j\omega : 0 \le \omega < \infty$

$$T(j\omega) = \frac{K\frac{1}{T}}{j\omega+\frac{1}{T}} \begin{cases} |T(j\omega)| = \dfrac{\frac{K}{T}}{\sqrt{\omega^2+\frac{1}{T^2}}} = \dfrac{K}{\sqrt{1+\omega^2T^2}} \\ \angle T(j\omega) = -\tan^{-1}(\omega T) \end{cases}$$

ω	$\lvert T(j\omega)\rvert$	$\angle T(j\omega)$
0	K	0
$\frac{1}{T}$	$\frac{K}{\sqrt{2}}$	-45^0
∞	$\to 0$	-90^0

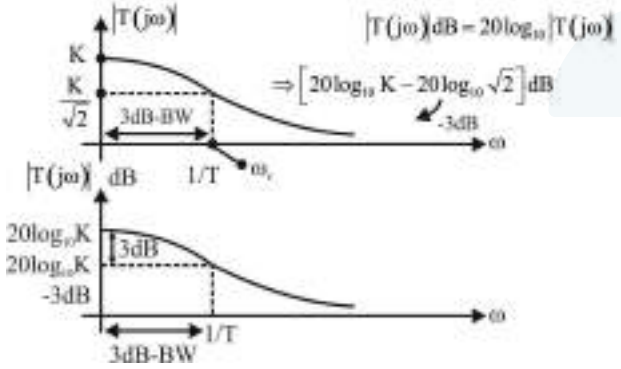


Fig 8.13

For K = 1:

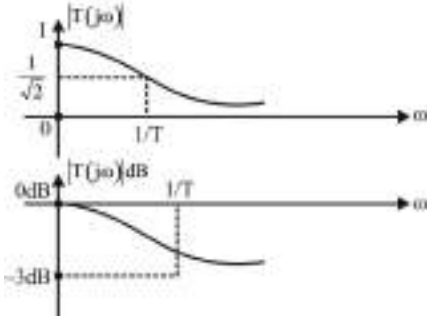


Fig 8.14

$$H(S) = \frac{\frac{K}{T}}{S+\frac{1}{T}} \leftrightarrow h(t) = \frac{K}{T}e^{-\frac{t}{T}}$$

$$3\,dB \;\; B.W. = \frac{1}{Time\ constant}$$

$$\omega = 0 \to |T(j\omega)|_{max} \Rightarrow peak\ value$$

$$\omega = \frac{1}{T} \to |T(j\omega)| = \frac{|T(j\omega)|_{max}}{\sqrt{2}}$$

Frequency Domain Analysis Of 2nd Order System

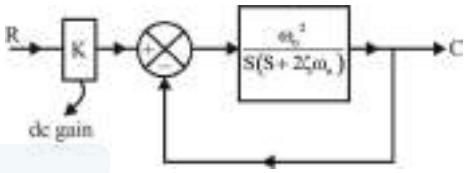


Fig. 8.15

$$\frac{C(S)}{R(S)} = \frac{K\omega_n^2}{S^2+2\zeta\omega_n S+\omega_n^2} = T(S)$$

For Positive Frequency Line Put

$$S = j\omega(0 \le \omega < \infty)$$

$$T(j\omega) = \frac{K\omega_n^2}{-\omega^2+2\zeta\omega_n j\omega+\omega_n^2}$$

$$T(j\omega) = \frac{K}{\left(1-\frac{\omega^2}{\omega_n^2}\right)+j\left\{\frac{2\zeta\omega}{\omega_n}\right\}}$$

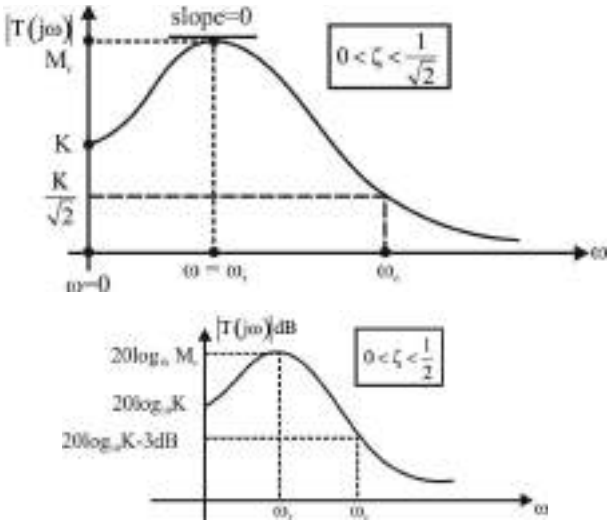
Let

$$u = \frac{\omega}{\omega_n} \to Normalized\ frequency$$

$$T(ju) == \frac{K}{(1-u^2)+j(2\zeta u)}$$

$$|T(ju)| = \frac{K}{\sqrt{(1-u^2)^2+(2\zeta u)^2}}$$

$$\angle T(ju) = -\tan^{-1}\left[\frac{2\zeta u}{(1-u^2)}\right]$$



For K = 1:

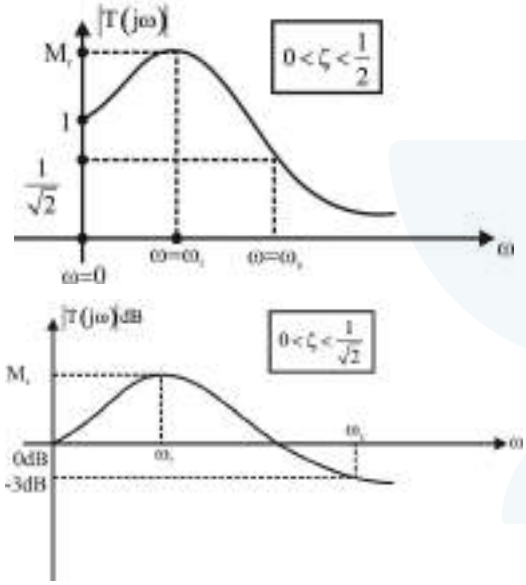


Fig. 8.16

Resonance frequency (ω_r)

It is the frequency at which the magnitude of the sinusoidal transfer function has a maximum value.

Calculation of resonant frequency: (ω_r)

$$\left|T(ju)\right| = \frac{K}{\sqrt{\left(1-u^2\right)^2 + \left(2\zeta u\right)^2}}$$

At u =u_r

$$\frac{d\left|T(ju)\right|}{du} = 0 = \frac{-1\left[2\left(1-u^2\right)(-2u) + 8\zeta^2 u\right]}{2\left[\left(1-u^2\right)^2 + \left(2\zeta u\right)^2\right]^{\frac{3}{2}}} = 0$$

$$u = \frac{\omega}{\omega_n} \qquad 4u\left(1-u^2\right) = 8\zeta^2 u$$

$$u = \sqrt{1-2\zeta^2} \Rightarrow \omega_r = \omega_n\sqrt{1-2\zeta^2}$$

Common mistake:

- Carefully read the question, what is being asked? resonant frequency or damped frequency.
- Students write the wrong formula for resonant frequency in a hurry.

$$\omega_r = \omega_n\sqrt{1-2\zeta^2} \rightarrow \text{Real when } 1-2\zeta^2 \geq 0$$

$$\Rightarrow \zeta \leq \frac{1}{\sqrt{2}}$$

$$\zeta = 0 \rightarrow \omega_r = \omega_n \Rightarrow \left|T\left(j\omega_r\right)\right| \Rightarrow \infty$$

$$0 < \zeta \leq \frac{1}{\sqrt{2}} \rightarrow \omega_r = \omega_n\sqrt{1-2\zeta^2}$$

$$\Rightarrow \left|T\left(j\omega_r\right)\right| = M_r + peak$$

$$\zeta > \frac{1}{\sqrt{2}} \rightarrow \omega_r \rightarrow Not\ calculated\ by\ formula,$$

does note exist

$$\Rightarrow \left|T\left(j\omega_r\right)\right| \rightarrow No\ peak$$

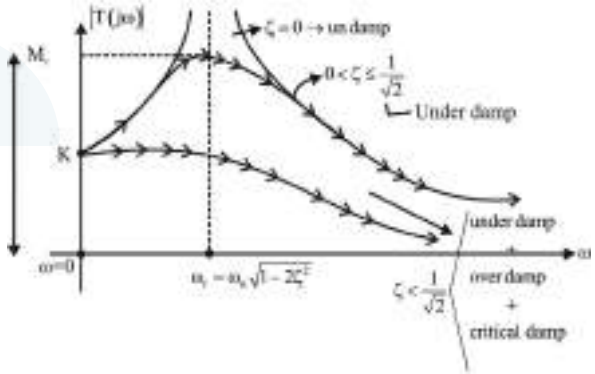


Fig. 8.17

Resonant peak (M_r):

It is the maximum value of magnitude M occurring at a resonant frequency ω_r.

$$M_r = \left|T\left(ju_r\right)\right| = \frac{K}{\sqrt{\left(1-u_r\right)^2 + \left(2\zeta u_r\right)^2}}$$

$$u_r = \sqrt{1-2\zeta^2}$$

$$M_r = \frac{K}{2\zeta\sqrt{1-\zeta^2}}$$

$$M_r = \begin{cases} \infty & \zeta = 0 \\ \dfrac{K}{2\zeta\sqrt{1-\zeta^2}} & 0 < \zeta \le \dfrac{1}{\sqrt{2}} \end{cases}$$

For $\zeta > \dfrac{1}{\sqrt{2}}$

- Not calculated from formula
- Not defined
- $\omega_r = 0 \rightarrow K$

Points To Remember

1. $T(S) = \dfrac{K\omega_n^2}{S^2 + 2\zeta\omega_n S + \omega_n^2} \rightarrow dc\ gain = K$

2. Resonant frequency $\Rightarrow \omega_r = \omega_n\sqrt{1-2\zeta^2}$

$$\omega_r = \begin{cases} \omega_r = \omega_n & \zeta = 0 \\ \omega_r = \omega_n\sqrt{1-2\zeta^2} & 0 < \zeta \le \dfrac{1}{\sqrt{2}} \\ 0 & \zeta > \dfrac{1}{\sqrt{2}} \end{cases}$$

3. **Resonant peak:** 'Don't forget dc gain'

$$M_r = \frac{K}{2\zeta\sqrt{1-\zeta^2}}$$

$$M_r = \begin{cases} \infty & \zeta = 0 \\ \dfrac{K}{2\zeta\sqrt{1-\zeta^2}} & 0 < \zeta \le \dfrac{1}{\sqrt{2}} \\ K & \zeta > \dfrac{1}{\sqrt{2}} \end{cases}$$

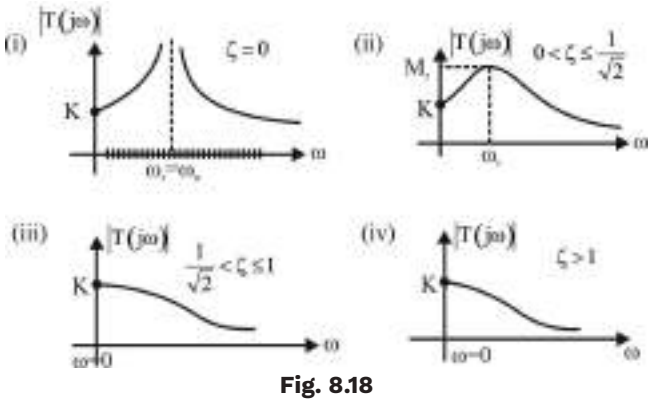


Fig. 8.18

3-DB Cut-Off Frequency

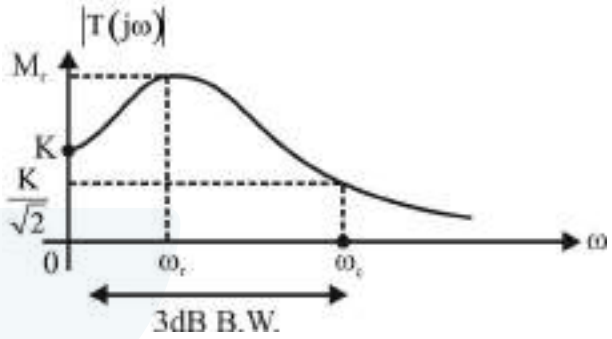


Fig. 8.19

$$|T(j\omega_c)| = \frac{|T(jo)|}{\sqrt{2}}$$

$$|T(j\omega_c)| \neq \frac{|T(j\omega)|_{max}}{\sqrt{2}}$$

Cut-off Frequency

It is the frequency at which the magnitude of the closed-loop transfer function has a value over $\dfrac{1}{\sqrt{2}}$. It indicates the ability of the system to distinguish the signal from noise.

$$|T(ju)| = \frac{K}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}}$$

$$|T(jo)| = K$$

$$\frac{K}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}} = \frac{K}{\sqrt{2}}$$

$$\left(1-u^2\right)^2+\left(2\zeta u\right)^2=2$$

$$1+u^4-2u^2+4\zeta^2u^2=2$$

$$u^4-u^2\left[2-4\zeta^2\right]-1=0 \qquad u^2=x$$

$$x^2-x\left(2-4\zeta^2\right)-1=0$$

$$x=\frac{\left(2-4\zeta^2\right)\pm\sqrt{\left(2-4\zeta^2\right)^2+4}}{2}$$

$$u^2=x=\left(1-2\zeta^2\right)\pm\sqrt{\left(1-2\zeta^2\right)^2+1}$$

$$u^2=\left(1-2\zeta^2\right)+\sqrt{\left(1-2\zeta^2\right)^2+1}$$

$$u^2=\left(1-2\zeta^2\right)+\sqrt{\left(1+4\zeta^4-4\zeta^2+1\right)} \quad u=\frac{\omega_c}{\omega_n}$$

$$u=\sqrt{\left(1-2\zeta^2\right)+\sqrt{4\zeta^4-4\zeta^2+2}}$$

$$\omega_c=\omega_n\sqrt{\left(1-2\zeta^2\right)+\sqrt{4\zeta^4-4\zeta^2+2}}$$

$\rightarrow 3-dB\,cut-off\,frequency$

Bandwidth (B.W.):

It is the range of frequencies over which the magnitude has a value of $\frac{1}{\sqrt{2}}$.It indicates the speed of response of the system.

Wider Bandwidth $\Rightarrow$ Faster response

$$BW\propto\frac{1}{t_r}\,where\,t_r=Rise\,time$$

3-dB bandwidth:

$$\omega_{BW}=\omega_c-0$$

$$\omega_{BW}=\omega_n\sqrt{\left(1-2\zeta^2\right)+\sqrt{4\zeta^4-4\zeta^2+2}}\ \frac{rad}{sec}$$

Case 1: $\omega_r=\omega_n\sqrt{1-2\zeta^2}$ $\omega_n=$ constant

As $\zeta\uparrow\left(0<\zeta\le\frac{1}{\sqrt{2}}\right)\rightarrow\omega_r\downarrow$

$\zeta\uparrow\left(\zeta>\frac{1}{\sqrt{2}}\right)\rightarrow\omega_r$=constant

Case 2:

$$M_r=\frac{K}{2\zeta\sqrt{1-\zeta^2}}$$

As

$\zeta\uparrow\left(0<\zeta\le\frac{1}{\sqrt{2}}\right)\rightarrow M_r\downarrow$

$\zeta\uparrow\left(0>\frac{1}{\sqrt{2}}\right)\rightarrow M_r=$ constant

Case 3: 3-dB Bandwidth:

$$\omega_c=\omega_n\sqrt{\left(1-2\zeta^2\right)+\sqrt{4\zeta^4-4\zeta^2+2}}$$

$\rightarrow\omega_c=$ constant

As $\zeta\uparrow\rightarrow\omega_c\downarrow$

$$\left\{\left[\left(1-2\zeta^2\right)>0\right]\cap\left[\left(1-2\zeta^2\right)+1>0\right]\right\}$$

Points To Remember

1. $\omega_r=f\left(\omega_n,\zeta\right)\begin{cases}\zeta=constant:\omega_r\alpha\,\omega_n\\ \omega_n=\text{constant}\ \zeta\uparrow\omega_r\downarrow\end{cases}$
2. $M_r=f\left(\zeta\right)\rightarrow\zeta\uparrow\rightarrow M_r\downarrow$
3. $B_\omega=f\left(\omega_n,\zeta\right)\begin{cases}\zeta=\text{constant}:BW\,\alpha\,\omega_n\\ \omega_n=\text{constant}\ \zeta\uparrow BW\downarrow\end{cases}$

Note: $\omega_c>\omega_n>\omega_r$

Correlation between time Domain and Frequency Domain Specifications

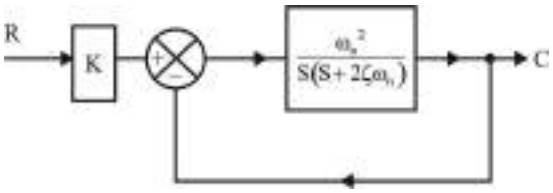


Fig. 8.20

$$\frac{C}{R}=\frac{K\omega_n^{\,2}}{S^2+2\zeta\omega_nS+\omega_n^{\,2}}$$

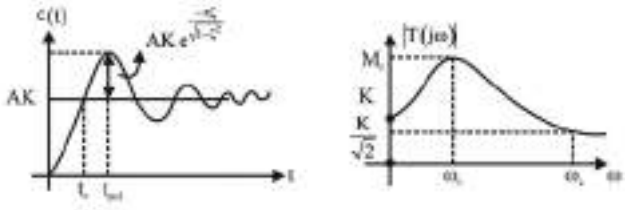


Fig. 8.21

$0 < \zeta < 1 \qquad 0 < \zeta \le \frac{1}{2}$

Time responseq Frequency domain analysis [Unit step response]

1. $t_r = \dfrac{\pi - \phi}{\omega_n \sqrt{1-\zeta^2}}$

   $\omega_r = \omega_n \sqrt{1 - 2\zeta^2 + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$

   $\zeta \to fix$

   $t_r \ \alpha \ \dfrac{1}{3dB - BW}$

2. Peak overshoot: $AKe^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}$

   $\%M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \to f(\zeta)$

   Resonant peak

   $M_r = \dfrac{K}{2\zeta\sqrt{1-\zeta^2}}$

   $f(\zeta)$

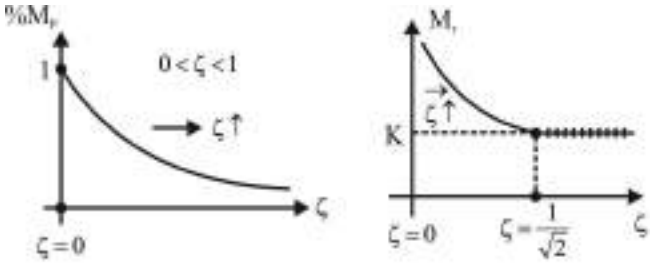
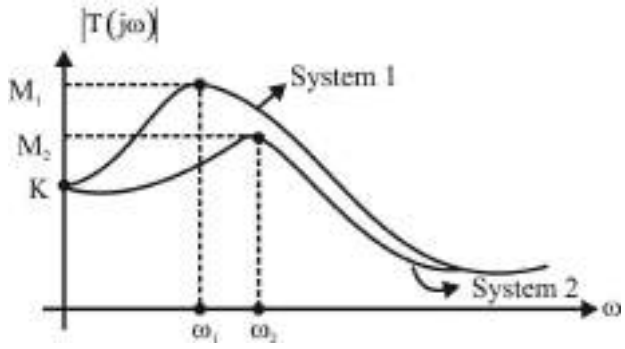


Fig. 8.22

2.



System 1 (vs)System 2 → Which is more stable for constant ω_n.

Solution:

$S\ 1 \to under - damp\left(0 < \zeta \le \dfrac{1}{\sqrt{2}}\right)$

$S\ 2 \to under - damp\left(0 < \zeta \le \dfrac{1}{\sqrt{2}}\right)$

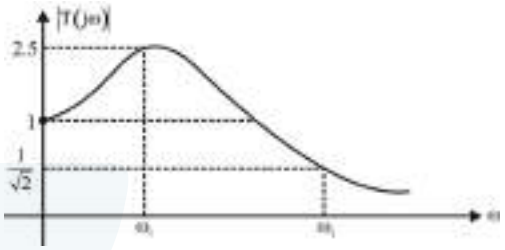
$M_1 > M_2 \qquad as\ \ \zeta \uparrow M_r \downarrow \omega_r \downarrow \omega_c \downarrow$

$\zeta_1 < \zeta_2$

S.1 → Practically achievable

$S.2 \to$ Is not practically achievable.

4.



Calculate ζ of the system whose magnitude spectrum is shown above.

Solution:

$M_r = 2.5$

d.c. gain, K = 1

$M_r = \dfrac{K}{2\zeta\sqrt{1-\zeta^2}} = \dfrac{1}{2\zeta\sqrt{1-\zeta^2}}$

Method 1:

$2\zeta\sqrt{1-\zeta^2} = \dfrac{1}{2.5}$

$\zeta^2\left(1-\zeta^2\right) = \dfrac{1}{25}$

$\zeta^4 - \zeta^2 + .04 = 0 \to x^2 - x + 0.04 = 0$

$\zeta^2 = x$

$x = 0.958$

$x = 0.041$

$\zeta^2 = 0.958 \to \zeta = \pm 0.978 \to \zeta = 0.978$

$\zeta^2 = 0.042 \to \zeta = \pm .204 \to \zeta = 0.204$

$\downarrow 0 < \zeta \le \dfrac{1}{\sqrt{2}}$ *or* $0 < \zeta \le 0.707$

$\zeta = 0.204$

Method 2:

$$\frac{1}{2\zeta\sqrt{1-\zeta^2}} = 2.5$$

$$2\zeta\sqrt{1-\zeta^2} = \frac{1}{2.5}$$

$$\sin 2\theta = \frac{1}{2.5}$$

$$2\theta = \sin^{-1}\left(\frac{1}{2.5}\right) \pm 2n\pi$$

$$\theta = n\pi \pm \frac{1}{2}\sin^{-1}\left(\frac{1}{2.5}\right)$$

$$\theta = 180 - 11.78°$$

$$0 < \zeta < \frac{1}{\sqrt{2}}$$

$\zeta = \text{Cos}\, 11.78°$

$\zeta = 0.978$

$\zeta = -ve$

Frequency Domain Analysis of OLTF

$T(S) = G(S)H(S)$

S.1 Put $S = j\omega$: $\quad 0 \le \omega \le \infty$

S.2 $T(j\omega) = G(j\omega)H(j\omega)$

$|T(j\omega)| = |G(j\omega)H(j\omega)|$

$\angle T(j\omega) = \angle G(j\omega)H(j\omega)$

S.3 $|T(j\omega)|$ vs $\omega \rightarrow$ *Magnitude plot*

$\angle T(j\omega)$ vs $\omega \rightarrow$ *phase plot*

10. $G(S)H(S) = \dfrac{K}{S(1+S)}$

Solution:

$$G(S)H(S) = \frac{K}{S(1+S)} = T(S)$$

$$S.1\ T(j\omega) = \frac{K}{j\omega(1+j\omega)}$$

$$S.2\ \overset{\text{Non linear}}{|T(j\omega)|} = \frac{|K|}{\omega\sqrt{1+\omega_2}}$$

$\rightarrow$ *Magnitudes are not being added*

S.3 $|T(j\omega)|$ vs $\omega \rightarrow$ *Exact Magnitude plot*

$\angle T(j\omega)$ vs $\omega \rightarrow$ *Exact Phase Plot*

$\angle T(j\omega) = -90^0 - \tan^{-1}\omega$

$\rightarrow$ *Phase are being added*

Modification Of Magnitude Calculation

$\ell n = \log_e$

$\log = \log_{10}$

$$|T(j\omega)| = \frac{K}{\omega\sqrt{1+\omega^2}}$$

$$\log_{10}|T(j\omega)| = \log_{10} K - \log_{10}\omega - \frac{1}{2}\log_{10}(1+\omega^2)$$

$\rightarrow$ *Non linear additive*

$$\log_{10}\underset{y}{|T(j\omega)|} = \underset{c}{\log_{10} K} - \underset{m}{2}\underset{x}{\log_{10}\omega}$$

If $\omega >>> 1$

$\log_{10}|T(j\omega)|$ vs $\log_{10}\omega \rightarrow$ *linear + additive*

Exercise Problems

1. The open-loop frequency response of a system at two particular frequencies is given by $1.2\angle -180^0$ *and* $1.0\angle -190^0$. The closed-loop unity feedback control system is

(A) Stable

(B) Unstable

(C) May be stable

(D) May be unstable

Solution: (B)

2. A system has poles at 0.01 Hz, 1 Hz and 80 Hz; zeroes at 5 Hz and 200 Hz. The approximate phase of the system response at 20 Hz is

(A) -90^0 (B) 0^0

(C) 90^0 (D) -180^0

Solution: (A)

3. Non-minimum phase transfer function is defined as the transfer function

(A) Which has zeroes in the right half s-plane

(B) Which has zeroes only in the left half s-plane

(C) Which has poles in the right half s-plane

(D) Which has poles in the left half s-plane

Solution: (A) and (C)

4. If the closed-loop transfer function of a control system is given as

$$T(s)=\frac{(s-5)}{(s+2)(s+3)}$$

Then it is **[ESE-2015]**

(A) An unstable system

(B) An uncontrollable system

(C) A minimum phase system

(D) A non-minimum phase system

Solution: (D)

5. A system with a transfer function $G(s)=\frac{1}{(s+1)(s+a)}, a>0$ is subjected to input 5cos(3t). The steady-state output of the system is $\frac{1}{\sqrt{10}}\cos(3t-1.892)$. The value of "a" is _____.

Solution: 4

6. For a unity feedback control system with the forward path transfer function

$$G(s)=\frac{K}{s(s+2)}$$

The peak resonant magnitude M_r of the closed-loop frequency response is 2. The corresponding value of the gain K (correct to two decimal places) is _____.

Solution: 14.92

7. The magnitude of the frequency response of an under-damped second-order system is 5 at 0 rad/sec and peaks to $\frac{10}{\sqrt{3}}$ at $5\sqrt{2}$ rad/sec. The transfer function of the system is

(A) $\frac{500}{s^2+10s+100}$

(B) $\frac{375}{s^2+5s+7s}$

(C) $\frac{720}{s^2+12s+144}$

(D) $\frac{1125}{s^2+25s+225}$

Solution: (A)

Exercise Problems

1. (B)

Given: Open-loop frequency response at two different frequencies are $1.2\angle -180^0$ and $1.0\angle -190^0$ respectively.

Let the two different frequencies are ω_1 and ω_2

ω	$\lvert G(j\omega)H(j\omega)\rvert$	$\angle G(j\omega)H(j\omega)$
ω_1	1.2	-180^0
ω_2	1.0	-190^0

Table 8.1

The frequency at which phase of the open-loop transfer function will be -180^0, is known as phase crossover frequency.

Hence, $\omega_{pc}=\omega_1$

From the table, gain of the open-loop transfer function at ω_{pc} is 1.2.

Gain margin is given by,

$$GM=\frac{1}{\lvert G(j\omega_{pc})H(j\omega_{pc})\rvert}=\frac{1}{1.2}$$

In dB, $GM=20\log GM$

$$GM = 20\log\frac{1}{1.2} = -1.58dB$$

The frequency at which gain of the open-loop transfer function will be unity, is known as gain crossover frequency.

Hence, $\omega_{gc} = \omega_2$

From the table, the phase angle of theopen-loop transfer function at ω_{gc} is -190^0.

Phase margin is given by,

$$PM = 180^0 + \angle G(j\omega_{gc})H(j\omega_{gc})$$

$$PM = 180^0 - 190^0 = -10^0$$

Since gain margin and phase margin both are negative, hence the system is unstable.

Hence, the correct option is (B).

2. (A)

Given :

(A) Location of poles: 0.01Hz, 1Hz and 80 Hz

(B) Location of zeroes: 5Hz, 100Hz and 200Hz

Hence transfer function of the system is given by,

$$T(s) = \frac{(s+2\pi\times5)(s+2\pi\times100)(s+2\pi\times200)}{(s+2\pi\times0.01)(s+2\pi\times1)(s+2\pi\times80)} \quad ..(i)$$

Put $s = j\omega$ in equation (i),

$$T(j\omega) = \frac{(j\omega+2\pi\times5)(j\omega+2\pi\times100)(j\omega+2\pi\times200)}{(j\omega+2\pi\times0.01)(j\omega+2\pi\times1)(j\omega+2\pi\times80)}$$

$$T(jf) = \frac{(jf+5)(jf+100)(jf+200)}{(jf+0.01)(jf+1)(jf+80)}$$

Phase angle of system is given by,

$$\phi = \tan^{-1}\left(\frac{f}{5}\right) + \tan^{-1}\left(\frac{f}{100}\right) + \tan^{-1}\left(\frac{f}{200}\right)$$

$$-\tan^{-1}\left(\frac{f}{0.01}\right) - \tan^{-1}\left(\frac{f}{1}\right) - \tan^{-1}\left(\frac{f}{80}\right)$$

At $f = 20Hz, \phi$ is given by,

$$\phi = \tan^{-1}\left(\frac{20}{5}\right) + \tan^{-1}\left(\frac{20}{100}\right) + \tan^{-1}\left(\frac{20}{200}\right)$$

$$-\tan^{-1}\left(\frac{20}{0.01}\right) - \tan^{-1}\left(\frac{20}{1}\right) - \tan^{-1}\left(\frac{20}{80}\right)$$

$$\phi = -98.2^0$$

The approximate phase of the system response at $20Hz$ is.

$$\phi \approx -90^0$$

Hence, the correct option is (A).

3. (A) and (C)

Non – minimum phase transfer function has atleast one pole and /or one zero in the right half of s-plane or outside the unit circle in z-plane.

Thus, options (A) and (C) both are correct. But if it is an objective type question, then we will prefer to option (A).

Hence, the correct options are (A) and (C).

4. (D)

Given: $T(s) = \frac{(s-5)}{(s+2)(s+3)}$

Option (A): Poles of the closed-loop transfer function lie in the left half of s-plane; hence, it is a stable system.

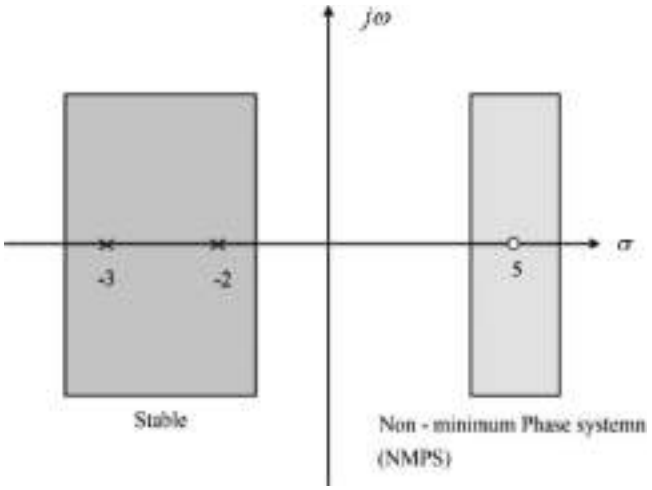
Option (C): Minimum phase transfer function has all poles and zeroes in the left half of s-plane or inside the unit circle in z-plane.

Hence, it is not a minimum phase system.

Option (D): Non – minimum phase transfer function has atleast one pole or one zero in the right half of s-plane.

Since the given closed-loop transfer function has one zero (i.e. s=5) in right of s-plane or outside the unit circle in z-plane.

Hence, it is non – minimum phase system.



Hence,T(s) represents a non – minimum phase and stable system.

Hence, the correct option is (D).

5. (4)

Method 1:

Given: $G(s) = \dfrac{1}{(s+1)(s+a)}, a > 0$

Response $= \dfrac{1}{\sqrt{10}}\cos(3t - 1.892^0)$

$G(j\omega) = \dfrac{1}{(j\omega+1)(j\omega+a)}$

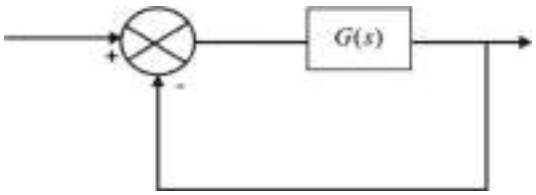
Input = $5\cos 3t$

$\omega = 3$ rad/s

$G(j3) = \dfrac{1}{(j3+1)(j3+a)}$

$|G(j3)| = \dfrac{1}{\sqrt{(9+1)}\sqrt{(9+a^2)}}$

Sinusoidal Response,



Comparing amplitudes,

$5 \times \dfrac{1}{\sqrt{3^2+1^1}\sqrt{9+a^2}} = \dfrac{1}{\sqrt{10}}$

$\dfrac{5}{\sqrt{9+a^2}} = 1$

$25 = 9 + a^2$

$a = 16$

$a = \pm 4$ (as $a > 0$)

So, $a = 4$

Hence,the value of a is 4.

Method 2:

Angle criteria, $-\tan^{-1}\dfrac{\omega}{1} - \tan^{-1}\dfrac{\omega}{a} = -1.892$

$(\omega = 3)$

$-\tan^{-1}\dfrac{3}{a} = -0.643$

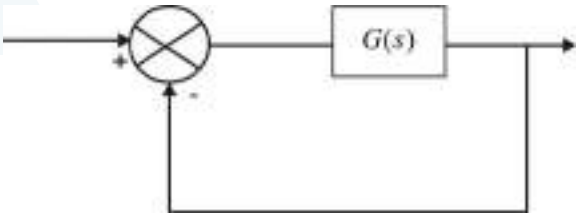
$\dfrac{3}{a} = \tan 0.643 = 0.74$

$a = 4.004 = 4$

Hence, the value of a is 4.

6. (14.92)

Given: For a unity feedback system $G(s) = \dfrac{K}{s(s+2)}$ and resonant peak $M_r, = 2$



Closed-loop transfer function of the unity feedback system is given by,

$\dfrac{C(s)}{R(s)} = \dfrac{G(s)}{1+G(s)} = \dfrac{K/s(s+2)}{1+\dfrac{K}{s(s+2)}}$

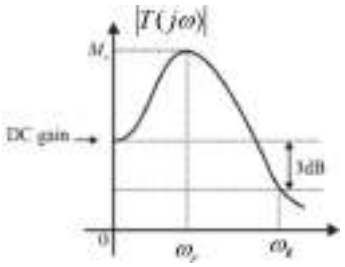
$\dfrac{C(s)}{R(s)} = \dfrac{K}{s^2+2s+K}$

Characteristic equation is,

$s^2 + 2s + K = 0$ ...(i)

One should note that the value of D.C. gain is l, not 'K'.

Resonant peak is given by,



$$M_r = \frac{\text{DC gain}}{2\xi\sqrt{1-\xi^2}}$$

$$\frac{1}{2\xi\sqrt{1-\xi^2}} = 2$$

$$4\xi\sqrt{1-\xi^2} = 1$$

$$16\xi^4 - 16\xi^2 + 1 = 0$$

If we put $\xi^2 = x$, then the equation becomes

$$16x^2 - 16x + 1 = 0$$

$$x = 0.067$$

We have $\xi^2 = x = 0.067$

$$\xi = \sqrt{0.067} = 0.258$$

The characteristic equation for second-order system is given by,

$$s^2 + 2\xi\omega_s s + \omega_n^2 = 0 \qquad \text{...(ii)}$$

On comparing equation (i) and (ii),

$$\omega_n^2 = K \quad \Rightarrow \quad \omega_n = \sqrt{K}$$

and $2\xi\omega_n = 2$

$$\xi = \frac{1}{\omega_n} = \frac{1}{\sqrt{K}}$$

$$K = \frac{1}{\xi^2} = 14.92$$

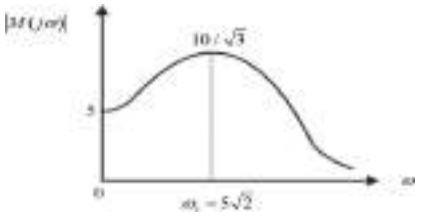
Hence, the value of the gain K is 14.92

7. (A)

Given: Magnitude of the frequency response of under damped second-order system $|M(j\omega)| = 5$ at 0 rad/sec

$$M_r = \frac{10}{\sqrt{3}} \text{ at } \omega_r = 5\sqrt{2}\,\text{rad/sec}$$

Method 1:



$K = M(j0) = 5$ gain = 5

Resonant peak is given by,

$$M_r = \frac{K}{2\xi\sqrt{1-\xi^2}}$$

$$\frac{10}{\sqrt{3}} = \frac{5}{2\xi\sqrt{1-\xi^2}}$$

$$\frac{2}{\sqrt{3}} = \frac{1}{2\xi\sqrt{1-\xi^2}}$$

$$4\xi\sqrt{1-\xi^2} = \sqrt{3}$$

$$16\xi^2\left(1-\xi^2\right) = 3$$

$$-16\xi^4 + 16\xi^2 - 3 = 0$$

Let $\xi^2 = x$,

Hence, $-16x^2 + 16x - 3 = 0$

$$x = \xi^2 = 0.25, 0.75$$

$$\xi = 0.5, 0.866$$

Resonant peak is valid only for $\xi < 0.707$.

Thus, $\xi = 0.5$

Resonant frequency is given by,

$$\omega_r = \omega_n\sqrt{1-2\xi^2}$$

$$\omega_n\sqrt{1-2(0.5)^2} = 5\sqrt{2}$$

$$\omega_n\sqrt{0.5} = 5\sqrt{2}$$

$$\omega_n = 10\,\text{rad/sec}$$

Closed-loop transfer function for second-order system is given by,

$$\frac{C(s)}{R(s)} = \frac{K\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$\frac{C(s)}{R(s)} = \frac{5\times 10^2}{s^2 + 10s + 100} = \frac{500}{s^2 + 10s + 100}$$

Hence, the correct option is (A).

9 Bode Plot

Objective

After completing this chapter, the user will be able to understand

1. Bodeplot of minimum phase system
2. Bodeplot of non–minimum phase system
3. Effect of transportation lag in bode plot
4. Recovery of transfer function from bode plot
5. Error analysis from bode plot
6. Bodeplot for complex corner frequency

Introduction

1. Bodeplots are graphical representations of the frequency responses and are used in solving design problems.
2. Bodeplot deals with the frequency response of a system simultaneously in terms of magnitude and phase.
3. Magnitude and phase functions are shown on separate graphs using a logarithmic frequency scale along the *x*-axis.
4. Logarithm of the frequency to base 10 is used for the *x*-axis of a graph.

Modification of Magnitude Calculation

$\ell n = \log_e$

$\log = \log_{10}$

$$\left|T\left(j\omega\right)\right| = \frac{K}{\omega\sqrt{1+\omega^2}}$$

$$\log_{10}\left|T\left(j\omega\right)\right| = \underbrace{\log_{10} K}_{\text{Exact plot}} - \underbrace{\log_{10}\omega - \frac{1}{2}\log_{10}\left(1+\omega^2\right)}_{\mapsto \text{Nonlinear} \to \text{Additive}}$$

$$\log_{10}\left|T\left(j\omega\right)\right| = \log_{10} K - 2\log_{10}\omega \qquad \text{If } \omega >>> 1$$

$$\log_{10}\left|T\left(j\omega\right)\right| vs \log_{10}\omega \to \textit{linear} + \textit{additive}$$

Purpose of Approximate Plot

$$\left|T\left(j\omega\right)\right| \to \underset{+\text{Rational}}{\text{Nonlinear}} \Rightarrow \textit{Exact Magnitude plot}$$

$$\downarrow$$

$$20\log_{10}\left|T\left(j\omega\right)\right| = \left|T\left(j\omega\right)\right| dB \; vs \; \underset{\mapsto \text{Nonlinear} + \text{Additive}}{\log_{10}\left[f\left(\omega\right)\right]}$$

$$\downarrow \textit{Approximation}$$

$$\left|T\left(j\omega\right)\right| dB \; vs \; \log_{10}\omega \to \textit{Linear} + \textit{Additive}$$

$$\Rightarrow \textit{Approximate Magnitude Plot}$$

Modification of Phase Calculation

$$\angle T\left(j\omega\right) = -90° - \tan^{-1}\omega \to \underset{+\text{Additive}}{\text{Nonlinear}}$$

$$\downarrow$$

Phase Exact Plot

Bodeplot or Logarithmic Plot

1. A sinusoidal transfer function may be represented by two separate plots, one giving magnitude versus frequency, and the otherphase angle (in degrees) versus frequency.
2. Bode plot consists of two graphs.
   a) Logarithmic of the magnitude of the sinusoidal transfer function.
   b) Phase angle.
      i) Both are plotted against frequency on a logarithmic scale.
      ii) The logarithmic representation is useful because it allows us to plot both low frequency and high-frequency components on the same graph.

Requirements

1. The loop transfer function G(s)H(s) should be in the time constant form

$$G(s)H(s) = \frac{k\left(1+sT_1\right)\left(1+sT_2\right)}{\left(1+sT_{p1}\right)\left(1+sT_{p2}\right)}$$

2. The transfer function must be a minimum phase transfer function.

Such plots are useful due to the following reasons:

1. For designing a lead compensator
2. For finding stability, gain margin and phase margin
3. For system identification from the frequency response

$$Bode\ Plot \begin{cases} Bode\ Magnitude \rightarrow Approximate\ Plot & Mag.Plot \\ Bode\ Phase\ Plot \rightarrow Exact\ Phase\ Plot \end{cases}$$

Bode Magnitude Plot

1. Also known as anapproximate magnitude plot
2. Graph between the magnitude of loop transfer function in dB, i.e. $|T(j\omega)|dB$ *vs* $\log_{10}\omega$
3. $|T(j\omega)|dB$ *vs* $\log_{10}\omega \rightarrow$ *Linear curve*
4. 

Fig. 9.1

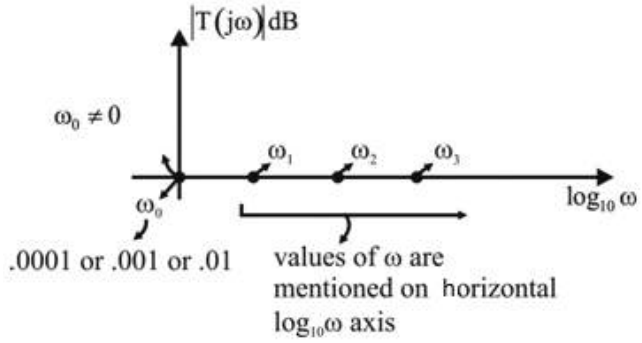
5. 

Fig. 9.2

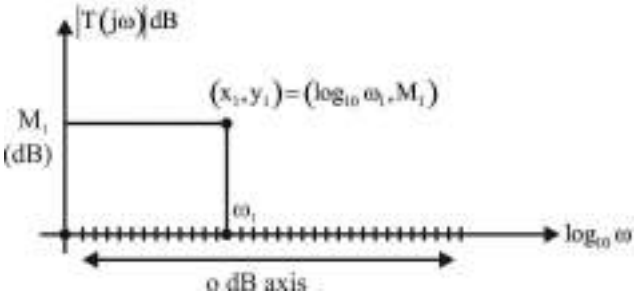
6. 

Fig. 9.3

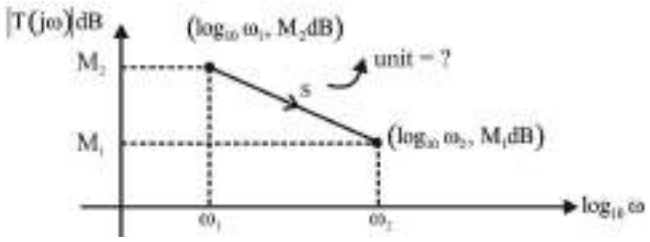
7. 

Fig. 9.4

$$S = \frac{(M_1 - M_2)}{\log_{10}\frac{\omega_2}{\omega_1}} \rightarrow unit = ?$$

8. Unit of slope =?

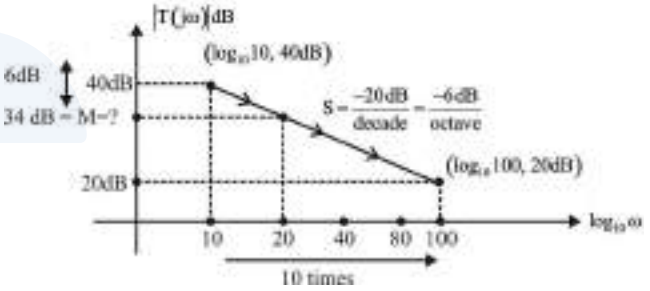


Fig. 9.5

$$S = \frac{[20 - 40]dB}{\log_{10} 100 - \log_{10} 10} = \frac{-20\,dB}{1\ decade}$$

$\frac{-20\ dB}{decade} \rightarrow$ "If frequency changes by 10 times, then the magnitude changes by –20 dB."

Determine M in dB at 10 rad/sec:

$$-20 = \frac{M - 40}{\log_{10} 20 - \log_{10} 10}$$

M = 34 dB

Consider points:

$(\log_{10} 10, 40\,dB)$

$(\log_{10} 20, 34\,dB)$

Slope, $S = \dfrac{-6\ dB}{\log_{10}\dfrac{20}{10}} \begin{cases} -20\ dB/decade \\ -6\ dB/octave \end{cases}$

Note:

1. 10 times change in frequency is called adecade.
2. 2 times change in frequency is called anoctave.
3. – 20 dB/decade = – 6 dB/octave

Unit:

$$S = \left\{ \frac{Change\ in\ magnitude\ in\ dB}{\left(\frac{\omega_2}{\omega_1}\right)} \right\}$$

$\dfrac{\omega_2}{\omega_1} = 10^1$ $\qquad$ $\dfrac{\omega_2}{\omega_1} = 2^1$

$S = \dfrac{dB}{1\ decade}$ $\qquad$ $S = \dfrac{dB}{1\ octave}$

9.20n dB/decade = 6ndB/octave; $n \varepsilon\ I$

–20n dB/decade = –6ndB/octave; $n \varepsilon\ I$

10. $\omega_1 = 10 \longrightarrow \omega_2 = 10,000 \Rightarrow \dfrac{\omega_2}{\omega_1}$

$= 10^3 \rightarrow 3\,decade$

$\omega_1 = 2 \longrightarrow \omega_2 = 2048 \Rightarrow \dfrac{\omega_2}{\omega_1} = 1024 = 2^{10}$

$10 \rightarrow octave$

11. $M\dfrac{dB}{octave} \rightarrow N\dfrac{dB}{decade}$

$36\dfrac{dB}{octave} \rightarrow 36\dfrac{dB}{.303\ decade} =$

$120\,dB/decade$

1 decade = 3.32 octave

1 octave = 0.030 decade

$2^n = 10$

$n\log_2 2 = \log_2 10$

$1\dfrac{dB}{octave} = 3.3\dfrac{1dB}{.303decade} = \dfrac{20}{6}\dfrac{dB}{decade}$

$M\dfrac{dB}{octave} = M\dfrac{dB}{.303\,decade} = \dfrac{MdB}{.0303\ decade} =$

$\dfrac{20}{6}\left\{M\dfrac{dB}{decade}\right\}$

$M\dfrac{dB}{octave} = \dfrac{20}{6}\left\{M\dfrac{dB}{decade}\right\}$

Important Observations

Case 1: Change in magnitude = M dB between 2 frequencies ω_1 *and* ω_2

Slope of line is $\pm 20n$ dB/decade ($N\,dB/decade \rightarrow NER$) between 2 frequencies ω_1 *and* ω_2

Factor affecting frequency $\Rightarrow \underbrace{\dfrac{M}{20n}}_{\frac{M}{N}=r} = r$

Change in frequency $\Rightarrow 10^r$ *times*

Case 2: Change in magnitude = M dB between 2 frequencies ω_1 *and* ω_2.

Slope of line = $\pm 6n$ dB/octave ($N\,dB/octave \rightarrow NER$) between 2 frequencies ω_1 *and* ω_2

Factor affecting frequency $\Rightarrow \underbrace{\dfrac{M}{6n}}_{\frac{M}{N}} = r$

Change in frequency $= 2^r$.

Example: Determine and

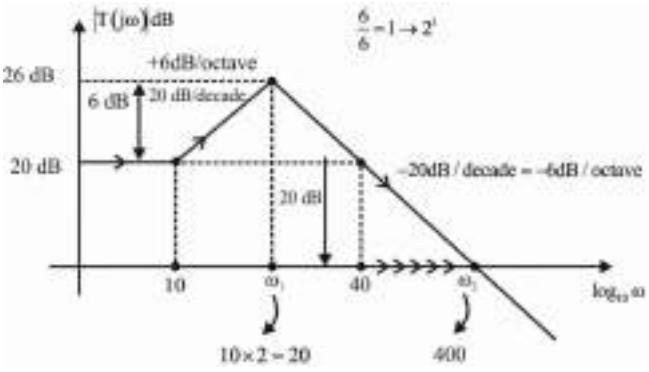


Fig. 9.6

Example: Determine, and

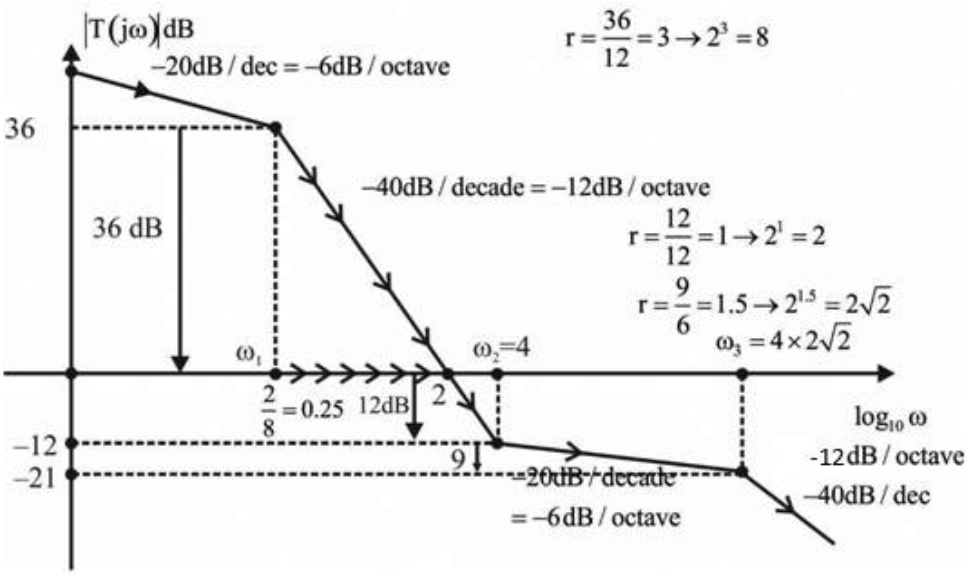


Fig. 9.7

Bode Phase Plot

1. Exact phase plot
2. $\angle T(j\omega)$ *vs* $\log_{10}\omega$
3. Linear or nonlinear
4.

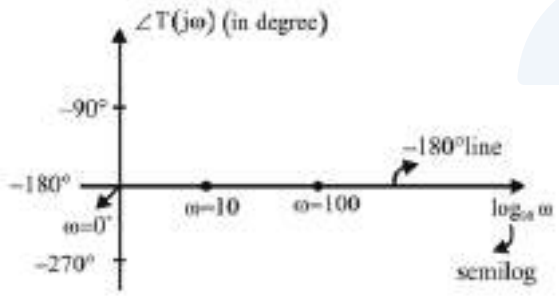


Fig. 9.8



Fig. 9.9

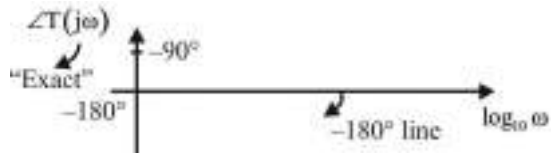


Fig. 9.10

1. Draw a bode plot for the following loop transfer function, $G(s)H(s) = 1$

Solution:

Step 1: T(s) = 1

Step 2: $s = j\omega \rightarrow T(j\omega) = 1$

$(0+ \le \omega < \infty)$

(or)

$0 < \omega < \infty$

Step 3: $T(j\omega) = 1$

Magnitude: $|T(j\omega)| = 1$

$|T(j\omega)|\,dB = 20\log_{10} 1 = 0dB$

Plot $|T(j\omega)|\,dB$ *vs* $\log\omega$ on semi log paper

Phase: $\angle T(j\omega) = 0^0$

Plot $\angle T(j\omega)$ *vs* $\log\omega$ on semi log paper.

Step4:

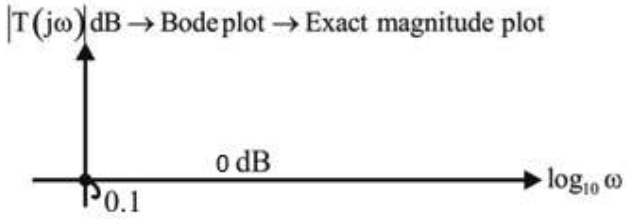


Fig. 9.11

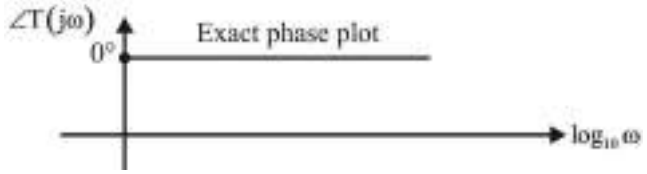


Fig. 9.12

2. Draw a bode plot for the following loop transfer function, G(s)H(s) = k

Solution:

S.1 T(s) = k

S.2 $s = j\omega \rightarrow T(j\omega) = k$

S.3 $|T(j\omega)|dB = 20\log_{10} k$ (0<k<1: -ve dB; k>1: +ve dB)

$\angle T(j\omega) = 0^0$

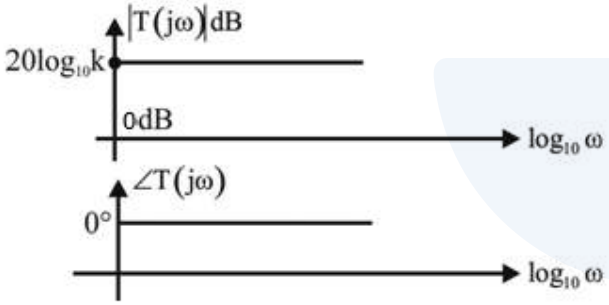


Fig. 9.13

Note: On multiplication of gain 'k'into G(s) H(s) →Bode Magnitude plot shifts parallel by $20\log_{10} k$ [Upwards or Downwards]

The bode phase plot remains unchanged.

3. Draw a bode plot for the following loop transfer function, G(s)H(s) = k.s

Solution:

S.1 T(s) = ks

S.2 $s = j\omega \rightarrow T(j\omega) = k(j\omega)$

S.3 $|T(j\omega)| = k\omega \rightarrow \underbrace{|T(j\omega)|dB}_{y} =$

$\underbrace{20}_{m}\underbrace{\log_{10}\omega}_{x} + \underbrace{20\log_{10} k}_{c}$

$\angle T(j\omega) = +90^0$

Intersection with 0 dB axis: At $\omega = \omega_0$

$|T(j\omega_0)| = 0dB$

$20\log_{10}\omega k = 0 \rightarrow \log_{10}\omega k = 0 \rightarrow \omega k = 10^0$

$\omega k = 1$

$\omega = \frac{1}{k}$

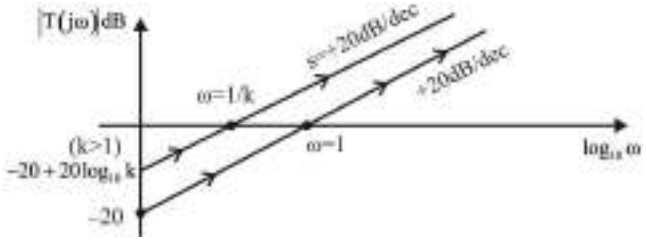


Fig. 9.14

4. Draw aBode plot for the following loop transfer function, G(s)H(s) = KS^P

Solution:

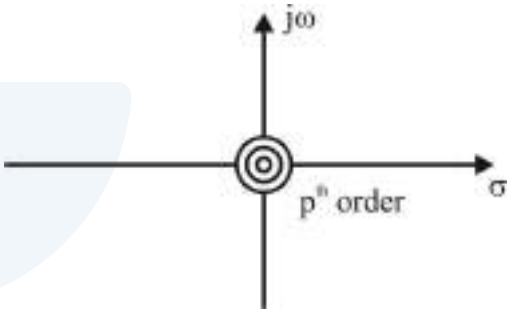


Fig. 9.15

S.1 $T(s) = kS^p$

S.2 $T(j\omega) = k(j\omega)^p = kj^p\omega^p$

S.3 $|T(j\omega)| = k\omega^p \rightarrow \underbrace{|T(j\omega)|dB}_{y} =$

$\underbrace{20p}_{m}\underbrace{\log_{10}\omega}_{x} + \underbrace{20\log_{10} k}_{c}$

$\angle T(j\omega) = +90^\circ p$

Intersection with 0 dB axis:

At $\omega = \omega_0$ $\quad |T(j\omega_0)|dB = 0dB$

$20p\log_{10}\omega_o + 20\log_{10} k = 0$

$20p\log_{10}\omega_o = -20\log_{10} k$

$\log_{10}\omega^p = \log_{10} k^{-1}$

$\omega = \left(\frac{1}{k}\right)^{\frac{1}{p}}$

ω	$\lvert T(j\omega)\rvert dB$	$\angle T(j\omega)$
0.1	$-20p + 20\log_{10} k$	$+90^0$p
1	$+20\log_{10} k$	$+90^0$p
10	$20p + 20\log_{10} k$	$+90^0$p
100	$40p + 20\log_{10} k$	$+90^0$p

Table 9.1

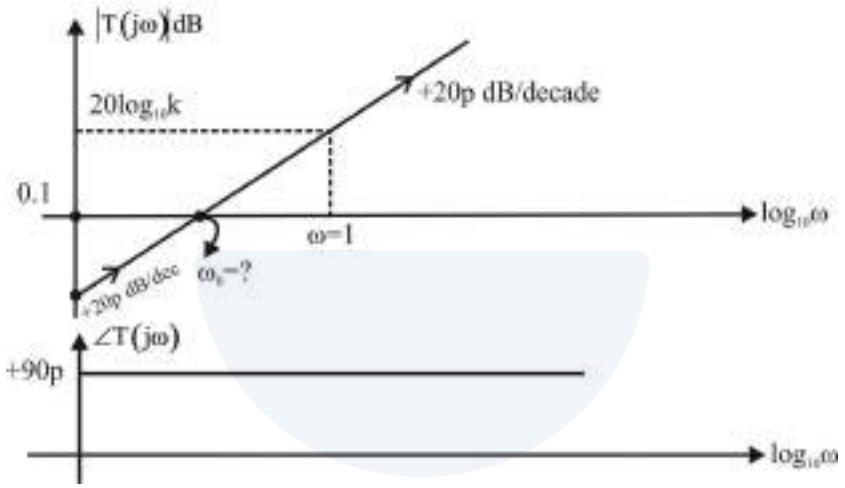


Fig. 9.16

Summary:

O.L.T.F.	Initial Slope in dB/dec	Intersection with 0dB Axis	Slope at $\omega \to \infty$ in dB/Decade	Phase
K	0dB/decade Value = $20\log_{10}$k	No Intersection	0dB/decade	0°
KS	+20dB/dec	$\left(\frac{1}{k}\right)^1$	+20dB/decade	+90°
KS^2	+40dB/dec	$\left(\frac{1}{k}\right)^{\frac{1}{2}}$	+40dB/dec	+180°
⋮	⋮	⋮	⋮	⋮
kS^P	+20p dB/dec	$\left(\frac{1}{k}\right)^{\frac{1}{p}}$	+20p dB/dec	+90°p

Table 9.2

5. Draw a bode plot for the following loop transfer function, G(s)H(s) = $\frac{k}{S^p}$

Solution:

S.1 $T(s) = \frac{k}{S^p}$

S.2 $S = j\omega \rightarrow T(j\omega) = \frac{k}{(j\omega)^p}$

S.3 $|T(j\omega)| = \frac{k}{\omega^p} \rightarrow \underbrace{|T(j\omega)|dB}_{y} =$

$\underbrace{-20p}_{m}\underbrace{\log_{10}\omega}_{x} + \underbrace{20\log_{10}k}_{+c}$

$\angle T(j\omega) = -90^0 p$

S.4 Intersection with 0dB axis:

At $\omega = \omega_0$

$|T(j\omega_0)|dB = 0\,dB$

$20p\log_{10}\omega = 20\log_{10}k$

$\omega^p = k$

$\omega = (k)^{\frac{1}{p}}$

ω	$\lvert T(j\omega)\rvert dB$	$\angle T(j\omega)$
0.1	$20p + 20\log_{10}k$	–90°p
1	$20\log_{10}k$	–90°p
10	$-20 + 20\log_{10}k$	–90°p
100	-40 + $20\log_{10}k$	–90°p

Table 9.3

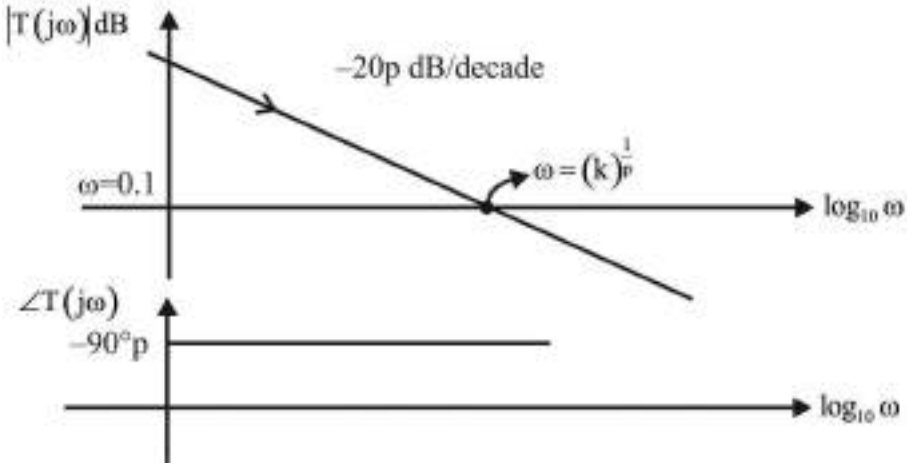


Fig. 9.17

Summary:

O.L.T.F.	Initial Slope in dB/dec	Intersection with 0dB Axis	Slope as $\omega \to \infty$ in dB/Decade	Phase
K/S	–20dB/dec	$\omega = (k)^1$	–20dB/dec	–90°
K/S^2	–40dB/dec	$\omega = (k)^{\frac{1}{2}}$	–40dB/dec	–180°
⋮	⋮	⋮	⋮	⋮
K/S^p	–20p dB/dec	$\omega = (k)^{\frac{1}{p}}$	–20p dB/dec	–90°p

Table 9.4

Important Conclusions

1.

Slope of Initial Line	Initial Phase	Type
+20p dB/dec	+90°p	0
0 dB/dec	0°	0
–20 dB/dec	–90°	1
–40 dB/dec	–180°	2
⋮	⋮	⋮
–20p dB/dec	–90°p	p

Table 9.5

2.

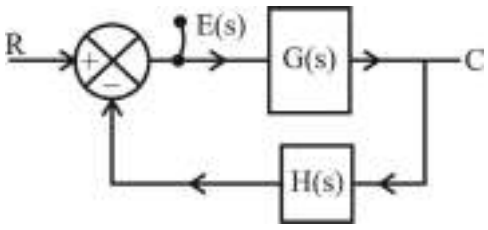


Fig. 9.18

Determine e_{ss} = ?

Case 1:

$G(s)H(s) = k$

$k_p = \lim_{s \to 0} G(s)H(s) = k$

$$e_{ss} = \frac{A}{1+k_p} = \frac{A}{1+k}$$

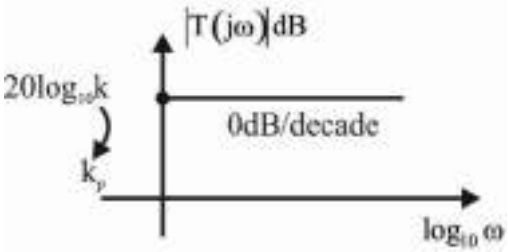


Fig. 9.19

Case 2:

G(s)H(s) = $\frac{k}{s}$

$$k_v = \lim_{s\to 0} sG(s)H(s) = k$$

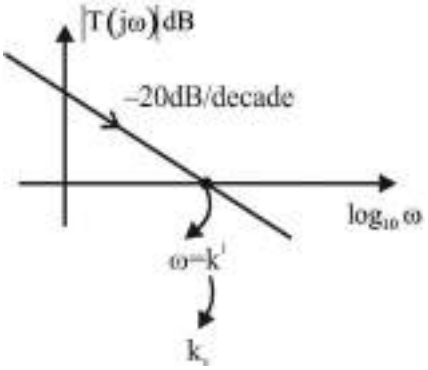


Fig. 9.20

Case 3:

$$G(s)H(s) = \frac{k}{s^2}$$

$$k_a = \lim_{s\to 0} s^2G(s)H(s) = k$$

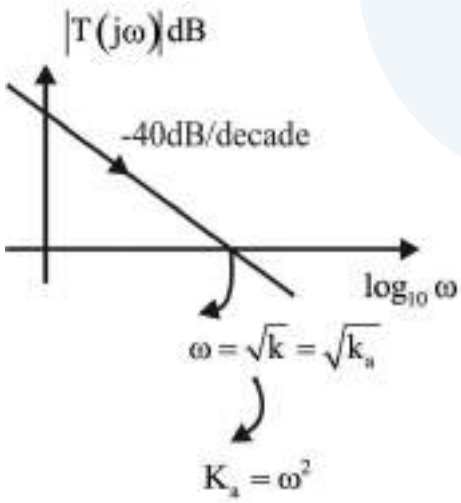


Fig. 9.21

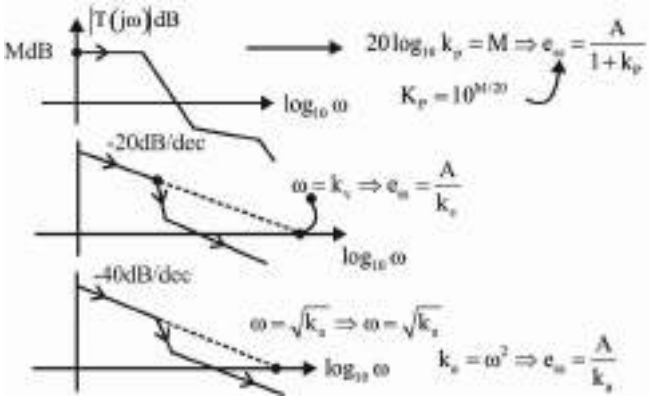


Fig. 9.22

Example:

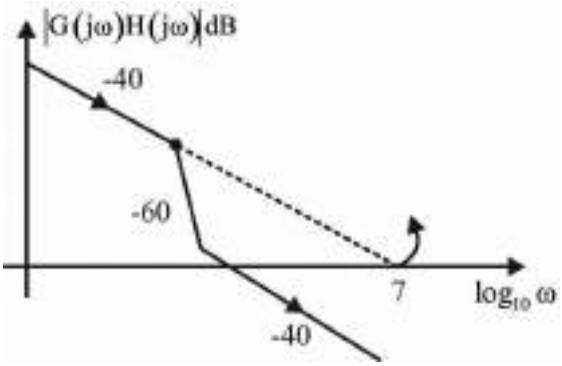


Fig. 9.23

Calculate e_{ss} for a unity feedback system whose O.L.T.F. has aBode plot as shown above for input $r(t) = 3t^2u(t)$

Solution:

$$r(t) = 3t^2u(t)$$

$$= 6\left(\frac{t^2}{2}\right)u(t)$$

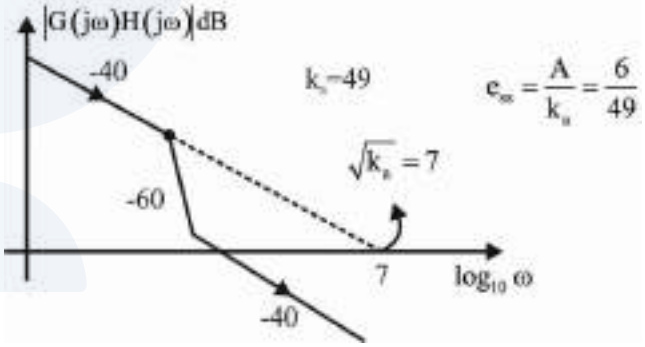


Fig. 9.24

Bodeplots for 1st Order Terms

Magnitude analysis:

$$\left|T(j\omega)\right| = k\left[1+\left(\frac{\omega}{\omega_c}\right)^2\right]^{\frac{1}{2}}$$

$$\left|T(j\omega)\right| dB = \frac{20}{2}\log_{10}\left[1+\left(\frac{\omega}{\omega_c}\right)^2\right] + 20\log_{10} k$$

$$\underbrace{\left|T(j\omega)\right| dB}_{y} = \underbrace{10\log_{10}\left[1+\left(\frac{\omega}{\omega_c}\right)^2\right]}_{mf(x)+c\to Nonlinear \xrightarrow{\text{should be converted into}} Linear} + 20\log_{10} k$$

Phase plot:

$$\angle T(j\omega) = \tan^{-1}\left(\frac{\omega}{\omega_c}\right) \rightarrow \begin{matrix} Non\,linear \\ Exact \end{matrix}$$

Lower Frequency Region (L.F.R.):

$\omega <<< \omega_c$

$\left(\frac{\omega}{\omega_c}\right) <<< 1$

$\left(\frac{\omega}{\omega_c}\right)^2 <<<<<<< 1$

$|T(j\omega)|dB = 20\log_{10} k \rightarrow Approx\ \ Analysis$

$\angle T(j\omega) = 0°$

Mid Frequency Region (M.F.R.): $\omega = \omega_c$

$|T(j\omega)|dB = 3dB + 20\log_{10} k$

$\angle T(j\omega) = 45°$

High Frequency Region: (H.F.R.)

$\omega >>> \omega_c$

$\left(\frac{\omega}{\omega_c}\right) >>> 1$

$\left(\frac{\omega}{\omega_c}\right)^2 >>>>> 1$

$|T(j\omega)|dB = +20\log_{10}\frac{\omega}{\omega_c} + 20\log_{10} k$

$\underbrace{20}_{m}\underbrace{\log_{10}\omega}_{x} + \underbrace{\left[20\log_{10} k - 20\log_{10}\omega_c\right]}_{+c}$

$\angle T(j\omega) = +90°$

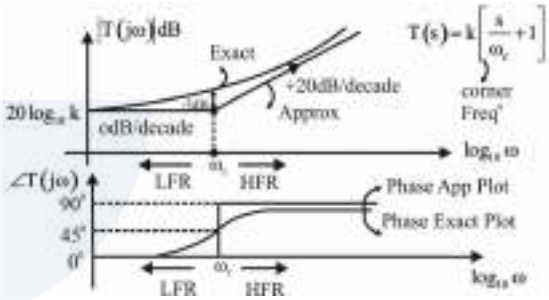


Fig. 9.25

Points to Remember

G(s)H(s)	Initial Line With Slope	Change In Slope At $\omega = \omega_c$	Error At Corner: Frequency	Slope At $\omega \rightarrow \infty$
$K\left[\frac{s}{\omega_c}+1\right]$	20 $\log_{10}$Kwith 0dB/decade	+20	3dB	+20
$K\left[\frac{s}{\omega_c}+1\right]^2$	20 $\log_{10}$Kwith 0dB/decade	+40	6 dB	+40
$K\left[\frac{s}{\omega_c}+1\right]^p$	20 $\log_{10}$Kwith 0dB/decade	+20p	3pdB	+20p

Table 9.6

6. Draw a bode plot for the loop transfer function, $G(s)H(s) = \dfrac{K}{\left[\dfrac{s}{\omega_c}+1\right]}$

Solution:

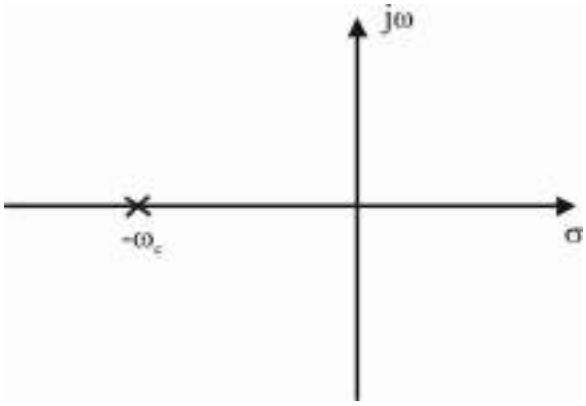


Fig. 9.26

S.1 $T(s) = \dfrac{K}{\left(\dfrac{s}{\omega_c}+1\right)}$

S.2

$$T(j\omega) = \frac{k}{\left(1+j\frac{\omega}{\omega_c}\right)} \left\langle \begin{array}{l} |T(j\omega)| = \dfrac{k}{\sqrt{1+(\omega/\omega_c)^2}} \\ |T(j\omega)| = -\tan^{-1}\dfrac{\omega}{\omega_c} \end{array} \right.$$

Magnitude:

$$|T(j\omega)|\,dB = -10\log_{10}\left[1+\left(\frac{\omega}{\omega_c}\right)^2\right] + 20\log_{10} k$$

Phase:

$$\angle T(j\omega) = -\tan^{-1}\frac{\omega}{\omega_c}$$

L.F.R.: $\omega <<< \omega_c \rightarrow \left(\dfrac{\omega}{\omega_c}\right)^2 <<<< 1$

$$|T(j\omega)| = 20\log_{10} k$$

$$\angle T(j\omega) = 0^0$$

M.F.R.: $\omega = \omega_c$

$$|T(j\omega)|\,dB = -3dB + 20\log_{10} k$$

$$\angle T(j\omega) = -45^0$$

H.F.R.: $\omega >>> \omega_c \rightarrow \left(\dfrac{\omega}{\omega_c}\right)^2 >>> 1$

$$\underbrace{|T(j\omega)|}_{y} = -\underbrace{20}_{m}\underbrace{\log_{10}\omega}_{x} + \underbrace{\left[20\log_{10} k + 20\log_{10}\omega_c\right]}_{c}$$

$$\angle T(j\omega) = -90^0$$

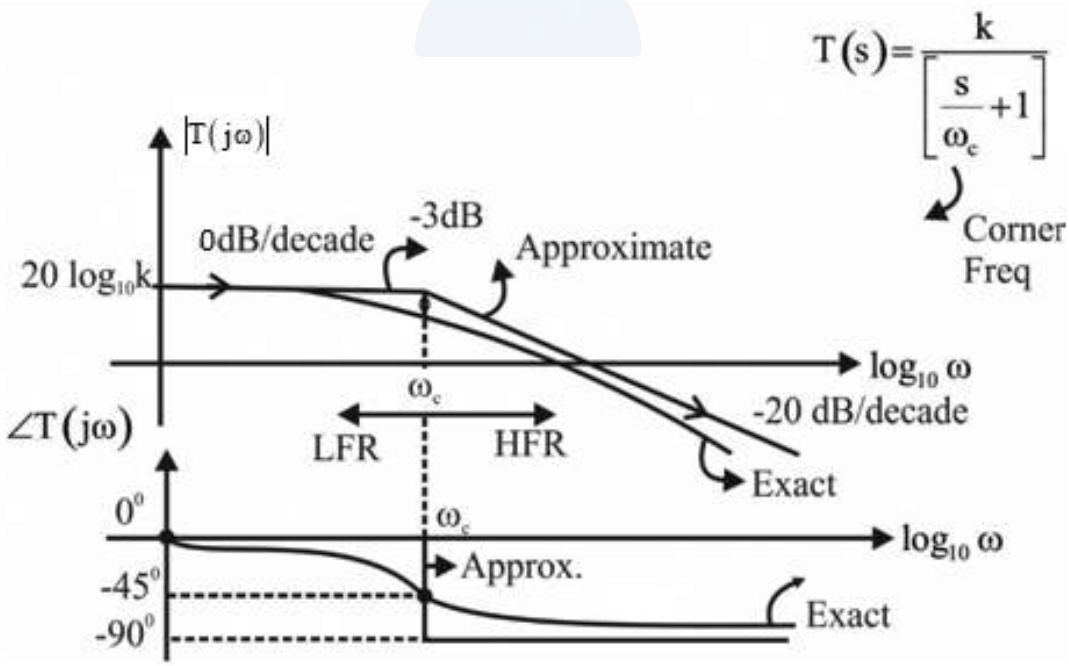


Fig. 9.27

Points to Remember

G(s)H(s)	Initial Line with Slope	Change in Slope at Corner Frequency	Error at Corner Frequency in dB	Slope at $\omega \to \infty$
$\frac{k}{\left[\frac{s}{\omega_c}+1\right]}$	20 $\log_{10}$K with slope 0dB/ decade	-20	-3dB	-20
$\frac{k}{\left[\frac{s}{\omega_c}+1\right]^2}$	20 $\log_{10}$Kwith slope 0dB/ decade	-40	-6 dB	-40
$\frac{k}{\left[\frac{s}{\omega_c}+1\right]^p}$	20 $\log_{10}$Kwith slope 0dB/ decade	-20p	-3p dB	-20p

Table 9.7

Important points:

1. $T(s)=\frac{1}{\left(1+\frac{s}{\omega_c}\right)}$

$s=j\omega :\to \left|T(j\omega)\right|=\frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_c}\right)^2}} \to LFR, HFR$

$\angle T(j\omega)=-\tan^{-1}\frac{\omega}{\omega_c} \to Exact$

2. $T(s)=\frac{1}{\left[1-\frac{s}{\omega_c}\right]}$

$s=j\omega, T(j\omega)=\frac{1}{1-j\frac{\omega}{\omega_c}} \to IV\ quad$

$\left|T(j\omega)\right|=\frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_c}\right)^2}} \to LFR, HFR$

$\angle T(j\omega)=+\tan^{-1}\frac{\omega}{\omega_c} \to Exact$

3. $T(s)=\frac{e^{-s}}{\left[1+\frac{s}{\omega_c}\right]}$

$s=j\omega$

$T(j\omega)=\frac{e^{-j\omega}}{1+j\frac{\omega}{\omega_c}}$

$\left|T(j\omega)\right|=\frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_c}\right)^2}} \to LFR, HFR$

$\angle T(j\omega)=-\omega-\tan^{-1}\left(\frac{\omega}{\omega_c}\right)$

Minimum phase system vs Non-minimum phase system:

$$T(s)=\frac{1}{s\left(\frac{s}{\omega_c}+1\right)} \rightarrow \textit{Minimum phase s/s}$$

$$T(s)=\frac{1}{s\left(1-\frac{s}{\omega_c}\right)} \rightarrow \textit{Non}\text{minimum phase s/s}$$

$$T(s)=\frac{1}{\left[\frac{-s}{\omega_c}-1\right]} \rightarrow \textit{Minimum phase s/s}$$

Important Conclusions

1. If the bode magnitude plot is given and the transfer function has to be recovered, then non-unique T.F. might exist.
2. If bode magnitude and phase plot is given, then unique T.F. is recovered.
3. If the bode magnitude plot of the minimum phase system is given, then transportation lag and non-minimum phase system should be ruled out.

Calculation of Magnitude at $\omega=1$ rad/sec, when the Initial Line is Extended.

Case 1:

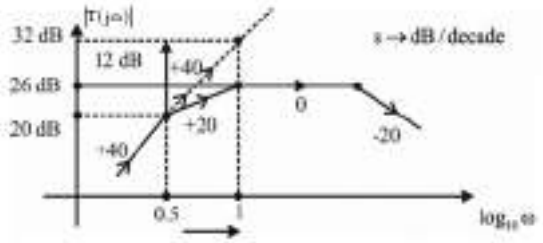


Fig. 9.27

$$\left|T(j\omega)\right|_{\omega=1}=26\,dB$$

Case 2:

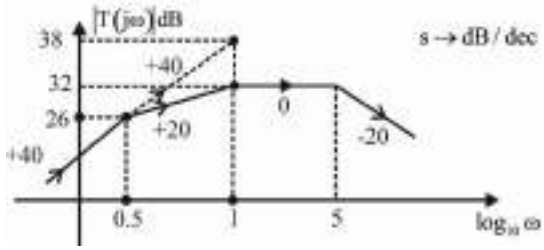


Fig. 9.29

Case 3:

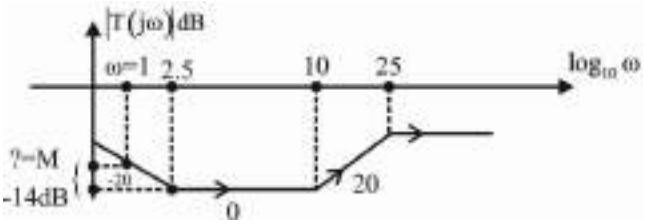


Fig. 9.30

$$slope=\frac{dB}{decade}=\frac{dB}{octave}$$

dB = slope x decade

dB = slope x octave = ?

$r = 2.5$

$2^{2.5} \rightarrow 5.656$

$10^{2.5} \rightarrow 316.22$

Approximation of T.F.

Case 1: $T(s)=\frac{T}{\left(1+\frac{s}{5}\right)} \rightarrow \textit{Time}$ constant

Corner frequency: $\omega=5$

$$0<\omega<5 \rightarrow T(s)=\frac{1}{1} \rightarrow T(j\omega)=1$$

$$\left|T(j\omega)\right|=1 \rightarrow \left|T(j\omega)\right|dB=0\,dB$$

$$\omega=5 \rightarrow T(s)=\frac{1}{1}\ or\ \frac{1}{s/5}$$

$$T(j\omega)=\frac{5}{j\omega}=\frac{5}{j5}=\frac{1}{1}$$

$$\omega>5 \quad T(s)=\frac{1}{s/5}=\frac{5}{s}$$

$$T(j\omega)=\frac{5}{j\omega}$$

$$\left|T(j\omega)\right|=5/\omega$$

$$\left|T(j\omega)\right|dB=20\log_{10}5-20\log_{10}\omega$$

Example: $T(s)=\frac{5(s+20)(s+50)}{(s+10)(s+100)}$

Solution:

$$T(s)=\frac{5\times 20\times 50\left(\frac{s}{20}+1\right)\left(\frac{s}{50}+1\right)}{10\times 100\left(\frac{s}{10}+1\right)\left(\frac{s}{100}+1\right)}$$

$$T(s)=\frac{5\left(1+\frac{s}{20}\right)\left(1+\frac{s}{50}\right)}{\left(1+\frac{s}{10}\right)\left(1+\frac{s}{100}\right)}\left\langle\begin{matrix}\omega=20,50\\ \omega=10,100\end{matrix}\right.$$

$\omega = 10, 20, 50, 100$

$0 < \omega < 10 \rightarrow T(s) = 5$

$\omega = 10 \rightarrow T(s) = 5$

$10 < \omega < 20 \rightarrow T(s) = \frac{5}{s/10} = \frac{50}{s}$

$\omega = 20 \rightarrow T(s) = \frac{50}{s}$

$20 < \omega < 50 \rightarrow T(s) = \frac{5\times s/20\times 1}{s/10\times 1} = \frac{5}{2}$

$50 < \omega < 100 \rightarrow \frac{5\times\frac{s}{20}\times\frac{s}{50}}{\frac{s}{10}\times 1} = \frac{s}{20}$

$\omega = 100 \rightarrow \frac{s}{20}$

$\omega > 100 \rightarrow \frac{5\times\frac{s}{20}\times\frac{s}{50}}{\frac{s}{10}\times\frac{s}{100}} = 5$

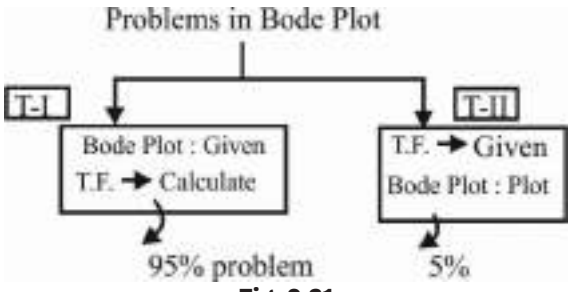


Fig. 9.31

Type 1: Inverse or Complementary Bode Plot: The bode plot is given. Determine the transfer function.

S.1 Identify initial slope →

$$\frac{0\,dB}{decade}\rightarrow k$$

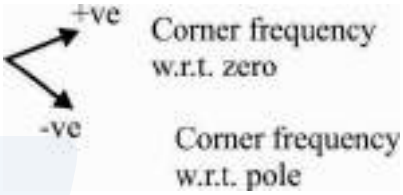
$$\frac{-20\,dB}{decade}\rightarrow\frac{k}{s}\Rightarrow -20p\,\frac{dB}{decade}\rightarrow\frac{k}{s^p}$$

$$\frac{+20\,dB}{decade}\rightarrow ks\Rightarrow +20p\,\frac{dB}{decade}\rightarrow ks^p$$

S.2 Identify the corner frequency and its nature

Corner frequency → Frequency where the slope changes

Nature: Change in slope



Change in slope at $\omega = \omega_c$ is +20p dB/decade

↓

$$\left[1+\frac{s}{\omega_c}\right]^p$$

Change in slope at $\omega = \omega_c$ is -20p dB/decade.

S.3 Calculation of k =?

If initial line is having slope.

Method 1: Use the approximate transfer function.

Or

Method 2: If initial line is having slope

1. Calculate magnitude at $\omega = 1$ (extended line magnitude)
2. $20\log_{10} k = |T(j\omega)|$ dB at $\omega = 1$ extended

Method 3: If an initialline has a slope = 0dB/decade

$20\log_{10} k = M\,dB \rightarrow$ Initial magnitude or magnitude of Horizontal line

Common mistakes:

1. Remember that x-axis is inlog formand the y-axis in dB (magnitude).

2. Carefully observe the question, find the unit of answer and carefully convert the answer into the required unit.
3. Take care of octave and decade.

Test Your Knowledge

1. The dB magnitude plot of the minimum phase system is shown below. The system transfer function is

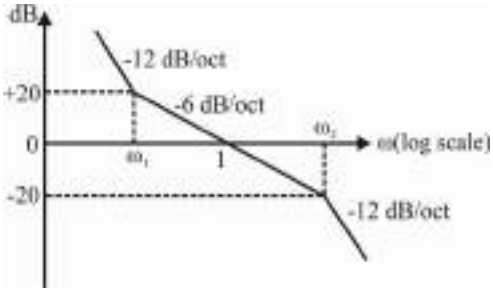


Fig. 9.32

(A) $\frac{10(s+0.1)}{s^2(s+10)}$

(B) $\frac{10(s+0.5)}{s^2(s+2)}$

(C) $\frac{(s+0.1)}{s^2(s+10)}$

(D) $\frac{4(s+0.5)}{s^2(s+2)}$

Solution: (A)

Bodeplot by transfer function:

Terms contained

1. A constant factor, k
2. A pole or zero at origin $(s)^{\pm p}$
3. Pole or zero's away from origin $(1+sT)^{\pm q}$
4. Complex poles or complex zeroes

$$\left(1+\frac{2\zeta\omega_n s}{\omega_n^2}+\frac{s^2}{\omega_n^2}\right)^{\pm r}$$

1. Draw the approximate bode plot for

$$T(s)=\frac{\omega_n^2}{s(s+2\zeta\omega_n)}$$

Solution:

1. Initial line having slope → –20dB/decade
2. ω_c corresponding to pole: $2\zeta\omega_n$

$$k_v=\lim_{s\to 0} sG(s)H(s)=\frac{\omega_n}{2\zeta}$$

3. Approximate analysis:

$\omega < 2\zeta\omega_n \qquad T(s)=\frac{\omega_n/2\zeta}{s}$

$$|T(j\omega)|\,dB = 20\log_{10}\frac{\omega_n}{2\zeta}-20\log_{10}\omega$$

$\omega > 2\zeta\omega_n \qquad T(s)=\frac{\omega_n/2\zeta}{s\times\frac{s}{2\zeta\omega_n}}=\frac{\omega_n^2}{s^2}$

$$|T(j\omega)|\,dB = 40\log_{10}\omega_n-40\log_{10}\omega$$

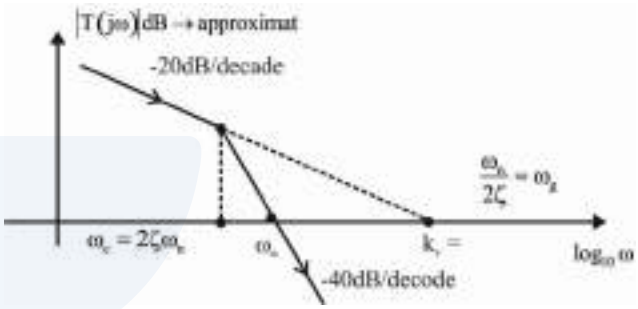


Fig. 9.33

$$T(s)=\frac{\omega_n/2\zeta}{s\left(\frac{s}{2\zeta\omega_n}+1\right)}$$

$\omega > 2\zeta\omega_n \rightarrow T(s)=\frac{\omega_n/2\zeta}{s\times\frac{s}{2\zeta\omega_n}}=\frac{\omega_n^2}{s^2}$

$|T(j\omega)|=\frac{\omega_n^2}{\omega^2}=1\Rightarrow |T(j\omega)|\,dB=0\,dB$

at $\omega=\omega_n$

$\omega_n=(\omega_g\omega_c)^{1/2}$ → G.M. of ω_g and ω_c

$\log_{10}\omega_n=\frac{\log\omega_g+\log_{\omega c}}{2}\Rightarrow$ Not A.M. of ω_g and ω_c

Bodeplot of Standard 2nd Order Closed-Loop System

$$\frac{C(s)}{R(s)}=\frac{k\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}=T(s)$$

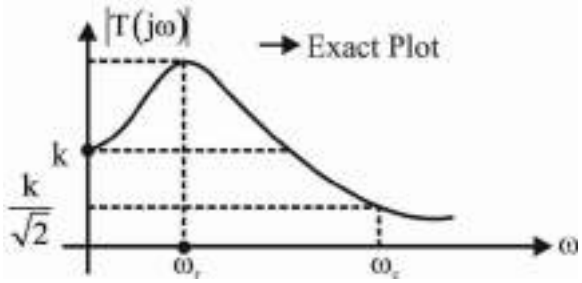


Fig. 9.34

Case 1:

$$\zeta = 1 \quad T(s) = \frac{k\omega_n^2}{(s+\omega_n)^2}$$

Critically damped:

$T(s) = \dfrac{k}{\left[\dfrac{s}{\omega_n}+1\right]^2}$; Corner frequency is real.

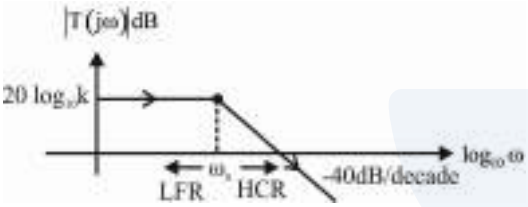


Fig. 9.35

Case 2:

$$T(s) = \frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad \zeta > 1 \rightarrow \text{over damp}$$

$$T(s) = \frac{k\omega_n^2}{(s+p_1)(s+p_2)} = \frac{k}{\left[\dfrac{s}{\omega_{c_1}}+1\right]\left[\dfrac{s}{\omega_{c_2}}+1\right]}$$

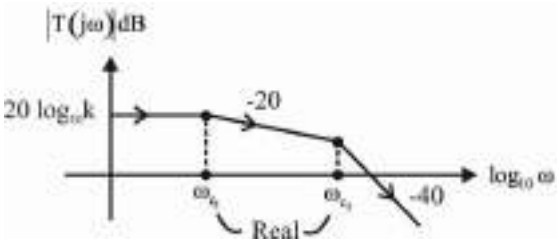


Fig. 9.36

Case 3: $0 < \zeta < 1$: Under–damped system

$$T(s) = \frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \text{poles} \rightarrow \text{complex}$$

$$s = j\omega \quad T(j\omega) = \frac{k\omega_n^2}{-\omega^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}$$

$$T(j\omega) = \frac{k}{\left(1-\dfrac{\omega^2}{\omega_n^2}\right) + j\left(2\zeta\dfrac{\omega}{\omega_n}\right)}$$

$$|T(j\omega)| = \frac{k}{\sqrt{\left(1-\dfrac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2\dfrac{\omega^2}{\omega_n^2}}}$$

$$|T(j\omega)|\text{dB} = 20\log_{10} k - 10\log_{10}\left\{\left(1-\frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2\frac{\omega^2}{\omega_n^2}\right\}$$

$$1 + \frac{\omega^4}{\omega_n^4} - \frac{2\omega^2}{\omega_n^2} + 4\zeta^2\frac{\omega^2}{\omega_n^2}$$

L.F.R.:

$\omega <<< \omega_n \Rightarrow |T(j\omega)|\text{dB} = 20\log_{10} k - 10\log_{10} 1 = 20\log_{10} k$

$$\frac{\omega}{\omega_n} <<< 1$$

M.F.R.:

$\omega = \omega_n \Rightarrow |T(j\omega)|\text{dB} = 20\log_{10} k - \{20\log_{10} 2\zeta\}$

$20\log_{10} 2\zeta$

$0 < \zeta < \frac{1}{2} \Rightarrow -ve$

$\zeta = \frac{1}{2} \Rightarrow 0$

$\frac{1}{2} < \zeta < 1 \Rightarrow +ve$

H.F.R.: $\omega >>> \omega_n$

$$\frac{\omega}{\omega_n} >>> 1$$

$$|T(j\omega)|\text{dB} = 20\log_{10} k - 40\log_{10}\frac{\omega}{\omega_n}$$

$$= 20\log_{10} k + 40\log_{10}\omega_n - 40\log_{10}\omega$$

$$y = \underset{m}{-40}\log_{10}\underset{x}{\omega} + \{c\}$$

$$0 < \zeta < \frac{1}{2} \rightarrow -20\log_{10} 2\zeta = +ve$$

Case 1:

$$-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2}$$

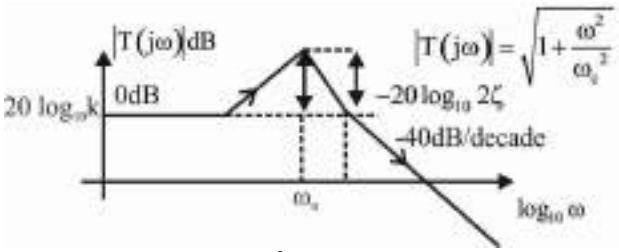


Fig. 9.37

Case 2:

$$\zeta = \frac{1}{2}$$

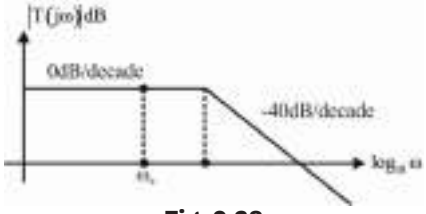


Fig. 9.38

Case 3:

$\frac{1}{2} < \zeta < 1 \Rightarrow -20\log_{10} 2\zeta = -ve$

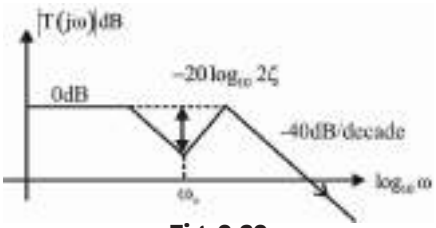


Fig. 9.39

Case 4:

$\zeta = 0 \qquad -20\log_{10} 2\zeta = -20\log_{10} 0 = \infty$

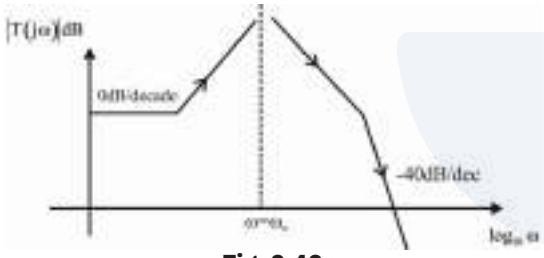


Fig. 9.40

1.

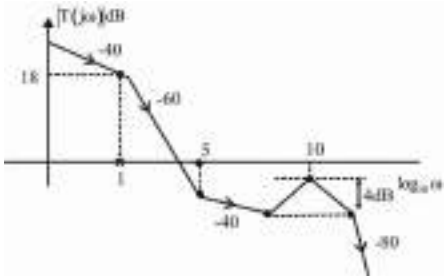


Fig. 9.41

Determine loop transfer function, T(s) =?

Solution:

$\omega_n = 10$

$-20\log_{10} 2\zeta = 4$

$\zeta = 0.315$

$20\log_{10} k = 18$

$k = 8$

$$T(s) = \frac{8\left(\frac{s}{5}+1\right)}{s^2\left(\frac{s}{1}+1\right)} \times \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$T(s) = \frac{8\left(\frac{s}{5}+1\right)}{s^2\left(\frac{s}{1}+1\right)} \times \frac{100}{s^2 + 6.3s + 100}$$

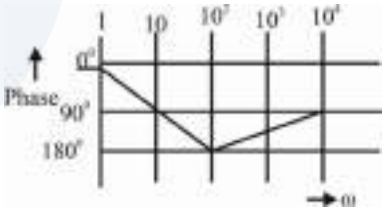
$$T(s) = \frac{8\left(\frac{s}{5}+1\right)}{s^2\left(\frac{s}{1}+1\right)} \times \frac{1}{\left\{\frac{s^2}{100} + \frac{6.3s}{100} + 1\right\}}$$

$$T(s) = \frac{160(s+5)}{s^2\left(s^2 + 6.3s + 100\right)(s+1)}$$

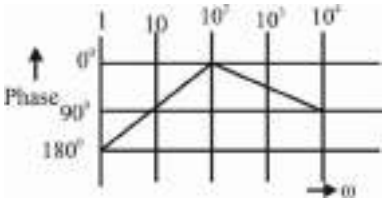
Test Your Knowledge

1. The straight-line approximated phase plot for a system with transfer function $G(s)H(s) = \frac{s+1000}{(s+10)^2}$, is

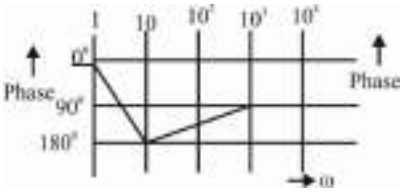
(A)



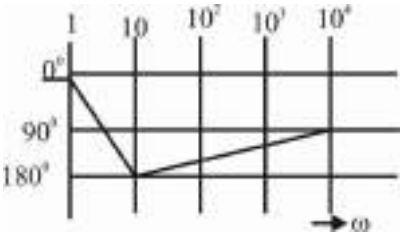
(B)



(C)



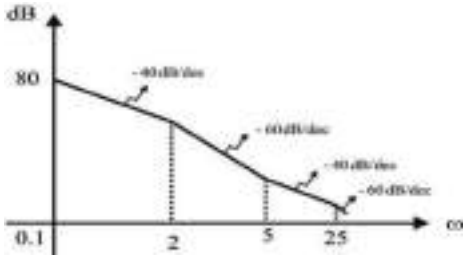
(D)



Solution: (A)

Exercise Problems

1. The asymptotic approximation of the log – magnitude frequency plot of a system containing only real poles and zeroes is shown. Its transfer function is



(A) $\frac{10(s+5)}{s(s+2)(s+25)}$

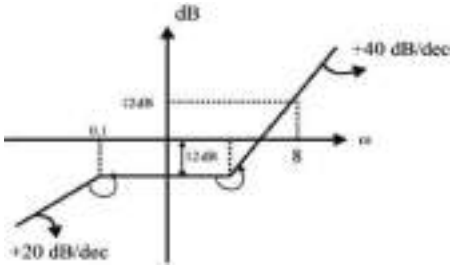
(B) $\frac{1000(s+5)}{s^2(s+2)(s+25)}$

(C) $\frac{100(s+5)}{s(s+2)(s+25)}$

(D) $\frac{80(s+5)}{s^2(s+2)(s+25)}$

Solution:(B)

2. The gain plot of a minimum phase function is given as



The T.F. of the system is (approximately)

(A) $\frac{1.6s(1+2s)^2}{(s+10)}$

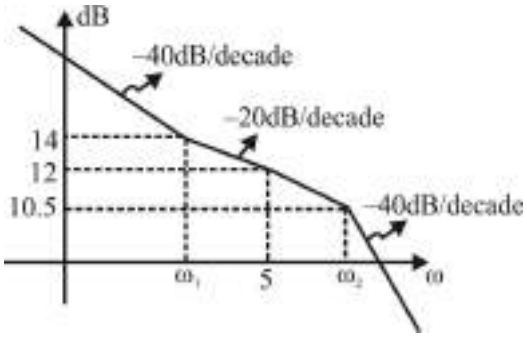
(B) $\frac{2.5(s+10)^2}{s(s+0.3)}$

(C) $\frac{0.625s(s+2)^2}{(10s+1)}$

(D) $\frac{2.5(s+0.5)^2}{s^2(s+10)}$

Solution:(C)

3.



The T.F. of minimum phase function

(A) $\frac{100(s+2)}{s^2(s+6)}$

(B) $\frac{120(s+2)}{s^2(s+3)}$

(C) $\frac{120(s+4)}{s^2(s+5)}$

(D) $\frac{120(s+4)}{s^2(s+6)}$

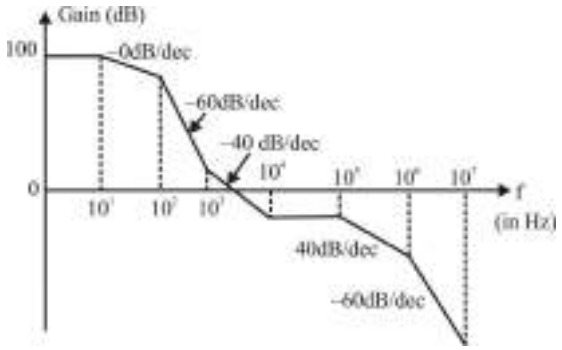
Solution:(D)

4. Draw the phase plots for the O.L.T.F.

$$G(s) = \frac{K}{s(1+0.1s)(1+0.05s)}$$

Solution: Corner frequencies = 10, 20

5. For an L.T.I. system, the bode plot for its gain is as illustrated in the figure shown. The number of system poles N_p, and the number of system zeroes N_z in the frequency range $1Hz \le f \le 10^7 Hz$ is



(A) $N_p = 5, N_z = 2$

(B) $N_p = 6, N_z = 3$

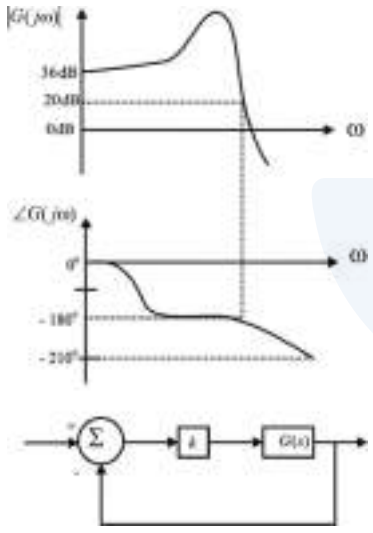
(C) $N_p = 7, N_z = 4$

(D) $N_p = 4, N_z = 2$

Solution: (B)

6. The figure below shows the bode magnitude and the phase plots of a stable transfer function.

$$G(s) = \frac{n_0}{s^3 + d_2 s^2 + d_1 s + d_0}$$



Consider the negative unity feedback configuration with gain k in the feed forward path. The closed-loop is stable for $k < k_0$. The maximum value of k_0 is

Solution: (0.1)

7. Consider a stable system with transfer function

$$G(s) = \frac{s^p + b_1 s^{p-1} + \ldots\ldots\ldots + b_p}{s^q + a_1 s^{q-1} + \ldots\ldots\ldots + a_q}$$

where $b_1, \ldots\ldots, b_p$, and $a_1, \ldots\ldots, a_q$, are real valued constants. The slope of the bode log magnitude curve of $G(s)$ converges to $-60\text{dB / decade as } \omega \to \infty$.

A possible pair of values for p and q is

(A) p = 0 and q = 3

(B) p = 1 and q = 7

(C) p = 2 and q = 3

(D) p = 3 and q = 5

Solution: (A)

Solutions

1. (B)

$cf = 2, 5, 25$

Least $cf = 2$

Slope at $\omega < \omega_{cl}$ i.e. ω<2

is $-40dB / dec$

Initial slope $= -40dB / dec$

Initial slope $= -20n = -40$

n=2, i.e., Type-2 system

$$G(s) = \frac{K\left(1 + \frac{s}{5}\right)}{s^2\left(1 + \frac{s}{2}\right)\left(1 + \frac{s}{25}\right)}$$

$At\ \omega = 0.1, \quad dB = 80$

$\omega_{cl} = 2, \quad \omega < \omega_{cl}$

$$G(s) = \frac{K}{s^2}, \quad G(j\omega) = \frac{K}{(j\omega)^2}$$

$$M = |G(j\omega)| = \frac{K}{\omega^2}$$

$dB = 20\log K - 40\log\omega$

$80 = 20\log K - 40\log(0.1)$

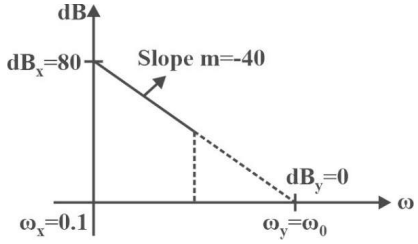
$\Rightarrow K = 100$

$$G(s) = \frac{100\left(1 + \frac{s}{5}\right)}{s^2\left(1 + \frac{s}{2}\right)\left(1 + \frac{s}{25}\right)} = \frac{1000(s+5)}{s^2(s+2)(s+25)}$$

For Type-2 system inter section of initial

slope line to 0-dB axis $K = \omega_o^2$

$\omega_o = \sqrt{K} = \sqrt{100} = 10$



$$dB_y - dB_x = m\log\frac{\omega_y}{\omega_x}$$

$$0 - 80 = -40\log\frac{\omega_o}{0.1}$$

$Type-2, \quad K = \omega_0^2 = 100$

7. (C)

Minimum phase function: All poles and zeroes are on the left half of s-plane

$c.f = 0.1, \omega_1$

Least $c.f = \omega_{cl} = 0.1$

Slope $\omega < \omega_{cl}$

Initial slope $-20n = -20$

$n = -1$

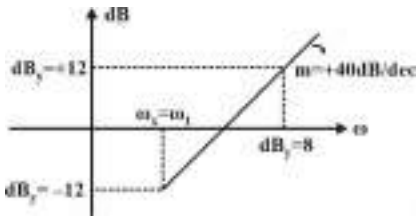
(it is not possible in general except differentiator G(s)=Ks)

$$\frac{K}{s^n} = \frac{K}{s^{-1}} = Ks$$

$$G(s) = Ks\frac{\left(1+\frac{s}{\omega_1}\right)^2}{\left(1+\frac{s}{0.1}\right)}$$

$$dB_y - dB_x = m\log\left(\frac{\omega_y}{\omega_x}\right)$$

$$12 - (-12) = m\log\left(\frac{8}{\omega_1}\right)$$



$\omega_x = \omega_1 \quad dB_x = -12$

$\omega_y = 8$

$$G(s) = Ks\frac{\left(1+\frac{s}{2}\right)^2}{\left(1+\frac{s}{0.1}\right)}$$

$\omega = 0.1 \quad dB = -12$

$\omega = \omega_1 = 2 \quad dB = -12$

$\omega = 8, \quad dB = +12$

$\omega = 0.1 \quad dB = -12$

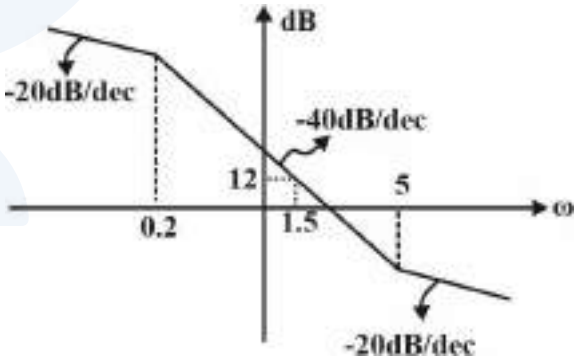
$G(s) = Ks$

$M = |G(j\omega)|_{dB}, \quad K = 20\log\omega + 20\log K$

$-12 = 20\log K + 20\log 0.1$

$K = 2.5$

$$= \frac{0.625s(s+2)^2}{10s+1}$$



Corner frequencies = 0.2, 5

$\omega_{cl} = 0.2$

Slope at $\omega < \omega_{cl}$

Initial slope = -20n = -20

n=1

Type-1 system

$$G(s) = \frac{K}{s}\frac{\left(1+\frac{s}{5}\right)}{\left(1+\frac{s}{0.2}\right)}$$

$At \quad G = 1.5 \quad dB = 12$

$$G(s) = \frac{K}{s\left(1+\frac{s}{0.2}\right)} = \frac{K}{j\omega(1+j5\omega)}$$

$$M = |G(j\omega)| = \frac{K}{\omega\sqrt{1+\left(\frac{\omega}{0.2}\right)^2}}$$

Gain

$$dB = 20\log M$$

$$= 20\log K - 20\log\omega - 20\log\left[1+\left(\frac{\omega}{0.2}\right)^2\right]^{\frac{1}{2}}$$

For $\omega > 0.2 \quad 1+\left(\frac{\omega}{0.2}\right)^2 = \left(\frac{\omega}{0.2}\right)^2$

$$dB = 20\log K - 20\log\omega - 20\log\left(\frac{\omega}{0.2}\right)$$

$$12 = 20\log K - 20\log 1.5 - 20\log\left(\frac{1.5}{0.2}\right)$$

$K = 44.7$

Slope $m = -40$

$dB_x = ? \quad \omega_x = 0.2$

$dB_y = 12 \quad \omega_y = 15$

$$12 - dB_x = -40\log\frac{1.5}{0.2}$$

$dB_x = 47$

At $\omega = 0.2 \quad dB = 47$

$$G(s) = \frac{K}{s}$$

$$dB = 20\log K - 20\log\omega$$

$K = 44.7$

$$G(s) = \frac{44.7\left(1+\frac{s}{5}\right)}{s\left(1+\frac{s}{0.2}\right)} = \frac{8.94(s+5)}{s(5s+1)}$$

8. (D)

Leaf c.f. $= \omega_1$

At $\dot{u} < \dot{u}_1$

Initial slope = -40

-20n = -40

n = 2

Type-2 system

$$G(s) = \frac{K}{s^2}\frac{\left(1+\frac{s}{\omega_1}\right)}{\left(1+\frac{s}{\omega_2}\right)}$$

$$dB_y - dB_x = m\log_{10}\frac{\omega_y}{\omega_x}$$

$$12 - 14 = -20\log\frac{5}{\omega_1}$$

$\omega_1 = 4$

$$10.5 - 14 = -20\log\frac{\omega_2}{4}$$

$\omega_2 = 6$

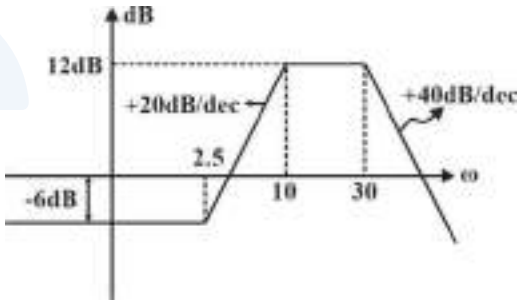
$$G(s) = \frac{K}{s^2}\frac{\left(1+\frac{s}{\omega_1}\right)}{\left(1+\frac{s}{\omega_2}\right)}$$

At $\omega = 4 \quad dB = 14, \quad G(s) = \frac{K}{s^2}$

$dB = 20\log K - 40\log\omega$

$14 = 20\log K - 40\log 4$

$K = 80.1$



Corner frequency = 2.5, 10, 30

$\omega_{cl} = 2.5$

$\omega < \omega_{cl}$

Initial slope

-20n = 0, n = 0, i.e., Type-0 system

$$G(s) = K\frac{\left(1+\frac{s}{2.5}\right)}{\left(1+\frac{s}{10}\right)\left(1+\frac{s}{30}\right)^2}$$

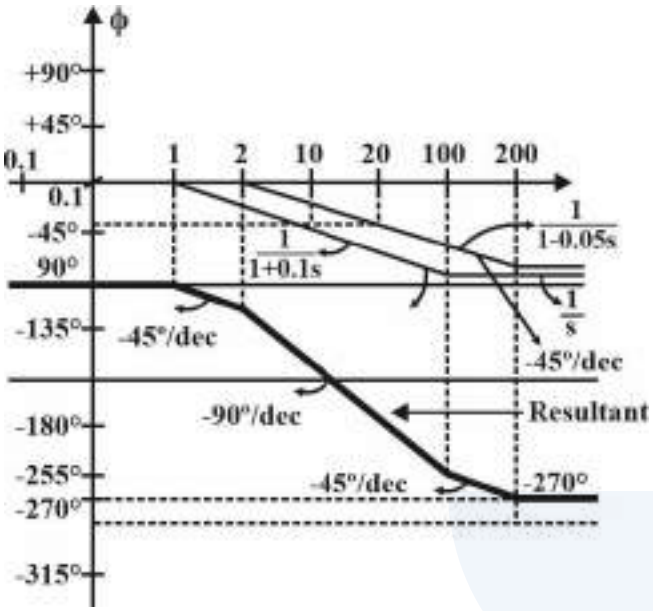
At $\omega = 2.5 \quad dB = -6$

$G(s) = K \quad M = K$

$dB = 20\log K$

$-6 = 20\log K, \quad K = 0.5$

9. Corner frequencies = 10, 20

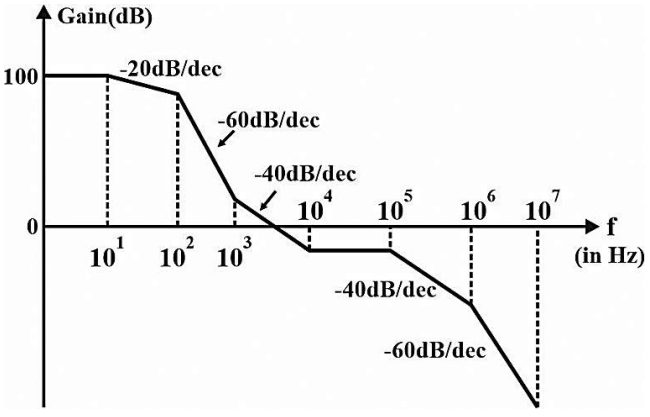


For $\frac{1}{1+0.1s} \quad c.f = 10$

For $\frac{1}{1+0.05s} c.f = 20$

10. (B)

The given bode plot is shown in the below figure



Observing the change in slope at each corner frequency to find number of poles/zeroes at that frequency:

At $f = 10^1 Hz$

Change in slope $= -20dB / dec$

$\rightarrow$ 1 pole

At $f = 10^2 Hz$

Change in slope $= -40dB / dec$

$\rightarrow$ 2 pole

At $f = 10^3 Hz$

Change in slope = +20 / dec

$\rightarrow$ l zero

At $f = 10^4 Hz$

Change in slope $= +40dB / deg$

$\rightarrow$ 2 zero

At $f = 10^5 Hz$

Change in slope $= -40dB / dec$

$\rightarrow$ 2 pole

At $f = 10^6 Hz$

Change in slope $= -20dB / dcc$

$\rightarrow$ 1 pole

So, the total number of poles

Total number of zeroes

Hence,

Where

Total number of systems poles

Hence, the correct option is (B).

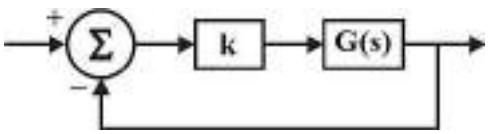
11. 0.1

Note: Given question is incorrect as the Bode Plot for the system represents an unstable system, but the system is mentioned as stable.

From the given bode magnitude plot, at gain is .

Gain margin

When the gain margin is less than 0dB, then the system is unstable.



Magnitude and phase angle plots for an unstable system with

Let

For the closed-loop system to be stable

Gain margin is given by

Hence, the maximum value of K, i.e., for which system is stable, is given as
Hence, the maximum value of is **0.1.**

12. (A)

Given:

Slope at is -60dB.

From G(s), slope at is given by,

Where q = number of poles

p = number or zeroes

a) **For option (A):** p=0, q=3
[correctly matched with given value]
b) **For Option (B):** p=1, q=7
[done not match with given value]
c) **For Option (C):** p=2, q=3
[done not match with given value]
d) **For Option (C):** p=3, q=5
[done not match with given value]
Hence, the correct option is (A).

10 Nyquist Plot

Objectives

After completing this chapter, the user will be able to:

- Map S-plane to Q(s)-plane.
- Map Nyquist contour to Q(s)–plane.
- Know Nyquist stability criteria and rules to draw a Nyquist plot.
- Identify encirclement and enclosement.
- Identify stability from the polar plot.

Introduction

- Frequency response methods were developed by Nyquist, Bode, Nichols, and many others.
- The Nyquist stability criterion enables us to investigate both the absolute and relative stabilities of linear closed–loop systems from the knowledge of their open–loop frequencyresponse characteristics.
- An advantage of the frequencyresponse approach is that frequencyresponse tests are, in general, simple and can be made accurately.

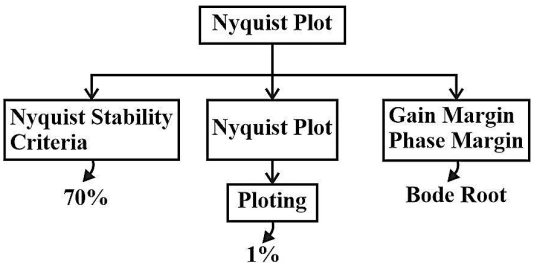


Fig. 10.1

Analysis of Complex Numbers

Complex no: $3+4j$

Complex variable $= x + jy = z$

$\rightarrow x, y \rightarrow$ variables.

Complex function $= f(x + jy) = f(z)$ $\quad i$ or j

Representation of Complex Number

1. Rectangular plane representation

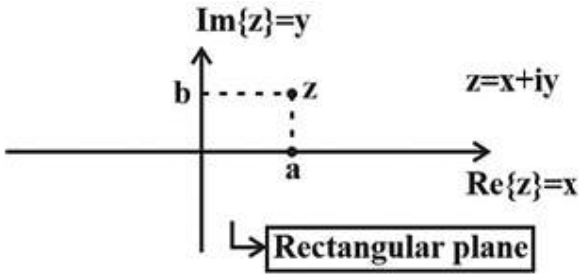


Fig. 10.2

2. Polar plane representation

$$Z = a + jb = |z| e^{j\angle z}$$

$$Z = r_0 e^{j\theta_0}, \theta_0 \text{ and } r_0 \rightarrow fix$$

$$r_0 = |z| = \sqrt{a^2 + b^2} \rightarrow fix$$

$$\theta_0 = \angle z = \tan^{-1}\frac{b}{a} \rightarrow fix$$

$n = n_0 \rightarrow$ *Radius circle*

$\theta = \theta_0 \rightarrow$ *Radial line* → Cut: Intersection will give z.

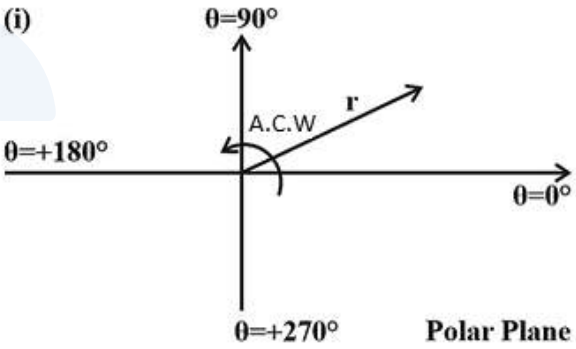


Fig. 10.3

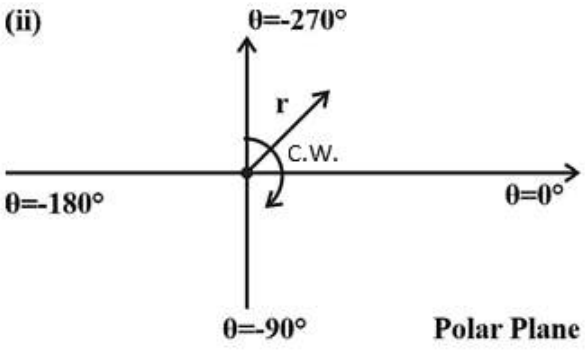


Fig. 10.4

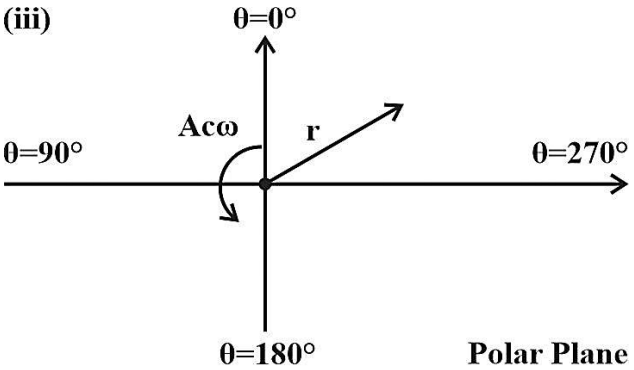


Fig. 10.5

Overlapping of i. and ii.

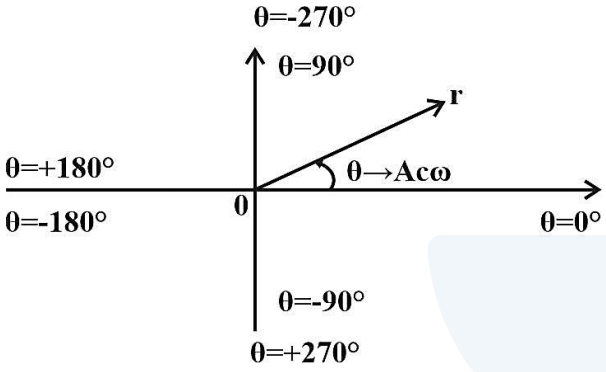


Fig. 10.6

Correlation between Rectangular and Polar Plane

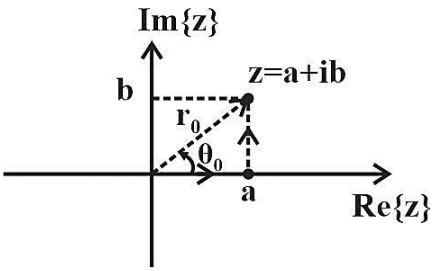


Fig. 10.7

$$z = a + jb = r_0 e^{j\theta_0}$$

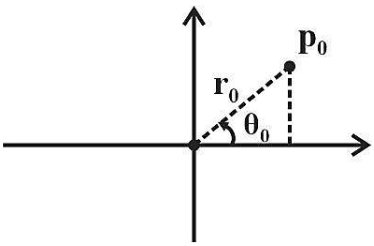
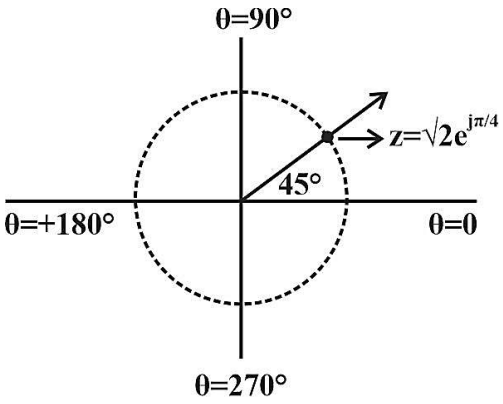


Fig. 10.8

$$r_0 \cos\theta_0 = a$$
$$r_0 \sin\theta_0 = b$$

Representation of a Point in the Complex Plane

$$|z| = \sqrt{2}$$
$$\angle z = 45^\circ$$
$$z = \sqrt{2}e^{j45^\circ}$$



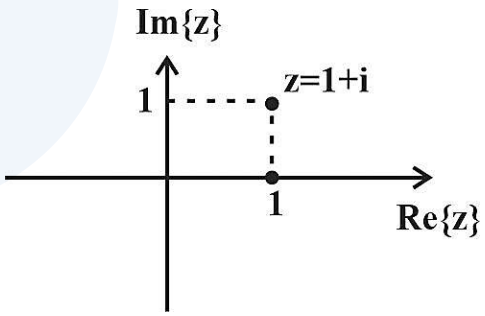


Fig. 10.9

Phase of a Complex Number

1. $P(x, y) = P(a, b) \rightarrow I\ quad$

$$\phi = \tan^{-1}\frac{b}{a} \pm 360^\circ$$
$$z = a + jb$$

2. $P(x, y) = P(-a, b) \rightarrow II\ quad$

$$\phi = \left(\tan^{-1}\frac{b}{-a}\right) \pm 180^\circ$$
$$z = -a + jb$$

3. $P(x, y) = P(-a, -b) \rightarrow III\ quad$

$$\phi = \tan^{-1}\left(\frac{-b}{-a}\right) \pm 180^\circ$$
$$z = -a - jb$$

4. $P(x,y) = P(a,-b) \rightarrow IV\ quad$

$$\phi = \tan^{-1}\left(\frac{-b}{a}\right) \pm 360^{\circ}$$

$$z = a - jb$$

Example: Represent the following point z = 1-i in a polar plane.

Solution: Z = 1-i = 1 + i (-1)

$$\phi = \tan^{-1}\left(\frac{-1}{1}\right) = -45^{\circ} \pm 360^{\circ} \begin{cases} \nearrow & -45^{\circ} \rightarrow cw \\ \searrow & +315^{\circ} \rightarrow Acw \end{cases}$$

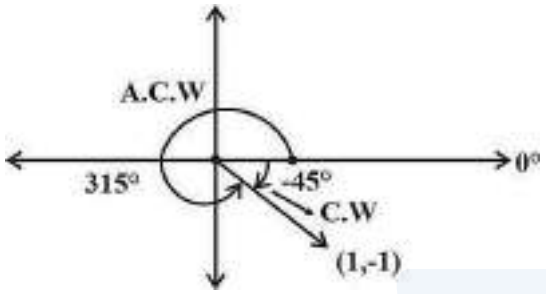


Fig. 10.10

Example: Represent the following point = -1-i in the polar plane.

Solution:

Z = -1-j

3 quadrant

$$\phi = \tan^{-1}\left(\frac{-1}{-1}\right) \pm 180^{\circ} = 45^{\circ} \pm 180^{\circ} \begin{cases} \nearrow & +225^{\circ} \rightarrow Acw \\ \searrow & -135^{\circ} \rightarrow cw \end{cases}$$

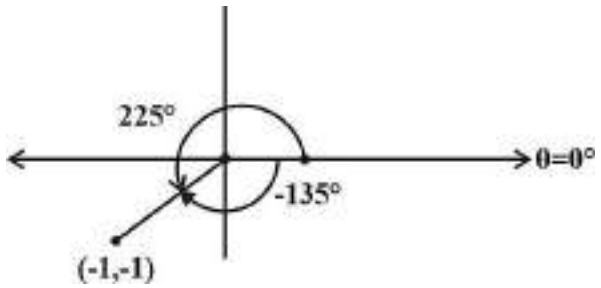


Fig. 10.11

Operations on Complex Numbers

$$z_1 = r_1 e^{j\theta_1}$$

$$z_2 = r_2 e^{j\theta_2}$$

1. $z = z_1 z_2 = r_1 r_2 e^{j(\theta_1+\theta_2)}$

$$|z| = r_1 r_2$$

$$\angle z = \theta_1 + \theta_2$$

2. $z = \frac{z_1}{z_2} = \frac{r_1}{r_2} e^{j(\theta_1-\theta_2)}$

$$|z| = \frac{r_1}{r_2}$$

$$\angle z = \theta_1 - \theta_2$$

Mapping

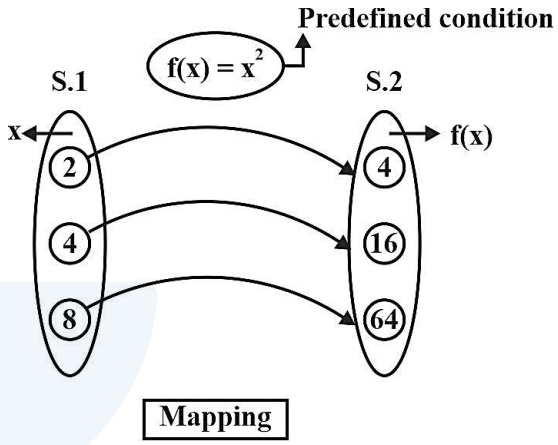


Fig. 10.12

Mapping between two Complex Planes

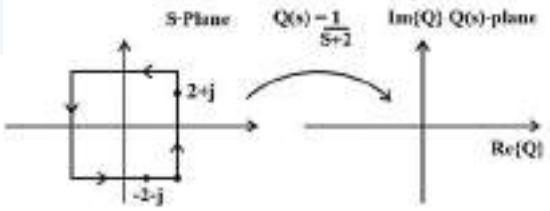


Fig. 10.13

Mapping between a Rectangular Plane and Complex Plane

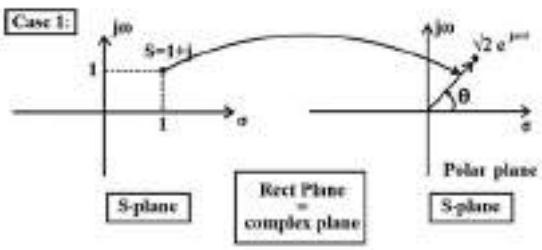


Fig. 10.14

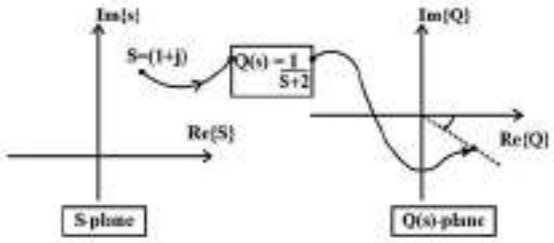


Fig. 10.15

Note: "In this chapter, mapping between a rectangular plane and polar has to be performed under predefined conditions."

1. Map the following s–plane trajectory in Q(s)–plane, where Q(s) = 1/(S+2)

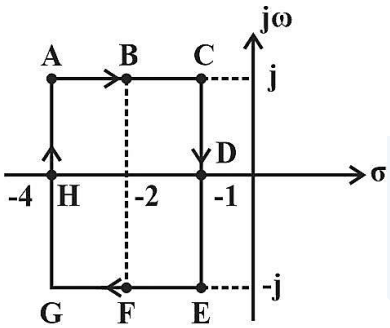


Fig. 10.16

Solution:

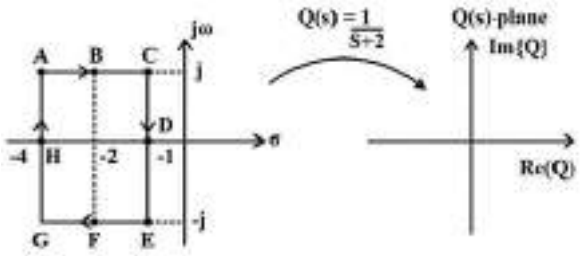


Fig. 10.17

S-plane

Closed curve in s–plane

or

Specified region in s–plane

or

Contour in s-plane

$A: \quad S = -4 + j \rightarrow Q(S) = Q(A)$

$$= \frac{1}{-4+2+j} = 0.4\angle -153°$$

$B: \quad S = -2 + j \rightarrow Q(S) = Q(B)$

$$= \frac{1}{-2+2+j} = 1\angle -90°$$

Point	S	Q(s)
A	$S = -4 + j$	$0.4\angle -153°$
B	$S = -2 + j$	$1\angle -90°$
C	$S = -1 + j$	$0.7\angle -45°$
D	$S = -1 + j0$	$1\angle 0°$
E	$S = -1 - j$	$0.7\angle 45°$
F	$S = -2 - j$	$1\angle 90°$
G	$S = -4 - j$	$0.4\angle 153°$
H	$S = -4$	$0.5\angle 180°$
A	$S = -4 + j$	$0.4\angle -153°$

Table 10.1

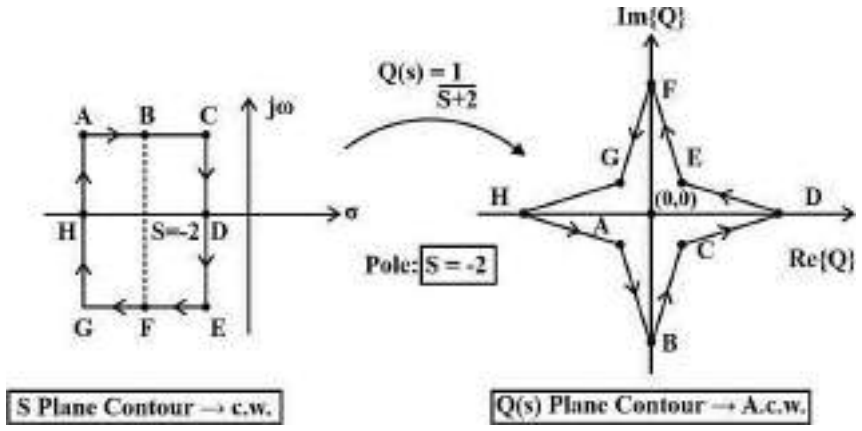
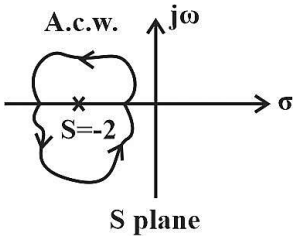


Fig. 10.17

Example:



Solution:

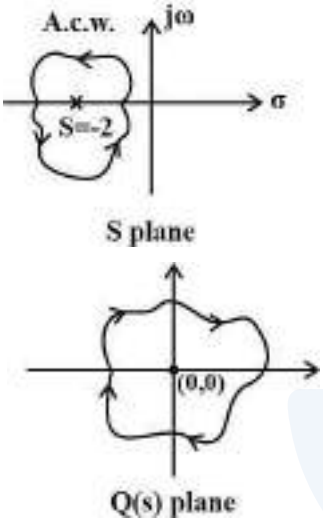


Fig. 10.18

$$(D) \rightarrow S = -1 \rightarrow Q(S) = Q(D) = \frac{1}{S+1} = \frac{1}{0} = \infty$$

Point	S	Q(s) = 1/s+2
A	$-4 + j$	$0.31\angle -161.5°$
B	$-2 + j$	$0.707\angle -135°$
C	$-1 + j$	$1\angle -90°$
D	$-1 + j0$	$\infty\angle 0°$
E	$-1 - j$	$1\angle 90°$
F	$-2 - j$	$0.707\angle 135°$
G	$-4 - j$	$0.31\angle 161.5°$
H	-4	$1/3\angle 180°$
A	$(-4 + j)$	$0.31\angle -161.5°$

Table 10.2

Example: Map the following s–plane trajectory in Q(s)–plane, where Q(s) = 1/(S+1)

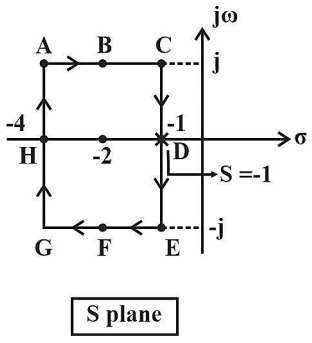


Fig. 10.19

Example:

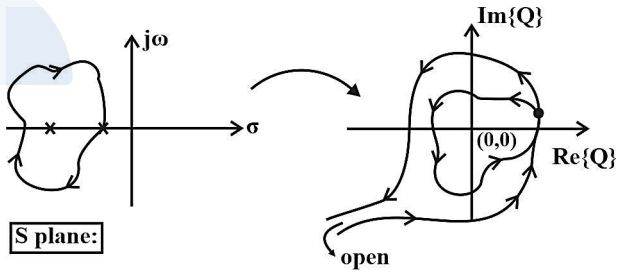


Fig. 10.20

Solution:

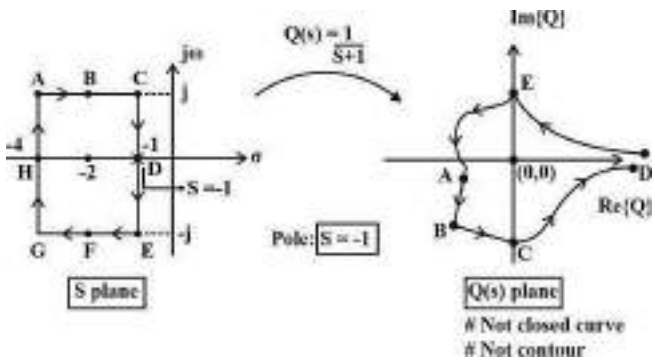


Fig. 10.21

Important Observations

1. Contour in the s–planecontains the m poles of Q(s) strictly inside it.	⟵ Opposite sense ⟶	Q(s) plot in Q(s)–plane will encircle the origin (0,0) m times.
2. If contour in s–plane passes through one or more poles of Q(s).	⟶	Q(s)plot in Q(s)–plane remains an open curve. Hence the concept of encirclement is Invalid.

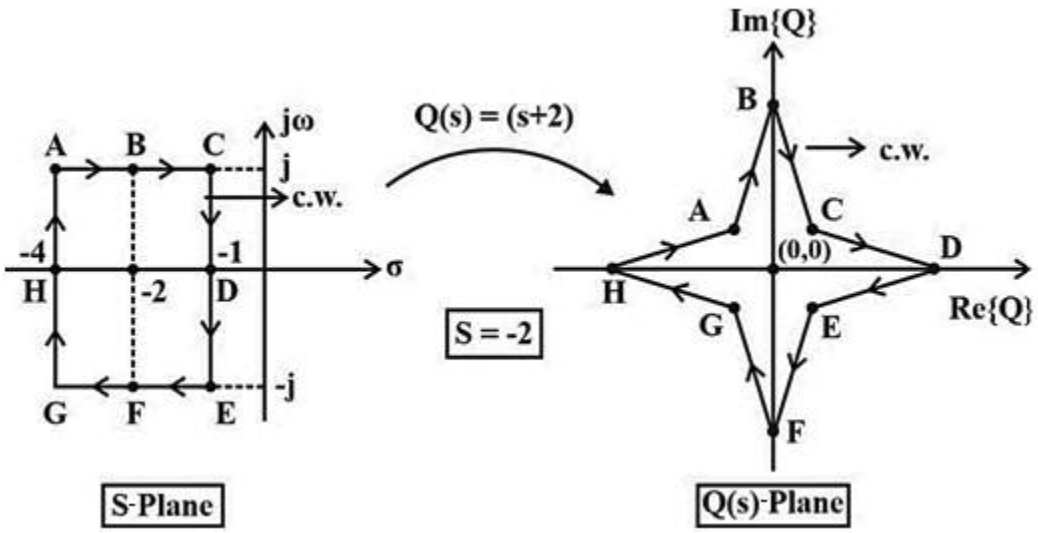


Fig. 10.22

Point	S	Q(s) = s+2
A	$-4+j$	$2.2\angle 153°$
B	$-2+j$	$1\angle 90°$
C	$-1+j$	$\sqrt{2}\angle 45°$
D	-1	$1\angle 0°$
E	$-1-j$	$\sqrt{2}\angle -45°$
F	$-2-j$	$1\angle -90°$
G	$-4-j$	$2.2\angle -153°$
H	-4	$2\angle 180°$
A	$(-4+j)$	$2.2\angle 163°$

Table 10.3

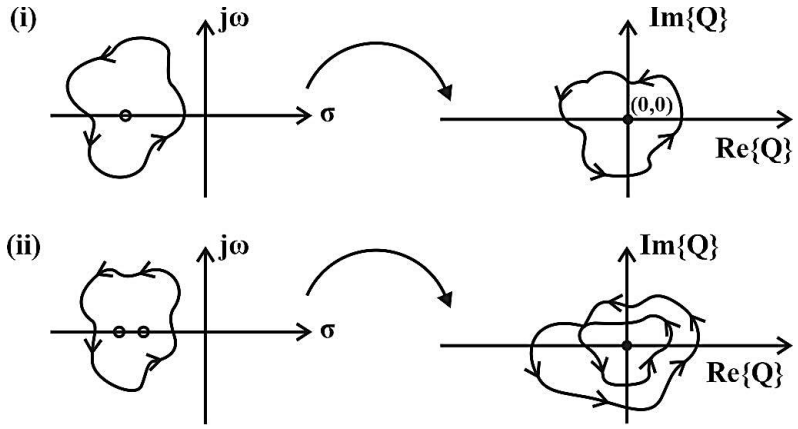


Fig. 10.23

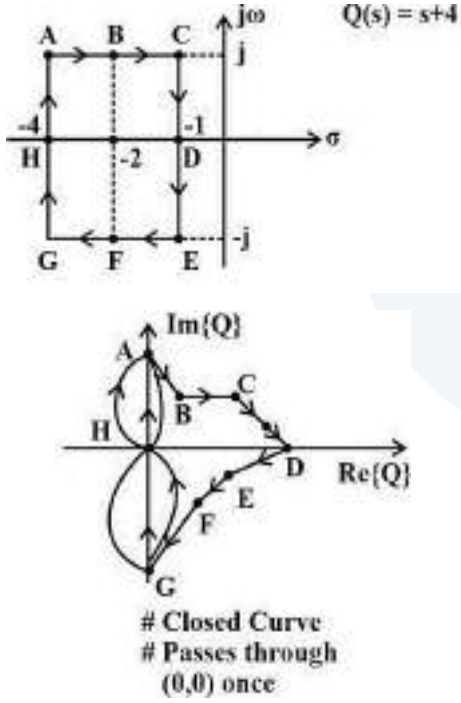


Fig. 10.24

Concept of encirciclement is Invalíd.

At H: S=-4 $Q(s) = s + 4 = 0\angle 0°$

Point	S	Q(s) = s+4
A	$-4 + j$	$1\angle 90°$
B	$-2 + j$	$2.23\angle 26°$
C	$-1 + j$	$3.16\angle 18°$
D	-1	$3\angle 0°$
E	$-1 - j$	$3.16\angle -18°$
F	$-2 - j$	$2.23\angle -26°$
G	$-4 - j$	$1\angle -90°$
H	-4	$0\angle 0°$
A	$(-4 + j)$	$1\angle 90°$

Table 10.4

Important Observations

1. Contour in s–plane contains m zeros of Q(s) strictly inside it.	⟵ Same direction ⟶	Q(s) plot in Q(s)–plane will encircle the origin (0,0) m times.
2. Contour in the s–plane has m zeros on the boundary of the contour.	⟵ Same sense ⟶	1. Q(s) plot in Q(s)–plane is closed 2. Q(s) plot crosses origin m times. 3. Concept of encirclement is invalid.

Example:

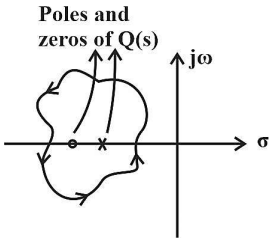


Fig. 10.25

Solution:

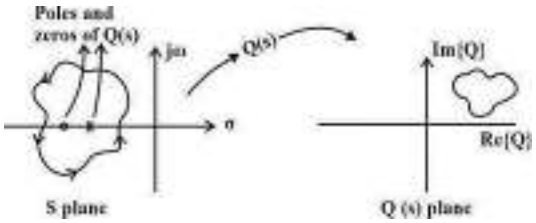


Fig. 10.26

Example:

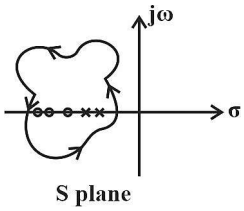


Fig. 10.27

Solution:

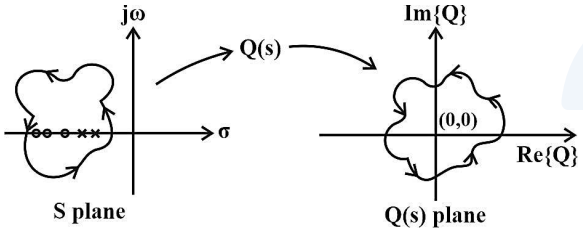


Fig. 10.28

No. of poles and zeros of Q(s) inside contour of s–plane

$p = l$

$z = m$

$m - l = 1$

Example:

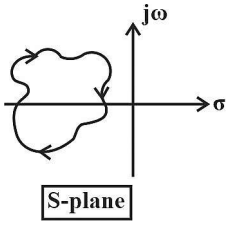


Fig. 10.29

Solution:

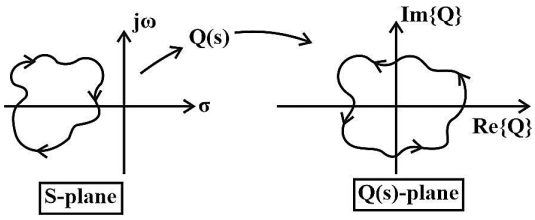


Fig. 10.30

No of poles and zeros of Q(s) inside the contour in s–plane:

P=l, Z=m & l-m = 1

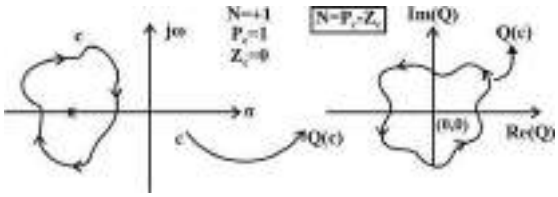


Fig. 10.31

Q(s) → Transfer function:

$$Q(s) = \frac{N(s) \to Z}{P(s) \to P}$$

P_c → No. of poles of T.F. Q(s) inside c.

Z_c→ No. of zeros of T.F. Q(s) inside c.

N = No. of encirclements of (0,0).

N → +ve : If "c" and Q(c) are in opposite direction

N → –ve : If c and Q(c) are in same direction

Rule of Mapping from S–plane to Q(s)-plane

$$Q(s) = \frac{N(s)}{D(s)} \to P_c, Z_c$$

$$\boxed{N = P_c - Z_c}$$

P_c = *No. of poles of T.F. Q(s) inside C*

z_c = No. of zeros of *T.F. Q(s)* inside C

N= No. of encirclements of origin by Q(s) plot.

$$N = \begin{cases} \nearrow & +ve \text{ if } c \text{ and } Q(c) \text{ are in opp direction} \\ \searrow & -ve \text{ if } c \text{ and } Q(c) \text{ are in same direction} \end{cases}$$

Limitations of the Above Rule

1. Poles of T.F. Q(s) should not lie on the boundary of s–plane contour.
2. Zeros of T.F. Q(s) should not lie on the boundary of s–plane contour.

Example:

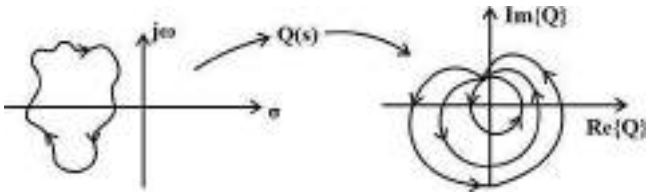


Fig. 10.32

No. of poles and zeros of Q(s) Inside the contour in the s–plane.

$$N = P_c - Z_c$$
$$3 = P_c - Z_c$$
$$N = +3$$

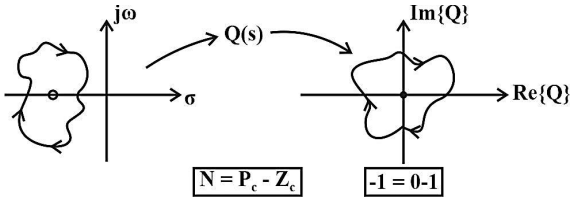


Fig. 10.33

$$N = -1$$
$$P_C = 0$$
$$Z_C = 1$$

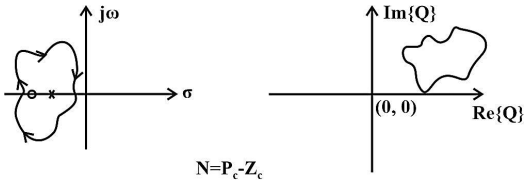


Fig. 10.34

$$N = 0$$
$$P_c = 1$$
$$Z_c = 1$$

$$N = P_c - Z_c$$
$$0 = 0$$

Relation between Plots in Q(s)–plane

Case 1: Q(s) plot in Q(s)–plane is given

Plot: KQ(s) plot in KQ(s)–plane

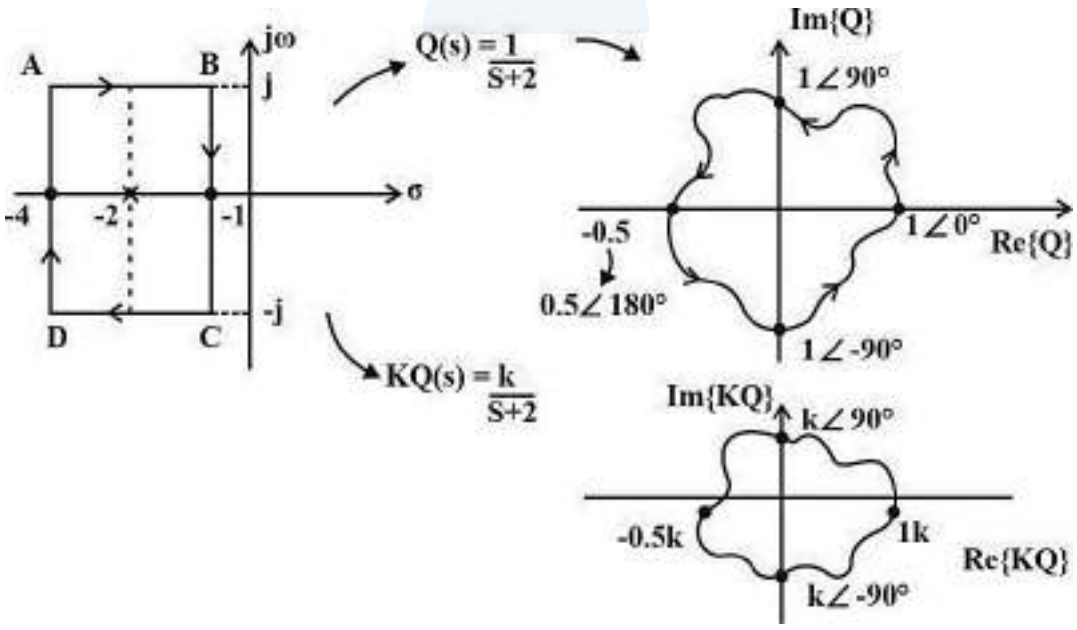


Fig. 10.35

Case 2: Given,

Q(s) plot in Q(s)–plane.

Plot: -Q(s) plot in -Q(s)–plane.

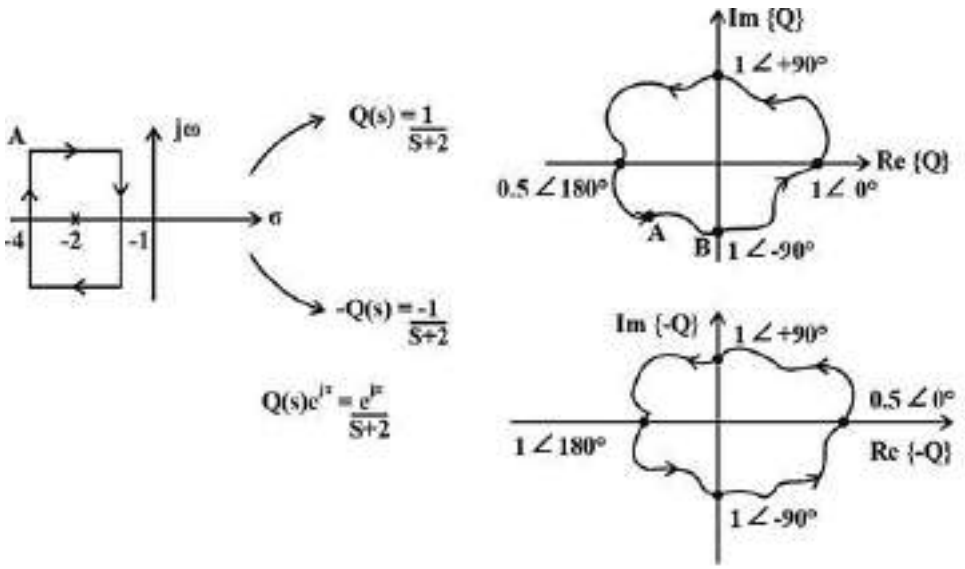


Fig. 10.36

Case 3: Given,

Q(s) plot in Q(s)–plane

Plot: $Ke^{j\theta}Q(s)$ plot in $Ke^{j\theta}Q(s)$ –plane

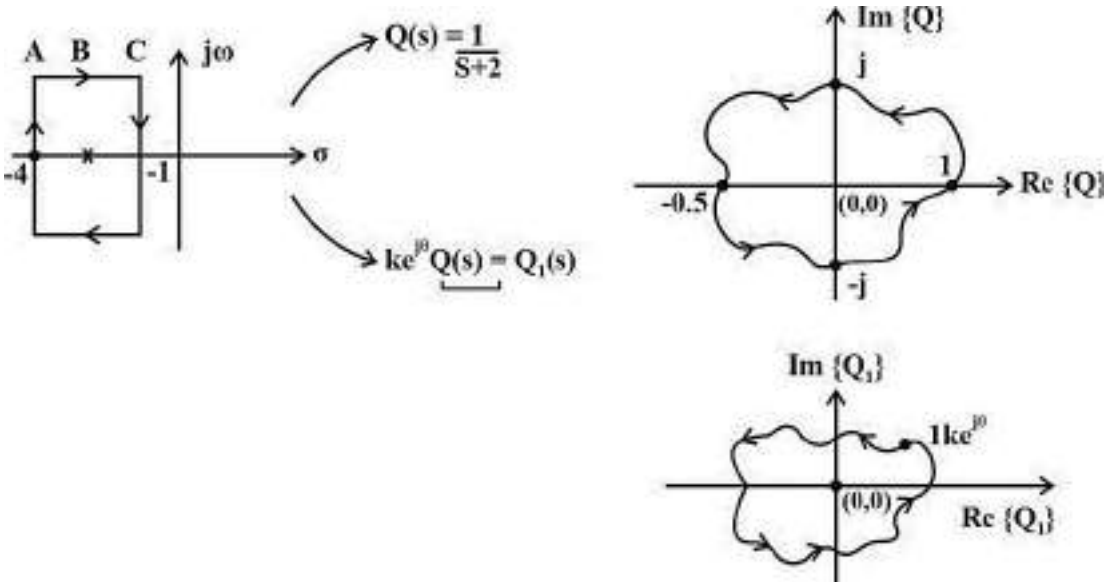


Fig. 10.37

Example:

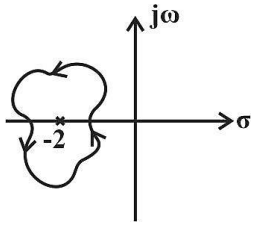


Fig. 10.38

Solution:

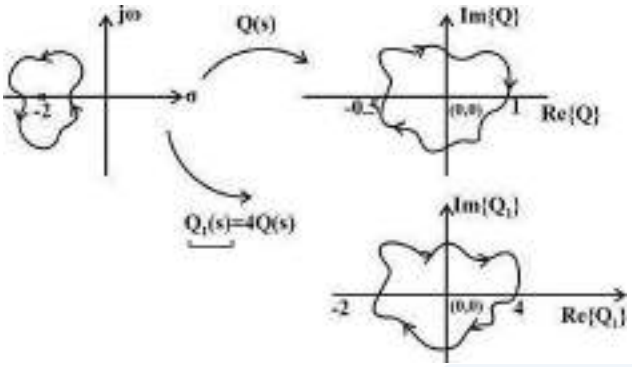


Fig. 10.39

Example:

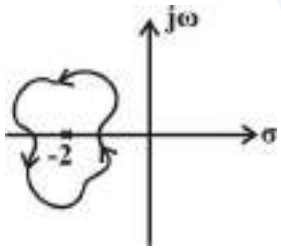


Fig. 10.40

Solution:

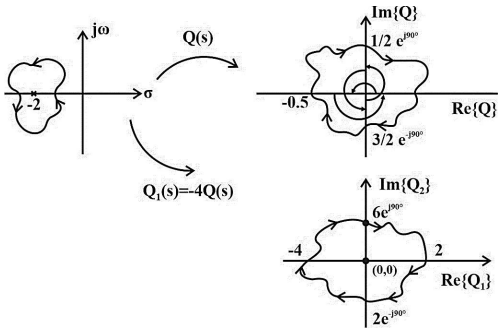


Fig. 10.41

Important Observations

If Q(s) plot in the Q(s)–plane is known for some contour in the s-plane.

$$Q_1(s) = \left(r_0 e^{j\theta_0}\right) Q(s)$$

#Then $Q_1(s)$ plot in the $Q_1(s)$–plane for the same contour will

1. Have the same sense of rotation.
2. Encirclement of origin does not change.
3. Plot of $Q_1(s)$ is Rotated by θ_0 in same sense of rotation & scaled by r_0.

Case 4: Given,

Q(s) plot in Q(s)–plane

Plot: $1+Q(s)$ plot in $1+Q(s)$ –plane

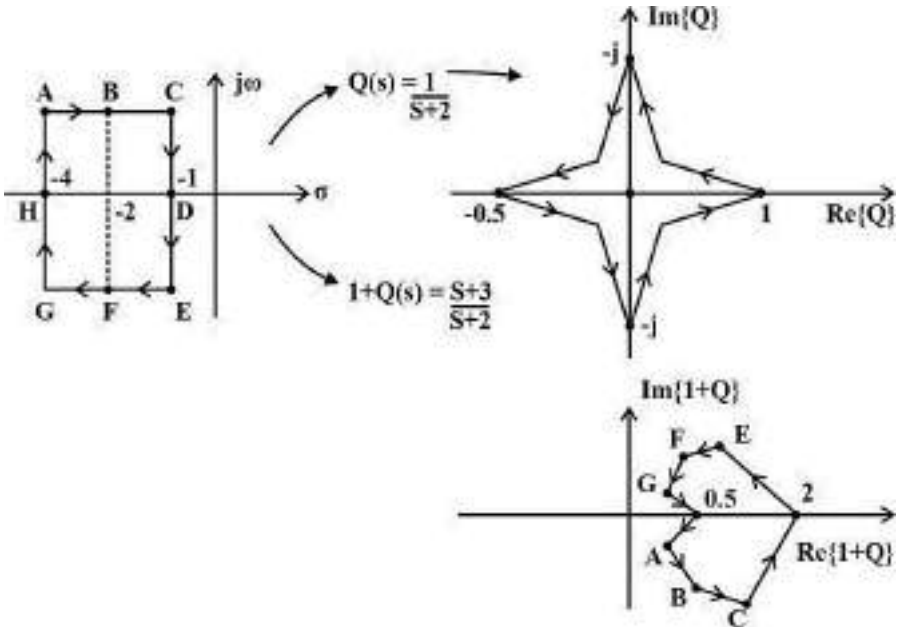


Fig. 10.42

Point	S	Q(s) = 1/S+2	1+Q(s) = S+3/S+2
A	$-4+j$	$0.4\angle -153°$	$0.6\angle -18.4°$
B	$-2+j$	$1\angle -90°$	$\sqrt{2}\angle -45°$
C	$-1+j$	$0.7\angle -45°$	$1.58\angle -18°$
D	$-1+j$	$1\angle 0°$	$2\angle 0°$
E	$-1-j$	$0.7\angle 45°$	$1.58\angle 18°$
F	$-2-j$	$1\angle 90°$	$\sqrt{2}\angle 45°$
G	$-4-j$	$0.4\angle 153°$	$0.6\angle 18.4°$
H	-4	$0.5\angle 180°$	$0.5\angle 0°$
A	$(-4+j)$	$0.4\angle -153°$	$0.6\angle -18.4°$

Table 10.5

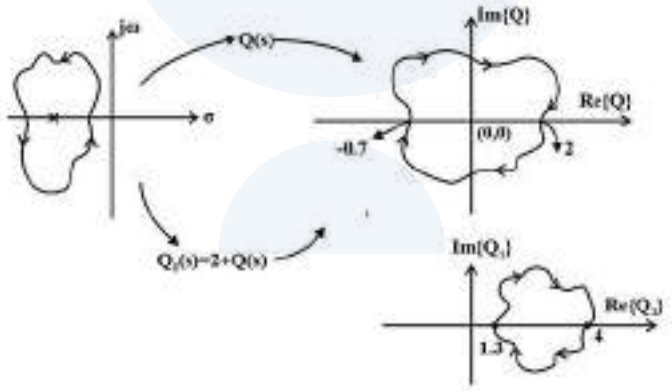


Fig. 10.43

2.

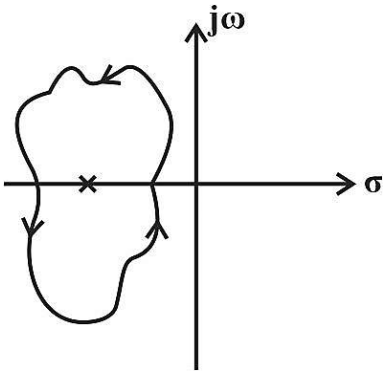


Fig. 10.44

Solution:

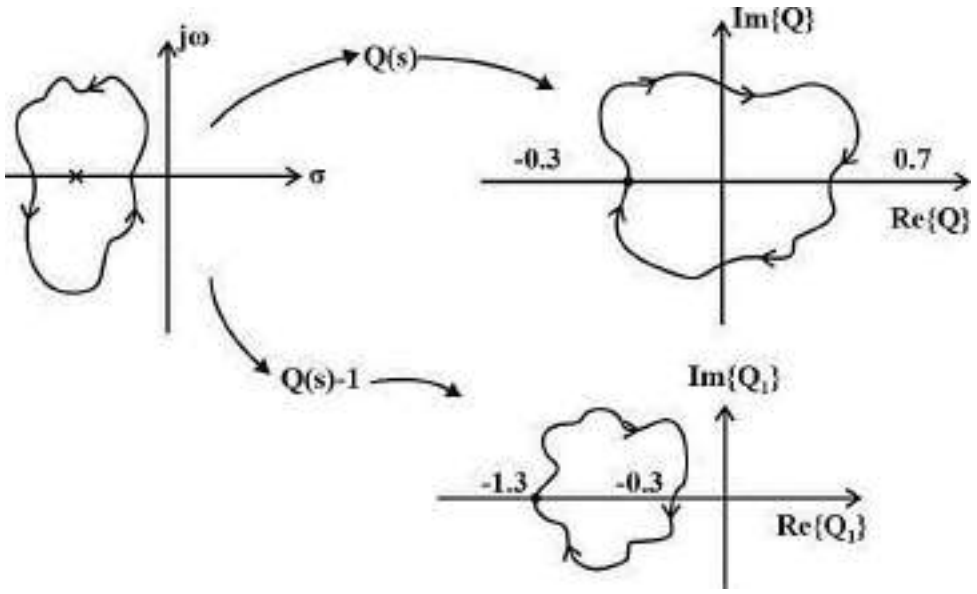


Fig. 10.45

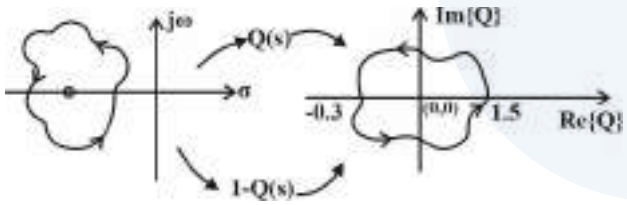


Fig. 10.46

$Q(s) \rightarrow -Q(s) \rightarrow -Q(s) + 1$

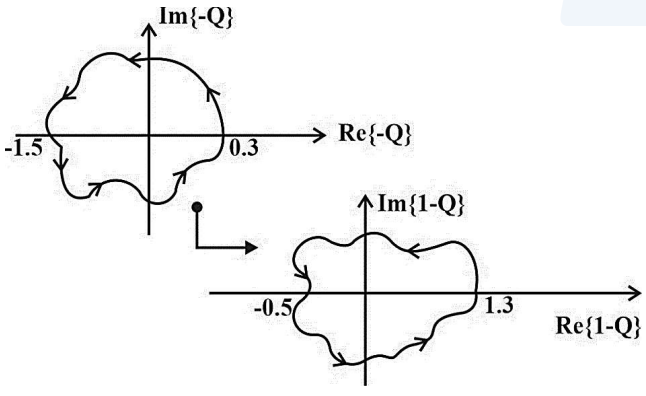


Fig. 10.47

Given: Q(s) plot in Q(s)–plane for a givencontour in s–plane.

Plot: -4+3Q(s) plot in -4+3Q(s)–plane for same contour.

$$Q(s) \rightarrow \underset{\text{Scaling every point of Q(s) by 3}}{3Q(s)} \rightarrow \underset{\text{Left shift by 4 units}}{3Q(s) - 4}$$

Important Observation:

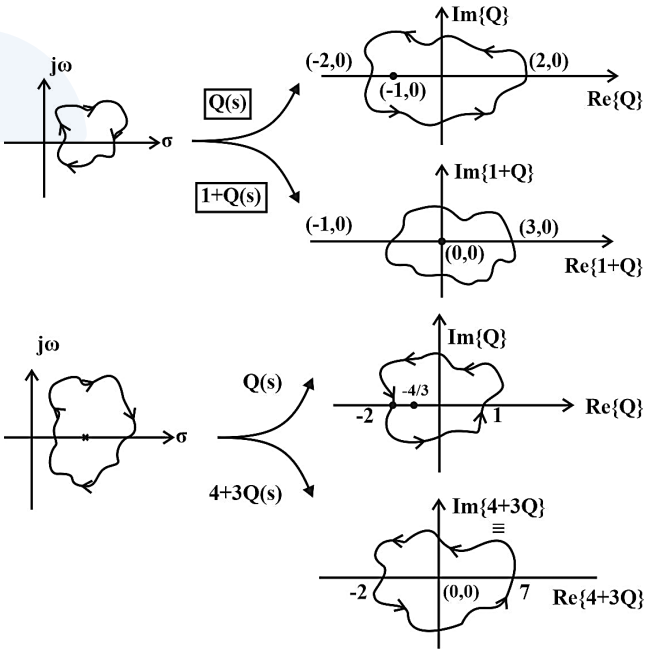


Fig. 10.48

$4 + 3Q(s) = 0$

$Q(s) = -4 / 3$

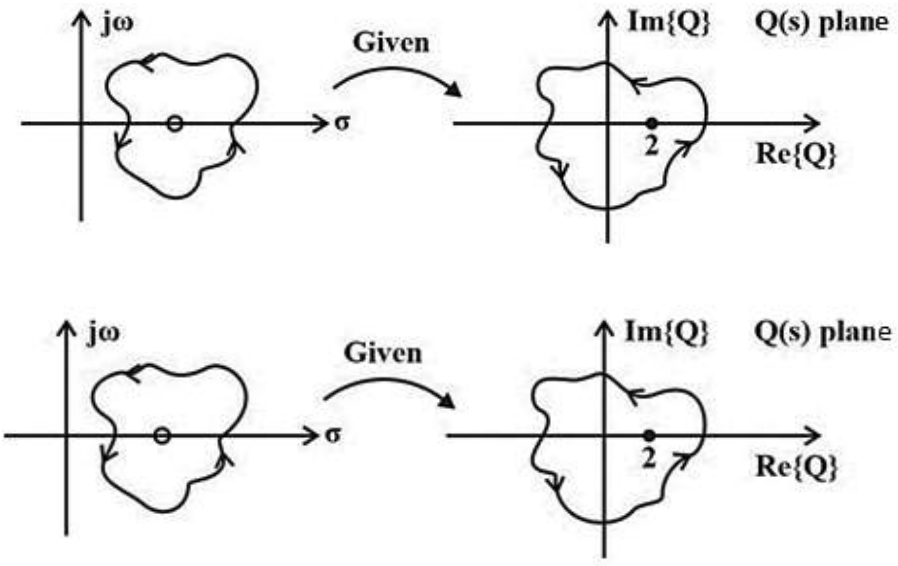


Fig. 10.49

Calculate the encirclement of (0,0) by 4-2Q(s) plot in 4-2Q(s)-plane for the same contour in s- plane → 1 time.

$\rightarrow 4 - 2Q(s) = 0 \rightarrow (0,0)$

$\rightarrow \quad Q(s) = 2 \rightarrow (2,0)$

Given		Plot
Q(s) plot	→	-Q(s) plot → 180° rotation in the same sense.
Q(s) plot	→	{1+Q(s)} plot → shift by 1 unit rightward.
Q(s) plot	→	{Q(s)-1} plot → shift by 1 unit leftward.

Table 10.6

Points To Remember

1. Principle of Arguments:

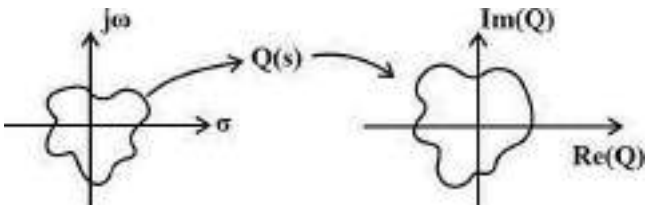


Fig. 10.51

$$Q(s) = \frac{N(s)}{D(s)}$$

If contour in S–plane contains P poles and Z zeros inside it, then

Q(s) plot in Q(s)–plane:

$P > Z \rightarrow$: Q(s) plot in the Q(s)-plane will encircle the origin (0,0), (P–Z) times inthe direction opposite to the contour in the S–plane.

$P < Z \rightarrow$: Q(s) plot in the Q(s)–plane will encircle the origin (0,0), (Z–P) times in the same direction as the contour in S-plane.

$P = Z \rightarrow$: Q(s) plot inQ(s)–plane does not encircle the origin.

2. Rule of Mapping:

It is applicable to T.F → $Q(s) = \frac{N(s)}{D(s)}$

$N = P_c - Z_c$

P_c = No. of poles of Q(s) lying inside contour in s–plane.

Z_c = No. of zeros of Q(s) lying inside contour in s–plane.

N = No. of encirclementsabout (0,0)

$N \nearrow +ve$: Sense of s and Q(s) opposite.

$N \searrow -ve$: Sense of s and Q(s) is same.

3.

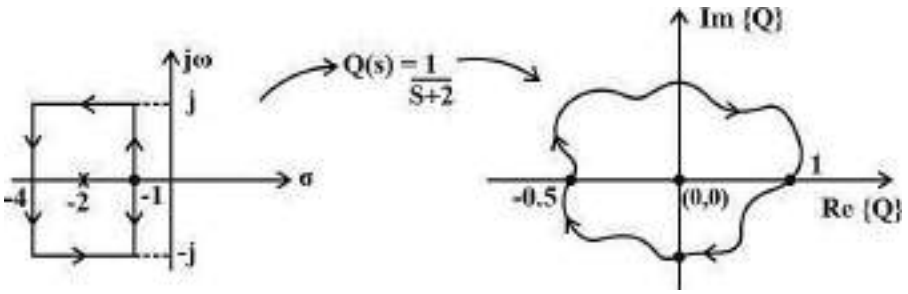


Fig. 10.52

Rule of Mapping

$$Q(s) = \frac{1}{s+2}$$

$$N = P_c - Z_c$$

$P_c = 1$

$Z_c = 0$

$N = +1$

1.

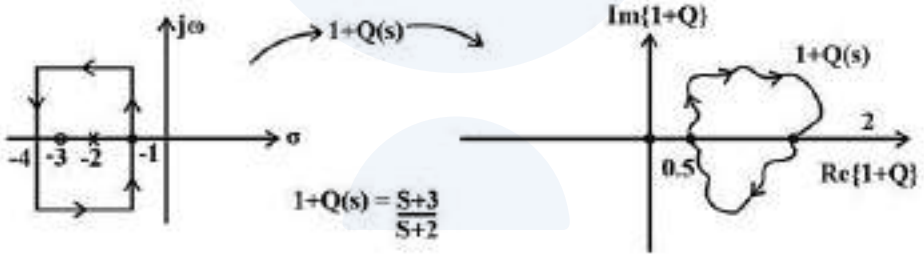


Fig. 10.53

$P_c = 1$

$Z_c = 1$

$N = 0$

$N = P_c - Z_c$

2.

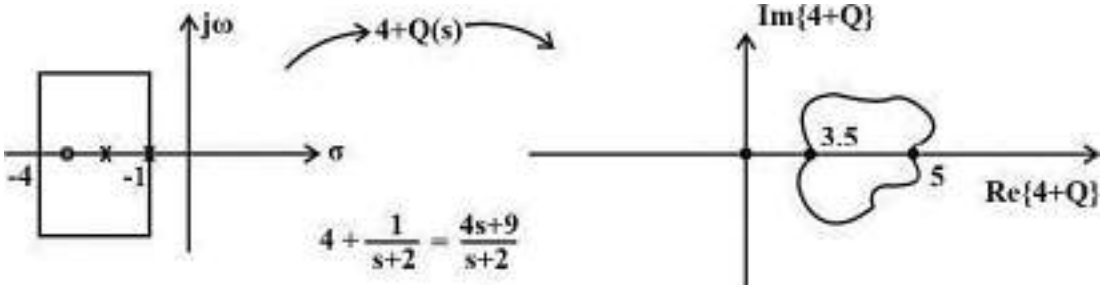


Fig. 10.54

Rule of mapping in 4+Q(s)

$N = P_c - Z_c$

T.F.: $Q(s) = \dfrac{N(s) \rightarrow Zeros}{D(s) \rightarrow poles}$

T.F.: $A + BQ(s) = A + \dfrac{BN(s)}{D(s)} = \dfrac{AD(s) + BN(S)}{D(s)}$

Poles of T.F. Q(s) and poles of A+BQ(s) will be the same.

If A+BQ(s) plot, encircles origin in A+BQ(s)–plane then:

$A + BQ(s) = 0$

$Q(s) = -A / B$

or

Q(s) plot in Q(s)–plane will encircle $(-A / B, 0)$ in Q(s)–plane.

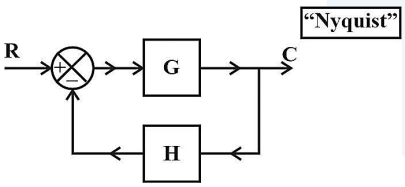


Fig. 10.55

** $OLTF: \quad G(s)H(s) \rightarrow T.F.$

C.E. : $1 + G(s)H(s) \rightarrow T.F.$

$CLTF: \dfrac{C(s)}{R(s)} = \dfrac{G(s)}{1 + G(s)H(s)} \rightarrow T.F.$

$\boxed{G(s)H(s) = \dfrac{N(s)}{D(s)}} \rightarrow T.F.$

$\boxed{1 + G(s)H(s) = \dfrac{N(s) + D(s)}{D(s)}} \rightarrow T.F.$

Roots of $T.F.(1 + G(s)H(s)) =$ *Zeros of* $T \cdot F \cdot (1 + G(s)H(s)) =$ Poles of C.L.T.F.

Poles of $T.F.(1 + G(s)H(s)) =$ Poles of T.F. G(s) H(s)

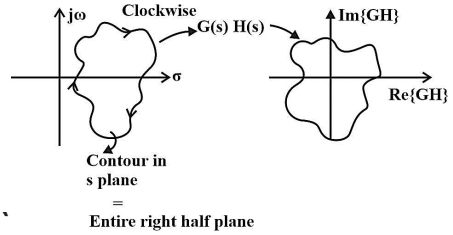


Fig. 10.56

Rule of Mapping

$N = P_c - Z_c \rightarrow T.F.$

$N =$ No. of encirclements of (0,0) by G(s) H(s) plot in G(s)H(s)–plane.

$N = +ve \rightarrow N : ACW$

$N = -Ve \rightarrow N : CW$

x=No. of zeros of T.F. G(s) H(s) lying in R.H.P.

For O.L.S → $G(s)H(s) = \dfrac{N(s)}{D(s)}$ to be stable:

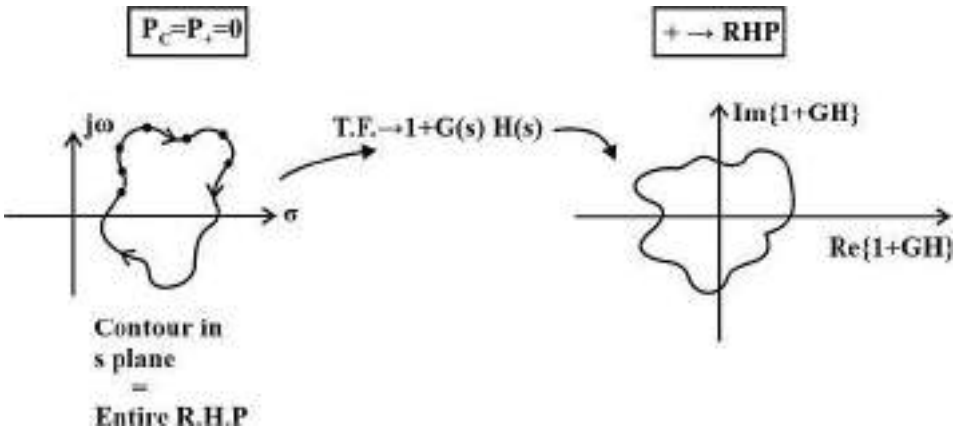


Fig. 10.57

Rule of Mapping

$$1+G(s)H(s)=\frac{N(s)+D(s)}{D(s)}\Rightarrow T\cdot F.$$

N = No. of encirclements of (0,0) by (1+G(s) H(s))plot in (1+G(s) H(s))–plane.

$$1+G(s)H(s)=0$$
$$G(s)H(s)=-1$$

Or

No. of encirclements of (-1,0) by G(s) H(s) plot in G(s) H(s)–plane.

$N=+ve:N:ACW$

$N=-ve \quad N:CW$

$P_c=P_+$ = No. of poles of 1+G(s)H(s) lying inside R.H.P.

or

No. of poles of G(s)H(s) lying inside R.H.P.

$Z_c=Z_+$ = No. of zeros of 1+G(s) H(s) lying inside R.H.P.

or

No. of poles of -ve f/b C.L.S. lying in R.H.P.

For T.F. (1+G(s)H(s)) to be stable → $\boxed{P_c=0=P_+}$

For C.L.T.F. with -ve feedback to be stable→ $\boxed{Z_c=0=Z_+}$

Standard Contour

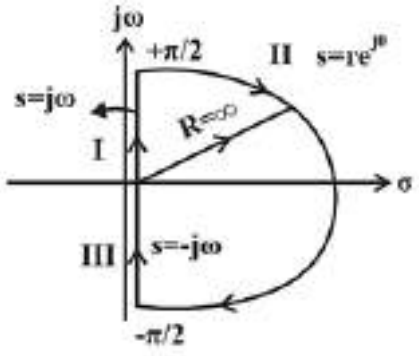


Fig. 10.58

1. $s=+j\omega:0<\omega<\infty$
2. $s=re^{j\theta}$

   $r=\infty$

   $\theta=+\frac{\pi}{2}\,to\,\frac{-\pi}{2}$
3. $s=-j\omega \quad 0<\omega<\infty$

1.

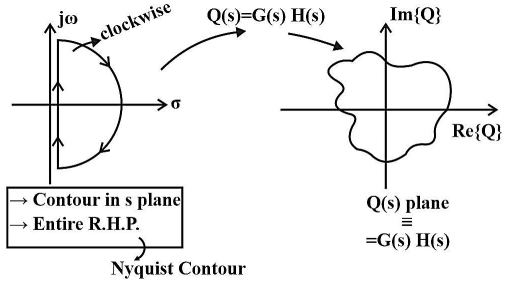


Fig. 10.59

2. Rule of mapping for T.F. Q(s) → G(s) H(s)

Nyquist Stability Criteria → NSC.

Case 1: NSC for G(s) H(s) = $\frac{N(s)}{D(s)}$

N = No. of encirclements of (0,0) by G(s) H(s) plot in G(s) H(s)–plane.

P_t = No. of poles of T.F. G(s) H(s) lying strictly in R.H.P.

Z_t = No. of zeros of T.F. G(s) H(s) lying strictly in R.H.P.

Case 2: N.S.C for T.F. 1+G(s)H(s)

N = No. of encirclements of (0,0) by 1+G(s)H(s) plot in (1+G(s)H(s))–plane.

$$1+G(s)H(s)=0$$
$$G(s)H(s)=-1$$

Or

No. of encirclements of (-1,0) by G(s)H(s) plot in G(s)H(s)-plane.

$N=+ve \quad N:ACW$

$N=-ve \quad N:CW$

P_+ = No. of poles of T.F. 1+G(s)H(s) lying strictly in R.H.P

Or

No. of poles of T.F. G(s)H(s) lying strictly in R.H.P.

Z_+ = No. of zeros of T.F. 1+G(s)H(s) lying strictly in R.H.P

OR

No. of poles of $C \cdot L \cdot T \cdot F. \dfrac{G(s)}{1+G(s)H(s)}$ lying strictly in R.H.P.

Case 3: N.S.C. for T.F. 1-G(s) H(s)

N = No. of encirclements of (0,0) by (1-G(s) H(s)) plot in (1-G(s)H(s))–plane.

$$1 - G(s)H(s) = 0$$
$$G(s)H(s) = 1$$

Or

No. of encirclements of (1,0) by G(s)H(s) plot in G(s)H(s)–plane.

$N = +ve \quad N : ACW$
$N = -ve \quad N : CW$

P_+ = No. of poles of (1-G(s)H(s)) lying strictly in R.H.P.

Or

No. of poles of G(s)H(s) lying strictly in R.H.P.

Z_+ = No. of zeros of 1-G(s)H(s) lying strictly in R.H.P.

Or

No. of poles of $CLTF = \dfrac{G(s)}{1-G(s)H(s)}$ lying in R.H.P

Case 4: N.S.C. for T.F. 4+3G(s) H(s)

N = No. of encirclements of (0,0) by 4+3G(s) H(s) plot in 4+3G(s) H(s)–plane.

$$4 + 3G(s)H(s) = 0$$
$$G(s)H(s) = -4/3$$

Or

No. of encirclements of (-4/3,0) by G(s)H(s) plot in G(s)H(s)–plane.

P_+ = No. of poles of 4+3G(s)H(s) lying strictly in R.H.P.

Or

No. of poles of G(s)H(s) lying strictly in R.H.P.

Z_+ = No. of zeros of 4+3G(s)H(s) lying strictly in R.H.P

Or

No. of poles of $C.L.T.F. \dfrac{G(s)}{4+3G(s)H(s)}$ lying strictly in R.H.P.

Case 5: N.S.C. for T.F. A+BG(s)H(s)

N = No. of encirclements of (0,0) by A+BG(s) H(s) plot in A+BG(s)H(s)–plane

Or

No. of encirclements of (-A/B, 0) by G(s)H(s) plot in G(s) H(s)–plane.

$N = +ve \rightarrow N : A.C.W$
$N = -ve \rightarrow N : C.W$

P_+ = No. of poles of A+BG(s)H(s) lying in R.H.P.

Or

No. of poles of G(s)H(s) lying in R.H.P.

Z_+ = No. of zeros of A+BG(s)H(s) lying in R.H.P.

Or

No. of poles of $\dfrac{G(s)}{A+BG(s)H(s)}$ lying in R.H.P.

Common Mistakes

Do not blindly calculate N; take care of the following points:

- Check the system: Open-loop or closed-loop
- Type of feedback
- Sign of N, to decide the direction of N

Problem Solving Approach

N.S.C for T.F. G(s)H(s) $= \dfrac{N(s)}{D(s)}$

$N = P_+ - Z_+$

1. $P_+ \rightarrow$ Roots of D(s) lying in R.H.P.
   a) By observation
   b) By Routh table using D(s)
   c) $P_+ = N + Z_+$
2. $Z_+ \rightarrow$ Roots of N(s) lying in R.H.P.
   a) By observation
   b) By Routh table using N(s)
   c) $Z_+ = P_+ - N$
3. $N \rightarrow$
   a) $N = P_+ - Z_+$
   b) From G(s)H(s) plot → or from Nyquist plot

N.S.C. for $1 + G(s)H(s) = \dfrac{N(s) + D(s)}{D(s)}$

N.S.C.:

Roots of D(s) in R.H.P.

OR

Poles of 1+G(s)H(s) in R.H.P.

1. By determining poles of G(s)H(s)

   Z_+ = Roots of (N(s)) + D(s)) = 0
2. By observation
3. By forming the Routh table using

Calculation of N

1. $N = P_+ - Z_+$
2. From Nyquist plot or G(s)H(s) plot.

Example The unity (-)vefeedback system has the O.L.T.F. $G(s) = \dfrac{1}{(s-1)(s+2)(s+5)}$.

The Nyquist plot encircles the origin (0,0) how manytimes.

Solution:

Nyquist plot → G(s)H(s) plot

N.S.C. in T.F. G(s)

$N = P_+ - Z_+$

$N = ?$

$P_+ = 1$

$Z_+ = 0$

$N = 1 - 0 \rightarrow$ Nyquist plot (GH plot) encircles origin once

ACW w.r.t clockwise N.C.

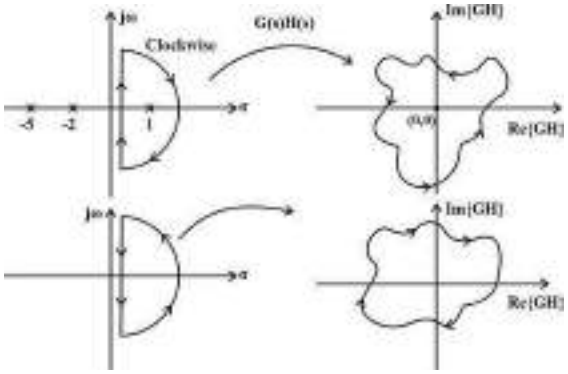


Fig. 10.61

1. How many timesdoes the Nyquist plot of {1+G(s)H(s)} encircles the origin?

   Or

   How many times does the Nyquist plot of {G(s)H(s)} encircles (-1, 0)?

   #N.S.C, for 1+G(s)H(s): $N = P_+ - Z_+$

   $N = ? \rightarrow Plot = ?$

   $P_+ = 1$

   $Z_+ = ? \rightarrow$

   $1 + G(s)H(s) = 1 + \dfrac{1}{(s-1)(s+2)(s+5)}$

   $1 + G(s)H(s) = 0$

   $(s-1)(s+2)(s+5) + 1 = 0 \rightarrow$ RH Table $Z_+ = 1$

   $N = P_+ - Z_+ = 0$

   $1 + G(s)H(s) = s^3 + 6s^2 + 3s - 9 \rightarrow$ 1 Root or 1 zero in R.H.P.

+	s^3	1	3
+	s^2	2	–3
+	s^1	9 / 2	0
–	s°	–3	

2. How many times Nyquist Plot of G(S)H(s) encircles the point (-4 / 3,0).

   $G(s)H(s) = -4/3$

   $4 + 3G(s)H(s) = 0$

$$4 + 3G(s)H(s)$$

N.S.C. → ↓ ↓ ↓

$$N = P_+ - Z_+$$

$P_+ = 1$

$Z_+ = 4 + 3G(s)H(s) \rightarrow Roots$

$$4 + 3GH = 4 + \frac{3}{(s-1)(s+2)(s+5)} = 0$$

$$4 + 3GH = 4(s-1)(s+2)(s+5) + 3 = 0$$

Using RH-criteria, $Z_+ = 1$

N.S.C.: $N = P_+ - Z_+ = 0$

3. No. of encirclements of (0,0) by Nyquist Plot 3-2G(s)H(s).

N.S.C. in $3 - 2G(s)H(s)$

$N = P_+ - Z_+$

$N = ?$

$P_+ = 1$

$Z_+ = R.H.$ -criteria

$$3 - 2GH = 3 - \frac{2}{(s-1)(s+2)(s+3)}$$

$$= 3(s-1)(s+2)(s+3) - 2 = 0$$

$Z_+ = 1$

$N = 0$

Variations of N.S.C.

Case 1:

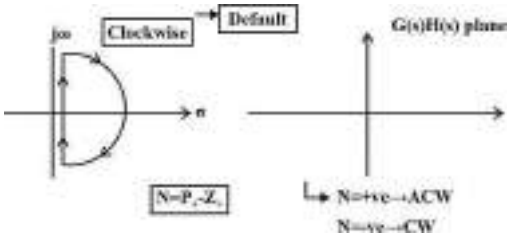


Fig. 10.62

Case 2:

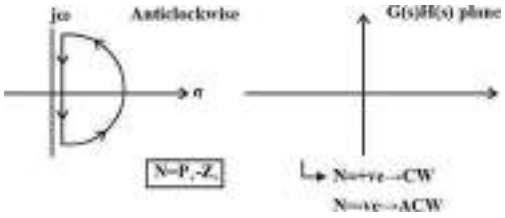


Fig. 10.63

Case 3:

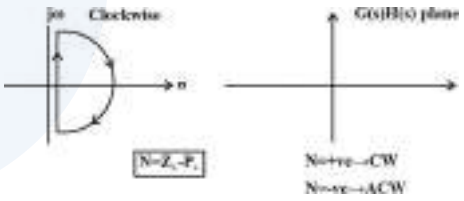


Fig. 10.64

Case 4:

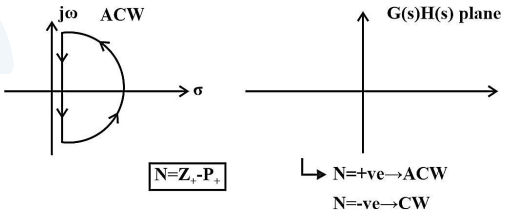


Fig. 10.65

Chapter Summary

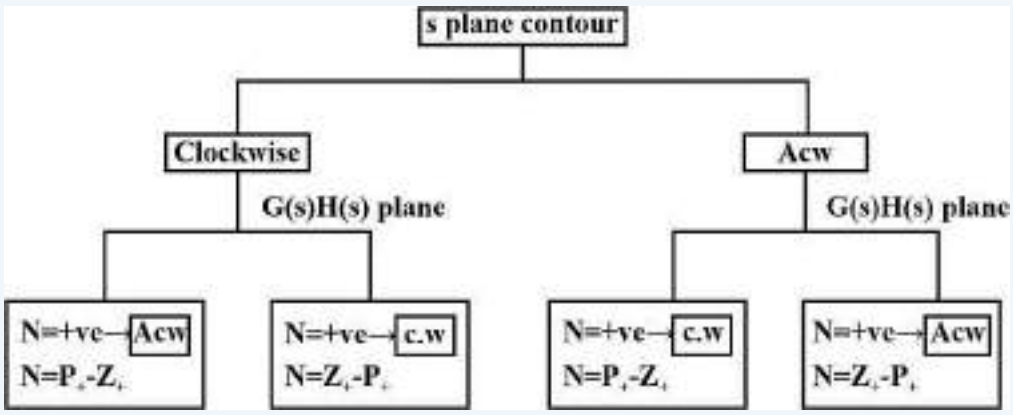


Fig. 10.66

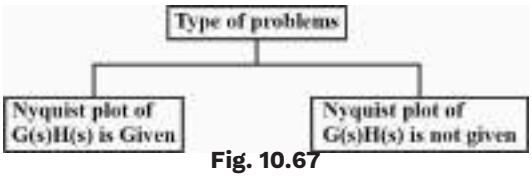


Fig. 10.67

$N = ?$

$P_+ = ?$ → Through plot

$Z_+ = ?$

Types of Nyquist Contour

Case 1: $T(s) = \dfrac{1}{s+1}$

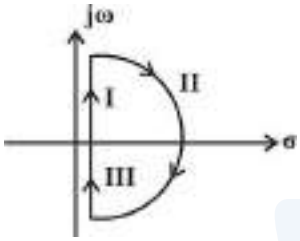


Fig. 10.68

Case 2: $T(s) = \dfrac{1}{s(1+s)}$

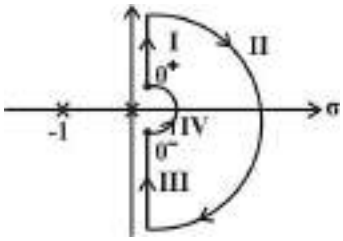


Fig. 10.69

1. $S = j\omega \quad \underbrace{0^+ < \omega < \infty}$

2. $S = \text{Re}^{j\theta}$

   $R = \infty$

   $\theta = +\dfrac{\pi}{2} \text{ to } \dfrac{-\pi}{2}$

   → c.w.

3. $S = +j\omega \quad -\infty < \omega < 0^-$

4. $S = \text{Re}^{j\theta}$

   $R = 0$

   $\theta = \dfrac{-\pi}{2} \text{ to } \dfrac{+\pi}{2}$

   → A.c.w.

Case 3:

$T(s) = \dfrac{1}{s(s^2+4)}$

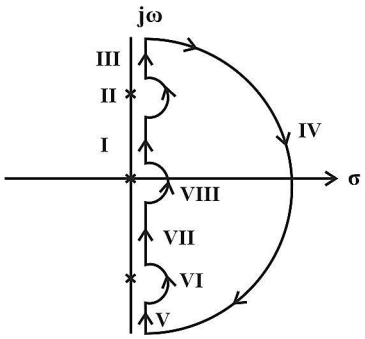


Fig. 10.70

Case 4:

$T(s) = s\left(s^2+4\right)$

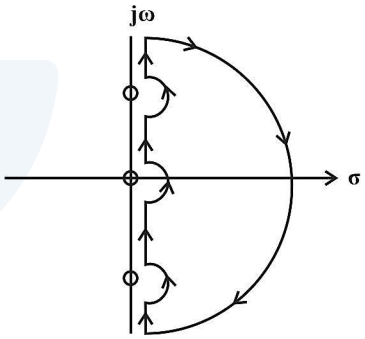
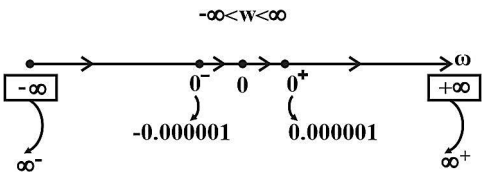


Fig. 10.71

Generalised Nyquist Contour



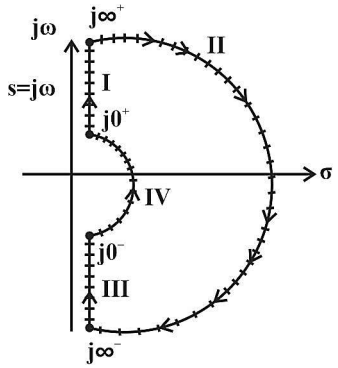


Fig. 10.72

1. $S = j\omega \quad 0^+ < \omega < \omega^+$

2. $S = Re^{j\theta}$

   $R \to \infty$

   $\infty^+ < \omega < \infty^-$

   $\theta \to \frac{\pi}{2}$ to $-\frac{\pi}{2}$

   C.W.

3. $S = j\omega \qquad \infty^- < \omega < 0^-$

4. $S = Re^{j\theta}$

   $R \to 0$

   $0^- < \omega < 0^+$

   $\theta \to \frac{-\pi}{2}$ to $\frac{\pi}{2}$

   A.C.W

Closing of Nyquist Plot

1. $S = j\omega \to 0^+ < \omega < \infty^+ \Rightarrow T(s) : T(j0^+)$ to $T(j\infty^+)$
2. $S = Re^{j\theta} \to \infty^+ < \omega < \infty^- \Rightarrow T(s) : T(j\infty^+)$ to $T(j\infty^-)$
3. $S = j\omega \to \infty^- < \omega < 0^- \Rightarrow T(s) : T(j\infty^-)$ to $T(j0^-)$
4. $S = Re^{j\theta} \to 0^- < \omega < 0^+ \Rightarrow T(s) : T(j0^-)$ to $T(j0^+)$

   ↓ ↓

   Closed contour in S – plane *Closed contour in $G(s)H(s)$ plane → Nyquist.*

$\theta \to -\frac{\pi}{2}$ to $+\frac{\pi}{2} \to$ A.c.w.

$-\theta \to \frac{\pi}{2}$ to $-\frac{\pi}{2} \to$ c.w.

$\theta = -\pi$ to $\pi \to$ clockwise

$-2\theta \to 2\pi$ to $-2\pi \to$ A.c.w.

$T(s) = \frac{1}{s+1}$

$T(j\omega) = \frac{1}{1+j\omega}$

$|T(j\omega)| = \frac{1}{\sqrt{1+\omega^2}}$ $\qquad T(j\omega) = \frac{1}{1+j\omega}$

Fig. 10.73

$\angle T(j\omega) = -\tan^{-1}\omega$

Method 1:

$\omega = 0^+ \quad |T(j0^+)| = 1$

$\angle T(j0^+) = 0°$

$\omega = \infty^+ \quad |T(j\infty^+)| = 0$

$\angle T(j\infty^+) = -90°$

Method 2:

$T(j\omega) = \frac{1}{1+j\omega} \to T(j0^+) = \frac{1}{1+j0^+} = 1 \Rightarrow 1\angle 0°$

$T(j\infty^+) = \frac{1}{1+j\infty^+} = \frac{1}{j\infty^+} \Rightarrow 0\angle -90°$

Polar Plot

It is a plot of absolute values of magnitude & phase angle in degrees of open-loop transfer functionG(jω)H(jω)vsω drawn on polar coordinates.

Mapping of segment (i) in s-plane ⇒ In G(s) H(s) plane → Polar plot

Mapping of segment (iii) in s-plane ⇒ In G(s)H(s) plane → Inverse polar plot or complimentary polar plot.

Note: Inverse polar plot is the mirror image of polar plot w.r.t. horizontal axis in the same direction of flow.

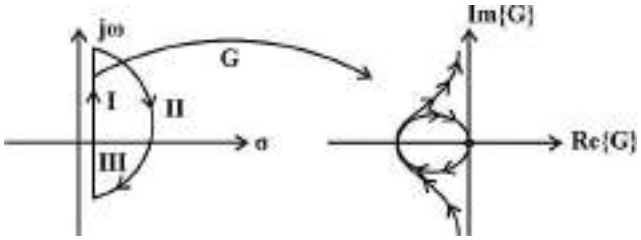


Fig. 10.74

Steps to draw polar plot:

Step 1: $T(s)|_{s=j\omega} = T(j\omega)$ $\qquad 0^+ < \omega < \infty^+$

Step 2:

$T\left(j0^{+}\right) = M_1 \angle \phi_1$

$T\left(j\infty^{+}\right) = M_2 \angle \phi_2$

Step 3: Rationalise

$$T(j\omega) = \text{Re}\{T(j\omega)\} + j\,\text{Im}\left\{T\left(j\omega\right)\right\}$$

Example: Draw polar plot for T(s)=s

Solution:

Step 1:→ $T(j\omega) = j\omega \quad 0^{+} < \omega < \infty^{+}$

Step 2:→ $T\left(j0^{+}\right) = j0^{+} = 0\angle + 90^{\circ}$

$T\left(j\infty^{+}\right) = j\infty^{+} = \infty\angle + 90^{\circ} \rightarrow$ "Asymptote"

Step 3:→ $T(j\omega) = 0 + j\omega$

$0^{+} < \omega < \infty^{+} \rightarrow$ Polar plot lies on +ve imaginary axis of the T(s)-plane.

$\text{Re}\left\{T\left(j\omega\right)\right\} = 0 \rightarrow$

$\text{Im}\left\{T\left(j\omega\right)\right\} = \omega \rightarrow$

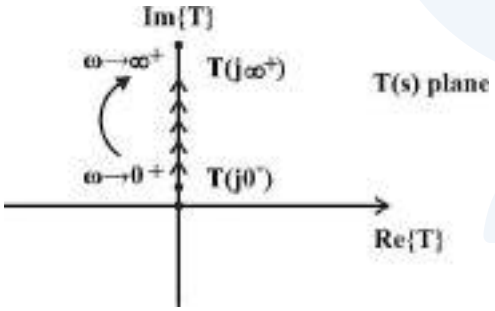


Fig. 10.75

Example: Draw a polar plot for $T\left(s\right) = \frac{1}{s}$.

Solution:

$S = j\omega \rightarrow 0^{+} < \omega < \infty^{+}$

Step 1: → $T(j\omega) = \frac{1}{j\omega}$

Step 2:→

$T\left(j0^{+}\right) = \frac{1}{j0^{+}} = \infty^{+}\angle - 90^{\circ} \rightarrow$ *Asymptotes*

$T\left(j\infty^{+}\right) = \frac{1}{j\infty^{+}} = 0^{+}\angle - 90^{\circ}$

Step 3:→ $T(j\omega) = 0 - j\,/\,\omega$

$0^{+} < \omega < \infty \rightarrow$ -ve imaginary axis

Example: Draw apolar plot for $T\left(s\right) = \frac{1}{s^4}$

Solution:

Step 1:→ $T(j\omega) = \frac{1}{(j\omega)^4}$

Step 2:→ $T\left(j0^{+}\right) = \frac{1}{\left(j0^{+}\right)^4} = \frac{1}{\left(0^{+}\right)^4 j^4} = \infty^{+}\angle 360^{\circ}$

$= \infty^{+}\angle 0^{\circ} \rightarrow$ *Asymptote*

$T\left(j\infty^{+}\right) = \frac{1}{\left(j\infty^{+}\right)^4} = \frac{1}{(\infty^{+})^4 j^4} = 0^{+}\angle - 360^{\circ} = 0^{+}\angle 0^{\circ}$

Step 3:→ $T(j\omega) = 0 + \frac{1}{\omega^4} \rightarrow 0^{+} < \omega < \infty^{+} \rightarrow$ *+ve* real axis

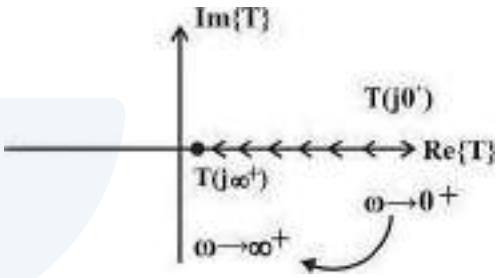


Fig. 10.76

1. Suppose we consider a Type-0 and Order-1 system for drawing a polar plot.

$$G(s)H(s) = \frac{1}{1+sT}$$

$$G(j\omega)H(j\omega) = \frac{1}{1+j\omega T}$$

$$|\,G(j\omega)H(j\omega)\,| = \frac{1}{\sqrt{1+\omega^2T^2}}$$

$$\angle GH = -\tan^{-1}(\omega T)$$

The value of magnitude and phase for different values of frequency are shown in the table below, and the polar plot is shown in the figure below:

ω	$\lvert GH \rvert$	$\angle GH$
0	1	0°
$\frac{1}{T}$	$\frac{1}{\sqrt{2}}$	−45°
∞	0	−90°

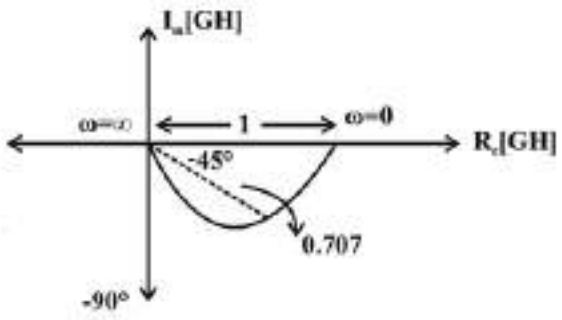


Fig. 10.77

a) Since the phase is always negative for all values of frequency and it varies between 0 and -90°, so the polar plot lies in 4th quadrant.

b) The point corresponding to $\omega = 0$ is called as thetail of thepolar plot, and the point corresponding to $\omega = \infty$ is called as head of the polar plot.

2. If we add a pole at the origin, then the system becomes Type-1 and Order- 2.

$$G(s)H(s) = \frac{1}{s(1+sT)}$$

$$G(j\omega)H(j\omega) = \frac{1}{j\omega(1+j\omega T)}$$

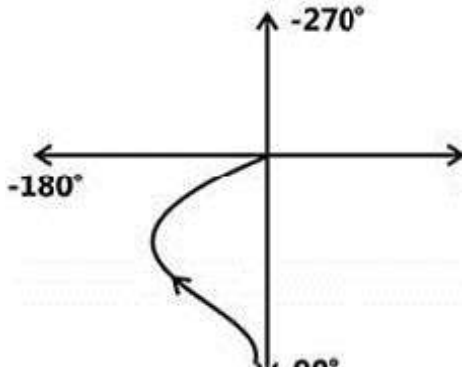


Fig. 10.78

$$|G(s)H(s)| = \frac{1}{\omega\sqrt{1+\omega^2T^2}}$$

$$\angle GH = -90^{\circ} - \tan^{-1}\omega T$$

ω	$\angle GH$	$\lvert GH\rvert$
0	-90°	∞
∞	-180°	0

Conclusion: When a pole is added at the origin, the head and tail of the polar plot shift by 90° in a clockwise direction.

3. If a pole is added in a transfer function away from the origin, then it becomes Type-0 and Order-2 system.

$$G(s)H(s) = \frac{1}{(1+sT_1)(1+sT_2)}$$

$$|G(s)H(s)| = \frac{1}{\sqrt{(1+\omega^2T_1^2)(1+\omega^2T_2^2)}}$$

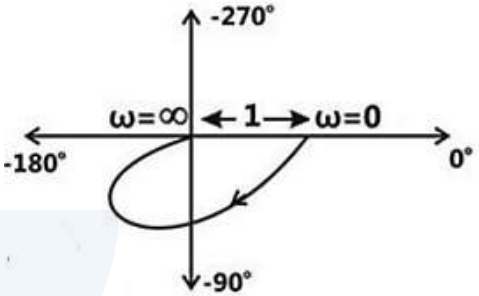


Fig. 10.79

$$\angle G(j\omega)H(j\omega) = -\tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2)$$

ω	$\angle GH$	$\lvert GH\rvert$
0	0	1
∞	-180°	0

$$If\ -\tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2) = -90^{\circ}$$

$$\tan^{-1}\left(\frac{\omega T_1 + \omega T_2}{1-\omega^2T_1T_2}\right) = 90^{\circ}$$

$$\omega = \frac{1}{\sqrt{T_1T_2}}$$

At this value of frequency, the polar plot intersects the negative imaginary axis.

Conclusion: When a pole on a real axis is added, the head of the polar plot shift by 90° in a clockwise direction, and the tail remain the same.

- Also, from the above examples, it is clear that the tail of the polar plot is determined by the type of the system, and the head of the polar plot is determined by order of the system.

- So, for the Order-2 system, the head is always oriented at 180° irrespective of the type of system.
- So, based on type and order, polar plots would lie in the following quadrants:

$$G(s)H(s) = \frac{K}{s^2(1+s)(2+s)} \rightarrow \begin{bmatrix} Type \rightarrow 2 \\ order \rightarrow 4 \end{bmatrix}$$

Common Mistake

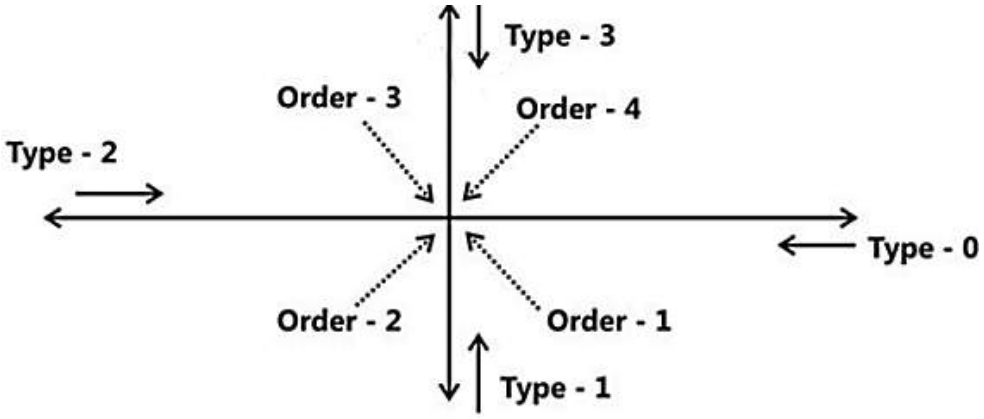


Fig. 10.80

Shortcut to Draw Polar Plot

1. No zeros should be present in G(s) H(s)
2. Poles should be present either at theorigin or in L.H.P.
3. Coefficient of each term in poles should be positive.

Example:

$$G(s)H(s) = \frac{K}{s^3(1+s)} \rightarrow \begin{bmatrix} Type \rightarrow 3 \\ order \rightarrow 4 \end{bmatrix}$$

$$G(s)H(s) = \frac{K}{s^2(1-s)} \rightarrow No \text{ shortcut}$$

$$G(s)H(s) = \frac{K}{s(-1-s)} \rightarrow No \text{ shortcut}$$

Points to Remember

1. Polar plot stars from type:

$Type \neq 0, 4, 8, \cdots\cdots \Rightarrow$ Shift starting point

2. Polar plot ends inorder

3. Polar plot circulates in the clockwise direction

General Shapes of Polar Plots

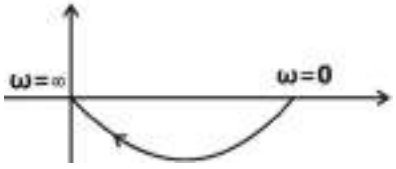
Type/order	Polar plot
• *Type* 0 / *order* 1, $G(s) = \frac{1}{1+sT}$	

Fig. 10.81

- *Type* 0 / *order* 2, $G(s) = \dfrac{1}{(1+T_1 s)(1+T_2 s)}$

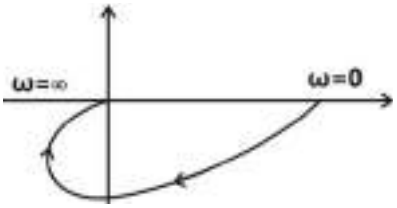


Fig. 10.82

- *Type* 0 / *order* 3, $G(s) = \dfrac{1}{(1+T_1 s)(1+T_2 s)(1+T_3 s)}$

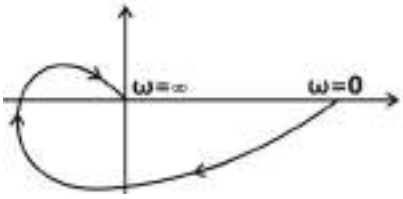


Fig. 10.83

- *Type* 1 / *order* 1, $G(s) = \dfrac{1}{s} = \dfrac{1}{j\omega} = \dfrac{1}{\omega} \angle -90^0$

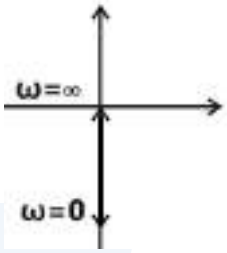


Fig. 10.84

- *Type* 0 / *order* 4

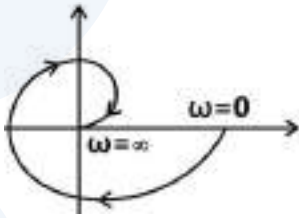


Fig. 10.85

$$G(s) = \frac{1}{(1+T_1 s)(1+T_2 s)(1+T_3 s)(1+T_4 s)}$$

- *Type* 1 / *order* 2, $G(s) = \dfrac{1}{s(1+T_1 s)}$

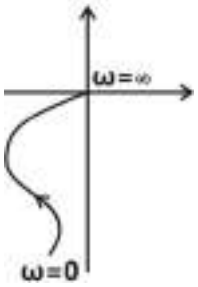


Fig. 10.86

- *Type* 1 / *order* 3, $G(s) = \dfrac{1}{s(1+T_1 s)(1+T_2 s)}$



Fig. 10.87

- *Type* 1 / *order* 4, $G(s) = \dfrac{1}{s(1+T_1 s)(1+T_2 s)(1+T_3 s)}$

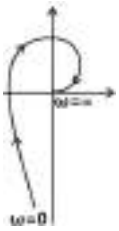


Fig. 10.88

- *Type* 2 / *order* 2, $G(s) = \dfrac{1}{s^2} = \dfrac{1}{(j\omega)^2} = \dfrac{1}{\omega^2} \angle -180°$



Fig. 10.89

- *Type* 2 / *order* 3, $G(s) = \dfrac{1}{s^2(1+Ts)}$

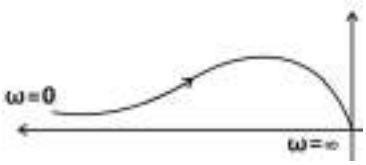


Fig. 10.90

- *Type* 2 / *order* 4, $G(s) = \dfrac{1}{s^2(1+sT_1)(1+sT_2)}$

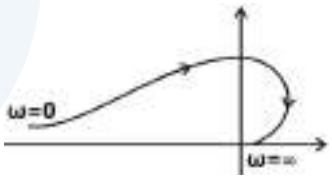


Fig. 10.91

- *Type* 3 / *order* 3, $G(s) = \dfrac{1}{s^3} = \dfrac{1}{\omega^3} \angle -270^0$

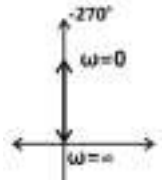


Fig. 10.92

- *Type* 3 / *order* 4, $G(s) = \dfrac{1}{s^3(1+Ts)}$



Fig. 10.92

Limitations of Polar Plot

1. Drawing a polar plot involves a number of calculations if it has multiple intersections with a coordinate axis.
2. Not feasible to find the magnitude and ϕ of the transfer function at any desired frequency using a polar plot.
3. Stability of a system can be analysed if the loop transfer function is of minimum phase and there is no information about the degree of instability which means we cannot identify the number of poles lying in the Right Half of the s-plane.

Concept of Enclosement and Encirclement

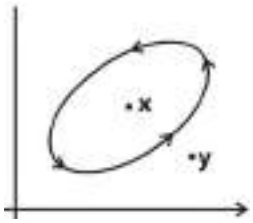


Fig. 10.93

- A point is said to be enclosed by a contour if it lies to the right side of the direction of the contour.
- A point is said to be encircled if it lies inside a contour which is a closed path.
- In the above figure, point 'y' is said to be enclosed, whereas point x is said to be encircled in an anti–clockwise direction.

Nyquist Plot

Step 1: Draw pole-zero diagram in s-plane.

Step 2: Choose a proper Nyquist contour.

Step 3: Draw a polar plot.

Step 4: Map the remaining segments of the contour.

Example: Draw Nyquist plot for $T(s) = s$

Solution:

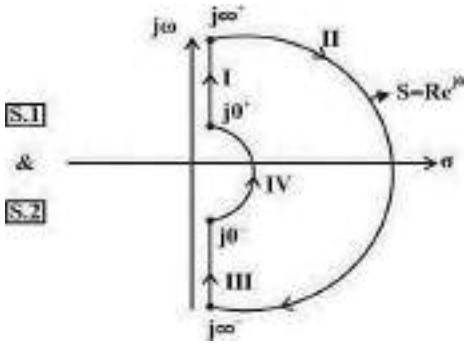


Fig. 10.94

Step 3

Mapping of segment (I)

1. $S = j\omega \quad 0^+ < \omega < \infty^+ \Rightarrow$ Polar plot

2. $T(j\omega) = j\omega \rightarrow \begin{matrix} T(j0^+) = j0^+ = 0^+ \angle 90^\circ \\ T(j\infty^+) = j\infty^+ = \infty^+ \angle 90^\circ \end{matrix}$

3. $T(j\omega) = 0 + j\omega$

$0^+ < \omega < \infty^+ \rightarrow$ +ve imaginary axis.

Mapping of segment (II)

$$S = Re^{j\theta} \quad R \rightarrow \infty \quad \theta \rightarrow +\pi/2 \text{ to } -\frac{\pi}{2} \rightarrow c.w.$$

$$T(s) = \lim_{R\to\infty} Re^{j\theta} = \infty e^{j\theta} + \frac{\pi}{2} \text{ to } \frac{-\pi}{2} \rightarrow c.w.$$

Mapping of segment (III)

$S = j\omega \quad -\infty < \omega < 0^- \rightarrow$ Inverse polar plot:

Mapping of segment (IV)

$$S = Re^{j\theta} \quad R = 0^+ \quad \theta = -\frac{\pi}{2} \text{ to } +\frac{\pi}{2} \rightarrow A.c.w.$$

$$T(s) = \lim_{R\to 0^+} Re^{j\theta} = 0^+ e^{j\theta}$$

$$\Rightarrow \theta \rightarrow -\frac{\pi}{2} \text{ to } \frac{+\pi}{2} \rightarrow A.c.w.$$

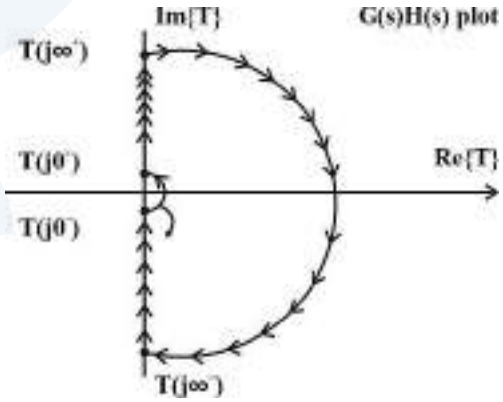


Fig. 10.95

$G(s)H(s) = T(s) = s$

N.S.C. in G(s)H(s)

$N = P_+ - Z_+$

L.H.S.: N = 0

R.H.S.: $0 = p_+$

$0 = z_+$

Example: Draw Nyquist plot for $T(s) = \frac{1}{(1+ST)}$

Solution:

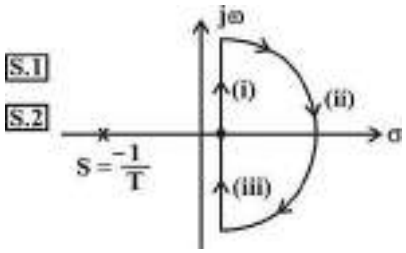


Fig. 10.96

S.3→

Mapping of segment (I)

1. $T(j\omega) = \dfrac{1}{1+j\omega T}$

2. $T\left(j0^+\right) = \dfrac{1}{1+j0^+(T)} = \dfrac{1}{1} = 1\angle 0^\circ$

$T\left(j\infty^+\right) = \dfrac{1}{1+j\infty^+(T)} = \dfrac{1}{j\infty^+} = 0^+\angle -90^\circ$

3. $T(j\omega) = \dfrac{1}{(1+j\omega T)}\dfrac{(1-j\omega T)}{(1-j\omega T)}$

$= \dfrac{1}{1+\omega^2T^2} - j\left\{\dfrac{\omega T}{1+\omega^2T^2}\right\}$

$0^+ < \omega < \infty^+$ $\quad \text{Re}\{T(j\omega\} = +ve \rightarrow$ (IV) quad

$\text{Im}\{T(j\omega\} = -ve$

Mapping of segment (II)

$S = \text{Re}^{j\theta} \quad R \to \infty \quad \theta = +\dfrac{\pi}{2} to -\dfrac{\pi}{2} \to c.w.$

$T(s) = \lim\limits_{R\to\infty^+} \dfrac{1}{1+T\,\text{Re}^{j\theta}} = \dfrac{1}{1+\infty^+e^{j\theta}} = 0^+e^{-j\theta}$

Mapping of segment (III)

I.P.P

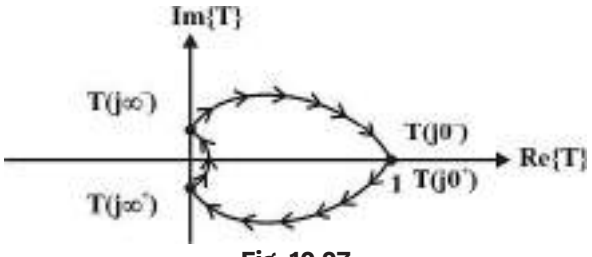


Fig. 10.97

N.S.C. in T(s) $N = P_+ - Z_+$

L.H.S. N = 0

R.H.S. $P_+ = 0, Z_+ = 0$

L.H.S. = R.H.S.

4. Draw Nyquist plot for $G(s)H(s) = \dfrac{(1-s)}{(1+s)}$

Solution:

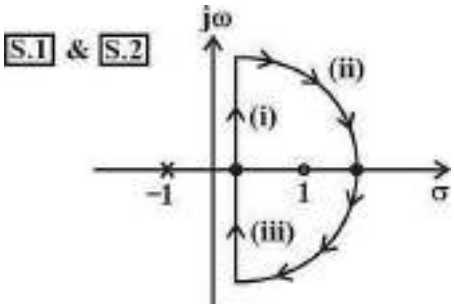


Fig. 10.98

S.3

Mapping of segment (I)

1. $T(j\omega) = \dfrac{1-j\omega}{1+j\omega}$

2. $T\left(j0^+\right) = \dfrac{1-j0^+}{1+j0^+} = 1\angle 0^\circ$

$T\left(j\infty^+\right) = \dfrac{1-j\infty^+}{1+j\infty^+} = \dfrac{-j\infty^+}{j\infty^+} = -1 = 1\angle 180^\circ$

3. $\dfrac{(1-j\omega)(1-j\omega)}{1+\omega^2} = \dfrac{\left(1-\omega^2\right)}{\left(1+\omega^2\right)} - j\dfrac{2\omega}{\left(1+\omega^2\right)}$

$\text{Re}\{T\} = \dfrac{1-\omega^2}{1+\omega^2} \qquad \text{Im}\{T\} = \dfrac{-2\omega}{1+\omega^2}$

$1-\omega^2 > 0 \rightarrow \omega < 1 \rightarrow \text{Re}\{T\} = +ve$

$\text{Im}\{T\} = -ve \rightarrow IV$ quad

$1-\omega^2 = 0 \rightarrow \omega = 1 \rightarrow \text{Re}\{T\} = 0$

$\text{Im}\{T\} = -1 \rightarrow$ Cuts –veimaginary axis

$1-\omega^2 < 0 \rightarrow \omega > 1 \rightarrow \text{Re}\{T\} = -ve$

$\text{Im}\{T\} = -ve \rightarrow III$ quad

Mapping of segment (II)

$S = \infty e^{j\theta}$

$\lim\limits_{R\to\infty} \dfrac{1-\infty e^{j\theta}}{1+\infty e^{j\theta}} = -1 = 1\angle 180°$

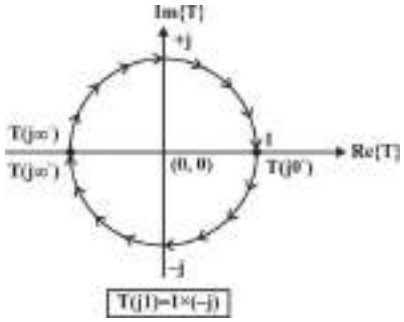


Fig. 10.99

$$G(s)H(s) = \frac{1-S}{1+S}$$

N.S.C. in G(s) H(s)

$N = -1$

$P_+ = 0 \quad Z_+ = 1$

Segement (III)

$$T(j\omega) = \frac{1-j\omega}{1+j\omega} \rightarrow |T(j\omega)| = 1$$

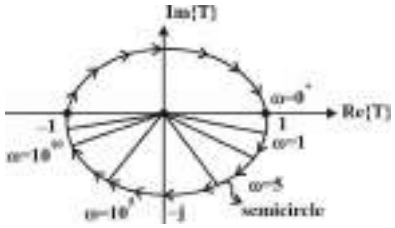


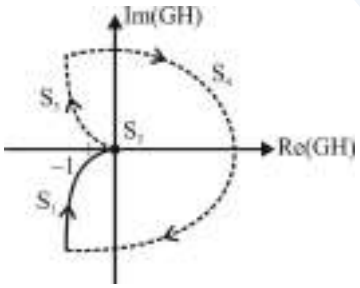
Fig. 10.100

Test your Understanding

1. Plot Nyquist plot for the system whose open-loop transfer function is given by

$$G(s) = \frac{10}{s(s+2)}$$

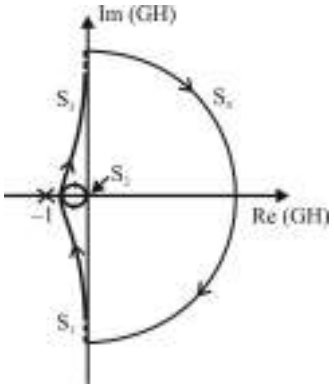
Solution:



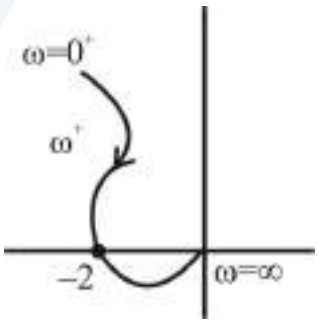
2. Plot Nyquist plot for the system whose open-loop transfer function is given by

$$G(s) = \frac{10}{s(s+2)(s+4)}$$

Solution:



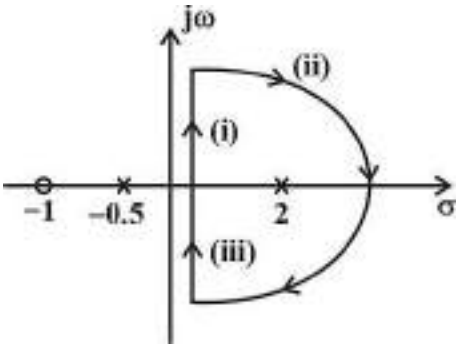
3. The Nyquist plot of a type 1 system which has one right half of the s-plane, for $\omega = 0^+$ *to* $\omega = \infty^+$ is given below. Discuss the stability of the closed-loop system.



Solution: System is stable

4. $G(s)H(s) = \frac{K(s+1)}{(s+0.5)(s-2)} \; 0 < k < \infty$ →Draw Nyquist plot

Solution:



Mapping of segment (I)

$S = j\omega : 0^+ < \omega < \infty^+$

$$T(j\omega) = \frac{K(1+j\omega)}{(0.5+j\omega)(-2+j\omega)}$$

$$T\left(j0^+\right) = \frac{K\left(1+j0^+\right)}{\left(0.5+j0^+\right)\left(-2+j0^+\right)} = -k = k\angle 180°$$

$$T(j\infty^+) = \frac{K(1+j\infty^+)}{\left(0.5+j\infty^+\right)\left(-2+j\infty^+\right)} = \frac{k\left(j\infty^+\right)}{\left(j\infty^+\right)^2}$$

$$= \frac{k}{j\infty^+} = 0^+\angle -90^\circ$$

$$T(j\omega) = \frac{K(1+j\omega)}{(0.5+j\omega)(-2+j\omega)}\frac{(0.5-j\omega)(-2-j\omega)}{(0.5-j\omega)(-2-j\omega)}$$

$$T\left(j\omega\right) = \frac{K(1+j\omega)\left(-1-0.5j\omega+2j\omega-\omega^2\right)}{\left(0.25+\omega^2\right)\left(4+\omega^2\right)}$$

$$T(j\omega) = \frac{-K\left(1+2.5\omega^2\right)}{\left(0.25+\omega^2\right)\left(4+\omega^2\right)} - \frac{j\omega K\left(\omega^2-0.5\right)}{\left(0.25+\omega^2\right)\left(4+\omega^2\right)}$$

$\rightarrow 0 < \omega < \sqrt{0.5}$

$\text{Re}\{T\} = -ve \qquad \text{Im}\{T\} = +ve \rightarrow IInd\ Quard$

$\text{Re}\{T\} = -\frac{2}{3}k \qquad \text{Im}\{T\} = 0 \rightarrow cuts - ve\ real\ axis$

$\rightarrow \sqrt{0.5} < \omega < \infty^+ \qquad \text{Re}\{T\} = -ve$

$\text{Im}\{T\} = -ve \rightarrow \text{III}^{rd}\ \text{quad}$

Mapping of segment (II)

$S = \text{Re}^{j\theta} \quad R \rightarrow \infty$

$\theta = +\pi/2\ to\ -\pi/2 \rightarrow C.W$

$$T\left(s\right) = \lim_{R\to\infty^+}$$

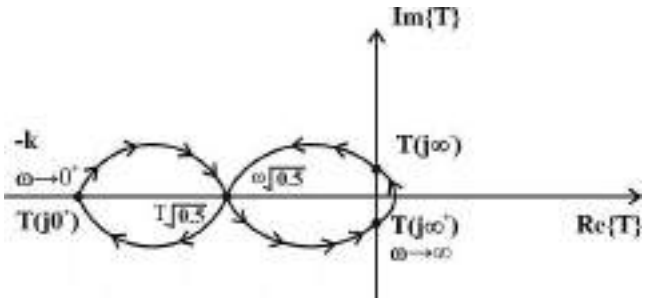
$$\frac{K\left(\text{Re}^{j\theta}+1\right)}{\left(\text{Re}^{j\theta}+0.5\right)\left(\text{Re}^{j\theta}-2\right)} = \frac{K\left(\infty^+e^{j\theta}+1\right)}{\left(\infty^+e^{j\theta}+0.5\right)\left(\infty^+e^{j\theta}-2\right)}$$

$$\frac{K\,\text{Re}^{j\theta}}{\text{Re}^{j\theta}\,\text{Re}^{j\theta}} = \frac{K}{\infty^+e^{j\theta}} = 0^+e^{-j\theta}$$

$-\theta \rightarrow \frac{-\pi}{2}\ to + \pi/2$ A.C.W

Mapping of segment (III)

I.P.P.



N.S.C. in G(s) H(s)

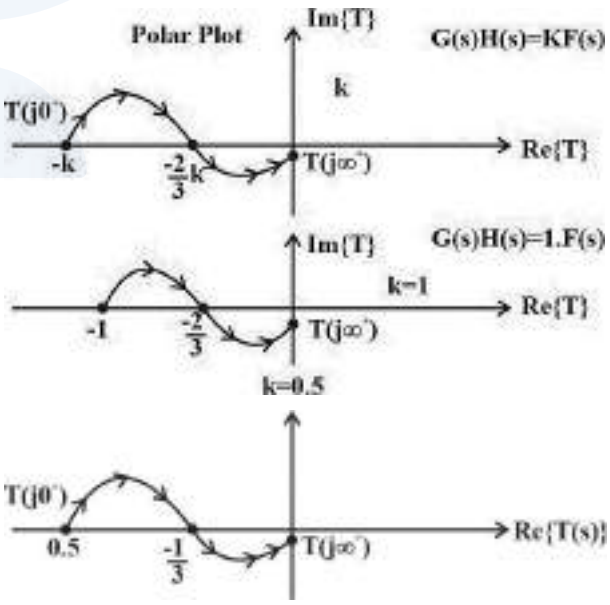
$N = 1$

$P_+ = 1$

$Z_+ = 0$

$\boxed{N = P_+ - Z_+}$

Observations



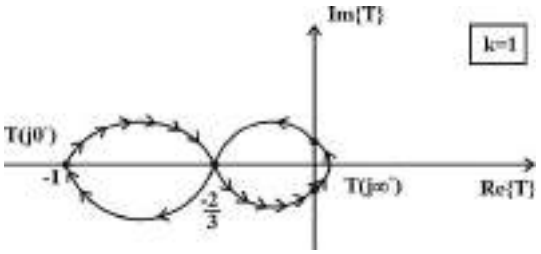
$T_1(s) = 0.5F(s) \rightarrow$ Plot given

$T_2(s) = kF(s) \rightarrow$ Plot =?

$T_1(s) \rightarrow T_1'(s) = 2 \times T_1(s)$

$T_2(s) = KT_1'(s)$

1. Polar Plot



2. Stability of O.L.S.

$$G(S)\cdot H(s)=\frac{1(1+s)}{(0\cdot 5+s)(s-2)}$$

Pole→R.H.P. →Unstable

Stability of C.L.S.

Method 1:

$1+G(s)H(s)=0$

$$1+\frac{(s+1)}{(s+0.5)(s-2)}=0$$

$s^2-2s+0.5s-1+s+1=0$

$s^2-0.5s=0$

$s(s-0.5)=0$

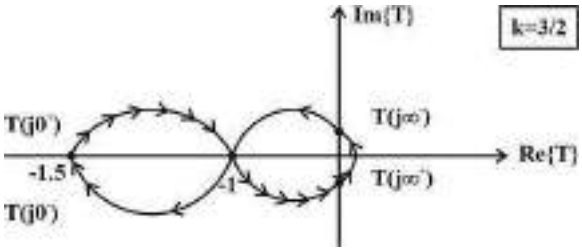
$s=0$
$s=0.5$ $\rightarrow$ f Poles

C.L.S. is unstable

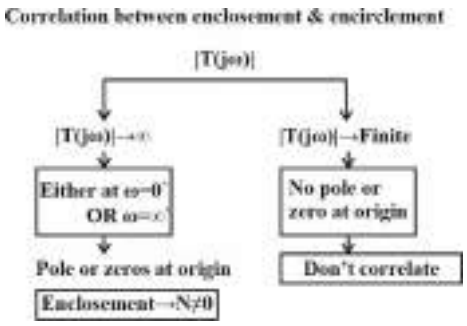
Method 2:

$N=P_+-Z_+$ in $1+G(s)H(s)$

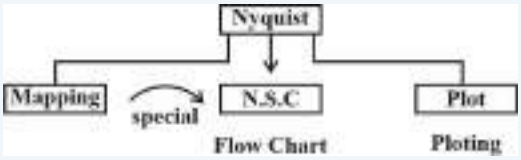
$N=?\rightarrow$ N.S.C. is invalid→



3.



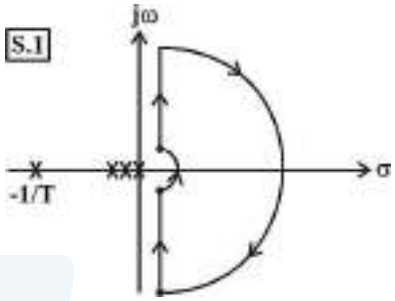
Chapter Summary



Example:

$$G(s)H(s)=\frac{K}{s^3(1+sT)} \qquad 0<K<\infty$$

Solution:



Mapping of segment (I)

$s=j\omega$

$$T(j\omega)=\frac{K}{(j\omega)^3(1+j\omega T)}$$

$$T\left(j0^+\right)=\frac{K}{\left(j0^+\right)^3\left(1+j0^+\right)}$$

$$T\left(j0^+\right)=\frac{K}{j^3\left(0^+\right)^3}=\infty^+\angle -270^\circ$$

$$T\left(j\infty^+\right)=\frac{K}{(j\infty^+)^3\left(1+j\infty^+\right)}=\frac{K}{j^4\left(\infty^+\right)^4}=0^+<0^\circ$$

Rationalization

$$T(j\omega)=\frac{k(1-j\omega T)}{-j\omega^3(1+j\omega T)(1-j\omega T)}$$

$$T(j\omega)=\frac{jk(1-j\omega T)}{\omega^3\left(1+\omega^2T^2\right)}$$

$$T(j\omega)=\frac{kT}{\omega^2\left(1+\omega^2T^2\right)}+\frac{jK}{\omega^3\left(1+\omega^2T^2\right)}$$

$Re\{T\}=+ve$
$Im\{T\}=+ve$ $\rightarrow$ I quad

$$T\left(j0^{+}\right) = \infty^{+} + j\infty^{+}$$

Mapping of segment (II)

$$S = Re^{j\theta}$$

$$\lim_{R\to\infty} \frac{K}{R^3 e^{j\theta}\left(1 + Re^{j\theta}\, T\right)} = \frac{K}{R^4 e^{j4\theta}} = 0^{+} e^{-j4\theta}$$

$$-4\theta \to -2\pi \text{ to } +2\pi \to \text{A.C.W}$$

Mapping of segment (III)

I.P.P.

Mapping of segment (IV)

$$S = Re^{j\theta} \quad R \to 0 \quad \theta \to -\frac{\pi}{2} \text{ to } +\pi/2 \to \text{A.c.w.}$$

$$\lim_{R\to 0} \frac{K}{R^3 e^{3j\theta}\left(1 + Re^{j\theta}\, T\right)} = \infty^{+} e^{-j3\theta}$$

$$\Delta\theta = 540^{\circ}$$

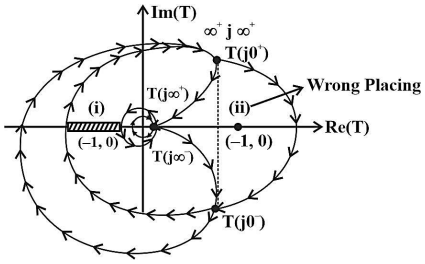
$$-3\theta \to +\frac{3\pi}{2} \text{ to } -\frac{3\pi}{2} \to \text{c.w.}$$

N.S.C. in G(s) H(s)
N=0

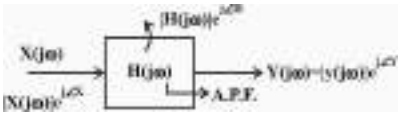
$$N = P_{+} - Z_{+}$$

$$P_{+} = 0$$

$$Z_{+} = 0$$



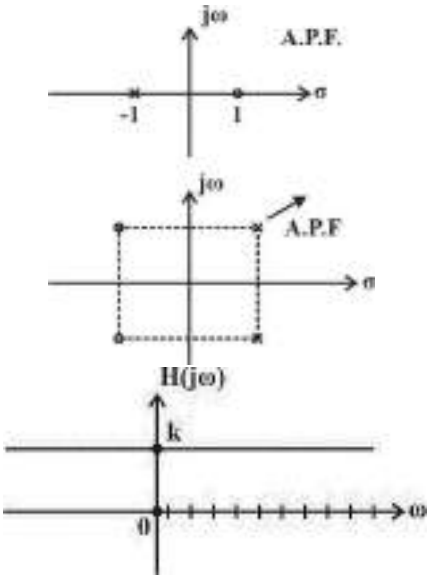
Discussion about all Pass Filter



$$Y(j\omega) = H(j\omega)x(j\omega)$$

$$|Y(j\omega)| = |H(j\omega)|\ |X(j\omega)|$$

$$\angle Y(\omega) = \angle H(j\omega) + \angle x\left(j\omega\right)$$



$$y(\omega) = x(\omega)H(\omega)$$

$$Y(\omega) = Kx(\omega) \to \text{A.P.F.}$$

All pass filter: Poles and zeros are at mirror image w.r.t -axis.

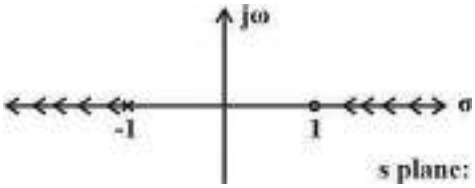
$$H(s) = k\left\{\frac{1-S}{1+S}\right\}$$

$$H(s) = \frac{K\left(s^2 + s + 1\right)}{\left(s^2 - s + 1\right)}$$

1. Root locus of A.P.F

$$H(s) = \frac{k(1-s)}{1+s} \text{ or } \frac{-k(s-1)}{s+1}$$

$$0 < k < \infty$$



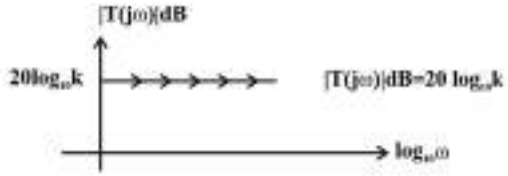
2. Bode plot of A.P.F

$$T(s) = \frac{K(1-s)}{(1+s)}$$

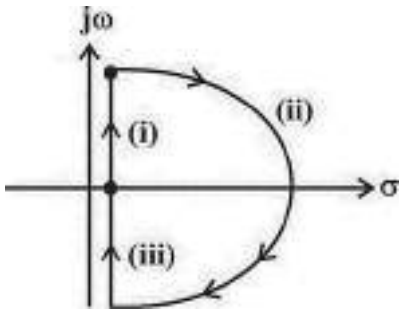
$$s = j\omega$$

$$T(j\omega) = \frac{K(1-j\omega)}{(1+j\omega)}$$

$$|T\left(j\omega\right)| = K$$



Nyquist plot of A.P.F.

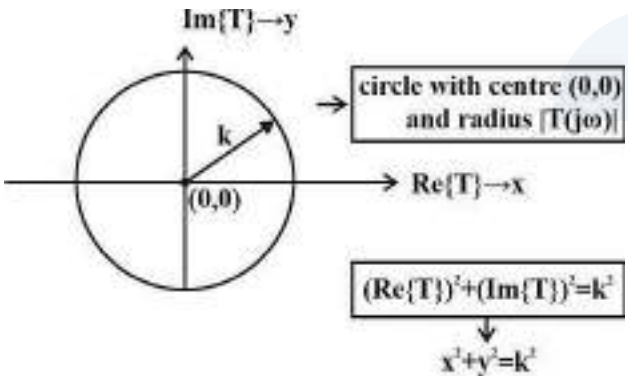


$$T(s)=\frac{K(1-s)}{(1+s)} \text{ or } T(s)=\frac{k\left(s^2-s+1\right)}{\left(s^2+s+3\right)}$$

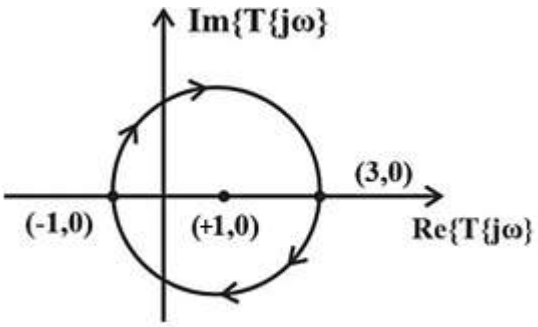
$\rightarrow S = j\omega$

쫣T(jω)쫣 K → "Distance from (0,0) in G(s) H(s)-plane."

or



Note:



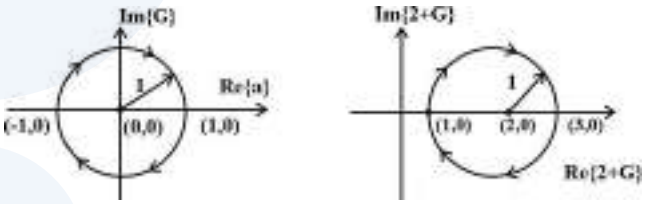
It is the Nyquist Plot of A.P.F.: True/False

Example:

$$G(s)H(S)=\frac{1-s}{1+s}$$

Solution:

$$G(s)H(S)=\frac{1-s}{1+s}$$

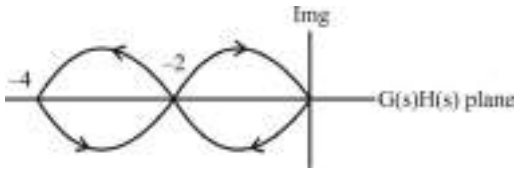


$$2+G(s)H(s)=\frac{2+2s+1-s}{1+s}=\frac{s+3}{1+5}$$

C.L.S. Unstable

Test your Understanding

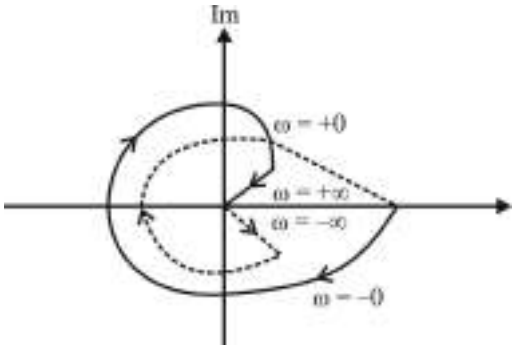
1. The Nyquist plot of a certain G(s) H(s)002C, which has no right-hand pole, is given below. Discuss the stability of the closed-loop system.



Solution:

Closed-loop system is unstable with th0065 pole in a right-sided plane.

2. The Nyquist plot for a control system is shown below.The type of the system is



(A) 0

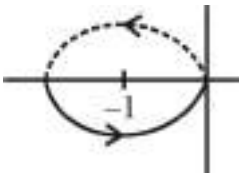
(B) 1

(C) 2

(D) 3

Solution: (A)

3. The Nyquist plot of a unity feedback system having one open-loop pole in the right half of the s- plane is shown. The closed-loop system is (take Nyquist contour in the clockwise direction)



(A) Stable

(B) Unstable

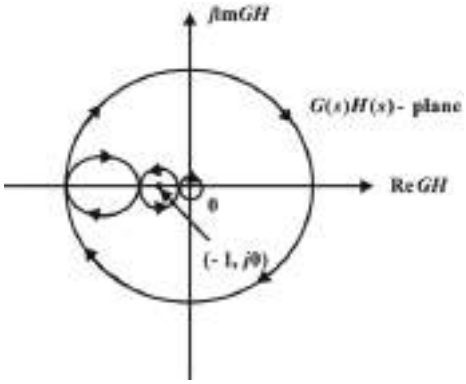
(C) Marginally stable

(D) Conditionally stable

Solution: (A)

Practice Questions

1. The complete Nyquist plot of the open–loop transfer function G(s)H(s) of a feedback control system is shown in the figure,



If G(s)H(s) has one zero in the right half of the s- plane, the number of poles that the closed–loop system will have in the right half of s–plane is

(A) 0

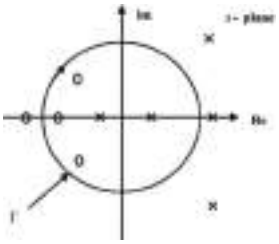
(B) 1

(C) 4

(D) 3

Solution: (D)

2. The pole–zero map of a rational function G(s) is shown below. When the closed contour is mapped into G(s)-plane, then the mapping encircles



(A) The pointof the G(s)–plane once in the clockwise direction.

(B) Theorigin of G(s)–plane once in the counter-clockwise direction.

(C) The point of the G(s)–plane once in the counter–clockwise direction.

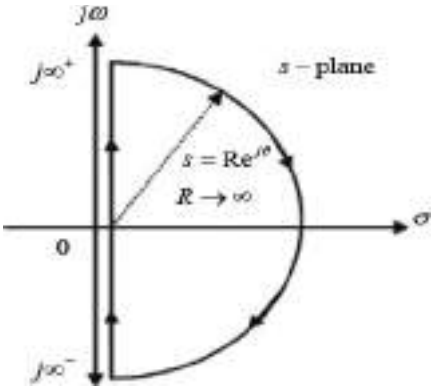
(D) The origin of G(s) – plane once in the clockwise direction.

Solution: (D)

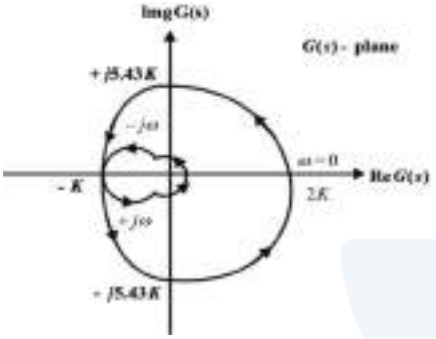
3. A unity feedback control system is characterised by the open–loop transfer function

$$G(s) = \frac{10K(s+2)}{s^3 + 3s^2 + 10}$$

The Nyquist path and the corresponding Nyquist plot of G(s) are shown in the figures below,



Nyquist path for G(s)



Nyquist plot for G(s)

If , then the number of poles of the closed–loop transfer function that lie in the right–half of the s–plane is

(A) 0

(B) 1

(C) 2

(D) 3

Solution: (C)

4. The Nyquist plot of the transfer function Does not encircle the point (-1+j0) for K=10 but does encircle the point (-1+j0) for K=100. Then the closed loop system (having unity gain feedback) is

(A) Stable for K=10 and stable for K=100.

(B) Stable for K=10 and unstable for K=100.

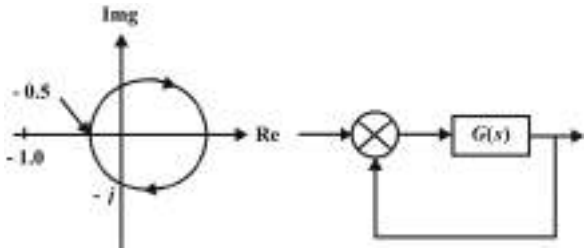
(C) Unstable for K=10 and stable for K=100

(D) Unstable for K=10 and unstable for K=100.

Solution: (B)

Common Data for Questions 5 and 6

The Nyquist plot of a stable transfer function G(s) is shown in the figure. We are interested in the stability of the closed-loop system in the feedback configuration shown below,



5. Which of the following statements is true?

(A) G(s) is an all–pass filter.

(B) G(s) has a zero in the right–half plane.

(C) G(s) is the impedance of a passive network.

(D) G(s) is marginally stable.

Solution: (B)

6. The gain and phase margins of G(s) for closed-loop system are

(A) 6dB and

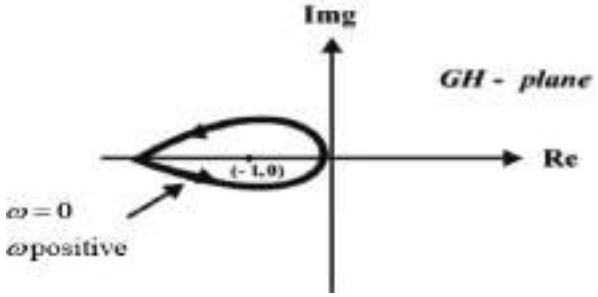
(B) 3dB and

(C) 6dB and

(D) 3dB and

Solution: (C)

7. Figure shows the Nyquist plot of the open–loop transfer function G(s)H(s) of a system. If G(s)H(s) has one right–hand pole, the closed–loop system is



(A) Always stable

(B) Unstable with one closed–loop right-hand pole

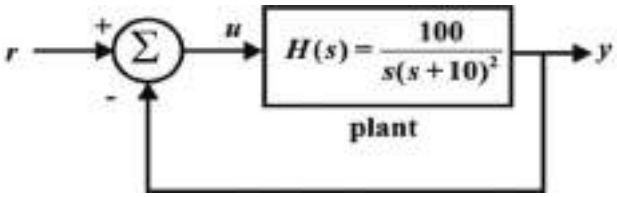
(C) Marginally Stable

(D) one

Solution: (A)

Common Data for Questions 8 and 9

8 & 9. The input-output transfer function of a plant. The plant is placed in a unity negative feedback configuration as shown in the figure below,



The gain margin of the system under closed-loop unity negative feedback is

(A) 0dB

(B) 20dB

(C) 26dB

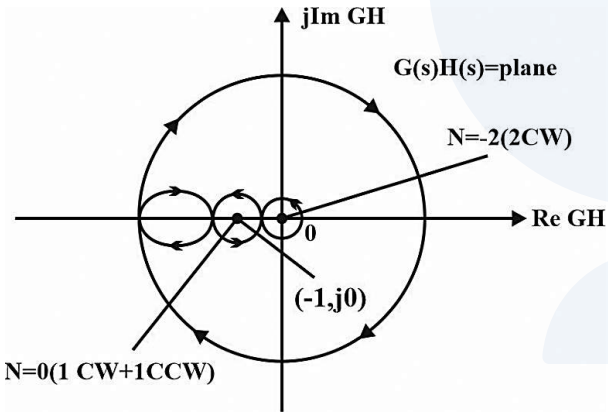
(D) 46dB

Solution: (C)

Key and Explanations

1. (D)

Given the Nyquist plot is shown below,



Case 1: If we assume Nyquist contour in a clockwise direction

Given: Open-loop transfer function has one zero at RHS.

Principle of argument is given by,

$N = P - Z$ ...(i)

Where, N= Number of encirclements of about origin in an anti-clockwise direction.

P = Number of right-handpoles of open-loop transfer function i.e., G(s) H(s).

Z = Number of right-handzeros of open-loop transfer function i.e., G(s) H(s).

From the figure, the number of encirclements about the origin is two times clockwise direction.

Thus, N =-2 (anti-clockwise direction)

Since, Z = 1

From equation (i),

$-2 = P - 1$

$P = -1$ ...(ii)

Number of right-hand poles of the open-loop transfer function cannot be negative indicates, Nyquist contour is drawn in clockwise.

Case 2: If we assume Nyquist contour in an anti-clockwise direction, then according to the principle of argument

$N = P - Z$ ...(iii)

where, N= Number of encirclements of Nyquist plot of G(s) H(s) about the origin in the clockwise direction.

P= Number of right-handpoles of open-loop transfer function, i.e., G(s) H(s).

Z= Number of right-handzeros of open-loop transfer function, i.e., G(s) H(s).

Thus, N=2 (Clockwise direction)

Since, Z=1

From equation (iii)

$2 = P - 1$

$P = 3$ ...(iv)

Hence, the number of right-handpoles of an open-loop transfer function is 3.

According to the Nyquist stability criteria:

$N = P - Z$ ...(v)

Where, N= Number of encirclements about (-1+0 j) in the clockwise direction

Z= Number of closed-loop poles in the right half of the s-plane.

P= Number of right-handpoles of the open-loop transfer function.

From the given the Nyquist plot number of encirclements about (-1+0j) is 1 time CW and 1 time CCW.

Hence, N=1-1=0

Using equation (v)

0 = P-Z

Z = P

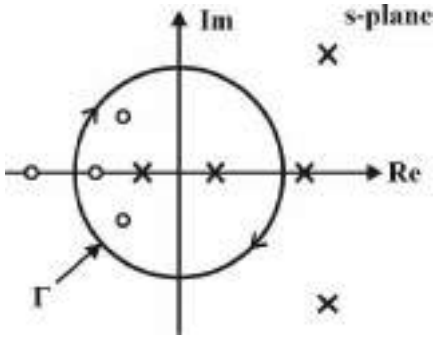
Z = 3

Thus, the number of closed-loop poles in the right half of the s-plane is 3.

Hence, the correct option is (D).

2. (D)

Given pole-zero plot is shown in the figure where the closed contour encloses 3 zeros and 2 poles in the s-plane.



From the principle of an argument of the Nyquist plot, if a closed contour encloses P poles and Z zero in the s-plane, then the origin of the G(s) plane is encircled P-Z times.

N = P-Z

N= Number of encirclements of origin around the origin in the counterclockwise direction.

N is negative for the clockwise direction.

Here, P=2, Z=3

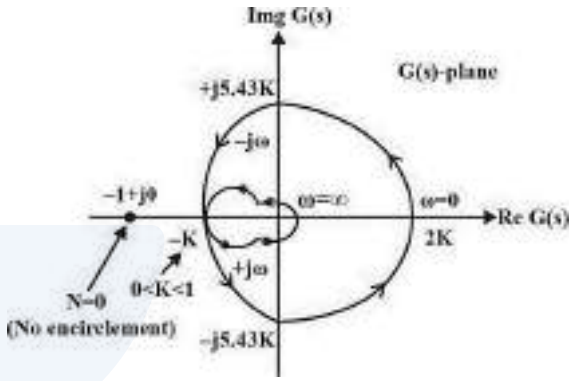
So, N=2-3=-1

Hence, the mapping will encircle the origin of the G(s) plane in once clockwise direction.

Hence, the correct option is (D).

3. (C)

$$G(s) = \frac{10K(s+2)}{s^3 + 3s^2 + 10}$$



Nyquist stability criterion for given contour is given by,

N=P-Z ...(i)

Where,

N= Number of encirclement about critical point (-1+j0) in counter clockwise (anti clockwise) direction.

P= Number of right-handpoles of open-loop transfer function i.e. G(s) H(s).

Z= Number of right-handpoles of the closed loop transfer function.

Characteristics equation of OLTF is given by,

$s^3 + 3s^2 + 10 = 0$

Routh tabulation:

s^3	1	0
s^2	3	10
s^1	$\frac{-10}{3}$	0
s^0	10	0

From the Routh table, two sign changes in the first column, hence the number of right-handpoles of OLTF, P=2.

From the figure, there is no encirclement about the critical point (-1+j0).

Thus, N=0

From equation (i),

0=2-Z

Z=2

Thus, two poles of the closed-loop transfer function will lie in RHS, and therefore the system is unstable.

Hence, the correct option is (C).

4. (B)

Given: $G(s)H(s) = \dfrac{K}{(s^2+2s+2)(s+2)}$

a) If K is 10, the Nyquist plot does not enclose -1+j0.

b) If K is 100, Nyquist plot enclose -1+j0.

Method 1: Nyquist plot

1. For K=10

$$G(s)H(s) = \frac{10}{(s^2+2s+2)(s+2)}$$

Put $s = j\omega$ in the above equation,

$$G(j\omega)H(j\omega) = \frac{10}{(-\omega^2+2j\omega+2)(j\omega+2)}$$

Magnitude of $G(j\omega)$ is given by,

$$|G(j\omega)H(j\omega)| = \frac{10}{\sqrt{(2-\omega^2)^2+(2\omega)^2}\sqrt{\omega^2+4}}$$

Phase angle of $G(j\omega)$ is given by,

$$\angle G(j\omega)H(j\omega) = -\tan^{-1}\frac{2\omega}{2-\omega^2} - \tan^{-1}\frac{\omega}{2}$$

ω	$\lvert G(j\omega)H(j\omega)\rvert$	$\angle G(j\omega)H(j\omega)$
0	2.5	0°
∞	0	–270°

To calculate ω_{pc}, equating the imaginary part of $G(j\omega)$ to zero.

$$\text{Img}\left[-j\omega^3+6j\omega\right] = 0$$

$$-\omega_{pc}^3 + 6\omega_{pc} = 0$$

$$\omega_{pc}\left[6-\omega_{pc}^2\right] = 0$$

$$\omega_{pc} = 0$$

and $\left(6-\omega_{pc}^2\right) = 0$

$$\omega_{pc}^2 = 6$$

$$\omega_{pc} = \pm\sqrt{6} rad/\sec$$

Since, ω_{pc} is a positive real frequency.

Therefore, $\omega_{pc} = \sqrt{6}\text{rad}/\sec$

Magnitude at phase crossover frequency is given by,

$$\left|G(j\omega_{pe})\right| = \frac{10}{4-4\omega_{pc}^2}$$

$$\left|G(j\omega_{pe})\right| = \frac{10}{4-4(6)} = -0.5$$

2. For K = 100

$$G(s)H(s) = \frac{100}{(s^2+2s+2)(s+2)}$$

Put $s = j\omega$ in the above equation,

$$G(j\omega)H(j\omega) = \frac{100}{(-\omega^2+2j\omega+2)(j\omega+2)}$$

Magnitude of $G(j\omega)$ is given by,

$$|G(j\omega)H(j\omega)| = \frac{100}{\sqrt{(2-\omega^2)^2+(2\omega)^2}\sqrt{\omega^2+4}}$$

Phase angle of $G(j\omega)$ is given by,

$$\angle G(j\omega)H(j\omega) = -\tan^{-1}\frac{2\omega}{2-\omega^2} - \tan^{-1}\frac{\omega}{2}$$

ω	$\lvert G(j\omega)H(j\omega)\rvert$	$\angle G(j\omega)H(j\omega)$
0	2.5	0°
∞	0	–270°

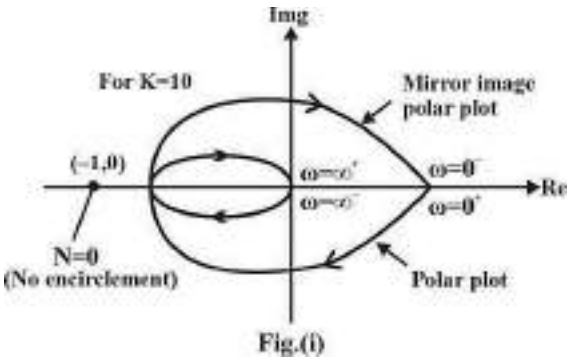


Fig.(i)

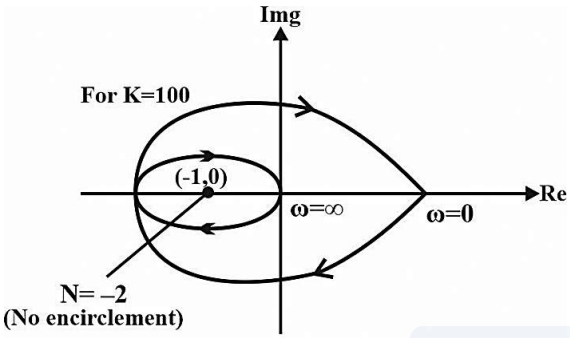


Fig.(ii)

Nyquist stability criterion is given by,

N=P-Z ...(i)

Where,

N= Number of encirclement about critical point (-1+j0) in counter clockwise (anti clockwise) direction.

P= Number of right-handpoles of open-loop transfer function i.e. G(s) H(s).

Z= Number of right-handpoles of the closed-loop transfer function.

For the system to be stable: Z=0

Characteristic equation of open-loop transfer function is given by,

$s^3 + 4s^2 + 6s + 4 = 0$

Routh tabulation:

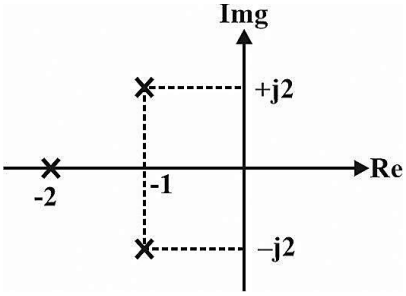
s^3	1	6
s^2	4	4
s^1	5	0
s^0	4	0

In the above Routh table, there is nosign of change in the first column.So the number of right-handpoles of OLTF, P=0.

OR

$$G(s) = \frac{K}{(s^2 + 2s + 2)(s + 2)}$$

Location of poles, $s = -2, -1 \pm j2$



As shown in the above figure, the number of right-handpoles of OLTF is zero; hence P=0.

From figure (i), there is no encirclement aboutthe critical point (-1+j 0).

Thus,N=0

From equation (i),

0=0-Z

Z = 0

Thus, no pole of the closed-loop transfer function will lie in RHS, and therefore the system is stable.

From figure (ii), the number of encirclements about the critical point (-1+j0) is two times in a clockwise direction.

Thus, N=-1-1=-2

From equation (i)

-2=0-Z

Z=2

Thus, two poles of the closed-loop transfer function will lie in RHS, and therefore the system is unstable.

Hence, the correct option is (B).

Method 2: Routh Hurwitz Criteria

Characteristic equation is given by,

$1 + G(s)H(s) = 0$

$$1 + \frac{K}{(s^2 + 2s + 2)(s + 2)} = 0$$

$s^3 + 4s^2 + 6s + 4 + K = 0$

Routh tabulation:

s^3	1	6
s^2	4	$4+K$
s^1	$\frac{20-K}{4}$	0
s^0	$4+K$	0

For the system to be stable,

$$\frac{20-K}{4} > 0 \text{ and } 4+K > 0$$

$$K < 20 \text{ and } K > -4$$

Hence, the system will be stable for -4<K<20 i.e., stable for K=10 and unstable for K>20 i.e., for K=100,

Hence, the correct option is (B).

5. (B)

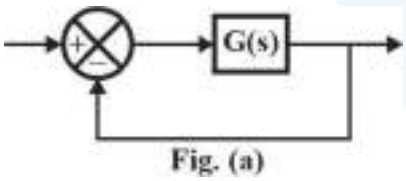


Fig. (a)

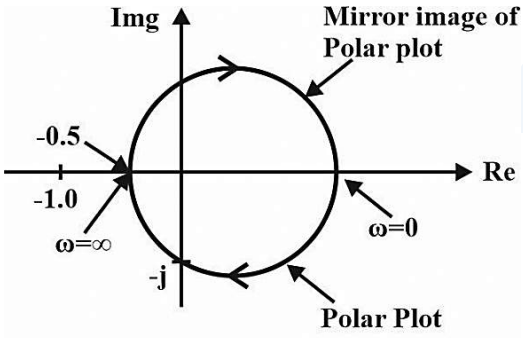


Fig. (b) Nyquist plot (first Possibility)

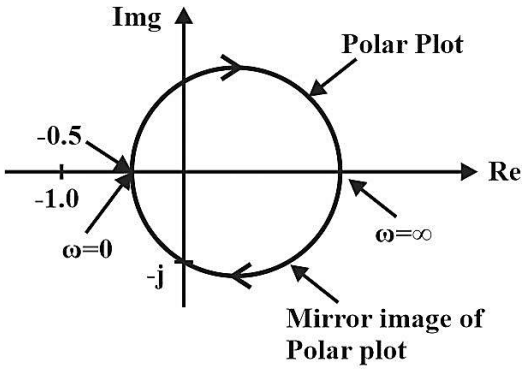


Fig. (c) Nyquist plot (Second Possibility)

(A): It is given that G(s) is all-pass filter, and we know that an all-pass filter has a constant (fixed) magnitude for all frequency, so its Nyquist plot should be a circle of the constant radius with center at the origin.

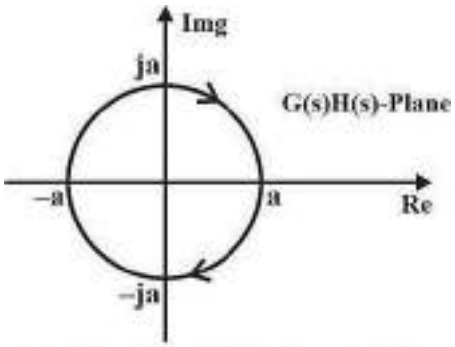


Fig. Nyquist plot of all pass filter

(B): It is given that G(s)has a zero in the right half of the s-plane (non-minimum phase system)

Assume $G(s) = K\left(\frac{s-z_1}{s+p_1}\right)$

where $z_1 > 0$ and $p_1 > 0$

Put $s = j\omega$ in the above equation,

$$G(j\omega) = K\left(\frac{j\omega - z_1}{j\omega + p_1}\right)$$

Magnitude of theopen-loop transfer function $G(j\omega)$ is given by,

$$|G(j\omega)| = K\left(\frac{\sqrt{\omega^2+z_1^2}}{\sqrt{\omega^2+p_1^2}}\right)$$

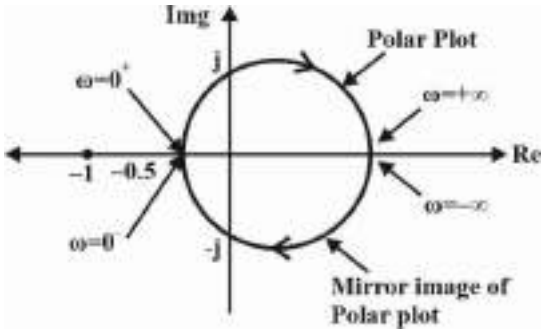
Phase angle of theopen-looptransfer function $G(j\omega)$ is given by,

$$\phi(\omega) = \angle G(j\omega)$$

$$\phi(\omega) = 180° - \tan^{-1}\left(\frac{\omega}{z_1}\right) - \tan^{-1}\left(\frac{\omega}{p_1}\right)$$

ω	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
0	$\frac{Kz_1}{p_1}$	180°
∞	K	0°

This will satisfy the Nyquist plot given in the question.



Comparing the table with the given Nyquist plot,

$$\frac{Kz_1}{p_1} = 0.5 \qquad \ldots(v)$$

(C): It is given that G(s) is the impedance of a passive network. Assuming a passive network having the series combination of R and C, its impedance will be

$$G(s) = R + \frac{1}{Cs} \quad where R > 0, C > 0$$

$$G(s) = \frac{RCs + 1}{Cs}$$

$$G(s) = \frac{RC\left[s + \frac{1}{RC}\right]}{Cs} = \frac{R\left[s + \frac{1}{RC}\right]}{s}$$

Put $s = j\omega$ in the above equation,

$$G(j\omega) = \frac{R\left[j\omega + \frac{1}{RC}\right]}{j\omega}$$

Magnitude of the open-loop transfer function $G(j\omega)$ is given by,

$$|G(j\omega)| = R\frac{\sqrt{\omega^2 + \left(\frac{1}{RC}\right)^2}}{\omega}$$

Phase angle of the open-loop transfer function $G(j\omega)$ is given by,

$$\phi(\omega) = \angle G(j\omega)$$

$$\phi(\omega) = \tan^{-1}\left(\frac{\omega}{\frac{1}{RC}}\right) - \tan^{-1}\left(\frac{\omega}{0}\right)$$

$$\phi(\omega) = -90° + \tan^{-1}(\omega RC)$$

ω	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
0	∞	-90°
∞	R	0°

Here at $\omega = 0$ magnitude is ∞ but, for given Nyquist plot, all the frequencies have finite magnitude; thus option (C) is incorrect.

(D): It is given that G(s) is marginally stable, but in the question, it is mentioned that G(s) is stable. Therefore, option (D) is incorrect.

Hence, the correct option is (B).

Key point:

Impedance of passive network represents minimum phase system i.e., left-hand poles and zeros in s-plane. In the minimum phase system, there is no possibility of the negative sign of $G(j\omega)$ at $\omega = 0$ or $\omega = \infty$.

6. (C)

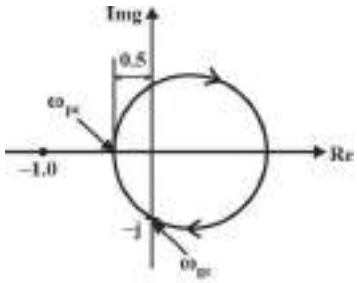


Fig. For stable system, $\omega_{pc} < \omega_{gc}$

From the figure magnitude at phase crossover frequency is 0.5.

$$\left|G\left(j\omega_{pc}\right)\right| = 0.5$$

Gain margin is given by,

$$GM = \frac{1}{\left|G\left(j\omega_{pc}\right)\right|} = \frac{1}{0.5} = 2$$

In dB, $GM = 20\log GM = 20\log 2$

GM = 6.02 dB

From the figure, the phase at gain crossover frequency is -90°.

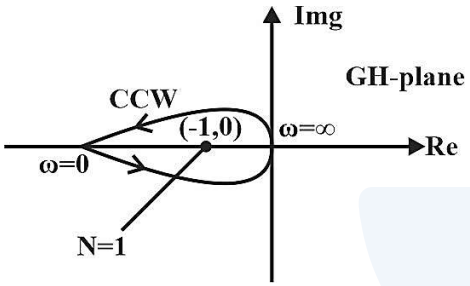
$$PM = 180° + \angle G\left(j\omega_{gc}\right) = 180° - 90°$$

PM = 90°

Hence, the correct option is (C).

7. (A)

Given Nyquist plot is shown below,



Nyquist stability criterion is given by,

N=P-Z ...(i)

where,

N= Number of encirclement about critical point (-1+j0) in counter clockwise (anti clockwise) direction.

P= Number of right-handpoles of open-loop transfer function i.e. G(s)H(s).

Z= Number of right-handpoles of the closed-loop transfer function.

From figure, number of the encirclement about the critical point (-1+j0) is one anticlockwise.

Thus, N=1

Since, number of right-handpole of OLTF

G(s) H(s), P=1 [Given]

Hence, from equation (i),

1=1-Z

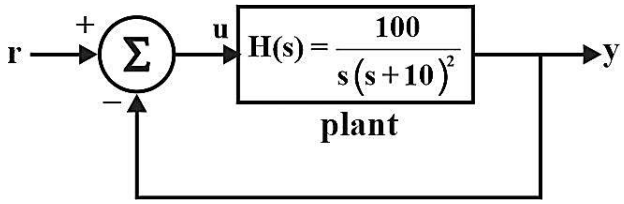
Z=0

Thus, no pole of closed transfer function will lie in RHS, and therefore the system is stable.

Hence, the correct option is (A).

8. (C)

The plant with unity feedback configuration is shown below,



Method 1:

$$OLTF, H(s) = \frac{100}{s(s+10)^2}$$

Put $s = j\omega$ in the above equation,

$$H(j\omega) = \frac{100}{j\omega(j\omega+10)^2}$$

$$H(j\omega) = \frac{100}{j\omega\left(-\omega^2+100+j20\omega\right)}$$

$$H(j\omega) = \frac{100}{-j\omega^3+j100\omega-20\omega^2} \quad \text{...(i)}$$

Magnitude of the open-loop transfer function is given by,

...(iii)

Phase angle of the open-loop transfer function is given by,

Calculation of :

Phase angle at phase crossover frequency is given by,

Or

Equating the imaginary part of to zero,

From equation (i),

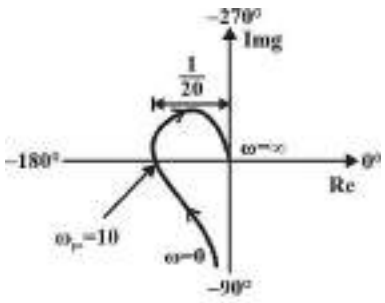


Fig. Type 1, order 3

Note: This method is the only valid, when the polar plot crosses -180°axis.

Fromthe equation (ii),

Gain margin is given by,

In dB, GM(dB) = 20log20=26dB

Hence, the correct option is (C)

Method 2:

The characteristic equation is given by,

For , replace gain with K in H(s).

For third order system to be marginal stable,

Internal product(IP) = External product (EP)

Gain margin is given by

In

Hence, the correct option is (C).

11 Gain Margin and Phase Margin

Objective

After completing this chapter user will be able to understand concept of:

1. Gain margin
2. Phase margin

Introduction

- We discussed how to sketch and interpret a Nyquist diagram to determine a closed-loop system's stability. Let us extend our discussion to the design of transient response characteristics via frequency response techniques.
- Using the Nyquist diagram, we define two quantitative measures of how stable a system is. These quantities are called gain margin and phase margin.
- Systems with greater gain and phase margins can withstand greater changes in system parameters before becoming unstable.
- In a sense, gain and phase margins can be qualitatively related to the root locus in that systems whose poles are farther from the imaginary axis have a greater degree of stability.

Minimum Phase System

'All poles and zeros must be on L.H.P.'

Pole or zero at origin is allowed.

$$T(s) = \frac{1}{(s+1)(s+2)}$$

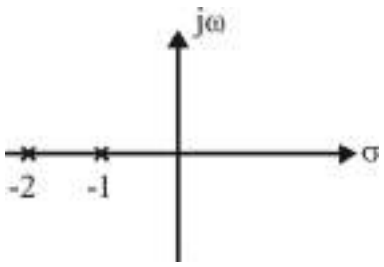


Fig. 11.1

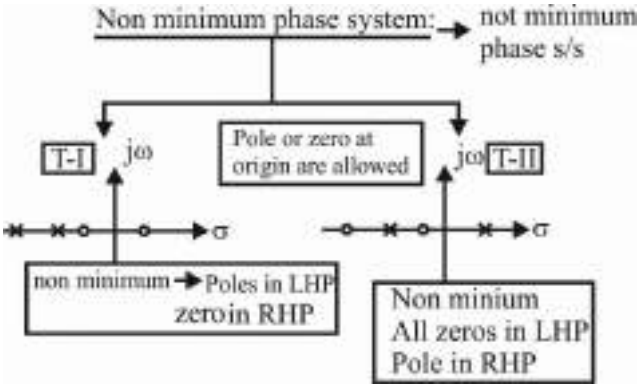


Fig. 11.2

NOTE:

'Poles and zeros present on $j\omega$-axis does not alter the definition of minimum or non-minimum or maximum phase system'.

Gain Margin

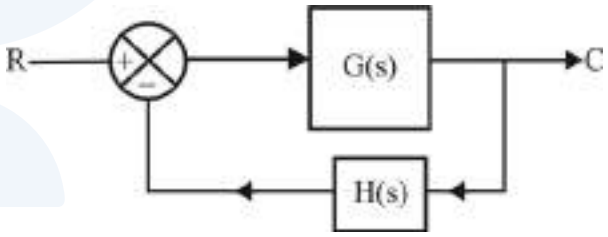


Fig. 11.3

$$C.L.T. = \frac{G(s)}{1+G(s)H(s)}$$

$$OLTF : G(s)H(s)$$

- Gain margin is calculated for C.L.S.
- It is the amount of gain (k_g) multiplied to O.L.T.F. G(s)H(s) such that corresponding C.L.S. becomes marginally stable.
- It is the amount of gain (in dB) added to O.L.T.F. G(s)H(s) such that corresponding C.L.S. becomes marginally stable (on the verge of instability).
- The gain margin (k_g) is the reciprocal of the magnitude $|G(j\omega)|$ at the phase crossover frequency.

Phase Crossover Frequency (Ω_p)

The frequency at which the phase angle of the open-loop transfer function equals -180°.

$$k_g = \frac{1}{\left|G\left(j\omega_p\right)\right|}$$

In terms of decibels,

$$k_g\left(dB\right) = 20\log k_g = -20\log\left|G\left(j\omega_p\right)\right|$$

Phase Margin (γ):

- Phase margin is calculated for C.L.S.
- It is the amount of phase (Angle) introduced in the phase of O.L.T.F. such that corresponding C.L.S. becomes marginally stable.
- P.M. → in degree
- It is the amount of phase introduced (added) in the phase of O.L.T.F. such that C.L.S. reaches on the verge of instability.

Gain Crossover Frequency ()

- It is the frequency at which the magnitude of open-loop transfer function $\left|G\left(j\omega_g\right)\right|$ is unity.
- The phase margin γ is 180° plus the phase angle ϕ of the open-loop transfer function at gain crossover frequency. $\gamma = 180 + \phi$

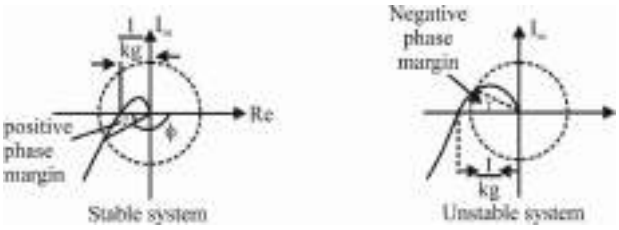


Fig. 11.4

- From the above figure we can observe that positive phase margin represents a stable system and a negative phase margin represents an unstable system.
- Higher phase margin indicates that higher phase shift is required to make the system unstable and thus the system is relatively more stable.

Important observations

Case 1: O.L.T.F. of a C.L.S. is given:

$$G\left(s\right)H\left(s\right) = \frac{k}{s\left(s+1\right)\left(s+3\right)}$$

Comment on the gain margin =?

Solution:

$$\frac{k}{s\left(s+1\right)\left(s+3\right)} \qquad 0 < k < \infty$$

–ve feedback

Root Locus

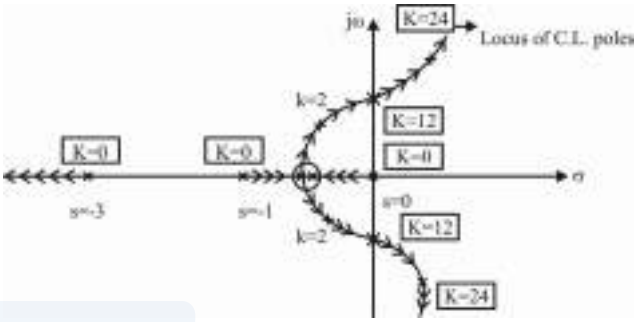


Fig. 11.5

$$G\left(s\right)H\left(s\right) = \underbrace{\frac{2\times k_1}{s\left(s+1\right)\left(s+3\right)}}_{\text{minimum phase s/s}} \Rightarrow G.M. = 6 \rightarrow 6 > 1$$

$$GM\left(dB\right) = 20\log_{10} 6 = +ve\ in\ dB$$

$$G\left(s\right)H\left(s\right) = \underbrace{\frac{12\times k_1}{s\left(s+1\right)\left(s+3\right)}}_{\text{min phase s/s}} \Rightarrow G.M. = 1 \rightarrow 1 = 1$$

$$GM\left(dB\right) = 0dB$$

$$G\left(s\right)H\left(s\right) = \underbrace{\frac{24\times k_1}{s\left(s+1\right)\left(s+3\right)}}_{\text{min phase s/s}} \Rightarrow G.M. = \frac{1}{2} \rightarrow \frac{1}{2} < 1$$

$$GM\left(dB\right) = -ve\ in\ dB$$

Case 2:

O.L.T.F. of a C.L.S. is given:

$$G\left(s\right)H\left(s\right) = \frac{k\left(1-s\right)}{\left(s+1\right)\left(s+2\right)} \qquad 0 < k < \infty$$

–ve feedback

Solution:

Locus of closed-loop poles

$$G\left(s\right)H\left(s\right) = \frac{-k\left(s-1\right)}{\left(s+1\right)\left(s+2\right) \rightarrow C.R.L.}$$

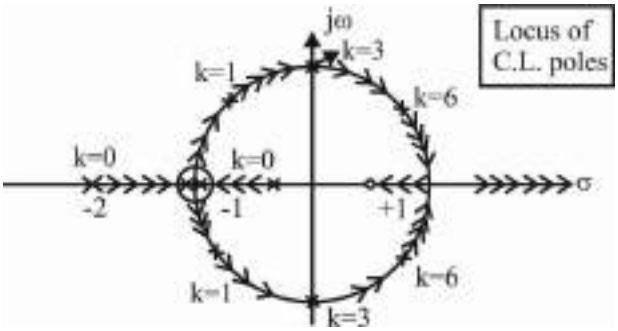


Fig. 11.6

$$G(s)H(s)=\frac{1(1-s)\times k_1}{(s+1)(s+2)}\Rightarrow G.M.=3\rightarrow 3>1$$

non minphase s/s Type I

$$G.M.(dB)=+ve\ in\ dB$$

$$G(s)H(s)=\frac{3(1-s)\times k_1}{(s+1)(s+2)}\Rightarrow G.M.=1\rightarrow 1=1$$

non minphase s/s type I

$$G.M.(dB)=0dB$$

$$G(s)H(s)=\frac{6(1-s)\times k_1}{(s+1)(s+2)}\Rightarrow G.M.=\frac{1}{2}\rightarrow\frac{1}{2}<1$$

$$GM(dB)=-ve\ dB$$

Common Observations Of Minimum Phase Or Non-Minimum Phase (Type – I) O.L.T.F G(S)H(S)

1. 'For C.L.S. to be stable'
   $\leftrightarrow GM>1$ *or* $GM>0dB$
2. 'For C.L.S. to be marginallystable
   $\leftrightarrow GM=1$ *or* $GM=0dB$,
3. 'For C.L.S. to be unstable'
   $\leftrightarrow GM<1$ *or* $GM<0dB$

Case 3:

$$G(s)H(s)=\frac{k(s+1)}{(s-1)(s-3)}$$

Locus of closed-loop poles as
$0<k<\infty$ *–ve feedback*

$$G(s)H(s)=\frac{k(s+1)}{(s-1)(s-3)}\rightarrow DRL$$

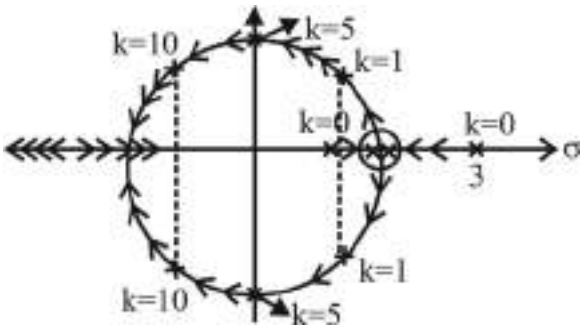


Fig. 11.7

$$G(s)H(s)=\frac{(s+1)\times k_1}{(s-1)(s-3)}\rightarrow G.M.=5\rightarrow 5>1$$

non minimum:T-2

$$GM(dB)=+ve\ in\ dB$$

$$G(s)H(s)=\frac{5(s+1)\times k_1}{(s-1)(s-3)}\rightarrow G.M.=1\rightarrow 1=1$$

non-minimum:T-2

$$GM(dB)=0dB$$

$$G(s)H(s)=\frac{10(s+1)\times k_1}{(s-1)(s-3)}\rightarrow G.M.=\frac{1}{2}$$

$$GM(dB)=-ve\ in\ dB$$

Note: Observation fornon-minimum phase system O.L.T.F. (Type – II) G(s)H(s)

1. For C.L.S to be stable
   $\leftrightarrow G.M.<1$ *or* $GM<0dB$
2. For C.L.S. to be marginally stable
   $\leftrightarrow GM=1$ *or* $GM=0dB$
3. For C.L.S. to be unstable
   $\leftrightarrow GM>1$ *or* $GM>0dB$

Points To Remember

C.L.T.F	O.L.T.F.	G.M.(dB)	P.M.(in^0)
Stable /unstable	Min. phase s/s	+ve (dB)/–ve(dB)	+ve in ^{0}/–ve in ^{0}

C.L.T.F	O.L.T.F.	G.M.(dB)	P.M.(in^0)
Stable/ unstable	Non min phase s/s T-1	+ve (dB) /–ve(dB)	+ve in ^{0}/–ve in ^{0}
Stable/ unstable	Non min phase s/s T-2	-ve (dB) /+ ve(dB)	-ve in ^{0}/+ve in ^{0}

Table 11.1

Mathematical Calculation Of Gain Margin

Method 1:

T(s) = F(s)

Step 1:

$s = j\omega$: Range of ω

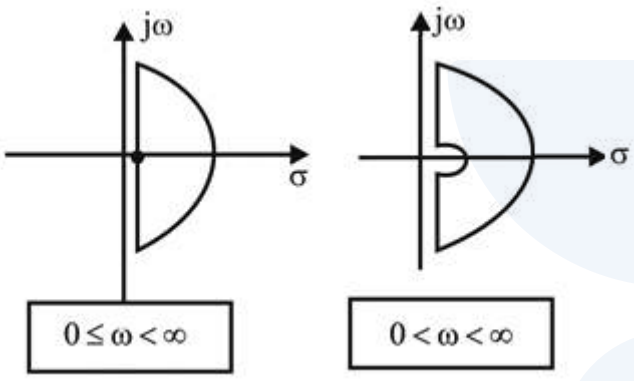


Fig. 11.8

Step 2:

$T(j\omega)$

$\angle T(j\omega) = -180^0 \rightarrow$ *Calculate* $\omega = ?$

$\omega \rightarrow$ Possible phase crossover frequency.

Step 3:

Validation of $\omega = ?$

(i) $\angle T(j\omega) = -180^0$

(ii) $\omega \rightarrow$ Real *and* $+ve$ (i) $\cap$ (ii)

Step 4:

If $\omega = \omega_{PC}$ is valid.

$|T(j\omega_{PC})|$

$$G.M.. = \frac{1}{|T(j\omega_{PC})|}$$

Note:

If ω_{PC} is invalid then

$GM = +\infty\, dB$

Or

$GM = -\infty\, dB$

Based on absolute stability of C.L.S. and type of O.L.T.F(R-H Table).

C.L.S.	O.L.T.F.	G.M.(dB)
Stable/ unstable	Min phase	$+\infty$ /- ∞
Stable/ unstable	Non Min phase I	$+\infty$ /- ∞
Stable/ unstable	Non Min phase II	- ∞ /+ ∞

Table 11.2

1. The O.L.T.F. of C.L.S. with –ve feedback is.

$$G(s)H(s) = \frac{1}{(s+1)(s+2)}$$

Calculate gain margin.

Solution:

S.1

$$T(j\omega) = \frac{1}{(1+j\omega)(2+j\omega)}: \quad 0 \le \omega < \infty$$

S.2

$$\angle T(j\omega) = -\tan^{-1}\omega - \tan^{-1}\frac{\omega}{2} = -180^0$$

$$\frac{3\omega}{2-\omega^2} = 0$$

$\omega = 0 \rightarrow$ *possible* ω_{PC}

$$\tan^{-1}\left[\frac{\omega+\omega/2}{1-\frac{\omega^2}{2}}\right] = 180^0$$

S.3

(i) $\left\{-\tan^{-1}\omega-\tan^{-1}\frac{\omega}{2}\right\}\neq -180^0$

$\omega=0$

(ii) $\omega=0$ Real and positive (i) ∩ (ii)

Not valid ω_{pc}

S.4

Absolute stability of

C.L.S. $\rightarrow 1+G(s)H(s)=1+\frac{1}{(s+1)(s+2)}$

and

$OLTF \rightarrow \min\ phase \Rightarrow GM=+\infty\, dB(stable)$

Method 2:

Step 1:

$s=j\omega \rightarrow T(j\omega) \rightarrow Range\ of\ "\omega"$

Step 2:

$T(j\omega)=T_R(j\omega)+jT_I(\omega) \rightarrow Rationalisation$

Step 3:

$T_I(\omega)=0 \rightarrow \omega=possible\ \omega_{pc}$.

Step 4:

Validity:

$\{\omega=\text{Real and positive}\}\cap\{T_R(j\omega)=-ve\}$

Step 5:

If $T_R(j\omega)=-ve$

$GM=\frac{1}{|T_R(j\omega)|}$

2:

$T(s)=\frac{1}{(s+1)(s+2)}$

Sol: S.1

$s=j\omega$: $\quad T(j\omega)=\frac{1}{(1+j\omega)(2+j\omega)}: 0\le\omega\le\infty$

S-2

$T(j\omega)=\frac{(1-j\omega)(2-j\omega)}{(1+\omega^2)(4+\omega^2)}$

$T(j\omega)=\frac{(2-\omega^2)}{(1+\omega^2)(4+\omega^2)}+j\left\{\frac{-3\omega}{(1+\omega^2)(4+\omega^2)}\right\}$

S-3

$\text{Im}\{T(j\omega)\}=0\rightarrow\omega=0\rightarrow possible\ \omega_{pc}$

$\text{Re}\{T(j\omega)\}=\frac{1}{2}\rightarrow +ve\quad \omega=0\rightarrow Invalid\ \omega_{pc}$

S-4

$G(s)H(s)=\frac{1}{(s+1)(s+2)}$

$1+G(s)H(s)\rightarrow C.L.S.\ stable\rightarrow GM=+\infty\, dB$

O.L.T.F. minimum phase system

$G(s)H(s)=\frac{1\times k_1}{(s+1)(s+2)}\qquad G(s)H(s)=\frac{k}{(s+1)(s+2)}$

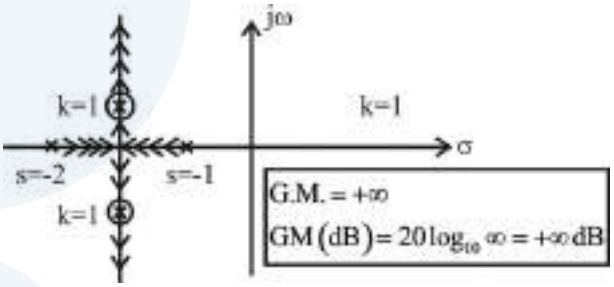


Fig. 11.9

3.

The O.L.T.F. of C.L.S. is.

$G(s)H(s)=\frac{1}{s(s+1)(s+3)}$

Calculate gain margin.

Solution:

S.1

$T(j\omega)=\frac{1}{j\omega(1+j\omega)(3+j\omega)}, 0<\omega<\infty$

S-2

$T(j\omega)=\frac{-j(1-j\omega)(3-j\omega)}{\omega(1+\omega^2)(9+\omega^2)}$

$T(j\omega)=\frac{-j(3-j\omega-3j\omega-\omega^2)}{\omega(1+\omega^2)(9+\omega^2)}$

$$= \frac{-4\omega}{\omega\left(1+\omega^2\right)\left(9+\omega^2\right)} + \frac{-j\left(3-\omega^2\right)}{\omega\left(1+\omega^2\right)\left(9+\omega^2\right)}$$

S-3

$T_m\left\{T\left(j\omega\right)\right\} = 0$

$\omega = \sqrt{3} \rightarrow Real\ and\ +ve \rightarrow possible$

$$\text{Re}\left\{T\left(j\omega\right)\right\} = \frac{-4}{4\left(12\right)} = \frac{-1}{12} \rightarrow -ve$$

$\omega = \sqrt{3} \rightarrow valid\ \omega_{pc}$

$$G.M. = \frac{1}{\left|\text{Re}\left\{T\left(j\omega\right)\right\}\right|} = 12$$

$GM\left(dB\right) = 20\log_{10} 12 = 21.58\,dB$

Some exceptional observations in Gain Margin:

Calculate G.M. of a C.L.S. with –ve feedback having O.L.T.F. as

$$G\left(s\right)H\left(s\right) = \frac{1}{\left(s+1\right)\left(s+2\right)}$$

Method 1:

1. $$T\left(j\omega\right) = \frac{1}{\left(1+j\omega\right)\left(2+j\omega\right)} : 0 \le \omega < \infty$$

2. $$\angle T\left(j\omega\right) = -\tan^{-1}\omega - \tan^{-1}\frac{\omega}{2} = -180^0$$

$$\frac{3\frac{\omega}{2}}{1-\frac{\omega^2}{2}} = 0$$

$\omega = 0$ or $\omega = \infty : \omega \rightarrow \infty$

Possible

$\omega_{pc} \rightarrow \omega = 0$

$\omega = \infty : \omega \rightarrow \infty$

$$-\tan^{-1}\omega - \tan^{-1}\frac{\omega}{2} = -180^0$$

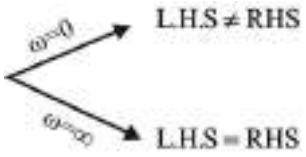


Fig. 11.10

$\omega_{pc} = \infty \rightarrow valid\ \omega_{pc}$

G.M.

$$\left|T\left(j\omega_{pc}\right)\right| = \left|T\left(j\infty\right)\right| = 0 \rightarrow G.M. = \frac{1}{\left|T\left(j\omega\right)\right|} = \infty$$

$\left|T\left(j\omega_{pc}\right)\right| dB = +\infty\, dB$

Point

$\rightarrow G.M. = \infty\ in\ Ratio$

$$G\left(s\right)H\left(s\right) = \frac{1}{\left(s+1\right)\left(s+2\right)}$$

$$k_1 G\left(s\right)H\left(s\right) = \frac{k_1}{\left(s+1\right)\left(s+2\right)}$$

$1 + k_1 G\left(s\right)H\left(s\right) = 0 = \left(s+1\right)\left(s+2\right) + \infty = 0 \rightarrow$

$Root\ \ locus\ (Invalid\ \omega_{pc})$

Example:

$$G\left(s\right)H\left(s\right) = \frac{1}{s\left(s+1\right)}$$

Solution:

G.M.

$$T\left(j\omega\right) = \frac{1}{j\omega\left(1+j\omega\right)} \quad 0 < \omega < \infty$$

Method 2:

$$T\left(j\omega\right) = \frac{1\left(1-j\omega\right)}{j\omega\left(1+\omega^2\right)} = \frac{1}{j\omega\left(1+\omega^2\right)} - \frac{j\omega}{j\omega\left(1+\omega^2\right)}$$

$$T\left(j\omega\right) = \frac{-1}{1+\omega^2} - \frac{-j}{\omega\left(1+\omega^2\right)}$$

$\text{Im}\left\{T\left(j\omega\right)\right\} = 0\ as\ \omega = \infty \rightarrow possible\ \omega_{pc}$

$$\text{Re}\left\{T\left(j\omega\right)\right\} = 0 \rightarrow GM = \frac{1}{|0|} = +\infty$$

$GM\left(dB\right) = 20\log_{10}\infty = \infty \rightarrow dB$

$$G\left(s\right)H\left(s\right) = \frac{1}{s\left(s+1\right)} \quad k_1 = \infty$$

$1 + G\left(s\right)H\left(s\right) = s\left(s+1\right) + k_1 = 0$

$s\left(s+1\right) + \infty = 0 \rightarrow$ "$Root\ \ Locus$"

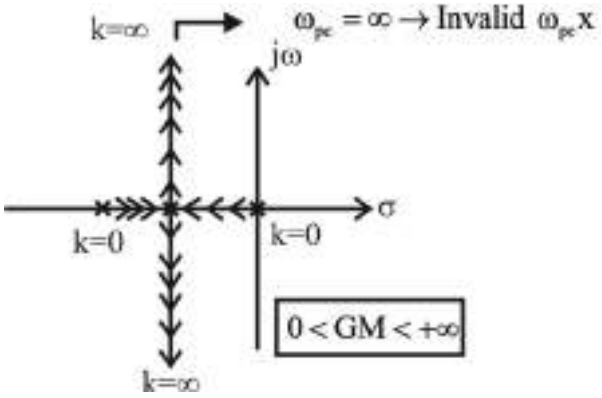


Fig. 11.11

$$G(s)H(s) = \frac{1}{s(s+1)}$$

$$G(s)H(s) = \frac{k_1}{s(s+1)}$$

$$1 + G(s)H(s) = 0$$

$$s(s+1) + \underset{k_1=0}{k_1} = 0 \rightarrow s = 0 \& s = -1 \rightarrow M.S.$$

Conclusions:

1. G.M. Range in Ratio $0 < GM < +\infty$
2. G.M. can never be exactly = 0 (in Ratio)
3. $GM \rightarrow +\infty$: 'Symbolic representation of absolute stable system'.
4. G.M. range in dB $(-\infty)dB < GM < +\infty(dB)$

Shortcut to calculate G.M.:

1. Given: G(s)H(s) = F(s)
2. $k_1 G(s)H(s) = k_1 F(s)$
3. $1 + \underset{\text{"Marginal stable"}}{k_1 G(s)H(s)} = 1 + k_1 F(s)$ *Non repeated*

Roots on $j\omega$ *axis or* Single *Root at origin as* $0 < k_1 < \infty$

Routh criteria of $1 + k_1 F(s)$

1. Odd row never zero for $0 < k_1 < \infty$
2. Odd row zero but $k_1 = -ve$
3. Odd row zero and $k_1 = +ve$ but Roots of A.E. not on $j\omega$-axis.
4. Odd row zero and $k_1 = +ve$, root of A.E. on $j\omega$-axis but repeated in nature.

   Last row of Routh table $\neq 0$ for $0 < k_1 < \infty$

   If odd row is zero $0 < k_1 < \infty$

And

Value of $k_1 = +ve$

And

Roots of A.E. $\rightarrow$ Non-repeated on $j\omega$-axis

$$\pm j\omega = \pm j\omega_{pc}$$

$$k_1 = G.M. \text{ in Ratio}$$

Last row of Routh table = 0 for $0 < k_1 < \infty$

$$k_1 = G.M. \qquad \omega_{pc} = 0$$

4.

$$G(s)H(s) = \frac{10(s+3)}{s(1+s)(s+2)}$$

Solution:

$$G(s)H(s) = \frac{10(s+3)}{s(1+s)(s+2)} = F(s)$$

$$1 + k_1 F(s) = s(1+s)(s+2) + 10k_1(s+3) = 0$$

$$1 + k_1 F(s) = s^3 + 3s^2 + (2 + 10k_1)s + 30k_1 = 0$$

Odd row never zero

s^3	1	$2 + 10k_1$
s^2	3	$30k_1$
s^1	$\frac{6 + 30k_1 - 30k_1}{3}$	0
s^0	$30k_1$ *not zero.*	

Absolute stability:

Stable

O.L.T.F.: Minimum phase system

GM = $+\infty$ dB

Important points:

1. Odd row is zero and last row non zero. (G.M. $\rightarrow$ Finite)
2. Odd row non zero and last row zero
3. Odd row non zero and last row non zero
4. Odd row zero and last row zero. (G.M. $\rightarrow$ Finite)

Phase Margin

Step 1:

Put **s =** $j\omega$ in $T(j\omega)$

Step 2:

$T(j\omega) = |T(j\omega)| e^{j\angle T(j\omega)}$

Step 3:

$|T(j\omega)| = 1 \rightarrow$ Possible gain crossover frequency (ω_{gc})

$\omega_{gc} \rightarrow$ Real and positive.

Step 4:

Valid ω_{gc}

$P.M. = \angle T(j\omega_{gc}) + 180^0$

Step 5:

Invalid $\omega_{gc} \rightarrow$ Absolute stability **+** Type of O.L.T.F.

$P.M. = \pm\infty^0$

5.

$G(s)H(s) = \dfrac{10}{s(s+1)}$ *Phase* margin

Solution:

S.1

$T(j\omega) = \dfrac{10}{j\omega(1+j\omega)}$

S.2

$|T(j\omega)| = \dfrac{10}{\omega\sqrt{1+\omega^2}} \Rightarrow |T(j\omega)| = 1$

$\angle T(j\omega) = -90^0 - \tan^{-1}\omega$

$100 = \omega^2(1+\omega^2)$

$x(1+x) = 100$

$x = ?$

$\omega = \sqrt{x} = 3.082$

$P.M. = \angle T(j\omega_{gc}) + 180^0$

$P.M. = -90^0 - \tan^{-1}(3.082) + 180^0$.

$P.M. = 90^0 - \tan^{-1}(3.082) = +18^0$

6.

$G(s)H(s) = \dfrac{10}{s^2(s+1)} \rightarrow G.M.$

Solution:

Method 1:

$T(j\omega) = \dfrac{10}{(j\omega)^2(1+j\omega)} \rightarrow 0 < \omega < \infty$

$\angle T(j\omega) = -180° - \tan^{-1}\omega = -180°$

$tan^{-1}\omega = 0°$

$\underset{Invalid}{\omega = 0} \rightarrow$ *possible* ω_{pc}

$|T(j\omega)| = \infty$

$GM = 0$ *in Ratio*

$G(s)H(s) = \dfrac{10}{s^2(1+s)}$

$1 + k_1 G(s)H(s) = \underset{k_1=0}{s^2(1+s)} + \underset{s=0,0,-1}{10k_1}$

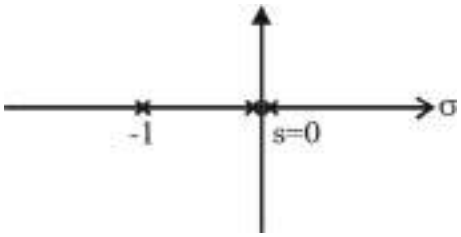


Fig. 11.12

Method 2:

$T(j\omega) = \dfrac{10(1-j\omega)}{(j\omega)^2(1+j\omega)(1-j\omega)}$

$T(j\omega) = \dfrac{10(1-j\omega)}{-\omega^2(1+\omega^2)}$

$T(j\omega) = \dfrac{-10}{\omega^2(1+\omega^2)} + \underset{\text{Im}\{T(j\omega)\}=0 \Rightarrow \omega \rightarrow \infty}{\underset{Invalid}{\dfrac{j10}{\omega(1+\omega^2)}}}$

$GM = +\infty\,(in\ Ratio)$ $\text{Re}\{T(j\omega)\} = 0$ x

Method 3:

$$T(s) = \frac{10}{s^2(1+s)}$$

$$s^3 + s^2 + 10k_1 = 0 \qquad 0 < k_1 < \infty$$

+	s^3	1	0
+	s^2	1	$10k_1$
+	s^1	$-10k_1$	0
+	s^0	$10k_1$	

$G.M. = -\infty dB \rightarrow$ "*symbolic representation*".

$\omega_{pc} \rightarrow$ *does not exist*.

Gain Margin and Phase Margin From Nyquist Plot

Let, $G(s)H(s) = \dfrac{k}{s(1+ST_1)(1+ST_2)}$

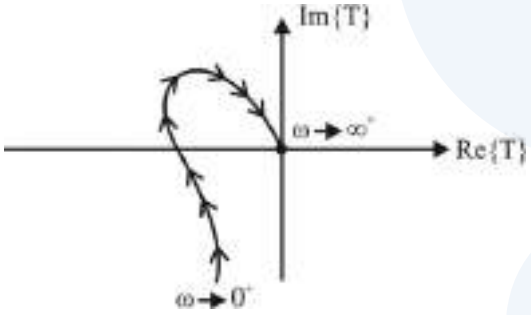


Fig. 11.13

Gain Margin

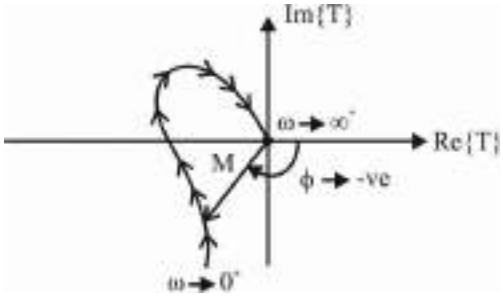


Fig. 11.14

$T(j\omega) = |T(j\omega)| e^{j\angle T(j\omega)}$

$= Me^{j\phi}$

$M \rightarrow$ Radial distance from origin

$\phi \rightarrow$ *Angle measured from* $+ve$ Real axis.

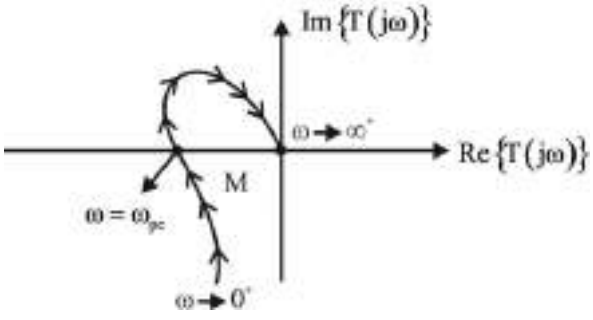


Fig. 11.15

For ω_{pc}

$\angle T(j\omega) = -180°$

$|T(j\omega_{pc})| = M$

$G.M. = \dfrac{1}{M}$

$GM(dB) = -20\log_{10} M$

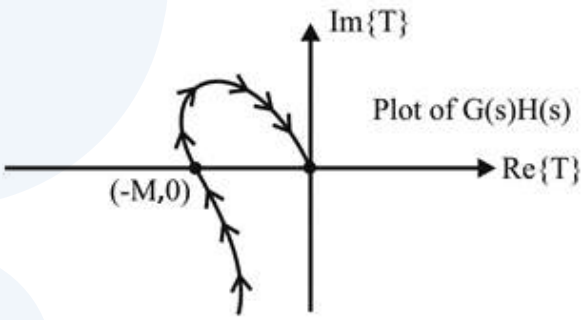


Fig. 11.16

$T(s) = G(s)H(s) \times k_1$

Example:

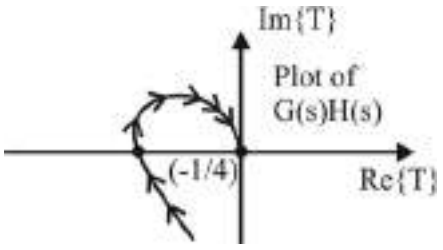


Fig. 11.17

$T(s) = kG(s)H(s)$

Calculate G.M. for k = 1

Solution:

G.M. = 4 (Ratio)

$GM(dB) = 20\log_{10} 4$

Note:

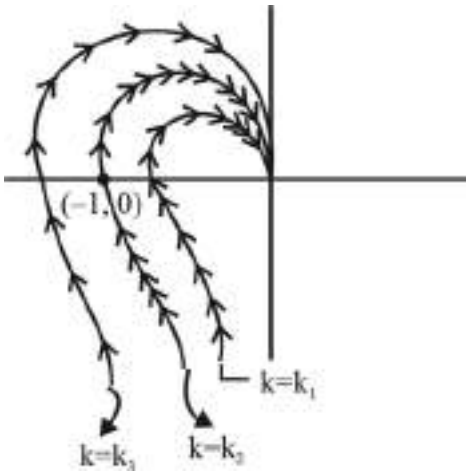


Fig. 11.18

$$G(s)H(s) = kF(s)$$

$$G(s)H(s) = \frac{k}{s(1+ST_1)(1+ST_2)}$$

$k = k_1 \rightarrow NSC \ in \ 1+G(s)H(s)$

$N = 0 \quad P_+ = 0$

$Z_+ = 0 \rightarrow C.L.S. \ stable$

$k = k_2 \rightarrow$? Marginal stable

$k = k_3$

$N \neq 0$

$P+ = 0$

Note:

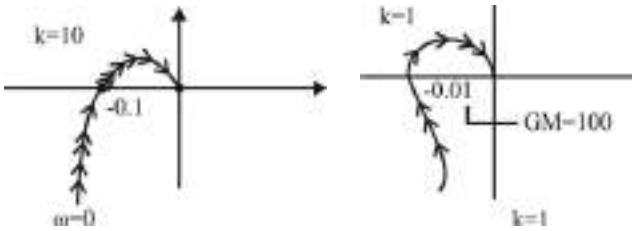
If poles or zeros are present at origin, concept of encirclement or enclosement is valid:

(–1, 0) point 'Boundary b/w stability and unstability'.

Gain Margin

The polar plot of G(s)H(s) for k = 10.

The gain k for a 40dB gain margin.



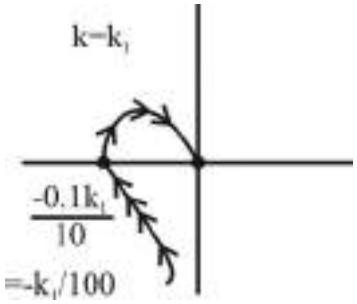
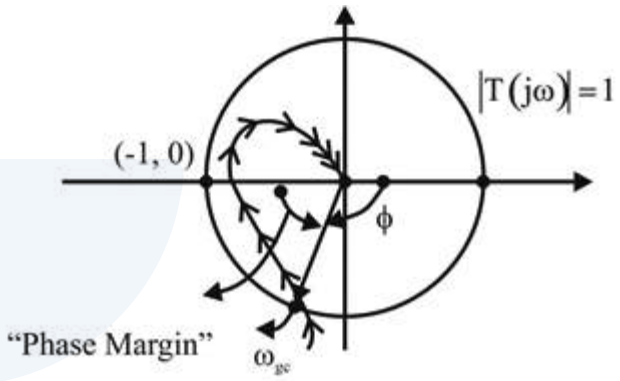


Fig. 11.19

G.M. = 40 = $20\log_{10}$ G.M.

G.M. = 100

Phase Margin



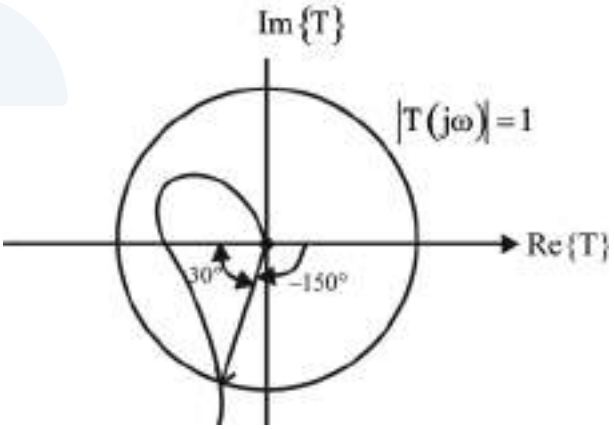


Fig. 11.20

Common mistake:

- While calculating the angle from polar plot, take care of sign (anti-clockwise or clockwise)
- While calculating angle for phase margin, carefully choose angle, sometimes students do the mistake and mistakenly add the phase margin (30° in above ex) to phase margin itself.

Case 1:

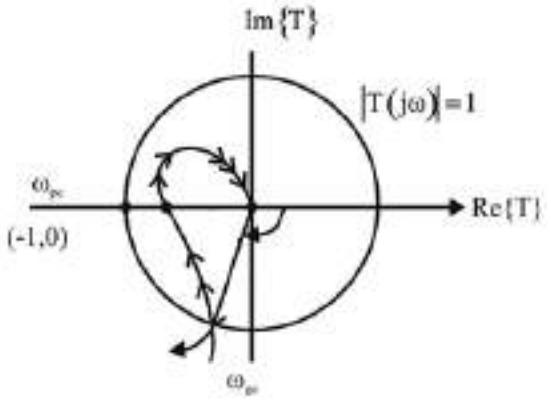


Fig. 11.21

G.M. > 1

Case 2:

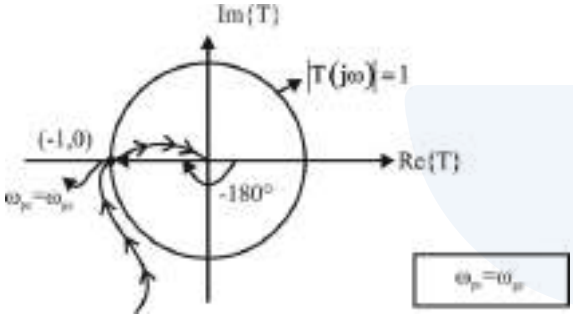


Fig. 11.22

Case 3:

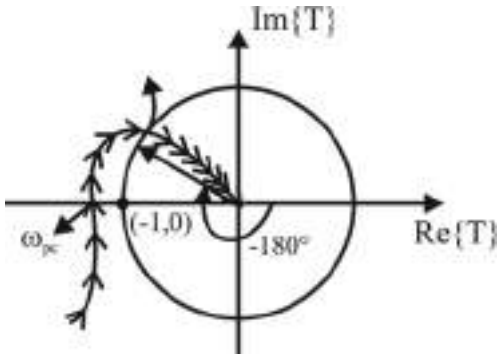


Fig. 11.23

System with Transportation LAG

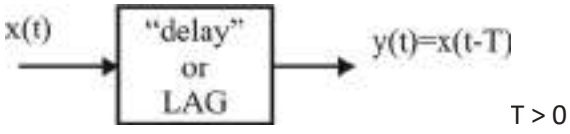


Fig. 11.24

Case 1:

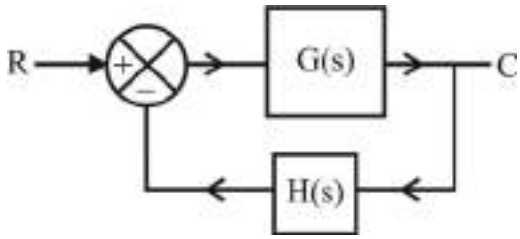


Fig. 11.25

G.M.

$$\angle T(j\omega) = \angle G(j\omega)H(j\omega) = -180^\circ$$

$$G.M. = \frac{1}{|T(j\omega_{pc})|}$$

P.M.

$$|T(j\omega)| = |G(j\omega)H(j\omega)| = 1$$

$$P.M. = 180^\circ + \angle T(j\omega_{gc})$$

$$T(s) = G(s)H(s)$$

1. $|T(j\omega)| = |G(j\omega)H(j\omega)|$
2. $\angle T(j\omega) = \angle G(j\omega)H(j\omega)$

Case 2:

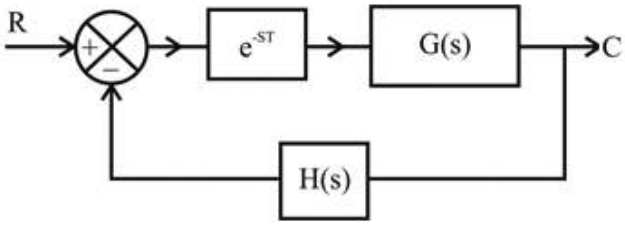


Fig. 11.26

$$F(s) = e^{-ST}G(s)H(s)$$

$$F(s) = e^{-ST}T(s) \qquad F(j\omega) = e^{-j\omega}T(j\omega)$$

1. Calculation of phase crossover frequency.

$$\angle F(j\omega) = -180^\circ$$

$$\angle F(j\omega) - \underset{\text{Radian}}{\omega} = -180^\circ \rightarrow \text{Phase cross Freq is different}$$

is different

2.

$$\text{Gain Margin}=\frac{1}{\left|F(j\omega_{pc})\right|}\rightarrow \text{G.M.is also different}$$

3. Calculation of gain crossover frequency:

$$\left|F(j\omega)\right|=\left|e^{-j\omega T}T(j\omega)\right|=1$$

$$\left|T(j\omega)\right|=1\rightarrow \omega_{gc}\text{remain same}$$

4. Phase margin: $PM=\angle F(j\omega)+180°$

$$PM=\angle T(j\omega_{gc})-\omega_{gc}+180°\rightarrow \text{P.M.is different}$$

P.M. is lesser → 'Corresponding C.L.S. relatively unstable'.

7. $G(s)H(s)=\frac{e^{-0.1s}}{s}$ calculate $\omega_{pc}=?$

Solution: $T(j\omega)=\frac{e^{-0.1j\omega}}{j\omega}$

$$\angle T(j\omega)=-180°$$

$$\leftarrow \frac{-\omega}{10}-90°\underset{\text{degree}}{=}-180°\Rightarrow \underset{\text{Radian}}{\frac{-\omega}{10}}-\underset{\text{Radian}}{\frac{\pi}{2}}=-\pi$$

$$\omega=\underset{\omega_{pc}}{5\pi}\text{ rad/sec}$$

8. $G(s)H(s)=\frac{3e^{-2s}}{s(s+2)}$

Determine Gain Margin&Phase Margin?

Solution:

Gain margin: $T(j\omega)=\frac{3e^{-2j\omega}}{j\omega(2+j\omega)}$

$$\omega_{pc}:-2\omega-\frac{\pi}{2}-\tan^{-1}\frac{\omega}{2}=-\pi$$

$$\tan^{-1}\frac{\omega}{2}+2\omega=\frac{\pi}{2}$$

$$\tan^{-1}(\theta)=\left\{\theta-\frac{\theta^3}{3}+.....\right\}$$

$$\left\{\frac{\omega}{2}-\frac{\left(\frac{\omega}{2}\right)^3}{3}+.....\right\}+2\omega=\frac{\pi}{2}$$

Approximation: $\frac{\omega}{2}+2\omega=\frac{\pi}{2}$

$$\omega=\frac{\pi}{5}\rightarrow \omega_{pc}=0.632$$

$$\text{G.M.}=\frac{1}{\left|T(j\omega_{pc})\right|}\Rightarrow GM(dB)=-7.33dB$$

Phase Margin

$$T(j\omega)=\frac{3e^{-2j\omega}}{j\omega(j\omega+2)}$$

$$\omega_{gc}\Rightarrow\left|T(j\omega)\right|=1\rightarrow\frac{3}{\omega\sqrt{4+\omega^2}}=1$$

$$\omega^1(4+\omega^2)=9 \qquad \omega^2=x$$

$$\omega_{gc}=1.267$$

$$\text{P.M.}=\pi+\left\{\frac{-\pi}{2}-2\omega-\tan^{-1}\frac{\omega}{2}\right\}$$

$$\text{P.M.}=1.528\text{rad}=-86.74°$$

Gain Margin, Phase Margin Through Bode Plot

$$T(j\omega)\rightarrow\left|T(j\omega_{pc})\right|$$

$$(dB)M=20\log_{10}\left|T(j\omega_{pc})\right|$$

$$\text{G.M.}=\frac{1}{\left|T(j\omega_{pc})\right|}$$

$$\text{G.M.}(dB)=-20\log_{10}\left|T(j\omega_{pc})\right|=-MdB$$

$$T(s)=\frac{k}{s\left[1+\frac{s}{\omega_1}\right]\left[1+\frac{s}{\omega_2}\right]\left[1+\frac{s}{\omega_3}\right]}$$

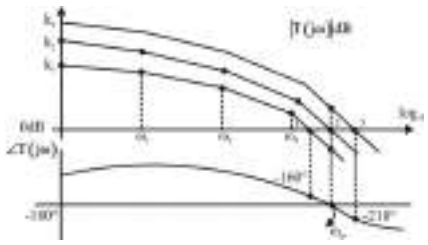


Fig. 11.27

$k = k_1 \quad \omega_{gc} < \omega_{pc} \Rightarrow G.M. = +ve \text{ in dB}$

$P.M. = +ve \text{ in degree}$

$k = k_2 \quad \omega_{pc} = \omega_{gc} \Rightarrow G.M. = 0dB$

$P.M. = 0°$

$k = k_3 \quad \omega_{gc} > \omega_{pc} \Rightarrow G.M. = -ve \text{ in dB}$

$P.M. = -ve \text{ in}°$

		G.M. (in dB)	P.M. (in°)	C.L.S.
$k = k_1$	$\omega_{pc} > \omega_{gc}$	+ve	+ve	Stable
$k = k_2$	$\omega_{pc} = \omega_{gc}$	0	0	M.S.
$k = k_3$	$\omega_{pc} < \omega_{gc}$	-ve	-ve	Unstable

Table 11.3

$k_3 > k_2 > k_1$

Polar Plot of Transportation LAG

$$G(s)H(s) = \frac{e^{-s}}{1+s}$$

S.1 $$T(j\omega) = \frac{e^{-j\omega}}{(1+j\omega)}$$

$$T(j0^+) = e^{-j0} = 1\angle 0°$$

$$T(j\infty^+) = \frac{e^{-j\infty^+}}{j\infty} = 0^+ e^{-j\left[\infty^+ + \frac{\pi}{2}\right]}$$

$$= 0^+ e^{-j\infty^+} = 0^+ e^{-j[360°]\times\infty}$$

Intersection with –ve Real axis

$$-\omega - \tan^{-1}\omega = -\pi$$

$$\tan(\pi - \omega) = -\tan\omega$$

$$\tan^{-1}\omega + \omega = \pi$$

$$\tan^{-1}\omega = (\pi - \omega)$$

$$\omega = -\tan\omega \rightarrow \tan\omega = -\omega \rightarrow \text{Infinite Roots}$$

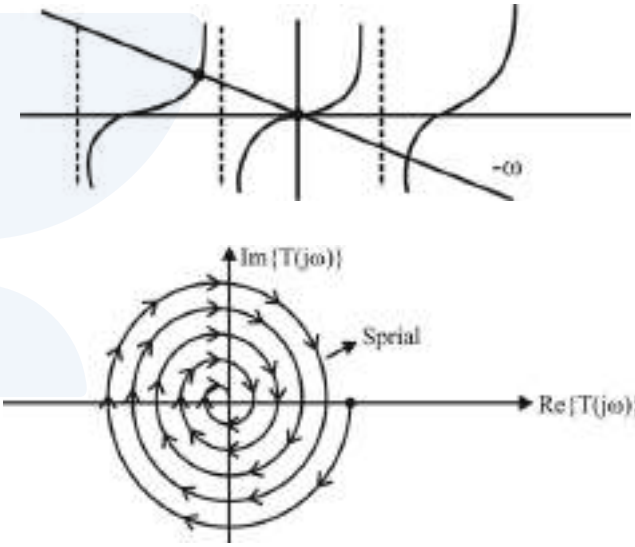


Fig. 11.28

Test your Understanding

1. TheNyquist plot for a system is shown below.

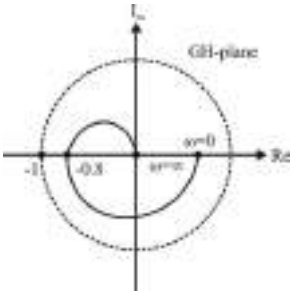


Fig. 11.29

Which of the following is correct for the system?

(A) $GM = \infty; PM = \infty$

(B) $GM > 1; PM = \infty$

(C) $GM > 1; PM = 0°$

(D) $GM = 0; PM = 0°$

Solution: (B)

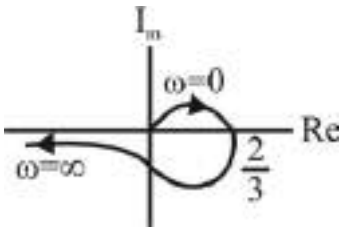
Common Data for Q 2, 3 and 4

A unity feedback system has open-loop transfer function,

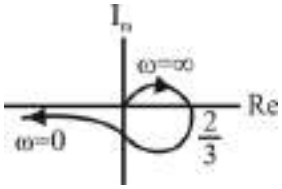
$$G(s)=\frac{1}{s(2s+1)(s+1)}$$

2. The polar plot for the system is

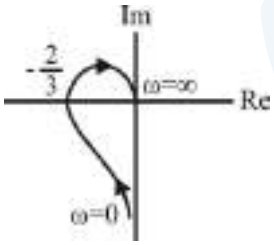
(A)



(B)



(C)



(D)

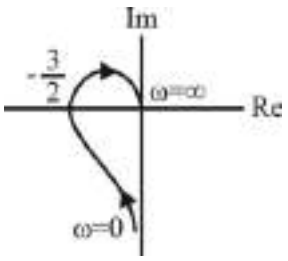


Fig. 11.30

Solution: (C)

3. The phase crossover and gain crossover frequencies are

(A) 1.414 rad/sec, 0.57 rad/sec

(B) 1.414 rad/sec, 1.38 rad/sec

(C) 0.707 rad/sec, 0.57 rad/sec

(D) 0.707 rad/sec, 1.38 rad/sec

Solution: (C)

4. The gain margin and phase margin are

(A) –3.52 dB, –168.5°

(B) –3.52 dB, 11.6°

(C) 3.52 dB, –168.5°

(D) 3.52 dB, 11.6°

Solution:(D)

Exercise Problems

1. In the feedback system shown below

$$G(s)=\frac{1}{(s+1)(s+2)(s+3)}$$

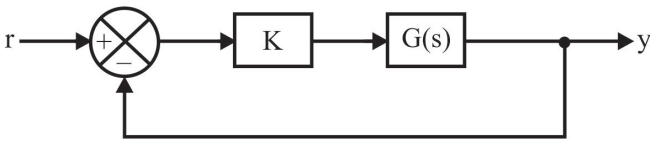


Fig. 11.31

The positive value of K for which the gain margin of the loop is exactly 0 dB and the phase margin of the loop is exactly 0 degree is _____.

Solution:60

2. The phase margin (in degree) of the system $G(s)=\frac{10}{s(s+10)}$ is _____

Solution: 84.35

3. The open-loop function of a unity-gain feedback control system is given by

$$G(s)=\frac{K}{(s+1)(s+2)}$$

The gain margin of the system in dB is given by

(A) 0

(B) 1

(C) 20

(D) ∞

Solution:(D)

Common data for questions 4 and 5

The open-loop transfer function of a unity feedback system is given by $G(s)=\frac{3e^{-2s}}{s(s+2)}$..

4. The gain and phase crossover frequencies in rad/sec are, respectively

(A) 0.632 and 1.26

(B) 0.632 and 0.485

(C) 0.485 and 0.632

(D) 1.26 and 0.632

Solution:(D)

5. Based on the above results, the gain and phase margins of the system will be

(A) –7.09 dB and 87.5°

(B) 7.09 dB and 87.5°

(C) 7.09 dB and –87.5°

(D) –7.09 dB and –87.5°

Solution:(D)

6. The phase margin (in degree) of a system having the loop transfer function is

$$G(s)H(s)=\frac{2\sqrt{3}}{s(s+1)}$$

(A) 45°

(B) –30°

(C) 60°

(D) 30°

Solution:(D)

Solutions

1. Solution: 60

a) Gain margin = 0 dB

b) Phase margin = 0°

Hence, system will be marginally stable,

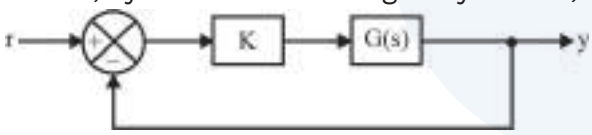


Fig. 11.32

$$G(s)=\frac{1}{(s+1)(s+2)(s+3)}$$

$$G'(s)=\frac{K}{(s+1)(s+2)(s+3)}$$

Method 1

We can calculate the value of K_{mar} by Routh Hurwitz criteria.

Characteristics equation is given by,

$$1+G'(s)=0$$

$$1+\frac{K}{(s+1)(s+2)(s+3)}=0$$

$$(s+1)(s+2)(s+3)+K=0$$

$$GM(dB)=-ve\ dB$$

$$s^3+5s^2+6s+s^2+5s+6+K=0$$

$$\leftrightarrow GM=1 \text{ or } GM=0dB$$

Routh tabulation:

$$\leftrightarrow GM<1 \text{ or } GM<0dB$$

For a marginally stable system,

$$\frac{66-6-K}{6}=0$$

$$60-K=0$$

K_{mar} = 60

Hence, the value of K_{mar} is 60.

2. Solution: 84.35

Given:

$$G(s)=\frac{10}{s(s+10)}$$

Put $s=j\omega$ in above equation,

$$G(j\omega)=\frac{10}{j\omega(j\omega+10)}$$

Magnitude of the open-loop transfer function $G(j\omega)$ is given by,

$$|G(j\omega)|=\frac{10}{\omega\sqrt{\omega^2+100}}$$

Phase angle of theopen-loop transfer function $G(j\omega)$ is given by,

$$\angle G(j\omega) = -90° - \tan^{-1}\frac{\omega}{10}$$

Magnitude at gain crossover frequency is given by,

$$|G(j\omega)H(j\omega)| = 1$$

$$1 = \frac{10}{\omega_{gc}\sqrt{\omega_{gc}^2 + 100}}$$

$$\leftrightarrow GM = 1 \text{ or } GM = 0$$

$$\omega_{gc}^2 + 100\omega_{gc}^2 - 100 = 0$$

Put $\omega_{gc}^2 = x$ in above equation,

$$x^2 + 100x - 100 = 0$$

$$x = 0.99, -100$$

Since ω_{gc} is a positive real frequency,

Therefore, $\omega_{gc}^2 = 0.99$

$$\omega_{gc} = 0.994\, rad/\sec$$

$$\angle G(j\omega_{gc}) = -90° - \tan^{-1}\left(\frac{0.99}{10}\right)$$

$$\angle G(j\omega_{gc}) = -95.65°$$

Phase margin is defined as,

$$PM = 180° + \angle G(j\omega_{gc})$$

$$PM = 180° - 95.65° = 84.35°$$

Hence, the phase margin is 84.35°.

3. Solution: (D)
Given:

$$G(s) = \frac{K}{(s+1)(s+2)}$$

Put s = $j\omega$ in above equation,

$$G(j\omega) = \frac{K}{(j\omega+1)(j\omega+2)}$$

Magnitude of the open-loop transfer function $G(j\omega)$ is given by,

$$|G(j\omega)| = \frac{K}{\sqrt{(\omega^2+1)(\omega^2+4)}}$$

Phase angle of the open-loop transfer function $G(j\omega)$ is given by,

$$\angle G(j\omega) = -\tan^{-1}\omega - \tan^{-1}\frac{\omega}{2}$$

ω	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
0	$\frac{K}{2}$	0°
∞	0	–180°

Polar plot for the open-loop transfer function can be drawn as shown in the figure,

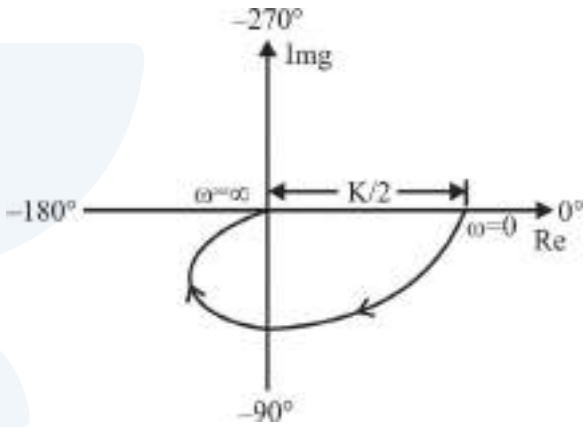


Fig. 11.33

From table,

$$\omega_{pc} = \infty$$

$$|G(j\omega_{pc}| = 0$$

Gain margin is given by,

$$GM = \frac{1}{|G(j\omega_{pc})|} = \frac{1}{0} = \infty$$

In dB, GM = 20 log (∞)

$$GM = 20\log\left(\frac{1}{0}\right) = (20\log 1 - 20\log 0)$$

$$GM = 0 - (-\infty) = \infty\, dB$$

Hence, the correct option is (D).

4. Solution: (D)

Given:

The open-loop transfer function of a unity feedback system is,

ω_{PC}

Put s = $j\omega$ in above equation,

$GM = -\infty\, dB$

Calculation of ω_{gc}:

Magnitude at gain crossover frequency is given by,

$$\left|G\left(j\omega_{gc}\right)H\left(j\omega_{gc}\right)\right| = 1$$

$$\left|G\left(j\omega_{gc}\right)\right| = \left|\frac{3e^{-2j\omega_{gc}}}{j\omega_{gc}\left(j\omega_{gc}+2\right)}\right| = 1$$

$$1 = \frac{3}{\omega_{gc}\sqrt{\left(\omega_{gc}^2+4\right)}}$$

$$\omega_{gc}^2\left(\omega_{gc}^2+4\right) = 9$$

$$\omega_{gc}^4 + 4\omega_{gc}^2 - 9 = 0$$

Let $\omega_{gc}^2 = x$,

$$x^2 + 4x - 9 = 0$$

$$x = 1.6, -5.6$$

∞

Since ω_{gc} is a positive real number.

Therefore, $G(s)H(s) = \frac{}{(s+1)(s+2)}$

Hence, only option (d) is satisfying.

Calculation of ω_{pc}:

Phase angle at phase crossover frequency is given by,

$$\angle G\left(j\omega_{pc}\right)H\left(j\omega_{pc}\right) = -180°$$

$$\angle G\left(j\omega_{pc}\right) = \left[-2\omega_{pc} - \frac{\pi}{2} - \tan^{-1}\left(\frac{\omega_{pc}}{2}\right)\right] rad$$

$$-\pi = \left[-2\omega_{pc} - \frac{\pi}{2} - \tan^{-1}\left(\frac{\omega_{pc}}{2}\right)\right] rad$$

$$\frac{\pi}{2} = \left[2\omega_{pc} + \tan^{-1}\left(\frac{\omega_{pc}}{2}\right)\right] rad$$

$$2\omega_{pc} + \tan^{-1}\left(\frac{\omega_{pc}}{2}\right) = \frac{\pi}{2}$$

[By using calculator]

$\omega_{pc} = 0.632\ rad/\sec$

Hence, the correct option is (D).

5. Solution: (D)

Method 1

Phase angle at gain crossover frequency is given by,

$$\angle G\left(j\omega_{pc}\right)H\left(j\omega_{pc}\right) = -180°$$

$$\angle G\left(j\omega_{gc}\right) = \left[-2\omega_{gc} - \frac{\pi}{2} - \tan^{-1}\left(\frac{\omega_{gc}}{2}\right)\right] rad$$

$$\angle G\left(j\omega_{gc} = j1.26\right) = -2\times 1.26 - \frac{\pi}{2} - 0.562$$

$$\angle G\left(j\omega_{gc}\right) = -4.6527\, rad$$

$$s = j\omega \rightarrow T\left(j\omega\right) \rightarrow Ra$$

Phase margin is defined as,

$$PM = 180° + \angle G\left(j\omega_{gc}\right)$$

$$PM = 180° - 266.585°$$

PM = – 86.585°

Gain margin is given by,

$$GM = \frac{1}{\left|G\left(j\omega_{pc}\right)H\left(j\omega_{pc}\right)\right|}$$

$$\left|G\left(j\omega_{pc}\right)H\left(j\omega_{pc}\right)\right| = \frac{3}{\omega_{pc}\sqrt{\omega_{pc}^2+4}}$$

$$\left|G\left(j\omega_{pc}\right)H\left(j\omega_{pc}\right)\right| = \frac{3}{\sqrt{\left(0.632\right)^2\left(0.632^2+4\right)}}$$

$$\left|G\left(j\omega_{pc}\right)H\left(j\omega_{pc}\right)\right| = 2.263$$

GM = –7.094 dB

Hence, the correct option is (D).

6. Solution: (D)

Given:

$$G(s)H(s)=\frac{2\sqrt{3}}{s(s+1)} \quad \text{...(i)}$$

Put s = $j\omega$ in equation (i),

$$G(j\omega)H(j\omega)=\frac{2\sqrt{3}}{j\omega(j\omega+1)}$$

Magnitude of the open-loop transfer function $G(j\omega)H(j\omega)$ is given by,

$$\left|G(j\omega)H(j\omega)\right|=\frac{2\sqrt{3}}{\omega\sqrt{\omega^2+1}} \quad \text{...(ii)}$$

Phase angle of the open-loop transfer function $G(j\omega)H(j\omega)$ is given by,

$$\angle G(j\omega)H(j\omega)=-90°-\tan^{-1}\omega \quad \text{...(iii)}$$

Calculation of phase margin:

Method 1:

ω	$\left\|G(j\omega)H(j\omega)\right\|$	$\angle G(j\omega)H(j\omega)$
0	∞	–90°
1	$\sqrt{6}$	–135°
$\sqrt{3}$	1	160°
10	$\sqrt{3}/50$	–174.28°
∞	0	–180°

The frequency at which gain $\left|G(j\omega)H(j\omega)\right|$ will be unity is known as gain crossover frequency.

From table, $\omega_{gc}=\sqrt{3}$

Phase angle at gain crossover frequency is,

$$\angle G(j\omega_{gc})H(j\omega_{gc})=-150^0$$

Phase margin is given by,

$$PM=180^0+\angle G(j\omega_{gc})H(j\omega_{gc})$$

$$PM=180^0-150^0=30^0$$

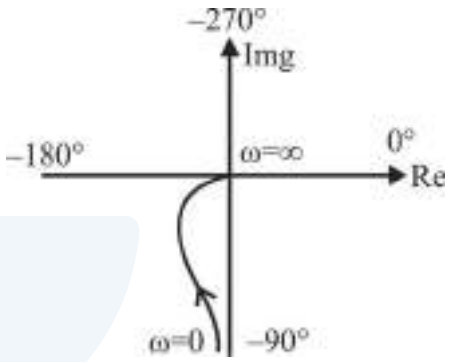


Fig. 11.34

Hence, the correct option is (D).